Praxia Update

Daily orthopaedic literature digest · Friday, 04 September 2026 · 546 new articles across 10 journals

AJSM

American Journal of Sports Medicine

Volume 54, Issue 11 · September 2026 · 32 new articles

Deeper Dive Into Methodology: Kaplan-Meier Curves.

Carey JL

PubMed · DOI

Adductor Canal Block and Local Anesthetic Versus Local Anesthetic Alone in ACL Reconstruction: A Double-Blind Randomized Controlled Trial.

Ojaghi R, Locke E, Elmi P, Pickell M

BACKGROUND: Effective postoperative analgesia is crucial for early recovery after anterior cruciate ligament reconstruction (ACLR). Local infiltration analgesia (LIA) and adductor canal block (ACB) are common regional techniques, but their combined efficacy remains unclear.

PURPOSE: To compare the effectiveness of LIA alone versus LIA combined with ACB in patients undergoing ACLR, with primary outcomes including postoperative opioid consumption and quadriceps function.

STUDY DESIGN: Randomized controlled trial; Level of evidence, 1.

METHODS: A double-blind randomized controlled trial enrolled 100 patients undergoing ACLR under general anesthesia. Patients were randomized into 2 groups: LIA + sham (saline injection) (n = 50) and LIA + ACB (n = 50). The primary outcome was postoperative opioid consumption in the first 24 hours. Secondary outcomes included visual analog scale (VAS) pain score, quadriceps function assessed by straight leg raise (SLR) at 3 hours, Quality of Recovery-15 (QoR-15) score, and Knee Injury and Osteoarthritis Outcome Score (KOOS) at 1 week. Statistical analysis was performed using t tests and chi-square tests with a P value <.05 considered significant.

RESULTS: There was no significant difference in 24-hour opioid consumption between the LIA + ACB and LIA-only groups (P = .109). Similarly, VAS pain scores at 24 hours postoperatively showed no significa […]

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Outcomes of Suture Tape-Augmented ACL Reconstruction in Patients With High Internal Rotational Tibial Subluxation: A Minimum 3-Year Cohort Study.

Wang Y, Zhang Y, Shi W, Wang J, Wang H, Liu Y, Yuan F, Ao Y et al.

BACKGROUND: Anterior cruciate ligament reconstruction (ACLR) using hamstring tendon (HT) autografts faces persistent challenges of graft failure, especially in patients with high internal rotational tibial subluxation (IRTS).

HYPOTHESIS: Patients with high IRTS who undergo HT autograft ACLR with suture tape augmentation (STA) will have lower graft failure rates and superior clinical outcomes as compared with those without STA.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: This retrospective cohort study included patients with high IRTS-defined as lateral minus medial anterior tibial subluxation >5.8 mm based on prior studies-who underwent primary ACLR using HT autografts with STA and had a minimum follow-up of 3 years. Propensity score matching (1:1) was performed to identify a control group of patients with similarly high IRTS who underwent HT ACLR without STA. Postoperative outcomes were assessed by the International Knee Documentation Committee score, Lysholm score, and Tegner activity scale, as well as by return-to-sport status and graft failure. Clinically meaningful improvements were evaluated by the minimal clinically important difference, Patient Acceptable Symptom State, and substantial clinical benefit.

RESULTS: This study included 62 patients with high IRTS: 31 with STA and 31 matched controls. The mean IRTS was 6.6 mm (range, 5.9-8.9) in the STA gro […]

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Primary ACL Repair and Reconstruction in Isolated ACL Ruptures: Forgotten Joint Scores and the Association Between Residual Laxity and Joint Awareness.

Altun O, Ergi■i Y, Alt■nta■ M, Özdemir E, Da■ar U, Bozkurt M

BACKGROUND: Primary anterior cruciate ligament (ACL) repair has recently reemerged as a treatment option for carefully selected proximal ACL tears. However, evidence regarding patient-reported joint awareness and the relationship between postoperative laxity and joint awareness remains limited.

PURPOSE: To compare joint awareness, clinical outcomes, and postoperative laxity between primary ACL repair and hamstring tendon autograft reconstruction in carefully selected patients with isolated ACL rupture, and to explore the association between residual laxity and postoperative joint awareness.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: This retrospective cohort study included 85 patients with isolated ACL rupture treated with either primary ACL repair (n = 26) or hamstring autograft reconstruction (n = 59), with a minimum follow-up of 24 months. Clinical outcomes, including visual analog scale score, Lysholm score, International Knee Documentation Committee (IKDC) score, Tegner activity scale score, postoperative knee laxity, return to sport, and rerupture rates, were evaluated. The authors also used the Forgotten Joint Score-12 (FJS-12), a measure of joint awareness, to evaluate patients. A higher score reflects lower joint awareness, suggesting function more similar to a native knee. Multivariable regression analyses were performed to evaluate variables associ […]

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Quadriceps Tendon Autografts Are Associated With Increased Short-term Reoperation Rates for Extension Deficit or Pain and Inferior Patient-Reported Outcomes Compared With Hamstring Tendon Autografts: An Analysis of 5653 Cases of Primary Anterior Cruciate Ligament Reconstruction.

Kekki C, Cristiani R, Stålman A, von Essen C

BACKGROUND: Anterior cruciate ligament reconstruction (ACLR) rates are rising, and quadriceps tendon (QT) autografts have gained popularity in recent years. Although QT autografts show objective and subjective outcomes comparable to hamstring tendon (HT) and bone-patellar tendon-bone (BPTB) autografts, nonrevision reoperations across graft types and their impact on patient-reported outcome measure (PROM) scores remain poorly studied.

PURPOSE: To compare nonrevision reoperation rates and PROM scores between primary ACLR using HT, BPTB, and QT autografts at 2 years' follow-up.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: In this retrospective cohort study, 5653 primary ACLR procedures (2015-2022) were categorized by autograft type. Reoperations for extension deficit or pain, meniscal injuries, or cartilage injuries within 2 years were identified through medical records. Revision ACLR and graft failure were not included. Patient-reported outcomes were assessed preoperatively and at 2 years postoperatively using the Knee Injury and Osteoarthritis Outcome Score (KOOS). The minimal important change, patient acceptable symptom state, and treatment failure were evaluated using the KOOS4. Subgroup analysis examined associations between graft diameter, sex, and reoperations for extension deficit or pain.

RESULTS: The QT group had a higher overall reoperation rate (19.2% […]

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Sociodemographic Predictors of Postpresentation Delay to Pediatric ACL Reconstruction and Associated Intra-articular Injury Severity.

Hamad CD, Wu S, Joachim K, Chung E, Liu T, Silva M, Bowen R, Baghdadi S

BACKGROUND: Delayed anterior cruciate ligament reconstruction (ACLR) in pediatric patients is associated with progressive intra-articular injury. Although disparities in access to orthopaedic care have been reported, it remains unclear whether structural determinants of health influence postpresentation surgical delay and downstream functional recovery.

PURPOSE: To determine whether sociodemographic and structural determinants are associated with prolonged postpresentation delay to pediatric ACLR and with intra-articular injury severity and postoperative functional outcomes.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: The records of 443 pediatric patients who underwent ACLR (2013-2025) at an urban academic referral center with a pediatric orthopaedic urgent care were reviewed. Exposures included race/ethnicity, insurance type, and neighborhood deprivation (Childhood Opportunity Index [COI]). Delay was defined as time from first orthopaedic presentation to surgery, with prolonged delay defined as >120 days. Outcomes included high-grade cartilage injury (Outerbridge grade 3 or 4), meniscal tear presence, failure to return to sport (≥365 days of follow-up), hop test symmetry, isometric strength symmetry, and graft failure. Multivariable regression models were adjusted for race/ethnicity, age, sex, insurance, and national COI.

RESULTS: The median time from injury […]

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Association of Early Knee Extension Range of Motion Deficits With Cartilage T2 Relaxation Following Anterior Cruciate Ligament Reconstruction.

Allison A, Harrington J, Weaver B, Manzer M, Sajja BR, Tao MA, Wellsandt E

BACKGROUND: Anterior cruciate ligament (ACL) injury significantly increases the risk for developing knee osteoarthritis (OA), yet the early contributors to cartilage degeneration remain poorly understood. While range of motion (ROM) deficits after ACL reconstruction (ACLR) are associated with long-term development of radiographic OA, the association between early ROM recovery and cartilage composition has not been established.

HYPOTHESIS: Knee extension ROM deficits at 2 months after ACLR would be associated with worsening tibiofemoral and patellofemoral cartilage T2 relaxation times at 6 months.

STUDY DESIGN: Cohort study; Level of evidence, 2.

METHODS: A total of 30 participants (15-35 years) were enrolled within 1 month of ACL injury and before ACLR. At 2 months post-ACLR, active knee extension ROM was measured in the supine position using a goniometer. An extension deficit was considered >3° less extension in the injured knee compared to the uninjured side. Quantitative magnetic resonance imaging-based T2 relaxometry of the injured knee was performed preoperatively and at 6 months post-ACLR to measure percent change in mean T2 relaxation times in predefined tibial and femoral cartilage regions. Independent t tests were used to compare changes in T2 values between participants with and without an extension deficit.

RESULTS: The mean age of the participants was 18.8 ± 3.9 […]

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Changes in ACL T2* Metrics Over the Course of the Female Menstrual Cycle: A New Biomarker for the ACL Injury Risk?

Argentieri E, Hamilton TW, Koff MF, Koch KM, Potter HG

BACKGROUND: Sex-based disparities in the anterior cruciate ligament (ACL) injury risk may be partly explained by cyclic variations in sex hormones that drive systemic shifts in tissue osmoregulation. Ultrashort echo time (UTE) T2* magnetic resonance imaging provides a noninvasive, quantitative means of evaluating the water content and collagen orientation in tissue.

HYPOTHESIS: Eumenorrheic female participants will exhibit significant ACL T2* changes across the cycle, while anovulatory control participants will demonstrate no significant changes over time.

STUDY DESIGN: Cohort study; Level of evidence, 2.

METHODS: A total of 25 women with no prior knee injuries were enrolled: 10 premenopausal, eumenorrheic participants and 15 anovulatory control participants (6 postmenopausal and 9 premenopausal using oral contraceptives). Bilateral knee magnetic resonance imaging was performed at 4 time points evenly spaced over 1 month, with the first visit within 24 hours of menses onset. Ovulation status was confirmed using commercially available ovulation predictor kits. Three-dimensional UTE sequences (11 echoes; 0.03-25 ms) were acquired to evaluate bicomponent ACL T2* metrics. Linear mixed-effects models assessed temporal differences in long and short T2* values, and the Cohen d quantified effect size.

RESULTS: Eumenorrheic participants demonstrated significantly shorter preovulator […]

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Longitudinal Quantitative Mapping of Meniscal Morphological and Signal Intensity Differences After ACL Surgery.

Barnes DA, Murray CJ, Movahhedi M, Beveridge JE, Kiapour AM, Murray MM, Fleming BC

BACKGROUND: Meniscal tears are frequently associated with anterior cruciate ligament (ACL) injuries. Evaluating longitudinal changes in meniscal integrity using quantitative magnetic resonance imaging after ACL surgery may provide valuable insight into joint recovery and subsequent reinjury risk.

HYPOTHESES: (1) The cross-sectional area (CSA) and normalized signal intensity (SI) profiles of the lateral and medial menisci for the surgical knees would be greater than those of the contralateral uninjured knees observed at 6, 12, and 24 months after surgery. (2) The CSA and Constructive Interference in Steady State (CISS) SI profiles obtained at 24 months would be closer to those of the contralateral limb when compared with those of the 6- and 12-month time points.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: Deidentified postsurgical CISS magnetic resonance imaging scans of patients (n = 82; median age 17.6 years; 42 females and 40 males; 57 ACL restoration and 25 ACL reconstruction) were analyzed. CSA and normalized CISS SI values were extracted from the lateral and medial menisci to create morphological profiles. Statistical parametric mapping was used to compare the CSA and normalized CISS SI profiles between the surgical and contralateral limbs to identify specific regions that were different among the menisci at 6, 12, and 24 months.

RESULTS: The CSA and nor […]

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Meniscal Allograft Transplantation Failure Is Associated With Female Sex: A 15-Year Survival Analysis.

Park YL, Wackerle AM, Jiang C, Vieider RP, Winkler PW, Irrgang JJ, Musahl V, Hughes JD

BACKGROUND: While short- and midterm outcomes after meniscal allograft transplantation (MAT) are well documented, there is a paucity of data regarding long-term graft survival.

PURPOSE: To assess mid- to long-term graft survivorship and patient-reported outcomes (PROs) after MAT.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: All patients who underwent MAT with bone block or soft tissue fixation between 1993 and 2020 at a single institution were identified. Individuals with <5 years of follow-up and untreated grade 4 cartilage lesions were excluded. Primary outcomes included graft survivorship and long-term PROs including International Knee Documentation Committee Subjective Knee Form (IKDC SKF) score, Knee injury and Osteoarthritis Outcome Score, and Marx Activity Scale score. Graft failure was defined as the need for meniscal allograft resection or repair, revision MAT, or conversion to total knee arthroplasty. Clinical failure was defined as an IKDC SKF score <56 at the final follow-up. Statistical analyses included chi-square/Fisher exact tests for categorical variables, the Mann Whitney U or t test for continuous variables, and Kaplan-Meier survival analysis with endpoints as graft failure. Statistical significance was defined as a P value <.05.

RESULTS: This study included 160 patients (57% of the screened population; 46% female; median age at surgery, 28 […]

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Length Changes of the Lateral Knee: Fibular Collateral Ligament, Anterolateral Ligament, and Iliotibial Band Fibers.

Augustin EJ, Oppenheim ZR, Vega J, Gomez Verdejo F, Casanova FJ, Escribano G, Koluman AC, Bi AS et al.

BACKGROUND: The lateral knee is stabilized by the fibular collateral ligament (FCL), anterolateral ligament (ALL), and iliotibial band (ITB) fibers. Understanding their flexion-dependent behavior is critical for guiding graft fixation during reconstruction.

HYPOTHESIS/PURPOSE: The hypothesis was that the FCL and ALL would shorten with flexion, while ITB fibers would show more minor, transient changes. The purpose was to characterize native length changes of these structures in cadaveric knees.

STUDY DESIGN: Descriptive laboratory study.

METHODS: Ten fresh-frozen knees were dissected with preservation of lateral soft tissue attachments. Bony landmarks were marked for the attachment sites: FCL (n = 10), ALL (n = 7), and ITB (n = 10 distal Kaplan fibers, n = 9 proximal). A 3-dimensional digitizer recorded coordinates at 0°, 15°, 30°, 45°, 60°, and 90° of flexion. Attachment distances were calculated, and paired t tests or Wilcoxon tests (based on Shapiro-Wilk normality) compared 0° with each flexion angle.

RESULTS: Distances between the FCL and ALL attachment sites progressively decreased with flexion, with mean reductions of -7.59 and -6.44 mm, respectively, from 0° to 90° (both P≤ .05). Distances from the Gerdy tubercle to the distal and proximal Kaplan attachments of the ITB increased significantly by 2 to 6 mm between 15° and 60° before returning toward baseline at 90°: di […]

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Quadriceps and Hamstring Muscle Strength Recovery After a Multiligament Knee Injury: A Long-term Follow-up.

Trøan I, Holm I, Moatshe G, LaPrade RF, Engebretsen L, Bere T

BACKGROUND: Multiligament knee injuries (MLKIs) are serious and heterogeneous injuries that often lead to long-term functional limitations. There is limited literature regarding objective muscle strength and functional recovery after surgical treatment of MLKIs.

PURPOSE: To investigate long-term quadriceps and hamstring muscle strength and functional recovery after surgical treatment of MLKI and to compare postoperative outcomes between single cruciate MLKI and bicruciate MLKIs.

STUDY DESIGN: Cross-sectional study; Level of evidence, 3.

METHODS: Patients surgically treated for an MLKI at a single level 1 trauma center between January 2013 and December 2020 were included. Isokinetic quadriceps and hamstring muscle strength were assessed using a Biodex dynamometer, and physical function was measured by single-leg hop, triple hop, crossover hop, and 6-m timed hop tests, with the limb symmetry index (LSI) calculated.

RESULTS: Of 124 eligible patients, 90 (72%) participated at a mean follow-up time of 7 years (SD, 2.1) since surgery to evaluation. The median age at the time of testing was 49 years (range, 21-72) and 59% were male. In total, 54% of the patients had not regained full quadriceps strength (LSI <90%), and 46% had persistent hamstring deficits for peak torque/body weight (PT/BW) at 60 deg/s. Across the entire cohort, there was a significant difference between the unin […]

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Selective Epiphysiodesis of the Anterior Distal Femoral Physis Alters Trochlear Morphology in a Rabbit Model.

Wills DJ, Dan MJ, Oliver R, Caldwell B, Cance N, Dejour D, Walsh WR

BACKGROUND: Trochlear dysplasia is a leading contributor to patellofemoral instability, but the developmental role of the anterior distal femoral physis in shaping trochlear morphology remains poorly understood.

HYPOTHESIS: Selective closure of different portions of the anterior distal femoral physis would alter trochlear morphology in a predictable manner based on the region of growth arrest.

STUDY DESIGN: Controlled laboratory study.

METHODS: A total of 12 New Zealand White rabbits underwent unilateral selective epiphysiodesis of the anterior distal femoral physis at either the medial, central, or lateral region (n = 4 per group). Contralateral limbs served as internal controls. At 6 months postoperatively, gross, micro-computed tomography (µCT), and histological assessments were performed. µCT parameters including sulcus angle, trochlear depth, trochlear inclination, and facet symmetry were analyzed. A caliper was used to measure trochlear width as well as trochlear ridge height and length. The paired t test compared treated versus control limbs within each group.

RESULTS: Selective epiphysiodesis altered trochlear morphology in a region-specific pattern. Lateral epiphysiodesis resulted in a decreased proximal sulcus angle (P = .005) and an increased lateral trochlear depth proximally (P = .021), centrally (P = .031), and distally (P = .036), as measured on axial µCT. Ce […]

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Platelet-Derived Growth Factor Synergistic Enhancement of Bone Marrow Stimulation for Enhanced Labral Repair and Functional Recovery in a Rat Bankart Model.

Mendiaz F, Vaish B, Rifat S, Gray N, Hoang L, Nguyen T, Borrelli J, Tang L

BACKGROUND: Bone marrow stimulation (BMS) is used to treat intra-articular defects, such as articular cartilage defects or to facilitate healing of soft tissue repair; however, BMS yields inconsistent clinical outcomes, potentially due to the inefficient delivery of endogenous progenitor cells.

HYPOTHESIS: Combining BMS with localized placement of platelet-derived growth factor (PDGF) would enhance in situ progenitor cell responses, leading to enhanced labral tissue repair and functional recovery in a rat model.

STUDY DESIGN: Controlled laboratory study.

METHOD: A Bankart glenoid labral tear was created in the right shoulder of 50 adult rats. The animals were randomly divided into 5 groups: (1) healthy (baseline), (2) suture only (control), (3) suture + PDGF, (4) suture + BMS, and (5) suture + PDGF + BMS. Healing was assessed at 1.5 (n = 3), 3, and 8 weeks (n = 4). Baseline groups were included at 3 (n = 4) and 8 weeks (n = 5). At each timepoint, labral healing was evaluated by quantifying progenitor cell recruitment, tissue differentiation, and general morphology. Functional recovery was assessed weekly via gait analysis.

RESULTS: BMS alone failed to elicit cell recruitment to the injured labral region. However, the combined BMS + PDGF treatment resulted in a 7-fold increase in progenitor cell accumulation at the tear site compared with BMS alone (27.46 ± 11.51 vs 3.5 ± 3. […]

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Role of Hypoxia-Inducible Factor-1α in Regulating Muscle Degeneration After Rotator Cuff Tears.

Zhang H, Lee A, Liu M, Diaz A, Zhang Y, Kim HT, Feeley BT, Liu X

BACKGROUND: Secondary muscle degeneration after a rotator cuff tear (RCT) critically affects clinical outcomes. Vascular compromise after a tendon injury creates a complex microenvironment that may be associated with the degeneration of rotator cuff muscle. The role of hypoxia-inducible factor-1α (HIF-1α), a master regulator of cellular stress responses to hypoxia, in modulating muscle abnormalities after an RCT remains undefined.

PURPOSE: To define the role of HIF-1α in stem cell differentiation and muscle degeneration after an RCT in a murine model.

STUDY DESIGN: Controlled laboratory study.

METHODS: A supraspinatus tendon transection and suprascapular nerve transection (TTDN) model was established in C57BL/6J, platelet-derived growth factor receptor α (PDGFRα)-green fluorescent protein (GFP) reporter, and inducible cell-specific HIF-1α knockout mice. Vascularity and HIF-1α colocalization with fibroadipogenic progenitor (FAP) cells and satellite cells were analyzed. Fibrosis, fatty infiltration, and myofiber cross-sectional area were assessed. In vitro, HIF-1α was modulated in isolated FAP cells via CRISPR-Cas9 or prolyl hydroxylase domain inhibitors to evaluate FAP cell differentiation.

RESULTS: TTDN induced significant capillary density reduction (CD31+) at 1, 2, and 6 weeks after an injury. Global HIF-1α expression decreased after TTDN compared to the sham side (1 week […]

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Preoperative Proximal Migration of the Radial Head as an Independent Predictor of Suboptimal Outcomes After Osteochondral Autograft Transplantation for Capitellar Osteochondritis Dissecans: A Retrospective Cohort Study.

Kamijo H, Takahashi N, Matsuki K, Ishii D, Akiyama T, Sugaya H, Kawashima I

BACKGROUND: Osteochondral autograft transplantation (OAT) is widely performed for capitellar osteochondritis dissecans (OCD). However, preoperative predictors of suboptimal postoperative outcomes remain unclear.

PURPOSE/HYPOTHESIS: The authors aimed to evaluate clinical outcomes after OAT for capitellar OCD and identify preoperative risk factors associated with suboptimal outcomes. They hypothesized that radiographic indicators of disease severity, including preoperative proximal migration of the radial head, lesion size, and lateral wall disruption, would be associated with suboptimal postoperative clinical outcomes.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: The records of adolescent athletes who underwent OAT for capitellar OCD with a minimum 2-year follow-up were retrospectively reviewed. Clinical outcomes included elbow range of motion (ROM) and Timmerman-Andrews (T-A) score. A suboptimal outcome was defined as a postoperative T-A score <160. Preoperative radiographs were used to measure proximal migration of the radial head relative to the coronoid process, hypertrophy of the radial head, OCD lesion area, and a 5-grade lateral wall disruption classification; measurement reliability was assessed. Multivariate logistic regression analysis was performed to identify independent predictors of a suboptimal outcome, and receiver operating characteristic (ROC) […]

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Higher Rates of PASS and SCB After Arthroscopic Subspine Decompression Are Associated With a Positive Diagnostic AIIS Injection: A Propensity Score-Matched Cohort Study.

Duan YP, Yang F, Liu RG, Sun H, Huang H, Xu Y, Ju XD, Wang JQ

BACKGROUND: Hip arthroscopy effectively treats femoroacetabular impingement syndrome (FAIS), but persistent pain may be related to concomitant extra-articular pathology such as subspine impingement syndrome (SSI). Standard diagnosis of SSI often relies on 3-dimensional computed tomography (3D-CT) morphology (Hetsroni type II/III), although this morphology is common in individuals who are asymptomatic and correlates poorly with symptoms.

PURPOSE: To compare minimum 2-year clinical outcomes after arthroscopic subspine decompression in patients with concurrent FAIS and type II/III anterior inferior iliac spine (AIIS) morphology, stratified by diagnostic method: 3D-CT morphology alone versus 3D-CT morphology plus a positive ultrasound-guided diagnostic injection.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: This study included patients aged 18 to 55 years with type II/III AIIS morphology who underwent primary hip arthroscopy for FAIS and SSI between January 2021 and November 2023 and had minimum 2-year follow-up. Patients diagnosed by CT morphology alone (CT classification group) were propensity score matched 1:1 to patients with a positive ultrasound-guided AIIS injection (injection group), with 57 patients per group. Matching variables were age, sex, body mass index, lateral center-edge angle, alpha angle, Tönnis grade, and Beighton score. All patients underwent […]

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Arthroscopic Correction of Pincer-Type FAI in the Presence of Acetabular Retroversion Results in Excellent Patient-Reported Outcomes With Low Risk of Reoperation or Conversion to Arthroplasty at Minimum 5 Years.

Filan D, Mullins K, Carton P

BACKGROUND: Acetabular retroversion is a distinct morphologic variation that may result in pincer-type femoroacetabular impingement (FAI). The role of arthroscopic management within this cohort is not fully understood.

PURPOSE: To assess patient-reported outcome measures (PROMs) and survivorship at 5-year follow-up in a series of cases undergoing arthroscopic correction of pincer-type FAI with acetabular retroversion.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: A single-center, prospective hip preservation registry was reviewed for all cases undergoing primary hip arthroscopy for symptomatic FAI between January 2014 and July 2020, with documented radiographic assessment of acetabular retroversion (ie, presence or absence of crossover sign, ischial spine sign, and/or posterior wall sign on a standing, standardized anteroposterior radiograph). Exclusion criteria were Tönnis grade >1, concomitant pathologies (protrusio, Perthes disease, avascular necrosis), excessive pelvic rotation or tilt on radiographs, and/or no crossover sign. Cases were assigned to 1 of 2 groups: moderate-global retroversion group (study group: crossover sign plus either or both ischial spine sign and posterior wall sign) or control group (crossover sign only). Case-control (1:1) fuzzy matching was performed based on age (±2 years), sex, and Tönnis grade (exact). Clinical outcome evaluation […]

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Artificial Intelligence Cannot Replace Peer Reviewers but May Help Editors Triage: A Comparative Analysis of a Large Language Model and Human Reviewer Recommendations at the American Journal of Sports Medicine.

Patel R, Shultz C, Ollivier M, Wascher DC

BACKGROUND: The peer review system faces increasing strain from rising manuscript volumes, reviewer fatigue, and well-documented interreviewer disagreement. Large language models (LLMs) have shown potential to support the peer review process, but their ability to replicate editorial decisions at high-impact medical journals and their utility as manuscript screening tools remain unknown.

PURPOSE: To compare the agreement between an LLM and the final editorial decision on manuscripts submitted to the American Journal of Sports Medicine and to evaluate the potential of LLMs as a manuscript screening tool.

STUDY DESIGN: Cross-sectional agreement study.

METHODS: Fifty-four manuscripts randomly selected from submissions to the American Journal of Sports Medicine (September 2024-October 2024) were reviewed by a locally deployed LLM (Ministral 3 14B; Mistral AI) using a standardized prompt. The artificial intelligence (AI) produced a categorical recommendation (reject, cascade, revision, or accept) and a numerical score (0-100) for each manuscript. Agreement with the final editorial decision was assessed by Cohen kappa (4-category model) for pooled human reviewers (n = 139 reviews) and the AI (n = 54). Screening performance was evaluated by positive predictive value (PPV), sensitivity, and specificity.

RESULTS: Pooled human reviewers demonstrated fair agreement with the final decis […]

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Prevalence of Psychiatric Diagnoses and Preoperative Opioid Use Among Patients Undergoing Hip Arthroscopy: A Systematic Review and Meta-analysis.

Villegas Meza AD, Nocek M, Morales Lugo B, Mitchell BC, Philippon MJ

BACKGROUND: Psychiatric morbidity (anxiety, depression, prior opioid use) is variably reported among patients undergoing hip arthroscopy, with prevalence estimates influenced by heterogeneous case definitions and ascertainment methods. Clarifying true prevalence and method-related differences is needed to inform preoperative screening and comparative research.

PURPOSE: To quantify the pooled prevalence of psychiatric diagnoses and prior preoperative opioid use among adults undergoing hip arthroscopy and to evaluate how prevalence varies by ascertainment method.

STUDY DESIGN: Systematic review and meta-analysis; Level of evidence, 3.

METHODS: The authors performed a PRISMA-compliant search of PubMed, Embase, PsycINFO, and Web of Science from inception to June 19, 2025. Observational studies reporting extractable prevalence of any psychiatric diagnosis, anxiety, depression, or prior opioid use among adult hip arthroscopy cohorts were eligible. Two reviewers independently screened articles, extracted data, and assessed risk of bias (Joanna Briggs Institute checklist). Pooled prevalences were estimated using random effects meta-analysis on logit-transformed proportions with continuity corrections; heterogeneity was summarized with $\tau2$ and I2. Prespecified subgroup and meta-regression examined the ascertainment method as a moderator.

RESULTS: Thirty-eight studies met inclusion cr […]

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Concomitant Intra-articular Injuries in 37,423 Primary and Revision ACL Reconstructions: A Systematic Review and Meta-analysis of Comparative Studies.

Vega TF, Thamrongskulsiri N, Brinkman JC, Pérez Lloveras GO, Casanova F, Cerezal Á, San Roman AV, Borace AF et al.

BACKGROUND: Intra-articular lesions of menisci and cartilage are common in anterior cruciate ligament reconstruction (ACLR) and may differ between primary and revision procedures. Clarifying their prevalence, morphology, and risk factors is important for prognosis and surgical planning.

PURPOSE: To perform a systematic review and meta-analysis comparing the prevalence and risk factors for intra-articular injuries in primary versus revision ACLR.

STUDY DESIGN: Systematic review and meta-analysis; Level of evidence, 4.

METHODS: Following PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines, the PubMed, MEDLINE, and Embase databases were searched, supplemented by citation review and author contact. Comparative studies reporting intraoperative meniscal and/or chondral findings in both primary and revision ACLRs were included. Pooled odds ratios with 95% confidence intervals were calculated using random- or fixed-effects models.

RESULTS: Nineteen studies consisting of 37,423 ACLRs were included. Revision ACLR was associated with higher odds of meniscal injury (OR, 1.64; 95% CI, 1.12-2.41). Meniscal treatment did not differ significantly between groups. Chondral lesions were >3-fold more prevalent in revisions (OR, 3.53; 95% CI, 2.71-4.60), especially at the trochlea (OR, 3.21; 95% CI, 2.59-3.97; I2 = 33%), medial tibial plateau (OR, 3.12; 95% CI […]

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Statistical Robustness of Randomized Controlled Trials Comparing Biceps Tenotomy Versus Tenodesis: A Reverse Continuous Fragility Index Analysis.

Hu EY, Althoff AD, Cervantes JE, Dave U, Kessler KG, Moran TE

BACKGROUND: The reverse continuous fragility index (rCFI) is a statistical measure utilized to evaluate the relative robustness or fragility of nonstatistically significant findings of randomized controlled trials. Such studies comparing biceps tenotomy versus tenodesis during shoulder arthroscopy have generally shown similar clinical and functional outcomes between the treatments.

PURPOSE: To evaluate the statistical robustness or fragility of nonsignificant outcomes of high-quality randomized controlled studies comparing clinical outcomes of biceps tenotomy versus tenodesis during shoulder arthroscopy.

STUDY DESIGN: Systematic review and meta-analysis; Level of evidence, 1.

METHODS: The PubMed, Embase, and Cochrane Library databases were searched according to the PRISMA guidelines from 2004 to September 2025 with the following search strategy: "('biceps' OR 'biceps brachii' OR 'long head biceps') AND ('tenodesis' OR 'tenotomy') AND ('Randomized controlled trial' OR 'RCT')." Studies were included if they were randomized controlled trials that compared clinical outcomes of biceps tenodesis versus biceps tenotomy during shoulder arthroscopy. rCFI was calculated on the primary nonsignificant outcomes of all studies and averaged to obtain the mean rCFI. Loss to follow-up was recorded for each study. The reverse continuous fragility quotient was calculated to account for the stu […]

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Return to Sport After Surgical Treatment of Capitellar Osteochondritis Dissecans Lesions in Baseball Players: A Systematic Review and Meta-analysis.

Paul RW, Anton B, Casalino G, Windsor JT, Tan X, Kohan E, Alberta FG

BACKGROUND: Because of the difficulty of baseball players with capitellar osteochondritis dissecans (OCD) lesions to return to sport (RTS), multiple operative strategies have been used for these lesions. However, there has yet to be a systematic review of RTS outcomes in specifically baseball players.

PURPOSE/HYPOTHESIS: The purpose of this study was to compare the RTS rates between surgical procedures for capitellar OCD lesions in youth baseball players, with a secondary purpose of comparing improvements in elbow range of motion (ROM) and improvements in Timmerman-Andrews scores between surgical procedures. The authors hypothesized that RTS rates would not differ based on the surgical procedure for capitellar OCD lesions in youth baseball players.

STUDY DESIGN: Systematic review; Level of evidence, 4.

METHODS: The authors performed a literature search in February 2025 to identify studies evaluating postoperative outcomes for the treatment of capitellar OCD lesions in youth baseball players. Surgical procedures were grouped into debridement/loose-body removal ± microfracture, fixation procedures, bone peg grafting, and autologous osteochondral transplantation. Preoperative and postoperative elbow flexion and extension ROM, Timmerman-Andrews scores, and RTS rate and time were collected. Meta-analysis included pooled effect estimates for improvements in flexion ROM, extension […]

PubMed · DOI

Vitamin D and the Risk of Stress Fractures in Athletes and Military Personnel: A Systematic Review and Meta-analysis.

Moraes BF, Leitão CVFS, Rodrigues VMS, Dressler HB, Paixão JC, da Silva ABN, de Oliveira GM, Medina G et al.

BACKGROUND: Stress fractures are common overuse injuries in military personnel and athletes. Although a low vitamin D level is a proposed risk factor, the evidence is conflicting, and the specific risk in athletes has not been systematically synthesized.

PURPOSE: To assess the association between vitamin D status and the risk of stress fractures in military personnel and athletes.

STUDY DESIGN: Systematic review and meta-analysis; Level of evidence, 3.

METHODS: Major databases were systematically searched through April 16, 2025 for studies that quantitatively measured serum levels of 25-hydroxyvitamin D in military or athlete populations with a stress fracture group and a control group. Data were pooled using a random-effects model to calculate the mean difference (MD) and 95% confidence interval (CI). Subgroup analyses were conducted based on population type (military vs athlete), sex, and vitamin D status. Meta-regression was also performed to investigate sources of heterogeneity.

RESULTS: A total of 15 studies (4183 participants) were included. Overall, patients with stress fractures had significantly lower vitamin D levels compared to controls (MD, -5.82 nmol/L [95% CI, -10.35 to -1.29]). The association was also significant for military personnel (MD, -7.33 nmol/L [95% CI, -11.77 to -2.89]) and male participants (MD, -9.14 nmol/L [95% CI, -14.23 to -4.06]). Subgroup an […]

PubMed · DOI

PASS Thresholds and Denominators After Nonoperative Management of Femoroacetabular Impingement Syndrome: Response.

Hoit G, Whelan DB, Lemieux V, Bates B, Dwyer T, Theodoropoulos J, Chahal J

PubMed · DOI

Rethinking Graft Equivalence After ACL Reconstruction With Lateral Extra-articular Tenodesis in Elite Athletes: Response.

Lynskey SJ, Ambrosanio A, Moteshari A, Jones M, Ball S, Church JS, Williams A

PubMed · DOI

Duration of Smoking Cessation Needed to Achieve Retear Rates Comparable to Those of Nonsmokers After Arthroscopic Rotator Cuff Repair: Response.

Chang HH, Chun YM, Kim SJ, Kim WM, Yoon TH

PubMed · DOI

Rethinking Graft Equivalence After ACL Reconstruction With Lateral Extra-articular Tenodesis in Elite Athletes: Letter to the Editor.

Qu X, Yi M, Tang X, Chen B

PubMed · DOI

Duration of Smoking Cessation Needed to Achieve Retear Rates Comparable to Those of Nonsmokers After Arthroscopic Rotator Cuff Repair: Letter to the Editor.

Mena L, Checchia C, Malavolta EA

PubMed · DOI

Methodological Concerns Regarding the Ankle-GO Score for Predicting Lateral Ankle Sprain in Elite Basketball Players: Letter to the Editor.

Wu Z, Liu C, Wu W

PubMed · DOI

PASS Thresholds and Denominators After Nonoperative Management of Femoroacetabular Impingement Syndrome: Letter to the Editor.

Duan YP

PubMed · DOI

Methodological Concerns Regarding the Ankle-GO Score for Predicting Lateral Ankle Sprain in Elite Basketball Players: Response.

Hardy A, Tassery F, Allaire T, Moussa MK, Valentin E, Fourchet F, Lopes R, Picot B

PubMed · DOI

BJJ

The Bone & Joint Journal

Volume 108-B, Issue 9 · September 2026 · 15 new articles

Beyond alignment : expanding the dimensionality of total knee arthroplasty.

Dunbar MJ, Haddad FS

PubMed · DOI

Necrotizing fasciitis.

White TO, Ablett AD, Neo C, Clement ND, Moran CG, Morgan MS, Duckworth AD

Necrotizing fasciitis is a rapidly progressive, life-threatening infection originating in a fascial plane which requires emergency surgery. Although uncommon, with an incidence of 0.63 per 100,000 per year, it affects the limbs in 40% of cases and carries a 22% 30-day mortality rate. The diagnosis is principally clinical. Severe pain, out of proportion to any visible abnormality of the skin, is the cardinal early feature, frequently accompanied by tenderness beyond the area of erythema, swelling, and local hypoaesthesia. Crucially, the skin may initially appear normal or nearly normal while rapid fascial destruction progresses beneath. Laboratory investigations, imaging, and scoring systems lack sufficient sensitivity and specificity to confirm or exclude the diagnosis and must not delay surgical exploration. The suspected diagnosis is confirmed or excluded by a diagnostic incision to the fascial plane. Necrotizing fasciitis is classified into four types based on the microbiology: polymicrobial (type I), Gram-positive monomicrobial (type II), Gram-negative monomicrobial (type III), and fungal (type IV). The focus of treatment is emergency radical excision of all necrotic tissue to healthy margins and broad-spectrum antibiotics. Surgery should not be delayed by medical management, imaging, or inter-hospital transfer. Failure to recognize and act within six hours of presentation […]

PubMed · DOI

Does transposition osteotomy of the acetabulum alter the natural history of developmental dysplasia of the hip?

Tanaka S, Fujii M, Kawano S, Ueno M, Oba Y, Morimoto T

AIMS: The aim of this study was to clarify the natural history of developmental dysplasia of the hip (DDH), borderline DDH (BDDH), and normal hips, and to determine whether transposition osteotomy of the acetabulum (TOA) alters the long-term joint survival in DDH and BDDH.

METHODS: We retrospectively reviewed 683 hips treated with TOA and 353 contralateral native hips. Hips were classified by the lateral centre-edge angle (LCEA) as DDH (< 20°), BDDH (20° to < 25°), or normal (≥ 25°). The median follow-up was 11 years (IQR 5.8 to 16) for contralateral native hips and 9.6 years (IQR 5.3 to 15) for TOA-treated hips. The cumulative probability of failure (progression to Tönnis grade 3, or conversion to total hip arthroplasty) was estimated using the Kaplan-Meier method. Cox proportional hazards models with patient-level cluster-robust standard errors identified predictors for failure.

RESULTS: In native hips, the 20-year survival was significantly lower in DDH than in BDDH and normal hips (69% (95% CI 55 to 80) vs 96% (95% CI 83 to 99) vs 100%, respectively; $p < 0.010$), with no significant difference between BDDH and normal hips. Increased BMI and decreased LCEA were independent predictors for failure. In DDH, TOA-treated hips had significantly increased 20-year survival compared with native hips (88% (95% CI 80 to 93) vs 69% (95% CI 55 to 80); $p < 0.001$), and native hip status, […]

PubMed · DOI

Patient, surgeon, and institutional variation in dual-mobility component use in primary total hip arthroplasty : a retrospective cohort study using data from the National Joint Registry Study.

Zucker BE, Howard JT, Whitehouse MR, Judge A

AIMS: The aim of this study was to quantify the contribution of patient-, surgeon-, and institution-level factors to variation in dual-mobility component use in primary total hip arthroplasty (DM-THA) in England. We sought to determine whether current patterns reflect appropriate patient selection or unwarranted practice variation. We hypothesized that surgeon- and institution-level factors account for a greater proportion of variation in DM-THA use than patient-level characteristics.

METHODS: We analyzed 238,455 primary THAs recorded in the National Joint Registry which were linked with Hospital Episode Statistics for England between 1 January 2018 and 31 December 2022. Using cross-classified multilevel logistic regression modelling, we partitioned variation in DM-THA use attributable to patient, surgeon, and institutional factors. Patient risk factors were included as fixed effects, with random effects specified for surgeons and hospitals to account for clustering.

RESULTS: Of the total cohort, 7,032 underwent DM-THA. These patients were older, more frail, and more frequently treated for neck of femur fractures. Surgeon-level factors explained 32.4% and institutional factors 22.9% of variation in DM-THA use. Patient-level factors accounted for approximately 10% of the variance (marginal $R^2 = 0.10$), while the full model explained 59.8% of variation (conditional $R^2 = 0.598$). […]

PubMed · DOI

Five-year survival and complications of more than 300 articulating high-dose antibiotic spacers in two-stage revision total hip arthroplasties for infection.

Weintraub MT, Midthun WR, Pagnano MW, Taunton MJ, Sierra RJ, Mabry TM, Perry KI, Berry DJ et al.

AIMS: Two-stage exchange arthroplasty with an antibiotic-eluting cement spacer remains the predominant strategy in the USA for managing chronic periprosthetic joint infection (PJI) following total hip arthroplasty (THA). The aim of this study was to evaluate a commercially available, partially intraoperatively fabricated articulating high-dose antibiotic cement spacer, focusing on survival, complications, and patient-reported outcomes.

METHODS: A total of 298 patients (302 THAs) underwent two-stage exchange arthroplasty for PJI using a specific articulating antibiotic spacer between August 2005 and May 2022. Patients with a previous failed two-stage exchange were excluded, although 121 THAs (302, 40%) had undergone previous PJI-related procedures. The mean age of the patients was 65 years (SD 11), the mean BMI was 32 kg/m2 (SD 7), and 116 (38%) were female. Most patients were classified as McPherson host B (158; 52%) and limb grade 2 (165; 55%), and 265 (88%) met the 2011 Musculoskeletal Infection Society criteria. The mean follow-up was six years (2 to 16). .

RESULTS: At five years after reimplantation, survival free of reinfection, aseptic rerevision, any rerevision, or any reoperation were 89% (95% CI 85.2 to 93.1), 93% (95% CI 89.3 to 96.1), 86% (95% CI 81.3 to 90.3), and 80% (95% CI 74.7 to 84.9), respectively. Dislocation accounted for most aseptic rerevisions (15/20). […]

PubMed · DOI

A prospective double-blinded randomized controlled trial comparing conventional jig-based versus robotic arm-assisted medial unicompartmental knee arthroplasty.

Kayani B, Fontalis A, Tahmassebi J, Konan S, Plastow R, Wazir U, Shah M, Oussedik S et al.

AIMS: This study reports the interim outcomes of a randomized controlled trial comparing the accuracy of component positions, early functional outcomes, patient-reported outcome measures, and complications in conventional jig-based unicompartmental knee arthroplasty with navigational control (CO UKA) with robotic arm-assisted unicompartmental knee arthroplasty (RO UKA) for patients who have medial compartment knee osteoarthritis (OA).

METHODS: In total, 107 patients with symptomatic medial compartment OA were prospectively randomized to undergo CO UKA (n = 52) or RO UKA (n = 55), representing the currently available recruited cohort. Patients in both treatment groups had CT-based surgical planning. All procedures were completed using the medial parapatellar approach, and all patients received a standardized postoperative rehabilitation programme. Predefined study outcomes were recorded at regular intervals for two years after surgery.

RESULTS: RO UKA was associated with increased accuracy in executing the planned femoral ($p < 0.001$) and tibial ($p < 0.001$) component positions compared with CO UKA. RO UKA was associated with reduced inpatient pain scores ($p < 0.001$), decreased inpatient opioid analgesia consumption ($p = 0.008$), and shorter length of hospital stay ($p = 0.004$) compared with CO UKA. There were no differences between the treatment groups in the Oxford Knee Score (p […]

PubMed · DOI

Impact of combined medial and lateral bone marrow lesions on clinical outcomes after medial unicompartmental knee arthroplasty : a retrospective cohort study.

Cao G, Wang A, Liu Q, Liu J, Liu Y, Huang Y

AIMS: Although the influence of medial bone marrow lesions (BMLs) on outcomes after medial unicompartmental knee arthroplasty (UKA) has been studied, the impact of concomitant lateral compartmental BMLs remains unclear. This study aims to determine whether the lateral BMLs, in addition to medial BMLs, influence clinical outcomes in patients undergoing medial UKA.

METHODS: We retrospectively reviewed patients who underwent medial UKA between January 2016 and December 2019 and had preoperative MRI. A total of 86 patients with isolated medial BMLs (BML-M) and 38 with combined medial and lateral BMLs (BML-M+L) were included. Clinical outcomes were assessed using the Numeric Rating Scale (NRS) and Oxford Knee Score (OKS). Radiological evaluation included pathological radiolucent lines, osteoarthritis progression in the lateral compartment, mechanical femorotibial angle (mFTA), and postoperative alignment zones. The proportion of patients achieving the minimal clinically important difference (MCID) for each score and a well-aligned mechanical axis was also calculated. Multivariable linear regression was employed to determine whether concomitant lateral BMLs independently influenced OKS improvement.

RESULTS: The mean follow-up was 6.4 years (SD 1.1). Both groups demonstrated significant improvements from baseline in NRS and OKS (both $p < 0.001$). No significant differences were found […]

PubMed · DOI

Secondary patella resurfacing following primary total knee arthroplasty : patient-reported outcomes, satisfaction, and patient-acceptable symptom state.

Clement ND, Parsons T, Botfield-Gray T, Plastow R, Afzal I, Kader N, Radha S, Kader DF

AIMS: Our aims were to assess joint-specific outcome (JSO), health-related quality of life (HRQoL), and patient satisfaction following secondary patella resurfacing (SPR), and to define a patient-acceptable symptom state (PASS).

METHODS: This retrospective study was conducted over 16 years (October 2006 to December 2022) and included 182 patients undergoing SPR. Patient demographic details, American Society of Anesthesiologists (ASA) grade, preoperative and postoperative (one year: n = 158 and two years: n = 143) Oxford Knee Score (OKS), EQ-5D, and EQ-VAS scores, as well as postoperative satisfaction, were obtained from a local arthroplasty register.

RESULTS: There were clinically meaningful and statistically significant ($p < 0.001$) improvements in the OKS, with a mean change of 6.8 (95% CI 4.7 to 8.9) and 8.1 (95% CI 5.9 to 10.2) points, and in the EQ-5D of 0.174 (95% CI 0.110 to 0.238) and 0.187 (95% CI 0.119 to 0.255) at both one and two years, respectively. There were 126 (79.7%) and 120 (85.7%) satisfied patients at one and two years, respectively. Prior satisfaction with the index total knee arthroplasty (TKA) (one year only: odds ratio (OR) 10.4 (95% CI 3.10 to 34.85); $p < 0.001$) and increasing time from primary TKA to the SPR (one year: OR 1.37 (for each additional year) (95% CI 1.03 to 1.81); $p = 0.028$, and two years: OR 2.32 (95% 1.23 to 4.38); $p = 0.009$) were indep […]

PubMed · DOI

Risk of venous thromboembolism after knee, hip, or shoulder arthroplasty : a retrospective cohort study of 476,620 procedures using linked hospital and primary care data.

Swain S, Leith N, Brown S, Wright Drakesmith C, Rees JL, Bankhead C, Powell J

AIMS: Arthroplasty is a widely used surgery for end-stage arthritis, but venous thromboembolism (VTE) remains a major postoperative complication. Most previous studies have relied solely on hospital data; no study has examined the effect of national clinical guidelines on the rates of VTE in the UK. This study aimed to assess VTE trends after arthroplasty using linked primary and secondary care data, and the impact of the National Institute for Health and Care Excellence (NICE) guidelines in the UK on VTE rates after arthroplasty.

METHODS: We used the Clinical Practice Research Datalink (CPRD) Aurum version, the largest primary care database in the UK, and linked hospital and death data for patients undergoing knee, hip, or shoulder arthroplasty between October 2007 and September 2019. The outcome was VTE event within six months of surgery. Trends in VTE were assessed using the Cox-Stuart trend test, and the impact of seasonality and clinical guidelines introduced in 2010, 2012, and 2018 via logistic regression with a polynomial function.

RESULTS: A total of 476,620 arthroplasty records were included. The overall VTE incidence was 1.5% (knee (1.6%), hip (1.4%), and shoulder arthroplasty (1%)). More than 30% of the VTEs were found through the linked data. A decline in VTE rates was observed over time for hip ($p = 0.004$) and knee arthroplasty ($p = 0.001$). Introduction of NICE g […]

PubMed · DOI

Poor mental health in patients with symptomatic osteochondral lesions of the talus.

Pijnacker JHM, Agyeman-Prempeh NO, Sierevelt IN, Emanuel KS, Amsterdam Ankle Cartilage Team, Broekman B, Kerkhoffs GMMJ, Stufkens SAS et al.

AIMS: Osteochondral lesions of the talus (OLT) can cause pain and functional limitations, potentially affecting mental health. This study aimed to compare mental health in patients with OLT to the general Dutch population and individuals with mood or anxiety disorders, and to identify factors associated with mental health in OLT patients.

METHODS: This prospective study included patients with symptomatic OLT prior to treatment. Data for the general population and individuals with mood or anxiety disorders were obtained from the Netherlands Mental Health Survey and Incidence Study-3. Mental health was assessed using the Mental Health Inventory-5 (MHI-5). The primary outcome was the mean difference in MHI-5 between OLT patients and the general population. Multivariate linear regression examined associations between mental health and age, sex, BMI, smoking, lesion characteristics, laterality, and Foot and Ankle Outcome Score (FAOS) pain.

RESULTS: Among the 358 included OLT patients, 87% (311 of 358) scored ≤ 76 on the MHI-5. OLT patients scored significantly lower than the general population (mean difference 20.2 (95% CI 18.9 to 21.5)) and showed similar scores to individuals with a mood or anxiety disorder (mean difference 2.0 (95% CI -0.3 to 4.3)). One and two years after treatment, 79% (170/214) and 78% (134/173) of patients, respectively, scored ≤ 76 on the MHI-5. Multivaria […]

PubMed · DOI

Total ankle arthroplasty via lateral transfibular approach with and without concomitant subtalar arthrodesis : radiological alignment, patient-reported outcomes, and range of motion at ≥ five-year follow-up.

Constantinescu SA, Barbero A, Carfi C, Crippa Orlandi N, Bonzano C, Indino C, Maccario C, Usuelli FG

AIMS: Concomitant subtalar pathology may require arthrodesis at the time of total ankle arthoplasty (TAA), yet direct comparisons of isolated TAA versus TAA + subtalar joint arthrodesis (STJA) via a lateral transfibular approach, using the same implant, remain limited. We compared ≥ five-year radiological, functional, and range of motion (ROM) outcomes.

METHODS: This retrospective single-centre cohort included 119 patients from January 2015 to December 2020 with ≥ five-year follow-up: 50 underwent TAA with concomitant subtalar arthrodesis and 69 underwent isolated TAA. Outcomes were Foot and Ankle Outcome Score (FAOS), visual analogue pain scale (VAS), 12-Item Short-Form Health Survey questionnaire physical component/mental component summary (SF-12 PCS/MCS), ankle ROM, radiological alignment (lateral distal tibial angle, α, tibiotalar surface (TTS), and Saltzman alignment), and complications. Analyses were performed using Welch's t-test or the Mann-Whitney U test for continuous variables and the chi-squared test or Fisher's exact test for categorical variables, together with multivariable regression, using a two-sided significance level of $\alpha = 0.05$ with Holm correction across angles. Post hoc minimum detectable effect (MDE) analysis at 80% power used observed SDs and group sizes.

RESULTS: Ankle osteoarthritis (OA) + STJ OA had worse baseline FAOS (24.8 (7.0) vs 42.6 (8.8); p […]

PubMed · DOI

Patient-evaluated functional outcome is systematically different from physician-evaluated outcome after proximal humerus fracture fixation.

Sahu D, Chaubey A, Shah D

AIMS: The study aimed to investigate the agreement between Constant-Murley score (CMS) and the shoulder subjective value (SSV); and the association of poor self-evaluated outcome with shoulder pain, range of motion (ROM) limitation, and shoulder strength after fixation of proximal humerus fractures.

METHODS: The study included 74 shoulders in 72 patients (mean age 56 years (SD 14), 40 male (56%)) who underwent locking plate fixation of displaced proximal humerus fractures between January 2015 and January 2024. We evaluated CMS, SSV, pain scores (visual analogue scale), and ROM measurements at the most recent (minimum one-year) follow-up. Agreement statistics between CMS and SSV were used descriptively, rather than to test true substitutability, because CMS and SSV are nonequivalent constructs.

RESULTS: The Bland-Altman analysis showed differences between CMS and SSV, as the mean difference between CMS and SSV (-14.21 (95% CI -17.25 to -11.18)) did not include the zero-difference line. The Pitman's test suggested that the variances were significantly different ($p = 0.029$). Firth's penalized method of logistic regression revealed that only the presence of shoulder pain was associated with an increased likelihood of a poor functional outcome (SSV ≤ 69.4) (coefficient 3.5 (95% CI 1.53 to 6.47); $p < 0.001$). The SSV threshold (69.4) was derived from the CMS ≤ 55 threshold on the sa […]

PubMed · DOI

Assessment of the plane of maximum curvature in adolescent idiopathic scoliosis using 3D ultrasound : an exploratory study.

Yang ZQ, Wu C, Zhou Y, Schlösser TPC, Gardner A, Castelein RM, Chu WC, Cheng JC et al.

AIMS: The size of the curves in patients with adolescent idiopathic scoliosis (AIS) is traditionally assessed using 2D Cobb angles, which may underestimate the extent of this complex 3D deformity. The plane of maximum curvature, measured using 3D reconstructions of the spine, including 3D ultrasound, offers more accurate 3D representation. However, the optimal 3D ultrasound-based method of measuring the plane of maximum curvature remains unknown. We evaluated the reliability and agreement of three methods of measuring the plane of maximum curvature (the maximum-curvature, endplate-apex-endplate, and maximum-axial-rotation methods) to provide radiation-free and severity-specific clinical recommendations for 3D ultrasound measurements of the plane of maximum curvature using the SRS-Lenke-Aubin 3D classification.

METHODS: A total of 88 patients with right major thoracic AIS, stratified by the severity of the curve (mild 29; moderate 34; severe 25), underwent 3D ultrasound. Conventional 2D parameters were quantified from the 3D ultrasound reconstructions. For each method of measurement of the plane of maximum curvature, we recorded the ultrasound curve angle (UCA) on the plane, its orientation, and the difference between it and the coronal UCA (ΔUCA). The reliability of all variables, the agreement between the three methods of measurement, comparisons between the severity groups, […]

PubMed · DOI

Locking plate fixation compared with tension-band wiring for patellar fractures: a multicentre randomized controlled trial.

Larsen P, Thorninger R, Severinsen RP, Beuke JC, Jensen SS, Rasmussen MK, Petruskevicius J, Barckman J et al.

AIMS: Patellar fractures are associated with considerable postoperative morbidity, and tension-band wiring has high complication rates and suboptimal outcomes. Locking plate fixation is a recent surgical innovation which confers superior biomechanical stability. However, there is little information about the clinical outcomes of this method of treatment. The aim of this study was to compare locking plate fixation with tension-band wiring in patients requiring surgical treatment for a patellar fracture.

METHODS: In this patient- and assessor-blinded, multicentre randomized trial, 122 adults with a displaced patellar fracture were assigned 1:1 to locking plate fixation or tension-band wiring. The primary outcome was patient-reported knee function measured by the five sub-scales of the Knee Injury and Osteoarthritis Outcome Score (KOOS) at 12 months. Superiority was predefined as a statistically significant difference in at least three KOOS sub-scales exceeding the minimal clinically important difference (pain 10, symptoms 9, activities of daily living 6, sport and recreation 10, quality of life 10). Secondary outcomes included additional clinical and functional outcomes. Analyses were performed according to the intention-to-treat principle.

RESULTS: A total of 63 patients were assigned to locking plate fixation and 59 to tension-band wiring; 57 (90.5%) and 52 (88.1%), respectiv […]

PubMed · DOI

Hip arthroplasty following proximal femoral tumour resection in children : the impact of age and articulation on implant survival and the outcomes following acetabular revision.

Baird CAE, Green N, Evans S, Kurisunkal V, Parry MC, Stevenson J

AIMS: Hip arthroplasty is rarely required before skeletal maturity. However, it is appropriate after resection of a proximal femoral bone tumour. This retrospective single-centre cohort study was conducted to evaluate how age and choice of articulation affects modes of implant failure and revision-free survival, and to understand the longer-term outcomes following inevitable revision.

METHODS: Between January 1982 and April 2023, 60 patients aged between two and 16 years underwent excision of a malignant bone tumour and proximal femoral endoprosthetic reconstruction (PFEPR) combined with either a hemiarthroplasty (HA) or total hip arthroplasty (THA) articulation. The median follow-up was 12.6 years (IQR 4 to 21).

RESULTS: There were 41 HAs and 19 THAs. Revision-free survival was poor in pre-adolescent patients (age ≤ 12 years) regardless of articulation (54% at five years (95% CI 38 to 77) and 21% at ten years (95% CI 8 to 54)). Adolescent patients (age > 12 years) undergoing THA (n = 12) had better median, five-year, and ten-year revision-free implant survival (21.9 years (IQR 11.9 to 22.7), 89% (95% CI 71 to 100), and 78% (55 to 100), respectively) than similarly aged patients undergoing HA (n = 15); (7.1 years (IQR 4.9 to 13.5), 68% (95% CI 46 to 100), 39% (18 to 82); p = 0.048), respectively). The leading indication for revision across all ages was an acetabular complicat […]

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Clinical Orthopaedics & Related Research

Volume 484, Issue 9 · September 2026 · 42 new articles

Editorial: All Orthopaedic Subspecialties, General-interest Orthopaedic Topics, Education, Culture, and Related Research Spells "CORR".

Gebhardt MC, Manner PA, Porcher R, Rimnac CM, Wongworawat MD, Leopold SS

PubMed · DOI

Editor's Spotlight/Take 5: Most Contemporary TKA Implant Systems Meet International Benchmark Survivorship Standards: Evidence From a US Registry.

Leopold SS

PubMed · DOI

Most Contemporary TKA Implant Systems Meet International Benchmark Survivorship Standards: Evidence From a US Registry.

Son SJ, Sambare TD, Sun H, Fasig BH, Reddy NC, Kelly MP, Paxton EW, Prentice HA

BACKGROUND: Benchmarking seeks to evaluate implant survivorship at certain follow-up time points against predetermined standards. The International Society of Arthroplasty Registries (ISAR) International Prosthesis Benchmarking Working Group (IPBWG) developed a predefined set of standards in which receiving a benchmark indicates that an implant has acceptable revision-free survivorship, whereas not receiving a benchmark suggests that the revision rate exceeded these established standards. Although benchmarking of TKA implant systems has been performed in select international countries, applications of benchmarks to TKA systems used in a US-based cohort are lacking.

QUESTIONS/PURPOSES: We sought to apply the ISAR IPBWG benchmarking standards to our US-based healthcare system's total joint replacement registry and identify TKA implant systems meeting (1) early 2- and 5-year benchmarks, (2) midpoint 10-year benchmarks, and (3) late 15-year benchmarks. As a secondary aim, we sought to evaluate the consistency of these benchmarks when stratified by patient gender and age.

METHODS: Data from our US-based healthcare system's Total Joint Replacement Registry were used to conduct a retrospective cohort study. The registry is well suited for benchmarking analyses because it captures all primary and revision TKAs performed within a closed, longitudinal healthcare system, enabling near-c […]

PubMed · DOI

CORR ® Curriculum-Orthopaedic Education: Residency Procedure Volumes Should Mirror Actual Practice.

Kellam J, Dougherty PJ

PubMed · DOI

Residency Diary: Saying Goodbye.

Friedman LGM

PubMed · DOI

ArtiFacts: Malgaigne's Patellar Hooks.

Hawk AJ

PubMed · DOI

Your Best Life: Look for the Good and You Will Find It.

Kelly JD

PubMed · DOI

Art in Science: Why Two Leicas?

Green SA

PubMed · DOI

Editorial Comment: Diversity and Disparities in Orthopaedic Surgery: Update 2026.

Templeton KJ, Hogan MV

PubMed · DOI

Which Neighborhood-level Metric Is Most Appropriate for Pediatric Sports Medicine Disparities Research?

Maxwell BE, Raffman ES, Navarro MB, Rosenberg SI, Merritt EH, Patel NM

BACKGROUND: Neighborhood conditions are associated with access to care and outcomes. The Child Opportunity Index (COI) and Area Deprivation Index (ADI) are commonly used in pediatric research but without a clear rationale as to why one is chosen over the other. Despite an increasing volume of health equity research, there is little evidence that has directly compared the ADI and COI's associations with previously established pediatric orthopaedic disparities. Thus, the most appropriate neighborhood-level measure for pediatric orthopaedic research remains unclear.

QUESTIONS/PURPOSES: (1) Do COI and ADI correlate with each other? (2) How do COI and ADI compare in their associations with previously established pediatric orthopaedic disparities, including time to ACL reconstruction (ACLR) and the presence of concomitant meniscal and chondral pathology?

METHODS: This is a retrospective, comparative study of patients aged 18 years or younger who underwent primary ACLR between 2010 and 2023 at one tertiary center. We excluded patients who were missing COI or ADI data. Patients who underwent multiligament reconstruction, revision ACLR, intentionally staged or delayed procedures, or previous surgery on either knee were also excluded. We initially considered 806 patients, of whom 9% (72) were excluded for prespecified reasons. Consequently, 734 patients were included in the study (mean […]

PubMed · DOI

CORR Insights®: Which Neighborhood-level Metric Is Most Appropriate for Pediatric Sports Medicine Disparities Research?

Kirchner GJ

PubMed · DOI

After ACL Injury in Children and Adolescents, Where in the Preoperative Timeline do Disparities in Timing and Clinical Course Originate?

Nutescu M, England P, Rosenberg SI, Merritt EH, Patel NM

BACKGROUND: Disparities in the timing of pediatric ACL reconstruction (ACLR) have been described along the lines of insurance and race. However, it is unclear where exactly in the preoperative timeline these inequities originate. Effective interventions to mitigate disparities cannot be designed without understanding this. This study aims to identify differences in the course and timing at each stage of the pre-ACLR process with respect to insurance and race or ethnicity. Such data can be used in the development of strategies to reduce delays for pediatric patients with ACL injuries.

QUESTIONS/PURPOSES: (1) When considering insurance and race or ethnicity, where in the preoperative timeline do disparate delays occur? (2) Do patients who are publicly insured or those in racially or ethnically minoritized groups experience differences in the type and number of clinicians seen before surgery?

METHODS: In this retrospective, comparative study, we identified 591 patients treated surgically for ACL injuries between 2011 and 2023. Of those, patients aged 18 years or younger who underwent primary ACLR were included. Fifty-seven were excluded due to intentionally delayed surgery, staged procedures, multiligamentous reconstruction, or prior surgery in either knee, leaving 534 for analysis. The mean ± SD age was 16 ± 2 years, and 51% (271) of participants were boys. Demographic data wer […]

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CORR Insights®: After ACL Injury in Children and Adolescents, Where in the Preoperative Timeline do Disparities in Timing and Clinical Course Originate?

Polousky JD

PubMed · DOI

Higher Area Deprivation Index Is Associated With Greater Practice-initiated Perioperative Communication Workload in Patients With Primary Total Joint Arthroplasty.

Lam AD, Leipman JH, Meacock SS, Parikh N, Sherman MB, Fillingham YA, Krueger CA

BACKGROUND: Although deficits in social determinants of health (SDOH) have been previously associated with adverse clinical outcomes after primary THA and TKA, their role in the perioperative communication workload remains poorly characterized. Even though it remains essential to appropriately identify and address modifiable SDOH before a procedure, orthopaedic practices must also have the resources to handle the coordination of care effectively. Understanding how deficiencies in SDOH can impact communication workload would help support effective resource planning and equitable patient engagement strategies, particularly as more perioperative management takes place outside the hospital setting.

QUESTIONS/PURPOSES: (1) What are the differences in touchpoint utilization in patients who live in locations with varying Area Deprivation Index (ADI) scores, a surrogate measure for social deprivation? (2) Does social deprivation have an association with the length of stay (LOS) during primary total joint arthroplasty? (3) How are readmission rates and patient-reported outcome measures (PROMs) different in patients living in areas with varying degrees of social deprivation?

METHODS: In this retrospective, comparative study, there were 92,801 patients who underwent primary, elective THA (43% [39,963]) or TKA (57% [52,837]) for osteoarthritis at one high-volume, urban, academic institut […]

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CORR Insights®: Higher Area Deprivation Index Is Associated With Greater Practice-initiated Perioperative Communication Workload in Patients With Primary Total Joint Arthroplasty.

Tyler WK

PubMed · DOI

Are Social Determinants of Health Disparities Associated With Worse Completion Rates for Patient-reported Outcome Measures After Total Joint Arthroplasty?

Meacock S, Hohmann A, Sellig MT, Sherman M, Fillingham YA

BACKGROUND: Patient-reported outcome measures (PROMs) are important markers to assess patient improvement after total joint arthroplasty (TJA). PROMs are increasingly relevant because of new PROM-reporting requirements for elective inpatient TJA from the Centers for Medicare & Medicaid Services. Social determinants of health (SDOH) disparities have been associated with various worse outcomes after TJA, but to our knowledge, it is not yet known how PROM completion may be affected by SDOH disparities.

QUESTIONS/PURPOSES: Among patients undergoing TJA, (1) are there SDOH disparities (such as insurance, transportation access, and living alone) that are associated with differences in PROM completion? (2) Are there neighborhood metrics, including Social Vulnerability Index (SVI) and the Area Deprivation Index (ADI), that are associated with differences in PROM completion?

METHODS: This study was a retrospective, comparative single-institution study of 12,842 patients who underwent primary, unilateral TJA for osteoarthritis between 2019 and 2022. Study participants had a mean ± SD age of 67 ± 10 years, 45% (5745) were men, and 86% (10,131 of 11,833) were White. Mean ± SD national ADI score was 34 ± 21, and SVI score was 0.4 ± 0.3. PROMs were collected within the first year preoperatively and at 6 months, 1 year, and 2 years postoperatively per institutional protocol. Patient demogra […]

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CORR Insights®: Are Social Determinants of Health Disparities Associated With Worse Completion Rates for Patient-reported Outcome Measures After Total Joint Arthroplasty?

Balkissoon R

PubMed · DOI

Are LGBTQ+ Allyship Symbols Associated With Patient Trust in the Orthopaedic Surgeon? A Dual Cohort Survey.

Catley CD, Payne ER, Ballatori SE, Pangelinan J, Chen L, O'Keefe R, Cahill SV

BACKGROUND: Lesbian, gay, bisexual, transgender, and queer (LGBTQ+) patients often face barriers to trust in healthcare settings because of historical and ongoing discrimination. In orthopaedic surgery-a specialty sometimes perceived as less inclusive-visible symbols of allyship such as rainbow flags may influence patient perceptions, yet this remains unstudied.

QUESTIONS/PURPOSES: (1) Is the display of LGBTQ+ allyship symbols (rainbow flags) associated with patient trust in an orthopaedic surgeon? (2) Which demographic factors are associated with positive or negative responses to these symbols? (3) What opinions do patients have regarding LGBTQ+ surgeons or displaying these symbols in an orthopaedic setting?

METHODS: We conducted two randomized, blinded surveys; the first recruited from a tertiary urban and academic orthopaedic clinic (362 of 477 patients contacted) and the second cohort recruited nationally via the online Amazon Mechanical Turk (MTurk) platform (439 of 1258 patients contacted). Clinic participants had a mean ± SD age of 48 ± 17 years, with 62% (223) women, while online participants had a mean ± SD age of 45 ± 17 years, with 49% (215) women. Online participants were more frequently White, men, urban dwelling, and highly educated, with 82% (358 of 439) holding a bachelor's degree or higher. Prior reported experience with orthopaedic care was common in both gr […]

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CORR Insights®: Are LGBTQ+ Allyship Symbols Associated With Patient Trust in the Orthopaedic Surgeon? A Dual Cohort Survey.

McAllister Nolan B

PubMed · DOI

Women Are Unequally Represented Among Clinical Trial Leadership by Orthopaedic Subspecialty.

Mao E, Glenn ER, Ryu D, Chang B, LaPorte DM, Aiyer A

BACKGROUND: Although men substantially outnumber women in orthopaedic surgery, prior studies have demonstrated that authorship disparities persist even after adjusting for this imbalance. If comparable patterns occur in clinical trial leadership, they may hinder women's academic advancement in our field. However, to our knowledge, no studies to date have examined proportional representation in this context.

QUESTIONS/PURPOSES: (1) Does the proportion of woman-led clinical trials differ among orthopaedic subspecialties? (2) Are women proportionally represented as PIs after adjusting for the percentage of women in each specialty? (3) Is the proportion of woman-led orthopaedic clinical trials associated with other characteristics, such as trial location or type of intervention?

METHODS: A retrospective analysis of orthopaedic surgery clinical trials registered on ClinicalTrials.gov from 2007 to 2025 was performed. Trials were manually reviewed for subspecialty relevance and PI identity. PI gender was determined via genderize.io, a validated online application that assigns a predicted gender to user-input names. Trials without orthopaedic surgeons as PIs and those with low-confidence gender predictions via genderize.io were excluded. Of the trials initially identified, 26% (1510 of 5842) met inclusion criteria, of which 55% (837 of 1510) were US-based and 42% (637 of 1510) were i […]

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CORR Insights®: Women Are Unequally Represented Among Clinical Trial Leadership by Orthopaedic Subspecialty.

Mener A

PubMed · DOI

Program Size and City Characteristics May Be Associated With Gender Diversity in Orthopaedic Surgery Residency Programs.

Park YL, Grossman M, Collier C, Christiansen MJ, Musahl V, Mulcahey MK

BACKGROUND: Orthopaedic surgery remains one of the least gender-diverse specialties in medicine. Prior studies have identified factors that influence a woman applicant's rank list, such as surgical experience, women faculty, and resident satisfaction; however, little is known about how the geographic location of a program is associated with the gender composition of residency classes.

QUESTIONS/PURPOSES: (1) Is the geographic location of an orthopaedic surgery residency program associated with the proportion of women in its residency class? (2) After adjusting for potential confounders like program size and type, what program- and city-level factors are associated with a higher percentage of women in orthopaedic residency programs?

METHODS: Accreditation Council for Graduate Medical Education (ACGME)-accredited orthopaedic residency programs with updated resident lists for the 2025 to 2026 academic year were included; we included 204 such programs from a total of 212. Resident gender was determined using photographs, biographies, and preferred pronouns when available. The proportions of women residents were compared across US geographic divisions as defined by the Electronic Residency Application Service (ERAS®). Secondary analyses explored whether program characteristics (program type, women faculty proportion, gender of chair and program director) and city-level factors (po […]

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CORR Insights®: Program Size and City Characteristics May Be Associated With Gender Diversity in Orthopaedic Surgery Residency Programs.

Lutnick E

PubMed · DOI

Minimum 10-year Results of Cementing a Polyethylene Liner Into an Acetabular Cup With a Deficient Locking Mechanism: Is It a Reliable Option?

Rhyu KH, Cho YJ, Chun YS, Lee MG

BACKGROUND: Cementing a new liner into a well-fixed acetabular cup with a deficient locking mechanism has been reported as a viable option, but concerns remain regarding complications such as dislocation and late loosening.

QUESTIONS/PURPOSES: What were the (1) durability of the cement-liner interface, (2) patient-reported outcome scores and complications, and (3) survivorship free from revision after liner cementation into a single acetabular cup design (Harris-Galante II, Zimmer) when the cup's locking mechanism was found to be deficient?

METHODS: Between May 2006 and January 2015, two surgeons performed 76 revision procedures with retention of the metallic shell by cementing a new polyethylene liner into the existing shell. During that time, this procedure was used when the acetabular shell was well fixed but the locking mechanism was found to be deficient, or when a compatible liner was unavailable. When the native locking mechanism appeared intact, liner exchange was performed using the original locking mechanism without cementation. All patients treated with this procedure were considered potentially eligible for inclusion in this retrospective study. Of those, 9% (7 of 76) were excluded because they had a shell other than the Harris-Galante II cup, which was the focus of this study. An additional 16% (12 of 76) were lost prior to the minimum study follow-up of 10 years […]

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CORR Insights®: Minimum 10-year Results of Cementing a Polyethylene Liner Into an Acetabular Cup With a Deficient Locking Mechanism: Is It a Reliable Option?

Chen KL

PubMed · DOI

Is Preoperative Moderate to Severe Varus Alignment Associated With Survivorship and Outcome Scores at a Minimum of 5 Years After Mobile-bearing Total Ankle Arthroplasty With the Hintegra Device?

Yoon YK, Park KH, Shim DW, Han SH, Lee JW

BACKGROUND: Analyses of long-term results after mobile-bearing total ankle arthroplasty (TAA) in ankles with preoperative moderate to severe varus deformity with adequate sample sizes remain limited. Because preoperative moderate to severe varus alignment may influence implant survival and functional outcomes, clarifying its long-term impact is important.

QUESTIONS/PURPOSES: (1) Do the clinical and radiologic outcomes of mobile-bearing TAA differ between ankles with preoperative neutral alignment and those with moderate to severe varus alignment? (2) What are the 5- and 10-year prosthesis survivorship, survivorship free from any revision including polyethylene exchange, and survivorship free from any unplanned reoperation in each group after mobile-bearing TAA? (3) Did the incidence of asymmetric polyethylene wear differ between the groups?

METHODS: Between September 2004 and May 2019, a single nondeveloper surgeon performed 417 TAAs. All procedures were performed using a single-design mobile-bearing device. The general indications for TAA were painful end-stage ankle arthritis. Of the 382 eligible ankles, 44% (168 ankles) were in patients with varus ankles (a preoperative coronal tibiotalar ankle ≥ 10° of varus) and 56% (214 ankles) were in neutral ankles (defined as < 10° of varus or valgus). Sixty-one percent (103 of 168) of the varus group and 61% (131 of 214) of the neut […]

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CORR Insights®: Is Preoperative Moderate to Severe Varus Alignment Associated With Survivorship and Outcome Scores at a Minimum of 5 Years After Mobile-bearing Total Ankle Arthroplasty With the Hintegra Device?

SooHoo NF

PubMed · DOI

What Is the Interversion Reliability and Agreement Between the Decision Tree Patient-rated Wrist Evaluation and the Full-length Version?

Peters ST, Wouters RM, Selles RW, Slijper HP, and the Hand-Wrist Study Group

BACKGROUND: A frequently used strategy to improve patient-reported outcome measure (PROM) response rates and reduce patients' burden with PROM completion is item reduction. In this context, a shortened decision tree version of the Patient-Rated Wrist Evaluation (DT-PRWE) was developed, reducing the number of items from 15 to 5. The DT-PRWE demonstrated excellent psychometric properties in simulated data; however, its psychometric properties have not yet been evaluated in a real-world clinical setting.

QUESTIONS/PURPOSES: (1) What is the interversion reliability and agreement between the DT-PRWE and the Patient-Rated Wrist Evaluation (PRWE) in a clinical setting? (2) What is the difference in completion time between the DT-PRWE and the PRWE?

METHODS: We conducted a prospective study at Xpert Clinics in the Netherlands, a multicenter, referral-based outpatient practice, with both urban and regional locations, that specializes in hand and wrist surgery and hand therapy to assess the interversion reliability and agreement between the PRWE and the DT-PRWE. Between January and April 2025, a total of 427 adult patients were treated for wrist-related conditions and completed the PRWE at baseline as part of routine outcome measurement at our clinic. Subsequently, we asked patients to complete the DT-PRWE again 5 to 10 days after the initial assessment. Of those, we considered adult pa […]

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CORR Insights®: What Is the Interversion Reliability and Agreement Between the Decision Tree Patient-rated Wrist Evaluation and the Full-length Version?

London DA

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Bilateral Open Adductor Longus Tenotomy Is Associated With Predictable Return to Play in Professional Soccer Players: A Case Series of 100 Patients.

Cugat R, Montaña F, Barastegui D, Barrueto D, Vázquez M, Ferré-Aniorte A, Láiz P, Seijas R

BACKGROUND: Dynamic pubic osteopathy is a clinical diagnosis characterized by chronic localized pain at the pubic insertion of the adductor longus that can limit high-level athletic performance in professional soccer players. Although open adductor longus tenotomy has been used for decades in selected athletes, little is known about return-to-play (RTP) timelines and outcomes in professional soccer players undergoing modern bilateral open release.

QUESTIONS/PURPOSES: We asked: (1) How long after bilateral open adductor longus tenotomy did professional soccer players return to unrestricted match participation? (2) What proportion returned to their preinjury competitive level? (3) Was player age associated with RTP duration? (4) What complications and reoperations occurred after surgery?

METHODS: We performed a retrospective case series of 100 professional men's soccer players (Tegner activity level 9 to 10) who underwent bilateral open adductor longus tenotomy between 2011 and 2024. All patients were diagnosed clinically with dynamic pubic osteopathy based on localized pubic insertion pain reproduced with resisted adduction. Radiographs were used to exclude fractures, stress fractures, tumors, or avulsion injuries. MRI was performed to evaluate associated findings such as bone edema or degenerative changes of the pubic symphysis; however, imaging findings were frequently withi […]

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CORR Insights®: Bilateral Open Adductor Longus Tenotomy Is Associated With Predictable Return to Play in Professional Soccer Players: A Case Series of 100 Patients.

Owusu-Akyaw K

PubMed · DOI

Can Personalized 3D Kinematic Modeling Predict Loss of Pronation and Supination in Diaphyseal Forearm Malunions? A Clinical Validation Study.

van Loon DFR, van Es EM, Siemensma MF, Eygendaal D, Stockmans F, Veeger DHEJ, Colaris JW

BACKGROUND: Despite the clinical relevance of forearm fractures and malunions and the impact of a functional limitation, the link between forearm malalignment and limited pronation and supination remains poorly understood and still relies on anatomical alignment expressed as angulation. Using recently developed technologies, mechanisms that limit function can be automatically detected by modeling individual forearm kinematics using three-dimensional (3D) bone models of the radius and ulna.

QUESTIONS/PURPOSES: We evaluated the accuracy of a personalized 3D kinematic model to identify limitations in forearm rotation in pronation and supination and to answer the following questions: (1) How accurately does the model-predicted ROM agree with the corresponding clinical measurements? (2) How accurately does the model classify malunited forearms according to the presence of clinically relevant functional limitations, defined as a range of pronation or supination less than 50°? (3) What is the frequency at which the model detects bone impingement and central band block during pronation and supination?

METHODS: This retrospective study evaluated a diagnostic model using the preoperative CT scans of 45 patients with unilateral diaphyseal forearm malunions, all of whom underwent corrective osteotomy due to a clinically relevant limitation in pronation or supination function. In all, 53% […]

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CORR Insights®: Can Personalized 3D Kinematic Modeling Predict Loss of Pronation and Supination in Diaphyseal Forearm Malunions? A Clinical Validation Study.

Beredjiklian PK

PubMed · DOI

Letter to the Editor: Guest Editorial: Recalling a Recall.

Ekinci S

PubMed · DOI

Reply to the Letter to the Editor: Guest Editorial: Recalling a Recall.

Cornell CN

PubMed · DOI

Letter to the Editor: Compensatory Activation of Periscapular Muscles Aids Active Abduction in Patients With Massive Rotator Cuff Tears.

Qin K, Xu Y, Liang W

PubMed · DOI

Letter to the Editor: Can Nutritional and Inflammatory Markers Be Combined to Develop an Accurate Survival Model for Surgical Decision-making in Patients With Lung Cancer Bone Metastases?

Eckerle L

PubMed · DOI

Reply to the Letter to the Editor: Can Nutritional and Inflammatory Markers Be Combined to Develop an Accurate Survival Model for Surgical Decision-making in Patients With Lung Cancer Bone Metastases?

Yan Y, Zhang Y

PubMed · DOI

Letter to the Editor: Editorial: AAOS Orthobiologics Registry-Sometimes, More is Less.

Wang X, Shen W

PubMed · DOI

Letter to the Editor: Does First Metatarsal Head Lowering During Minimally Invasive Chevron Akin Osteotomy Yield No Significant Difference in Patient-reported Outcomes Compared With Second Distal Metatarsal Minimally Invasive Osteotomy for Treating Intractable Plantar Keratosis?

Zhao J, Gao J

PubMed · DOI

Reply to the Letter to the Editor: Does First Metatarsal Head Lowering During Minimally Invasive Chevron Akin Osteotomy Yield No Significant Difference in Patient-reported Outcomes Compared With Second Distal Metatarsal Minimally Invasive Osteotomy for Treating Intractable Plantar Keratosis?

Choi JY, Yi Y, Choi SM, Suh JS

PubMed · DOI

Reply to the Letter to the Editor: Editorial: AI-assisted Letters to the Editor-Scope of a Growing Ethical and Practical Concern, and CORR 's Approach to Managing It.

Daly T, Teixeira da Silva JA

PubMed · DOI

JBJS (American)

Volume 108, Issue 17 · September 2026 · 18 new articles

The Next Era of Orthobiologics: From Promise to Precision.

Piuzzi NS, Goodman SB

PubMed · DOI

Combined Allograft-Prosthetic Composite Reverse Shoulder Arthroplasty and Lower Trapezius Transfer: Promise Amid Complexity: Commentary on an article by Joaquin Sanchez-Sotelo, MD, PhD, et al.: " Lower Trapezius Transfer at the Time of Reverse Shoulder Arthroplasty with an Allograft-Prosthetic Composite Can Restore External Rotation ".

Hantouly A, Khan M

PubMed · DOI

Clinical Impact of VEGFi Reinitiation Timing on Wound Complications After Spinal Metastatic Tumor Surgery: Commentary on an article by Chih-Chi Su, MD, et al.: " Timing of Vascular Endothelial Growth Factor Inhibitor Reinitiation and Risk of Wound Complications in Spinal Metastasis Surgery ".

Shigematsu H

PubMed · DOI

Can We Reproduce the Tendon-Bone Interface After Rotator Cuff Repair Surgery?: Commentary on an article by Yexin Li, MMed, et al.: " Prevascularized Bone Marrow-Derived Mesenchymal Stem Cell Sheets Promote Tendon-Bone Integration in Rotator Cuff Repair ".

Kenter K

PubMed · DOI

What's New in Orthopaedic Rehabilitation.

Riddle DL

PubMed · DOI

A Precision Medicine Framework to Improve Patient Care Using Orthobiologics.

de la Huerta Meza D, Singh P, Otero M, Kirschner JS, Rodeo SA

The lack of consistent responses to treatment is a considerable challenge in the medical field, in part driven by the one-size-fits-all approach traditionally applied to developing and testing therapies. This approach has resulted in treatments that work for some patients but fail to work for others and has led to the development of precision medicine initiatives aiming to refine treatment allocation. This is of particular interest for orthopaedic conditions, for which patient populations are often heterogeneous. The rapid implementation of orthobiologics and their lack of consistent clinical efficacy, for which the heterogeneity and variability stem from both the patients and the treatments, further underline the need to implement precision medicine approaches in orthopaedics. This article will discuss proposed steps, challenges, and opportunities to enable the deployment of a precision medicine framework to improve patient care using orthobiologics.

PubMed · DOI

Orthobiologics at a Crossroads: From Biological Promise to Evidence-Based Orthopaedic Practice.

Anzillotti G, Conte P, Dragoo JL, Kon E

PubMed · DOI

Cellular Bone Matrices Reconsidered: Immunogenicity, Experimental and Clinical Evidence of Enhanced Effectiveness, Safety, and Value Proposition.

Marino JF

PubMed · DOI

Orthobiologic Treatment: A Patient's Experience with MFAT Injection for Shoulder Osteoarthritis.

Oliveira LP, Holmes M

PubMed · DOI

A Century of Orthopaedic Excellence with Global Impact: Instituto de Ortopedia y Traumatología "Carlos E. Ottolenghi" at Hospital Italiano de Buenos Aires.

Buttaro M, Ranalletta M, Slullitel PA, Rossi LA, Aponte LA, Ayerza MA, Farfalli G, Piuzzi NS et al.

➢ Argentine orthopaedics was shaped by the influence of Dr. Vittorio Putti (Rizzoli Institute, Bologna, Italy). Dr. Carlos E. Ottolenghi, one of Dr. Putti's most distinguished disciples, together with Dr. José Valls, founded the Instituto de Ortopedia y Traumatología at Hospital Italiano de Buenos Aires in 1926. Prof. Ottolenghi transformed the Orthopaedic Service into a leading academic center and pioneer of subspecialization, serving as a foundational driver of orthopaedic development in Argentina and Latin America, while establishing strong international academic connections and collaborations. ➢ Major milestones include the establishment of one of the earliest institutional bone banks and the performance of one of the world's first reported massive bone allograft reconstructions (1948); the performance of the first Charnley total hip replacement in the Americas (1967) and the first total knee replacement in Argentina (1970); and the early adoption of intramedullary nailing (1972), knee arthroscopy (1976), and femoral bone impaction allografting (1987). These contributions helped to define modern orthopaedic practice across Latin America and influenced global reconstructive strategies. ➢ Advances such as surgical navigation (2010) and augmented reality (2025) have strengthened its role as a regional and globally connected center of excellence, particularly in musculoskeletal […]

PubMed · DOI

Regulatory Frameworks and Clinical Development of Orthobiologics Across Asia and Europe.

Lee JK, Kon E, Anzillotti G, Nakamura N

Asia and Europe are rapidly expanding hubs for orthobiologics, driven by aging populations, advancing infrastructure, and growing evidence for regenerative therapies. Regulatory heterogeneity across both regions remains a major barrier, highlighting the need for international collaboration to support safe, responsible innovation.

PubMed · DOI

An International Expert Consensus Statement Defining the Best Practices and Areas of Uncertainty Concerning the Use of Orthobiologics.

Kunze KN, Morgan JT, Gerhold C, Nishioka T, Piuzzi NS, Chahla J, the Delphi Study Group

BACKGROUND: The use of orthobiologics has received increasing interest given the potential positive treatment effects on symptoms and tissue regeneration. However, there remains substantial variability surrounding the uses of orthobiologics in clinical practice. The purpose of this study was to create an international Delphi expert consensus statement on the use of orthobiologics in musculoskeletal medicine.

METHODS: A working group consisting of 6 members developed 77 initial statements for review by a 24-member panel of international experts. Surveys were conducted in 3 rounds and administered electronically to participants. Participants were asked to respond to each statement using a Likert scale of "strongly disagree," "disagree," "neither disagree nor agree," "agree," and "strongly agree." The final consensus statement included those statements that achieved ≥70% agreement and <20% disagreement. Statements that did not initially achieve consensus were iteratively revised between rounds, which included the incorporation of qualifying language to reflect areas of clinical uncertainty.

RESULTS: Sixty-two final statements were utilized in round 3. Each round had 100% participation. Consensus increased sequentially from 48.1% (37 of 77 statements) in round 1 to 96.8% (60 of 62 statements) in round 3. Complete consensus (≥70% agreement and <20% disagreement) was achieved in th […]

PubMed · DOI

Osteochondral Autograft Transfer for Focal Osteochondral Defects of the Knee: Indications, Technique, Outcomes, and Future Directions.

Strony JT, Ina JG, Apostolakos JM, Calcei JG, Karns MR, Ode GE, Salata MJ, Voos JE et al.

➢ Osteochondral autograft transfer (OAT) restores hyaline cartilage and subchondral bone in a single stage, offering a durable joint-preserving option for focal full-thickness articular cartilage defects in young, active patients. ➢ Optimal candidates have unipolar femoral condyle, patellofemoral, or tibial plateau lesions measuring 1 to 4 cm 2 ; outcomes become less predictable once the defect size exceeds 3 cm 2 . Outcomes are further influenced by age, activity level, sex, alignment, instability, and meniscal deficiency, with older, lower-demand patients and those with larger lesions demonstrating comparatively inferior results. ➢ Technical success requires meticulous recipient-site preparation, perpendicular graft harvest, and flush implantation; both arthroscopic and open approaches achieve reliable results. ➢ OAT demonstrates high return-to-sport rates (often >85% within 6 months), significant functional improvements, and superiority over microfracture. Outcomes may be comparable or superior to cellular resurfacing or allograft techniques in appropriately selected patients. ➢ Emerging biologic adjuncts, recess-filling strategies, and donor-site substitutes may enhance graft integration and reduce morbidity, although long-term clinical benefits remain unproven.

PubMed · DOI

Peptide Therapeutics in Orthopaedics: Current Evidence and Future Directions.

Urish KL, Osifo SE, Steckbeck JD

➢ Peptides lack a rigid tertiary structure, allowing them to adopt a variety of conformations, and can be engineered for specific targets. This article examines the design principles underlying orthopaedic peptide applications, from preclinical candidates to clinically validated drugs. ➢ The use of peptides in orthopaedics began with naturally occurring agents, such as bone morphogenetic proteins, and has evolved toward rationally engineered synthetics. ➢ Teriparatide, a synthetic parathyroid hormone-derived peptide, is used for bone regeneration; PLG0206, a fully engineered antimicrobial peptide, has demonstrated activity against biofilm-forming and multidrug-resistant organisms that conventional antibiotics cannot eradicate, and is advancing through Phase-II and III clinical evaluations. ➢ Peptide engineering has produced candidates with potential for cartilage-penetrating, sustained local drug delivery, although no peptide-based disease-modifying osteoarthritis drug has yet achieved clinical validation. ➢ Artificial intelligence-driven peptide design is accelerating discovery, but the translation of computationally designed sequences to clinical efficacy remains an active challenge.

PubMed · DOI

Ethical Challenges in the Clinical Use and Commercialization of Orthobiologics: A Global Perspective.

Rossi LA, Shapiro SA

PubMed · DOI

Lower Trapezius Transfer at the Time of Reverse Shoulder Arthroplasty with an Allograft-Prosthetic Composite Can Restore External Rotation.

Sanchez-Sotelo J, Hart CM, Weir TB, Graden N, Gulzar M, ElHassan B, Wagner E

BACKGROUND: The purpose of this study was to describe, and report the outcomes of, a surgical technique that combines reverse shoulder arthroplasty (RSA) using an allograft-prosthetic composite (APC) with a lower trapezius transfer (LTT) to the infraspinatus of the allograft extended with the inferior capsule.

METHODS: Between 2019 and 2023, 2 surgeons performed a total of 27 RSAs using an APC in conjunction with an LTT. Once the RSA APC reconstruction had been performed, cuff-capsule allograft tissue was passed posterior to the glenosphere and secured to the harvested lower trapezius. One shoulder was resected for periprosthetic joint infection; the remaining 26 shoulders were followed for a minimum of 2 years (mean, 2.7 years; range, 2 to 5 years).

RESULTS: The rate of revision or reoperation for humeral complications was 15% and included secondary bone grafting (2 shoulders), resection for infection (1 shoulder), and internal fixation of a periprosthetic fracture (1 shoulder). Other complications requiring surgery included hematoma treated with evacuation (2 shoulders), dislocation (1 shoulder), and glenosphere disassociation (1 shoulder). Preoperatively, 11 shoulders presented with pseudoparalysis, and the mean active elevation of the remaining 16 shoulders was 43°, with only 1 shoulder demonstrating >90° of active elevation. At the most recent follow-up, 3 shoulders were […]

PubMed · DOI

Timing of Vascular Endothelial Growth Factor Inhibitor Reinitiation and Risk of Wound Complications in Spinal Metastasis Surgery.

Su CC, Pan YT, Lin YP, Chou CC, Yen HK, Tsai HC, Hsieh HC, Lin WH et al.

BACKGROUND: Vascular endothelial growth factor inhibitors (VEGFi) are widely used for metastatic cancer management but are known to impair wound-healing. In spinal metastasis surgery, postoperative wound complications remain a clinical challenge, and the perioperative management of systemic antineoplastic therapies remains poorly defined. Current recommendations regarding postoperative VEGFi reinitiation are largely consensus-based and range from 28 to 40 days. The aims of this study were to evaluate the association between perioperative antineoplastic therapies and wound complications and to define the optimal timing of postoperative VEGFi reinitiation.

METHODS: This retrospective cohort study included 1,164 Taiwanese patients (684 male and 480 female; median age, 60.7 years [interquartile range, 51.5 to 68.8 years]) who underwent spinal metastasis surgery at a single tertiary center between 2010 and 2022. The outcome was postoperative wound complications within 90 days. Multivariable logistic regression was used to identify independent risk factors. The association between the timing of postoperative VEGFi reinitiation and wound complications was assessed using receiver operating characteristic curve analysis and the Fisher exact test.

RESULTS: Wound complications occurred in 301 patients (25.9%). Multivariable analysis identified diabetes mellitus (odds ratio [OR] = 4.58; […]

PubMed · DOI

Prevascularized Bone Marrow-Derived Mesenchymal Stem Cell Sheets Promote Tendon-Bone Integration in Rotator Cuff Repair.

Li Y, Chen Y, Zhu Y, Li D, Liao L, Li W, Liu Q

BACKGROUND: Limited vascularization at the tendon-bone interface (TBI) hinders rotator cuff (RC) healing. Although cell sheet technology has shown promise for interfacial repair, prevascularization strategies remain underexplored.

METHODS: Twenty female New Zealand rabbits underwent bilateral infraspinatus tendon repair and were randomized to receive either bone marrow-derived mesenchymal stem cell (BMSC) sheets or prevascularized BMSC sheets generated by coculture with endothelial cells, implanted at the TBI. An age- and weight-matched uninjured group served as a control. Healing at 6 weeks was assessed by gross observation, histology, immunohistochemistry, gene expression, and biomechanical testing.

RESULTS: Prevascularization of the BMSC sheets enhanced TBI vascularization, indicated by greater density of α-smooth muscle actin-positive vessels (16.16 ± 2.81 versus $10.63 \pm 2.79/mm^2$, $p = 0.0079$). Immunohistochemistry demonstrated greater areas positive for collagen type II alpha 1 (86.96 ± 29.95 versus

40.25 ± 11.96 µm 2 , p = 0.0079) and interleukin 10 (14.93 ± 4.79 versus 7.43 ± 2.48 µm 2 , p = 0.0159). Biomechanically, prevascularization of the sheets yielded greater ultimate failure load (156.89 ± 51.92 versus 111.67 ± 27.51 N, p = 0.0364) and stiffness (37.27 ± 12.16 versus 27.16 ± 7.33 N/mm, p = 0.0486).

CONCLUSIONS: Prevascularization of BMSC sheets was able to pro […]

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Letter to the Editor on "Physiologic and Clinical Sequelae After Pneumatic Tourniquet Release in Frail Patients Undergoing Total Knee Arthroplasty Under Regional Anesthesia: A Prospective Observational Study".

Sun X, Fei J

PubMed · DOI

Reply to the Letter to the Editor on "Physiologic and Clinical Sequelae After Pneumatic Tourniquet Release in Frail Patients Undergoing Total Knee Arthroplasty Under Regional Anesthesia: A Prospective Observational Study".

Patel G, Kahlon S, Singh A

PubMed · DOI

Leadership Begins With the Small Things: Lessons for Early Career Arthroplasty Surgeons.

Griffin D, Scuderi GR, Mont MA

PubMed · DOI

Advocacy 101 for the Early-Career Arthroplasty Surgeon: Payment, Policy, and Professional Sustainability.

Lum ZC, Herndon CL, Wenzlick TS, Dallas-Orr D, Courtney PM, Rana AJ

Modern hip and knee arthroplasty surgeons graduate prepared to manage complex deformity, revision surgery, complications, and high-volume clinical practice but often receive little formal preparation for the payment and policy systems that determine whether that care remains accessible and sustainable. For early-career surgeons, this gap between clinical work and reimbursement reality can feel confusing, frustrating, and outside their control. Yet, payment policy directly shapes patient access, practice viability, workforce morale, and professional autonomy. This primer provides an "Advocacy 101" primer for arthroplasty surgeons in their first five years of practice, focusing on the core mechanics of physician payment: relative value units, the Medicare conversion factor, budget neutrality, 90-day global periods, and the distinction between physician professional fees and facility payments. It helps with understanding how declining inflation-adjusted reimbursement, increasing postoperative workload, outpatient migration, consolidation, and alternative payment models are reshaping the arthroplasty practice environment. We also discuss opportunities within bundled payment models, Transforming Episode Accountability Model, and gainsharing, emphasizing that surgeons who understand implant costs, postacute pathways, and episode spending are better positioned to lead value-based care […]

PubMed · DOI

Balanced Peer Review in the Era of Artificial Intelligence.

Taunton MJ, Browne JA, Scuderi GR, Smitterberg CW, Mont MA

PubMed · DOI

Statistical Choices in Propensity Score Matching Influence the Conclusions in Arthroplasty Outcomes Research.

Verhey JT, Tarabichi S, Novicoff WM, Mont MA, Browne JA, Bingham JS, Spangehl MJ, Clarke HD

PubMed · DOI

Outpatient Arthroplasty: The Site of Service Has Changed, but the Work and Financial Burden Have Not.

Mont MA, Scuderi GR

PubMed · DOI

The Shift to Outpatient Total Hip and Knee Arthroplasty: Perioperative Work Has Not Changed.

Warwick H, Salehi N, Kugelman DN, Yakkanti R, Krueger CA, Courtney PM

BACKGROUND: Total hip arthroplasty (THA) and knee arthroplasty (TKA) are increasingly being performed on an outpatient basis. With this shift in site of service, the volume and complexity of work required for these procedures may be affected. The purpose of this study was to quantify the amount of perioperative work performed by the surgeon and advanced practice providers for same-day THA and TKA.

METHODS: We prospectively timed preservice, immediate postservice, and discharge coordination activities for a consecutive series of 96 outpatient cases over a 4-week period. Timing data for activities performed in the operating room were obtained retrospectively from our institutional electronic medical record for 500 patients who underwent THA and 500 who underwent TKA. Results were compared with the current approved values reviewed by the Relative Value Scale Update Committee (RUC) in 2019. Cases were also separated into those performed in tertiary care hospitals, specialty hospitals, and ambulatory surgical centers to compare perioperative work by surgical setting.

RESULTS: The average pre-evaluation time was 43.9 ± 7.3 minutes. The average time from the patient entering the operating room to incision was 39.0 ± 9.8 minutes. Immediate post-service tasks took 26.6 ± 7.2 minutes, and discharge coordination tasks took 122.1 ± 33.9 minutes. Overall, tasks measured took 231.6 minutes […]

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Quantifying Clinical Encounters for Orthopaedic Hip and Knee Surgeries: A Retrospective Analysis of Provider Workload.

Rana AJ, Sewell JK, Sirisoma AV, Walsh ZA, Faour KNW, McGrory BJ

BACKGROUND: The time spent providing postoperative care for total hip arthroplasty (THA) and total knee arthroplasty (TKA) has increased provider workload, drawing attention to the Relative Value Scale Update Committee's (RUC) current estimation of work relative value units. Our aim for this study was to quantify the postoperative work performed by the surgeon and their team for THA and TKA during the 90-day global period. We hypothesized that the work intensity and time spent on postoperative communication were higher than estimated by current work relative value units.

METHODS: We retrospectively evaluated all patients undergoing primary total joint arthroplasty at our institution between January 1, 2019, and December 31, 2024. Primary outcomes included the number of postoperative interactions from discharge to 90 days after surgery. These included office visits, telehealth visits, phone calls, and patient portal messages.

RESULTS: From 2019 to 2024, the average number of postoperative total joint arthroplasty interactions per patient increased across all modalities. Telephone encounters spiked during the coronovirus disease 2019 pandemic and remain elevated, while portal messages rose more than fivefold. Office visits averaged 2.3 per patient, exceeding the two currently recognized in RUC valuations. Administrative tasks and telehealth also showed steady annual growth. Mea […]

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There Is No Difference in Outpatient Costs Within One Year Post-Total Hip Arthroplasty when Comparing Surgical Approach.

Barton KI, Kooner P, Petis SM, Somerville LE, Marsh JD, Howard JL, Lanting BA, Vasarhelyi EM

BACKGROUND: Total hip arthroplasty (THA) continues to be a substantial financial burden to the Canadian health care system. The purpose of this study was to determine the impact of surgical approach on the associated outpatient cost of THA.

METHODS: Patients who underwent a THA via an anterior, posterior, or lateral approach were recruited (not randomized). All patients prospectively completed a cost diary (self-reported use of allied health services, investigations, clinic visits, patient expenses, and number of employment days lost to THA) to estimate total resource use and patient expenses incurred after discharge. Group comparisons were performed using Pearson's Chi-squares and one-way analyses of variance; significance was accepted at $P \leq 0.05$.

RESULTS: A prospective, microcosting analysis was performed on 104 patients undergoing a THA through either an anterior (n = 37), posterior (n = 32), or lateral (n = 35) approach for one year postoperatively. The overall costs of outpatient resources and expenses were comparable between anterior ($1,321.00 CDN), posterior ($1,094.92 CDN), and lateral ($1,153.18 CDN) approaches (P = 0.114). When the overall costs were stratified by specific expenses, the costs of allied health services were significantly higher in the anterior group ($361.32 CDN; 95% confidence interval, 257.58 to 465.07) than the posterior group ($206.50 CDN; 95% […]

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Implant Prices and Physician Reimbursement Have Declined More Than Total Costs and Hospital Payments in Total Joint Arthroplasty.

Yu JS, Dykhouse GL, Heo KY, Mao YV, Christ AB, Premkumar A

BACKGROUND: Understanding implant price trends is critical amid growing demand for total joint arthroplasty (TJA) and increasing cost containment pressures. Previous studies have documented trends in costs, reimbursements, and volume for TJA. However, the relationship between implant prices, hospital and physician reimbursement, and patient financial burden remains poorly defined. This study evaluated inflation-adjusted implant pricing trends in TJA and their alignment with physician and hospital reimbursement and patient out-of-pocket (OOP) costs.

METHODS: Implant prices for primary total knee arthroplasty (TKA), total hip arthroplasty (THA), revision TKA (rTKA), and revision THA (rTHA) from 2009 to 2021 were obtained from a large publicly available implant registry. Cost, reimbursement, and patient OOP spending data were sourced from a commercial insurance claims database. There were 629,651 total procedures analyzed. All costs, reimbursements, and prices were adjusted for inflation. Trends were analyzed using linear regressions.

RESULTS: The average price for TKA implants was $5,899, $6,776 for THA, $11,576 for rTKA, and $7,419 for rTHA. Between 2009 and 2021, implant prices declined markedly for TKA (-38%), THA (-37%), and rTHA (-28%) and remained stable for rTKA (+8%). Overall costs and hospital reimbursement remained stable or modestly decreased, whereas physician reimb […]

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Implant Costs Consume a Large Proportion of the Reimbursement in Revision Hip and Knee Arthroplasties.

SmithOram AL, Mears SC, Barnes CL, Stronach BM, Stambough JB

BACKGROUND: Revision total hip arthroplasty (rTHA) and revision total knee arthroplasty (rTKA) burdens continue to increase while overall reimbursement decreases. We sought to evaluate the percentage and variability of the diagnosis-related group (DRG) reimbursement spent on implants in revision arthroplasty.

METHODS: A consecutive series of 199 rTHAs and 187 rTKAs performed between June 1, 2019, and June 1, 2021, was reviewed at one academic medical center. Patient characteristics, preoperative diagnosis, implant records, and billing data were recorded for DRG 466 (revision hip or knee arthroplasty with major complication), 467 (revision with complication or comorbidity), and 468 (revision without comorbidity or complication). Data were stratified by DRG and diagnosis. Implant, surgery type, and patient comorbidity factors were analyzed for association with increased costs.

RESULTS: Implant costs comprised an average of 24% of reimbursements for DRG 466 (range, 2.4 to 133), 36.7% for DRG 467 (range, 3.5 to 118), and 35% for DRG 468 (range, 2.5 to 175). When stratified by diagnoses, groups with the largest ranges in implant costs as a percentage of the reimbursement were aseptic loosening (range, 6.6 to 178.2), infection (range, 3.2 to 129.3), and metallosis (range, 2.4 to 108.6%). Implants for rTKA were significantly more of the overall reimbursement than rTHA across all DRG […]

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Safety Net Hospitals Are at Risk for Major Financial Penalties With New Centers for Medicare & Medicaid Services Patient-Reported Outcome Measure Reporting Requirements for Primary Hip and Knee Arthroplasty.

Salehi N, DeAnnuntis J, Gorleski M, Warwick H, Kugelman DN, Courtney PM

BACKGROUND: Starting in 2027, the Centers for Medicare & Medicaid Services (CMS) will penalize hospitals that do not have complete paired patient-reported outcome measures (PROMs) data on at least 50% of patients undergoing inpatient total hip (THA) and knee arthroplasty (TKA). Patients must also meet the minimum substantial clinical benefit (SCB) threshold for Hip Disability and Osteoarthritis Outcome Score Joint Replacement and Knee Injury and Osteoarthritis Outcome Score Joint Replacement scores (22 and 20 points, respectively). The purpose of this study was to identify the risk factors for patients who did not have complete paired PROMs and did not meet the minimum SCB following surgery.

METHODS: We included 5,165 primary THA and TKA patients between 2022 and 2023 at 22 hospitals. We collected data on demographics, medical comorbidities, insurance types, and socioeconomic status. Surveys were collected 90 to zero days preoperatively and 300 to 425 days postoperatively, consistent with CMS rules. A multivariate regression analysis was performed to identify independent risk factors for not meeting new CMS PROMs requirements.

RESULTS: There were 764 patients (14.8%) who met the CMS paired PROMs reporting requirement, whereas 554 (72.5%) met the SCB threshold. Risk factors for failing to meet paired PROMs collections included higher Charlson Comorbidity Index (odds ratio (OR) […]

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Socioeconomic Deprivation and Outcomes After Total Joint Arthroplasty: An Analysis of the American Joint Replacement Registry Using the Area Deprivation Index.

Zalikha L, Amanatullah DF, Jahan M, Donnelly P, Cohen EM, El-Othmani MM

BACKGROUND: There is an increasing interest in socioeconomic deprivation and its influence on outcomes in orthopaedics. This study aimed to examine the relationship between neighborhood-level social deprivation and its impact on 90-day readmission, length of stay, and all-cause revision after primary total knee arthroplasty (TKA) and total hip arthroplasty (THA).

METHODS: A retrospective cohort study was conducted using data from the American Joint Replacement Registry merged with the Centers for Medicare and Medicaid Services claims for patients aged 18 years and over who had undergone primary unilateral TKAs and THAs. Comparisons were performed between patients grouped into Area Deprivation Index (ADI) quartiles by postal code (Quartiles 1 to 4, with Quartile 1 being the least disadvantaged). Subanalyses were performed using multinomial logistic regression and Cox proportional hazard models for outcomes with Quartile 1 serving as a reference. A total of 1,654,680 TKA and 1,051,311 THA procedures met the inclusion criteria.

RESULTS: Unadjusted analysis demonstrated significant differences in all outcomes based on ADI quartile ($P < 0.001$ for all outcomes except TKA all-cause revision, $P = 0.003$). In a subanalysis, Quartile 4 was significantly more likely to undergo readmission within 90 days for TKA (hazard ratio 1.631; 95% CI [confidence interval]: 1.528 to 1.741, $P < 0.0001$ […]

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Social Determinants of Health Disparities Are Associated With Prolonged Opioid Utilization After Total Joint Arthroplasty.

Hohmann AL, Fellheimer HS, Meacock SS, Lizcano JD, Purtill JJ, Fillingham YA

BACKGROUND: Social determinants of health disparities (SDHDs) have been associated with an increased risk of poor outcomes after total joint arthroplasty (TJA). The purpose of this study was to assess whether SDHDs are associated with prolonged postoperative opioid use after primary TJA.

METHODS: This study was a single-institution, retrospective cohort study of patients undergoing primary, unilateral TJA. Opioid prescriptions for these patients were queried in a Prescription Drug Monitoring Program. Patients who had an opioid prescription filled within one year preoperatively were excluded. Patients were divided by procedure (total hip arthroplasty [THA] versus total knee arthroplasty [TKA]) and split into five cohorts: no postoperative opioid prescription filled, or the last prescription filled within zero to 30, 31 to 90, 91 to 180, or 181 to 365 days postoperatively. Patients were excluded if they did not fill postoperative opioid prescriptions in consecutive time frames. We identified 7,671 eligible patients who underwent THA and 7,707 who underwent TKA.

RESULTS: The mean overall Social Vulnerability Index (SVI) score increased, indicating greater socioeconomic vulnerability, with a later last fill date, with significant differences across cohorts for THA ($P < 0.001$) and TKA ($P = 0.026$) patients. In multilinear regression analyses, the last fill date in the 31- to 90-day […]

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Substantial Clinical Benefit Thresholds for Hip Disability and Osteoarthritis Outcome Score Subscales and Associated Risk Factors for Nonattainment After Total Hip Arthroplasty: A Prospective Cohort Study of 9,229 Patients.

Pasqualini I, Elmenawi KA, Khan ST, Hadad MJ, Cleveland Clinic Adult Reconstruction Research Group (CCARR), Piuzzi NS

BACKGROUND: The Centers for Medicare & Medicaid Services (CMS) now requires patient-reported outcome measures (PROMs) collection after total hip arthroplasty (THA), emphasizing substantial clinical benefit (SCB) in the Hip Disability and Osteoarthritis Outcome Score - Joint Replacement (HOOS-JR). This study aimed to establish SCB thresholds for HOOS-Pain, Physical Function Short Form (PS), and JR and to identify factors influencing SCB achievement after THA.

METHODS: A prospective analysis included 13,324 primary THAs performed between 2016 and 2022 at a tertiary health care institution. Of these, baseline PROMs were completed by 11,766 patients (88.3%), and 1-year follow-up data were available for 9,229 patients (78.4%). Demographic characteristics, comorbidities, and PROM data were collected, and patients were stratified into eight phenotypes based on pain, function, and mental health status. The SCB thresholds were calculated using anchor-based methods and anchored by the Veterans RAND-12 physical health question at one year. Multivariable logistic regression was conducted to determine factors linked to not achieving SCB.

RESULTS: Established SCB thresholds were 40 for HOOS-pain (83.7% sensitivity, 45.3% specificity), 29.3 for HOOS-PS (75.4% sensitivity, 49.5% specificity), and 32.3 for HOOS-JR (79.3% sensitivity, 50.3% specificity). Overall, 72.8% of patients reached SCB. […]

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Standard of Care and Its Implications on Joint Arthroplasty: A Primer.

Hershfeld BE, Tarazi JM, Cohn RM, Scuderi GR, Mont MA, Bitterman AD

"Standard of care" refers to the level of clinical practice that a reasonably competent physician is expected to provide under similar circumstances, based on evidence, specialty guidelines, and professional consensus. This article reviews current literature on the standard of care in total joint arthroplasty and orthopaedic surgery, clarifying its definition, application, and role in reducing malpractice risk through evidence-based protocols. Specifically, the authors examined (1) the definition and evolution of standard of care; (2) the role of clinical guidelines, including arthroplasty specific; (3) legal implications in malpractice; (4) differences in elective versus emergent arthroplasty; (5) postoperative protocols central to practice; and (6) challenges and open questions in defining and updating standards of care.

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General, Spinal, and Regional Nerve Blocks: Do Different Anesthesia Practices Affect Same-Day Discharge in Primary Total Hip and Knee Arthroplasty?

Wing CW, Hatami B, Oseili AT, Ubl DS, Springer BD, Ardon AE, Ledford CK

BACKGROUND: The purpose of this study was to compare the effect of spinal anesthesia (SA) versus general anesthesia (GA) on acute 90-day outcomes in same-day discharge (SDD) total hip (THA) and total knee arthroplasty (TKA) patients, including subgroup analysis of peripheral nerve block usage.

METHODS: A retrospective review of 1,002 SDD patients (402 THAs, 600 TKAs) was performed, including 868 and 134 undergoing SA and GA, respectively. All patients received periarticular injections and multimodal analgesia; 80% of THAs received a paravertebral block (PVB), and 58% of TKAs received an adductor canal block (ACB). Intraoperative and postoperative narcotic usage (morphine milligram equivalents, MMEs) through SDD, numerical pain scores, first ambulation distance, hospital length of stay, and 90-day complications (readmissions or reoperations) were compared.

RESULTS: The SA performed in SDD THA resulted in decreased intraoperative MME (0.1 versus 8.7, $P < 0.01$), postoperative MME (12.3 versus 16.1, $P < 0.02$), and pain scores (1.6 versus 3.8, $P < 0.01$). A similar result was found in TKA, as SA patients had decreased intraoperative MME (0.1 versus 13.1, respectively, $P < 0.01$), postoperative MME (9.7 versus 15.5, $P < 0.01$), and pain scores (1.5 versus 3.2, $P < 0.01$). There was no difference in hospital length of stay between groups, but THA patients undergoing GA showed an increas […]

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Uncontrolled Diabetes Mellitus in the Year Prior to Total Hip or Knee Arthroplasty Is Associated with Similar Perioperative Glucose Control, Early Complication Rates, and 5-Year Reoperation Rates after Preoperative Diabetes Mellitus Optimization.

Umelo J, Jubril A, Burger PU, Tran JN, Balkissoon R, Kaplan NB, Thirukumaran CP, Ricciardi BF

BACKGROUND: Perioperative glycemic control is considered a modifiable risk factor prior to primary total hip and total knee arthroplasty; however, it is still unknown whether preoperatively optimizing hemoglobin A1C values affects perioperative glucose control or postoperative complications. The purpose of this study was to compare patients who required preoperative diabetes mellitus (DM) optimization in the year prior to surgery (defined as hemoglobin A1C ≥ 8%) versus patients who had DM who did not require preoperative DM optimization to compare (1) postoperative glycemic control and (2) 90-day complications and incidence of revision arthroplasty.

METHODS: This was a retrospective single-center analysis. Patients undergoing primary total joint arthroplasty who had a diagnosis of DM (hemoglobin A1C greater than 6.5%) were eligible for inclusion ($N = 585$). Patients who had previously uncontrolled DM requiring optimization (hemoglobin A1C ≥ 8.0%) in the year prior to surgery ($N = 164$) were compared to patients who had controlled DM ($N = 421$). All patients had hemoglobin A1C less than 8% preoperatively. Outcomes included median and peak perioperative glycemic control (postoperative day zero to two), 90-day complications, and incidence of revision arthroplasty at the final follow-up (mean 64 months). Multivariable logistic regressions evaluated associations of underlying preopera […]

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Associations Between Preoperative Seasonal Vaccination and Complications After Total Joint Arthroplasty.

Bcharah G, Wininger AE, Elsabbagh Z, Van Schuyver PR, Moore LL, Bingham JS

BACKGROUND: Vaccinations against common respiratory pathogens are recommended for the general population. However, associations between respiratory vaccinations and outcomes following total joint arthroplasty (TJA) have not been investigated.

METHODS: Using a national registry, 595,339 adults who underwent TJA were identified. Patients were stratified by influenza vaccination within three months before surgery and COVID-19 vaccination within three months of surgery or any time prior. Propensity-score matching controlled for demographics, comorbidities, and recent infections. Primary outcomes included 90-day and 2-, 3-, and 4-year rates of all-cause revision, periprosthetic joint infection (PJI), aseptic loosening, and periprosthetic fracture. The secondary outcomes included 90-day medical complications and mortality.

RESULTS: Influenza vaccination within three months of TJA was associated with higher 90-day medical complications, including deep vein thrombosis (odds ratio (OR) 1.38, 95% confidence interval (CI) 1.13 to 1.69) and pneumonia (OR 1.51, 95% CI 1.18 to 1.95), as well as PJI (OR 1.24, 95% CI 1.03 to 1.48) at four years. In contrast, COVID-19 vaccination within three months of TJA was associated with lower odds of pneumonia (OR 0.62, 95% CI 0.39 to 0.97) at 90 days compared to unvaccinated controls. At two years postoperatively, COVID-19 vaccination any time before T […]

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Caring for the Caregiver: Caregiver Preparation and Stress Following Total Joint Arthroplasty.

Sontag-Milobsky IL, Selph TJ, Madhan A, Pagadala MS, Adelani MA, Edelstein AI, Schwarzkopf R, Suleiman LI

BACKGROUND: Social support improves outcomes after total hip and knee arthroplasties (THA/TKA), but the demands on informal caregivers, especially as surgeries transition to outpatient care, are understudied. This study strived to assess caregiver burden, predictors, and implications following joint arthroplasty.

METHODS: This prospective cohort study enrolled 185 patient-caregiver dyads undergoing primary total hip arthroplasty or total knee arthroplasty for osteoarthritis at a tertiary academic center. Caregivers completed assessments at four weeks postoperatively, including the Caregiver Strain Index (CSI) and Appraisal of Caregiving Scale (ACS), which measure perceived benefit, threat, and stress. Demographic, socioeconomic, and caregiving-related variables were collected. Multivariate linear regression identified factors associated with caregiver strain and experiences. Caregivers had a mean age of 64 years (range, 52.3 to 76.3), and 60% were women. Most (72.4%) were spouses, and 46.5% were retired.

RESULTS: The CSI scores showed considerable strain, especially among women caregivers ($\beta = 1.29$, $P = 0.001$), those who had higher daily time commitment postoperatively, and those who had lower preoperative preparedness. Regarding employment status, 7% worked part-time, and 3.2% were homemakers. Among ACS subscales, non-White race ($\beta = 0.31$, $P = 0.035$) and homemaker status (β […]

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National Trends in Unicompartmental Knee Arthroplasty During a Period of Major Survivorship Improvements: Insights From the Dutch Arthroplasty Register.

Burger JA, Vossen RJM, Bayoumi T, Sierevelt IN, Ten Noever de Brauw GV, Spekenbrink-Spooren A, Zuiderbaan HA

BACKGROUND: The rising use of unicompartmental knee arthroplasty (UKA) among various Western countries raises questions regarding the multifaceted determinants driving this trend. Therefore, this study aimed to examine trends and their associations with 3-year survivorship, focusing on patient, implant, hospital, and revision-related factors.

METHODS: All primary medial UKAs for osteoarthritis registered in the Dutch Arthroplasty Register from 2014 to 2021 were included. Age, sex, BMI, prior ipsilateral knee surgery, implant type (cemented and uncemented mobile-bearing; cemented fixed-bearing), hospital practice (annual volume and usage), reasons for revision, and revision type were analyzed across four consecutive 2-year intervals. The 3-year survivorship was evaluated, and factors associated with revision risk were identified across these intervals.

RESULTS: A total of 27,234 medial UKAs were analyzed, revealing an increased patient age, a higher proportion of men, and fewer prior ipsilateral knee surgeries over time ($P < 0.001$). A profound increase occurred in uncemented mobile-bearing use and high-volume and usage hospitals ($P < 0.001$). The 3-year survivorship increased from 93.9 to 96.2% ($P < 0.001$). Higher hospital volume and usage, older patient age, and fewer prior ipsilateral knee surgeries were associated with reduced 3-year revision risk ($P < 0.001$) across various […]

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Factor XI and XIa Inhibitors for Venous Thromboembolism Prophylaxis After Total Knee Arthroplasty: A Review From Mechanism to Phase III Trials.

Smitterberg CW, Ando W, Chen AF, Garvin KL, Hamilton WG, Lee GC, Lieberman JR, Lombardi AV et al.

BACKGROUND: Venous thromboembolism (VTE) prophylaxis after total knee arthroplasty (TKA) has improved substantially, yet the field continues to balance two competing priorities: reduction of symptomatic deep vein thrombosis and pulmonary embolism while minimizing bleeding, wound complications, and downstream infection risk. Contemporary practice has shifted toward aspirin as prophylaxis for many low-risk patients after TKA, with direct oral anticoagulants (DOAC) and low-molecular-weight heparin (LMWH) reserved for higher-risk patients; however, these agents still affect common-pathway thrombin generation and can increase bleeding or compromise wound healing. Aspirin is not an anticoagulant and, despite its widespread use, does not directly address coagulation-driven thrombosis while still increasing bleeding risk, particularly gastrointestinal bleeding.

METHODS: This review summarizes the biologic rationale for targeting factor XI (FXI) and activated FXI (FXIa), synthesizes completed orthopaedic clinical evidence across various FXI/XIa modalities, and details the design features of ongoing Phase III trials evaluating anti-FXI molecules for post-TKA prophylaxis.

RESULTS: The FXI and FXIa sit upstream in the intrinsic pathway and may contribute disproportionately to thrombosis relative to hemostasis. In orthopaedic surgery, multiple phase II programs, including antisense oligon […]

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Recognizing Sarcopenia in Total Knee Arthroplasty Patients: A Call for Greater Awareness.

Ta■c■ M, Özer M, Çamur S

BACKGROUND: Suspected sarcopenia, marked by early muscle weakness and functional decline, is increasingly recognized in elderly patients and may negatively affect total knee arthroplasty (TKA) outcomes. Although formal diagnosis may not always be practical, early identification through simple screening tools such as the Strength, Assistance with walking, Rising from a chair, Climbing stairs, and Falls (SARC-F) questionnaire and basic functional tests can help detect at-risk patients and improve postoperative management and prognosis.

METHODS: This retrospective study included 97 patients who underwent unilateral TKA. Sarcopenia was assessed using the European Working Group on Sarcopenia in Older People 2 (EWGSOP2) algorithm. All patients completed the SARC-F and Sarcopenia Quality■of■Life (SarQoL), Oxford, and Knee Society Score (KSS), as well as pre- and postoperative Visual Analog Scale (VAS) scores.

RESULTS: Suspected sarcopenia was present in 34 patients (35.1%), while 63 patients (64.9%) had no suspicion. Probable sarcopenia based on handgrip strength was identified in eight patients, whereas no patient met the muscle mass cutoff for confirmed sarcopenia. Patients who had suspected sarcopenia were older (P= 0.05), had higher postoperative VAS scores (P = 0.05), and more frequently fell below handgrip cutoff values (P = 0.03). Gait speed was significantly slower in this g […]

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Intraosseous Sustained-Release Anesthetics Promote Early Ambulation and Functional Recovery After Total Knee Arthroplasty.

Huang J, Abdullah Ezzi SH, Wu S, Ali Alshabi MM, Zhang W, Malik MA, Cao X

BACKGROUND: Postoperative pain is a primary impediment to functional recovery in patients who undergo total knee arthroplasty (TKA).This work aimed to evaluate the efficacy of intraosseous sustained-release anesthetics, administered into the subchondral trabecular space before the insertion of a prosthesis during TKA, in promoting early ambulation and functional recovery.

METHODS: This randomized controlled open-label study was carried out on 110 patients who had a mean age of 75 years (range, 65 to 90). Patients were subdivided into two groups: local infiltration anesthesia (LIA) group: 10 mL of sustained-release anesthetic injection solution was taken, and infiltration anesthesia was applied in the joint and around the incision (two mL was injected into the posterior joint capsule and lateral collateral ligaments, two mL was injected into the knee-extension device and the parapatellar, and six mL was injected into the periarticular soft tissues); and LIA + intraosseous infiltration anesthesia (TIA) or group LIA plus TIA: a sustained-release anesthetic injection solution of 20 mL, 10 mL as described before plus intraosseous infiltration anesthesia on the osteotomy surface of the tibia (10 mL).

RESULTS: Visual analog scale, at rest and during walking, was significantly lower at days one and three in the group LIA plus TIA than in the group LIA ($P < 0.05$). Patient-controlled e […]

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A Comparison of the Efficacy and Complications Between Intraosseous and Periarticular Multimodal Analgesic Cocktail Injections After Primary Total Knee Arthroplasty: A Randomized Controlled Trial.

Suttikadsanee W, Pinsornsak P, Pinsornsak P

BACKGROUND: Early postoperative pain control after total knee arthroplasty (TKA) plays a crucial role in patient satisfaction following surgery. Despite current multimodal pain management strategies, postoperative pain remains a concern for patients and can hinder the decision to undergo TKA. With the advent of intraosseous (IO) analgesic injections, the efficacy of IO multimodal analgesic cocktail administration compared to standard periarticular (PA) injections remains controversial.

METHODS: A prospective randomized controlled trial was conducted comparing 45 patients who received an IO multimodal cocktail injection (five mg of morphine, 30 mg of ketorolac, 20 mL of 0.5% bupivacaine, and 0.6 mL of 1:1,000 epinephrine) with 45 patients who received a standard PA cocktail injection. A total of 43 patients in each group completed the study. Postoperative visual analog scale for pain at rest and during motion, morphine consumption, functional outcomes, and postoperative complications were recorded.

RESULTS: In the IO group, visual analog scale for pain was significantly lower both at rest and during motion at four, six, and 12 hours, as well as two weeks postoperatively. The Timed Up and Go test was faster in the IO group at 48 hours, but not at two weeks. However, morphine consumption, time to start walking, time to discharge, postoperative range of motion, and complications […]

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Physiologic and Clinical Sequelae After Pneumatic Tourniquet Release in Frail Patients Undergoing Total Knee Arthroplasty Under Regional Anesthesia: A Prospective Observational Study.

Patel G, Kahlon S, Singh A, Chipre S, Anand V

BACKGROUND: Tourniquet deflation during total knee arthroplasty (TKA) produces abrupt reperfusion physiology, including hyperkalemia, acidosis, and hemodynamic instability. Frail patients are particularly vulnerable; however, this population has not been prospectively studied.

METHODS: We conducted a prospective, observational study of frail adults (Clinical Frailty Scale ≥ 4) undergoing elective, unilateral, cemented TKA under spinal anesthesia with a pneumatic thigh tourniquet. There were 40 frail patients (mean age, 76 years [range, 65 to 88]; 65% women) included. Tourniquet duration averaged 84 minutes (range, 70 to 96). Patients with baseline hyperkalemia, severe renal dysfunction, significant arrhythmias, or planned revision or bilateral arthroplasty were excluded. Hemodynamics and arterial blood gases (pH, $PaCO_2$, lactate, potassium, and electrolytes) were recorded at baseline, predeflation, and 1- to 30-minute postdeflation. The primary endpoint was the change in serum potassium after tourniquet release, and hemodynamic instability was analyzed as a key clinical outcome. The secondary outcomes included the incidence of hyperkalemia (≥ 5.5 mmol/L), acidosis (pH less than 7.30), hypotension, arrhythmias, postanesthesia care unit (PACU) interventions, acute kidney injury, and hospital lengths of stay.

RESULTS: After deflation, serum potassium rose from 4.3 ± 0.3 to 4.8 ± […]

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Osteonecrosis of the Femoral Head: A Dysbiotic Condition?

Ferrini A, He M, Mont MA, Goodman SB, Parvizi J

Osteonecrosis of the femoral head (ONFH) is a progressive and disabling condition of the hip joint that primarily affects young and active individuals, leading to progressive collapse of subchondral bone and often secondary arthritis. Despite extensive investigation, the precise etiology often remains unclear. While high-dose corticosteroids, chronic alcohol ingestion, and smoking are known associated risk factors, approximately 20 to 30% of ONFH cases are classified as idiopathic. Recently, the concept of gut dysbiosis, i.e., disruption of the normal intestinal microbial balance, has gained increasing attention due to its systemic immunologic and metabolic implications. Dysbiosis is associated with an increase in gut permeability, leading to the translocation of bacteria and their metabolic products, including lipopolysaccharides and short-chain fatty acids, into the systemic circulation. This may stimulate proinflammatory cascades throughout the body, including the joints, initiating a bone remodeling process. Emerging evidence from preclinical and human research suggests that specific gut microbiota taxa may influence key mechanisms involved in the pathogenesis of ONFH. In addition, early findings support the therapeutic potential of microbiota-targeted therapies such as probiotics, short-chain fatty acids-enriched diets, and fecal microbiota transplantation. Although a grow […]

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Cemented Femoral Stem Design Is Not Associated With Risk of Revision After Total Hip Arthroplasty in Patients Aged ≥ 65 Years: An Analysis of the American Joint Replacement Registry.

Kelly M, Hegde V, Zaniletti I, Anderson L, Gililland JM, De A, Kagan R

BACKGROUND: Cemented femoral fixation for total hip arthroplasty (THA) in those aged ≥ 65 years has potential benefits. However, few resources exist to assist in selecting cemented femoral implant designs. We examined the associated risk for all-cause revision, risk of periprosthetic femoral fracture, and aseptic loosening based on a modern classification of cemented femoral stem designs.

METHODS: An analysis of primary THA cases in patients aged ≥ 65 years was performed with American Joint Replacement Registry (AJRR) data linked to Centers for Medicare and Medicaid Services data from 2012 to 2023. Patient demographics, revision or open reduction and internal fixation (ORIF) for periprosthetic femoral fracture, and revision for aseptic loosening were recorded. We identified 20,484 primary THAs with cemented stems that were classified via a modern classification. There were 5,437 (26.5%) type I (taper-slip) and 14,796 (72.2%) type II (composite beam) stem designs. Cause-specific Cox proportional hazard models with competing risk of death were used to evaluate the association of cemented stem design with all-cause revision, revision or ORIF due to periprosthetic femoral fracture, and revision for aseptic loosening while adjusting for potential confounders.

RESULTS: Compared to those who had type I (taper-slip) designs, those with type II (composite beam) designs were not associ […]

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Low Follow-Up Rates After a Total Hip Arthroplasty Prosthesis Recall.

Cooper LH, Wahl SD, Stambough JB, Siegel ER, Mears SC, Barnes CL, Stronach BM

BACKGROUND: A modern hip arthroplasty liner was recalled for premature polyethylene wear with the potential for catastrophic osteolysis and failure. Physicians and hospitals are responsible for communicating recall information to patients. Our center instituted a protocol to reach patients who had recalled implants for follow-up. Our study looked at the effectiveness of this effort and risk factors for nonresponse.

METHODS: A retrospective study was performed to measure follow-up after total hip arthroplasty device recall. A standardized letter was used to reach out to all patients to notify them of the implant recall. Demographic information was collected, and an analysis was performed to determine risk factors for nonresponse.

RESULTS: Of the 763 patients, 58.8% (n = 449) followed up after recall letters were sent. There were significant differences in the follow-up rate found among the four surgeons who performed the total hip arthroplasties. Patient proximity to the performing institution (n = 264), performing surgeon, and adequate health literacy (n = 677) were significant predictors of follow-up ($P < 0.05$). Age, sex, body mass index, and race were not associated with follow-up.

CONCLUSION: The overall rate of follow-up was less than 60% despite an organized process for contacting patients to return for evaluation after implant recall. Our findings raise concerns for th […]

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Can External Rotator Preservation Reduce Dislocation Risk Following Posterior Approach Total Hip Arthroplasty in High-Risk Patients? A Propensity-Matched Comparative Study.

Won SH, Joh Y, Lim YW, Kwon SY, Kim YS, Kim SC

BACKGROUND: Although multiple studies have examined the efficacy of muscle-sparing techniques in reducing dislocation after posterior approach THA, their findings are limited by small sample sizes and short follow-up durations, and to our knowledge, none have specifically evaluated outcomes in high-risk patients. This study compared the overall and 90-day dislocation risks of the previously described external rotator preservation (ERP) and standard posterior approach THA in high-risk patients.

METHODS: We retrospectively reviewed patients who underwent THA between January 2008 and December 2022 at three tertiary centers, including those who had osteonecrosis of the femoral head (ONFH), inflammatory arthritis (IA), or a history of lumbar spinal fusion. Patients who underwent the ERP approach were identified from medical records and matched 1:1 to controls (standard posterior approach) by age, sex, body mass index, American Society of Anesthesiologists score, year of surgery, and surgical indication, yielding 501 patients per group. Binary logistic regressions were used to compare overall and 90-day dislocation risk. Kaplan-Meier analysis evaluated revision due to dislocation and all-cause revision. Surgical complications, Harris Hip Score (HHS), and postoperative radiographic parameters were compared. The mean follow-up was 10.7 years.

RESULTS: The ERP group showed lower overa […]

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Primary Total Hip Arthroplasty in Patients Who Have a Body Mass Index > 50: Is the Risk Worth the Reward?

Righolt CH, Finney C, Turgeon TR, Bohm ER, Sniderman J

BACKGROUND: The obesity epidemic has given rise to a growing number of patients who have a body mass index (BMI) exceeding 50. As BMI cutoffs are no longer recommended, arthroplasty surgeons continue to push the limits of surgical feasibility in total hip arthroplasty (THA) despite possible risks.

METHODS: A retrospective cohort study of patients who underwent primary THA (N = 7,458) was developed, comparing outcomes between patients who had a BMI ≥ 50 (N = 147) and those of other weight classes. We used Cox proportional hazard models to estimate the association between patient BMI and revision risk using overweight patients (BMI 25 to 30) as the reference group. Patients entered the study cohort on their date of surgery and exited on the earliest of the date of revision, date of death, or the end of the study. Patient-reported outcome measures were compared pre- and postoperatively.

RESULTS: Increased obesity class led to a gradient of elevated revision risk. In the first year after surgery, THA patients who had a BMI ≥ 50 demonstrated an adjusted hazard ratio (for revision of 11.8 (95% confidence interval [CI] 5.3 to 26.2). This risk was also elevated in patients who had a BMI of 40 to 50 (N = 635, hazard ratio = 3.8, 95% CI = 1.9 to 7.7). Although the patients who had a BMI ≥ 50 had a significantly worse preoperative Oxford-12 Hip score than other overweight patients, at f […]

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Is the Use of New Ceramic Heads With Titanium Sleeves on Retained Femoral Stems in Revision Total Hip Arthroplasty Associated With Femoral Head or Neck Junction Failure at Mean Eight-Year Follow-Up?

Seah MKT, Howard LC, Horwood NJ, Masri BA, Garbuz DS, Neufeld ME

BACKGROUND: Ceramic heads used in revision total hip arthroplasty (rTHA) typically utilize a titanium sleeve adapter to prevent damage by the retained stem taper; however, there is sparse mid-to long-term data on this practice, and risk factors for failure remain relatively unknown. This study aimed to determine revision-free survival, specifically for failure of the head and/or neck junction (HNJ) and all-cause re-revision in this patient population, and to identify factors associated with failure.

METHODS: We identified all consecutive aseptic rTHA that used a new ceramic head with a titanium sleeve on a retained stem at our institution between 2011 and 2022. A total of 316 rTHA (295 patients) were included with a mean follow-up of 7.7 years (range, 2.1 to 14.3). The mean age was 65 years, and 57% were women. The main indications for rTHA were adverse local tissue reaction (46%), instability (29%), and acetabular aseptic loosening (12%). Isolated modular component exchange was performed in 38% of revisions. Kaplan-Meier analysis was used to determine survivorship. We aimed to use multivariate logistic regressions to identify risk factors for HNJ failure and all-cause rerevision.

RESULTS: There were no rerevisions for HNJ failure. There were 66 hips (21%) that underwent rerevision at a mean of 1.8 years, most commonly for instability (39 of 66), infection (17 of 66), and ase […]

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Aseptic Revision Total Hip Arthroplasty in the Elderly: Is There a Higher Incidence of Early Complications?

Dandamudi S, Jones CM, Majji I, Yadav AS, Dunleavy ML, Levine BR, Behery OA

BACKGROUND: There is conflicting evidence on whether revision total hip arthroplasty (rTHA) in elderly patients (age ≥ 80 years) predisposes them to increased early complications compared to younger patients. This study compared 90-day complications, readmissions, and reoperations following aseptic rTHA between these two cohorts.

METHODS: We retrospectively identified all patients who underwent aseptic rTHA at a single academic institution from 2002 to 2023. Patients who had a history of periprosthetic joint infection, less than 90-day follow-up, or missing data were excluded. A total of 124 patients aged ≥ 80 years were eligible and matched to 124 rTHA patients aged less than 80 years using 1:1 propensity score-matching based on revision type (modular exchange, single-component, or two-component revision), body mass index (BMI), sex, and American Society of Anesthesiologists (ASA) score. Demographic data and outcomes of interest were analyzed using nonparametric tests and multivariate regression models.

RESULTS: The Charlson comorbidity index (CCI) was significantly higher in the ≥ 80 group ($P < 0.001$). No differences were observed in rTHA indications ($P > 0.132$) or 90-day medical or surgical complications ($P > 0.052$). Patients of ≥ 80 years were more likely to undergo reoperation for dislocation ($P = 0.035$), though the difference was small and likely not clinically relevant […]

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Subsidence of Modular Fluted Tapered Stems After Femoral Revision Surgery: Risk Factors and a Novel Classification System.

Mahlouly J, Terrier A, Borens O, Meylan A, Wegrzyn J, Steinmetz S

BACKGROUND: Modular fluted tapered (MFT) stems are widely used in femoral revision surgery for their ability to achieve diaphyseal fixation in the setting of bone loss. This study aimed to identify factors influencing MFT stem subsidence.

METHODS: We retrospectively analyzed a cohort of 443 femoral revision procedures performed at a single institution, all using the same model of MFT stem. A total of 180 procedures met the inclusion criteria (≥ four radiographs) and were analyzed for subsidence during follow-up using a radiographic analysis. The mean radiological follow-up was 28.5 months (range, one to 120). Subsidence ≥ five mm was defined as excessive. Implant survivorship was assessed by rerevision rate. The relationship between clinical variables and excessive subsidence was examined using logistic regression and Bayesian modeling, guided by a directed acyclic graph.

RESULTS: The mean stem subsidence at 12 months was 1.6 ± 2.3 mm. Among the full cohort, 18 stems (5.0%) underwent rerevision at a mean of 71.4 months. Excessive subsidence (≥ five mm) was observed in 8.3% of cases. Associative analysis identified severe bone defects (Paprosky IIIA to IIIB) as the main factor associated with excessive subsidence (odds ratio = 6.35, $P = 0.001$). Short MFT stems, extended trochanteric osteotomies, and men showed weaker, nonsignificant associations. Causal modeling confirmed the […]

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Identifying Risk Factors for Aseptic Loosening and Infection Following Distal Femoral Replacement for Periprosthetic Fracture in the Nononcologic Patient.

Brady TT, Bera SR, Grammatopoulos G, Adamczyk AP

BACKGROUND: Distal femoral replacement (DFR) is increasingly used for periprosthetic distal femoral fractures (PDFFs). Although infection and aseptic loosening are known complications, predictors, particularly of loosening, remain poorly defined in nononcologic populations. This study aimed to identify patient risk factors for aseptic loosening and infection following DFR for PDFF in a nononcologic cohort. Additionally, DFR utilization trends for PDFF were assessed over time.

METHODS: A retrospective study was conducted using an insurance claims database (2015 to 2023). Aseptic loosening, periprosthetic joint infection (PJI), and DFR for PDFF were identified using side-specific diagnostic and procedural codes. Oncologic patients were excluded. Kaplan-Meier endpoints were diagnosis of aseptic loosening or PJI. Failure was considered a reoperation. Multivariable logistic regressions identified independent risk factors. A total of 1,002 DFRs were included; 85% of patients were women, and 82% were over 65 years of age.

RESULTS: After excluding cases with ipsilateral PJI, aseptic loosening occurred in 6.6% of patients (n = 57), of which 39 (4.6%) underwent reoperation. Men (odds ratio [OR] 4.11) and obesity (OR 2.67) were independently associated with increased odds of loosening. A PJI occurred in 159 patients (15.9%), of whom 103 (10.3%) underwent reoperation. Men (OR 3.50), obes […]

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Draining Sinus Tracts and Periprosthetic Joint Infections: Traditional Synovial Fluid Counts May Be Misleading.

Alder KD, Dugdale EM, Osmon DR, Bedard NA, Berry DJ, Abdel MP

BACKGROUND: Serologic and synovial laboratory values guiding the diagnosis of periprosthetic joint infection (PJI) are well-established. We hypothesized that these values would be lower in patients who had a draining sinus tract. The purpose of this study was to determine if patients who had sinus tracts had lower values for serum erythrocyte sedimentation rate (ESR), serum C-reactive protein (CRP), synovial white blood cell (WBC) count, and synovial percent neutrophils.

METHODS: We retrospectively reviewed 611 primary total joint arthroplasties treated for PJI between 2000 and 2020 at a high-volume academic center. Only 191 patients (94 hips, 97 knees) who met the 2011 Musculoskeletal Infection Society (MSIS) major criteria for PJI and did not receive antibiotics within two weeks before laboratory evaluation were included. Patients were divided into those who had sinus tracts (n = 58) and those who did not (n = 133). Laboratory values for early (less than 90 days) and late (≥ 90 days) postoperative PJI cases were analyzed separately. False negative rates for detecting PJI were compared based on the 2018 International Consensus Meeting (ICM) cutoffs for each diagnostic test.

RESULTS: Median synovial WBC count was significantly lower in early (9,978 versus 76,068 cells/μL; P = 0.01) and late (20,365 versus 63,356 cells/μL; P = 0.03) PJI in those who had a sinus tract. False ne […]

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Multiple Primary Joint Arthroplasties and the Risk of Periprosthetic Joint Infection: Evidence From a Large Retrospective Cohort.

Schaffler BC, Prinos A, Kennedy MF, Ehlers M, Rozell JC, Schwarzkopf R

BACKGROUND: There is a growing number of patients who undergo multiple primary hip and knee joint arthroplasties during their lifetime. Whether patients who have multiple replaced joints are at an increased long-term risk of periprosthetic joint infection (PJI) is not known. The purpose of this study was to compare rates of PJI in patients who have more than one primary arthroplasty.

METHODS: We reviewed 36,125 patients who underwent primary total joint arthroplasty at a single institution from 2011 to 2024. Patients were categorized as having one to four primary hip or knee arthroplasties. The PJI incidence was compared using Chi-square testing and binary logistic regressions, and multivariate models adjusted for sex, body mass index, diabetes, renal disease, smoking status, and Charlson Comorbidity Index. Subanalyses compared patients who had one versus two, three, and four arthroplasties.

RESULTS: When comparing patients who had one, two, three, or four primary joint arthroplasties, there was no significant difference in the rates of PJI between groups (P = 0.112). Multivariate analyses showed no statistically significant association between the number of arthroplasties and PJI (adjusted odds ratio for two, three, and four arthroplasties versus one: 1.34, 95% confidence interval (CI) 0.96 to 1.74, P = 0.083; 1.98, 95% CI 0.77 to 4.12, P = 0.105; 1.57, 95% CI 0.09 to 7.24, […]

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Artificial Intelligence-Driven Decision-Making for Knee Joint Manipulation Following Primary Total Knee Arthroplasty.

Ghirardelli S, Chan KCA, Valpiana P, Indelli P, Sculco PK, Lee GC

BACKGROUND: Manipulation under anesthesia (MUA) is a commonly performed procedure to address postoperative stiffness after total knee arthroplasty (TKA), yet optimal timing, motion thresholds, and outcome expectations remain poorly defined due to limited high-quality evidence and heterogeneous patient presentations.

METHODS: This study evaluated the capacity of an artificial intelligence (AI) platform to synthesize a data-driven decision-making framework for MUA following primary TKA. A systematic literature review on the range of motion (ROM) trajectory and manipulation following primary TKA was performed, and the data were used to train the AI model. Scenarios iterating variables, including time from surgery, knee flexion angle, extension deficit, preoperative ROM, and body mass index, were modeled to define the appropriateness of MUA and expected ROM gains.

RESULTS: According to our AI model, MUA is indicated for patients who have knee flexion < 80 degrees and/or extension deficits greater than 20 degrees at six weeks postoperatively, predicting mean improvements of 26 degrees in flexion and three degrees in extension. Delaying MUA beyond 90 days reduced expected flexion gains to 17 degrees. For persistent stiffness beyond three months, the model advised combining MUA with arthroscopic lysis of adhesions, particularly for flexion less than 90 degrees. A body mass index gre […]

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Predicting Acetabular Fixation Failure and Bone Loss in Total Hip Arthroplasty: A Combined In Silico and In Vitro Approach.

Ramos A, Matos M

BACKGROUND: Total hip arthroplasty continues to rise in prevalence in both the United States and Europe. The leading cause of revision surgery remains acetabular failure, primarily associated with instability and peri-implant bone loss. This study aimed to investigate bone loss around acetabular components following total hip arthroplasty.

METHODS: A combined in silico (finite element method, FEM) and in vitro approach was employed to predict short- to mid-term bone loss and assess acetabular implant stability. The FEM model incorporated anatomically realistic geometries of the femur, ilium, and implants (acetabular shell, polyethylene liner, and stem). The in vitro model utilized composite bones instrumented with strain gauges to measure cortical strains. Bone loss and acetabular loosening were simulated in four progressive steps, guided by FEM strain analysis, and reproduced through sequential computer-controlled manufacturing in the in vitro setup to evaluate implant stability. Loading conditions reproduced hip joint forces during gait.

RESULTS: Cortical strain measurements demonstrated strong agreement between the in silico and experimental models (R2 = 0.91). Acetabular implantation reduced cortical bone strains by approximately 60% posteriorly and 80% anteriorly. Bone loss was most pronounced in the posterior-superior region and at cortical areas adjacent to the implant […]

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The Fate of the Patient Who Has Early Dislocation After Contemporary Primary Total Hip Arthroplasty.

Fuqua AA, Hrudka BT, Collins EC, Ross BJ, Premkumar A, Wilson JM

INTRODUCTION: Recurrent instability remains a leading indication for revision following primary total hip arthroplasty (THA). However, data regarding outcomes following early dislocation are dated and limited in scope. Whether these existing data apply to contemporary practice with the broad adaptation of high-stability bearings and modern surgical techniques is unknown. Therefore, we assessed revision and redislocation rates following early dislocation in a contemporary group of patients following primary THA.

METHODS: A national database was used to identify patients undergoing primary THA from 2009 to 2022. Patients were divided into two cohorts: those with one or more dislocation events within 90 days and those who did not dislocate as the control group. Landmark survivorship analysis was utilized, given the potential for immortal time bias. The 2-year cumulative incidence of aseptic revision, septic revision, and recurrent dislocation after 90 days was determined by competing risk analysis.

RESULTS: At two years, the dislocation rate in all patients undergoing a single primary THA (n = 420,053) was 1.2%. After landmark analysis, the incidence of aseptic revision or dislocation after 90 days in patients with an unrevised early dislocation event (n = 1,942) was 29.0 versus 1.3% in controls (n = 360,493) (P < 0.001). The incidence of either dislocation or aseptic revision a […]

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Concentration, Antibacterial Effect, and Safety of Combining Intra-articular Epsilon-Aminocaproic Acid and Vancomycin in Primary Total Knee Arthroplasty: A Randomized Study.

Zhang Y, Wei J, Su C, Hu M, Xu H, Xiang S

BACKGROUND: In this study, we aimed to investigate the intra-articular concentration, antibacterial effect, and safety of a combination of vancomycin and epsilon-aminocaproic acid (EACA) in primary total knee arthroplasty (TKA).

METHODS: There were 94 patients who underwent unilateral primary TKA and were randomized into the vancomycin plus EACA (VE) group and the vancomycin (V) group. There were 47 contemporary TKAs propensity score-matched to form the EACA (E) group. The VE group received vancomycin 30 mL (1,000 mg) plus EACA 20 mL, the V group received vancomycin plus saline, and the E group received saline plus EACA. Postoperative blood and drainage samples were collected for vancomycin analysis using liquid chromatography-tandem mass spectrometry. Perioperative blood loss, renal function, and bacterial tests for common periprosthetic joint infection pathogens were evaluated.

RESULTS: Perioperative blood loss in the VE group was equivalent to that in the E group and significantly lower than that in the V group (P < 0.01). The intra-articular vancomycin levels in the VE group were higher than those in the V group at eight and 24 hours, while the serum levels were similar. Antimicrobial effects were comparable in the VE and the V groups, both of which were superior to those in the E group. There were no cases of acute renal injury, ototoxicity, or anaphylaxis.

CONCLUSIONS: […]

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Moderate-to-Severe Varus Deformity Is Associated With Conversion to Arthroplasty in Patients Who Have Subchondral Insufficiency Fracture of the Knee.

Park JY, Cho BW, Kim TH, Park KK, Lee WS, Moon J, Kwon HM

BACKGROUND: Although untreated subchondral insufficiency fractures of the knee may result in progression of osteoarthritis, leading to arthroplasty in approximately 30% of cases, the factors associated with the requirement for arthroplasty remain unknown. We hypothesized that varus deformity would be associated with increased risk of nonoperative treatment failure and conversion to arthroplasty in patients who have subchondral insufficiency fractures with early osteoarthritis.

METHODS: A retrospective cohort study was conducted on 162 patients aged greater than 60 years diagnosed with subchondral insufficiency fracture and early osteoarthritis (Kellgren-Lawrence grades 1 to 2) between March 2015 and December 2021. All patients received initial nonoperative treatment and were followed for a minimum of 36 months. Radiographic evaluation included hip-knee-ankle (HKA) angle measurement and osteoarthritis grading. Magnetic resonance imaging assessment evaluated subchondral insufficiency fracture location, meniscal tears, cartilage defects, and bone marrow edema. Kaplan-Meier survival analyses and Cox proportional hazards regressions were used to identify risk factors for progression to arthroplasty. At a median follow-up of 44 months, 50 patients (30.9%) required conversion to arthroplasty due to osteoarthritis progression.

RESULTS: Patients who had moderate-to-severe varus deform […]

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Excision Versus Preservation of the Infrapatellar Fat Pad During Total Knee Arthroplasty: A Systematic Review and Meta-Analysis.

Skaik K, Oulousian S, Lameire DL, Abbas A, Khalik HA, Al-Obaedi O, Ravi B

BACKGROUND: This study aimed to evaluate whether preservation versus resection of the infrapatellar fat pad (IPFP) during total knee arthroplasty (TKA) affects clinical outcomes.

METHODS: A systematic review and meta-analysis, searching multiple databases up to November 2024 for comparative studies of infrapatellar fat pad-preservation (IPFP-P) versus resection in primary TKA, was conducted. Outcomes assessed included rates of complications, visual analog scale for pain, and the rate of anterior knee pain, Knee Society Score, patellar tendon length, range of motion, and operative time. Data were pooled using random-effects meta-analysis. There were 21 studies (3,573 patients, 4,107 TKAs: 2,298 preserved, and 1,809 resected IPFP) included.

RESULTS: The IPFP-P had a 76% reduction in the rate of complications within six weeks (relative risk = 0.24, 95% confidence interval [CI]: 0.12 to 0.48, P < 0.01), but not at later follow-up. The visual analog scale pain scores did not differ significantly between groups at any time point. The IPFP-P reduced the risk of anterior knee pain at three months (relative risk = 0.12, 95% CI: 0.02 to 0.94, P = 0.04), but this was not sustained at later follow-up. There were no significant differences found for operative time and Knee Society Score. However, infrapatellar fat pad-resection resulted in greater patellar tendon shortening at six and 12 […]

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Prevalence, Progression, and Clinical Impact of Stem Notching in Ceramic-on-Ceramic Total Hip Arthroplasty: A Minimum 15-Year Follow-Up Study.

Kang SY, Loh LL, Jeon YJ, Kim HS, Yoo JJ

BACKGROUND: Stem notching, resulting from impingement between the femoral stem and ceramic liner in ceramic-on-ceramic (CoC) total hip arthroplasty (THA), has been linked to adverse outcomes such as ceramic-related noise and ceramic component fractures. Despite its potential significance, its true incidence is infrequently documented, and the long-term clinical impact remains uncertain. This study aimed to assess the minimum 15-year outcomes and complications associated with stem notching in CoC THA.

METHODS: We performed a retrospective cohort analysis of patients who received CoC THA between November 1997 and December 2003, with a minimum follow-up period of 15 years. Stem notching was diagnosed using radiographic evaluation and was observed in 21.5% of cases (63 of 293). The lesions were monitored for an average of 12.5 years from detection (range, one to 23.2). The primary endpoints included changes in notch depth and the occurrence of ceramic-related complications such as ceramic component fractures and noise generation. The secondary outcomes involved functional evaluation via the modified Harris Hip Score.

RESULTS: Notch depth did not demonstrate meaningful progression beyond five years after identification. Lower cup inclination and higher anteversion were associated with the presence of notching. The incidence of ceramic component fractures was 6.3% in notched hips c […]

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Does Traumatic Brain Injury Increase Opioid Utilization After Primary Total Knee Arthroplasty?

Young B, Shankar D, Sabet CJ, Leopold AP, Fernando ND, Hernandez NM

BACKGROUND: Traumatic brain injury (TBI) is a common neurological injury with widespread systemic effects, leading to increased pain and opioid utilization. In this study, we investigated how a prior TBI affects perioperative opioid utilization in patients undergoing a total knee arthroplasty (TKA) and the risk of prolonged opioid usage.

METHODS: Using an administrative claims database, we identified patients undergoing a primary TKA from 2010 to 2022. Following inclusion and exclusion criteria, patients were categorized based on a prior diagnosis of a TBI, leading to a final TBI cohort of 127,369 patients and a control cohort of 1,116,605 patients. Our primary outcome was perioperative opioid utilization, defined as any opioid prescription between 30 days before and after surgery. Our secondary outcome was persistent opioid usage or continued opioid prescriptions 90 to 180 days after surgery. Multivariate regression models were used to assess the risk of persistent opioid usage based on TBI history, adjusting for demographics and comorbidities.

RESULTS: Patients who had a prior TBI had greater perioperative opioid utilization compared to patients who did not have a prior TBI, with 987.5 versus 896.7 morphine milligram equivalents (P < 0.001). The TBI cohort also showed greater rates of persistent opioid usage compared to the control cohort: 22,258 patients (17.5%) versus 162 […]

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Minimum 10-Year Outcomes of Total Hip Arthroplasty With Highly Cross-Linked Polyethylene in Patients Under 50 Years: A Systematic Review and Meta-Analysis.

Randall ZD, Mologne MS, Gaziano DJ, Clohisy JC, Bendich I

BACKGROUND: Total hip arthroplasty (THA) is increasing in utilization among young patients. While THA markedly improves outcomes in those who have degenerative hip disease, patients under 50 years face an extended lifetime risk for complications and revisions. Highly cross-linked polyethylene (HXLPE) is associated with reduced wear, revision rates, and osteolysis compared to conventional polyethylene, potentially enhancing implant longevity in this younger, active population. This study systematically reviewed and meta-analyzed the outcomes of THA using HXLPE in patients under 50 years of age at a minimum 10-year follow-up.

METHODS: A systematic review was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. Searches were performed in Embase, Ovid Medline, Scopus, and Cochrane databases. Inclusion criteria targeted studies with patients under 50 years reporting original data on THA using HXLPE with a mean follow-up of ≥ 10 years. Data extraction was performed independently by two reviewers, and study quality was evaluated using the Methodological Index for Non-Randomized Studies (MINORS) criteria. Multiple random-effects meta-analyses were performed to summarize outcomes. There were 18 studies included (n = 1,587 hips) with a weighted mean follow-up of 14.1 years.

RESULTS: Overall, revision-free survivorship was 98%, […]

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The Impact of Major Depressive Disorder on Postoperative Somatic and Psychiatric Complications Following Total Knee Arthroplasty: A Database Study.

Wahid M, Zaidi Z, Syed S, Nasser E, Meza C, Chen AF

BACKGROUND: Major depressive disorder (MDD) is a psychiatric condition characterized by persistent low mood, anhedonia, sleep disturbances, guilt, fatigue, impaired concentration, appetite changes, psychomotor abnormalities, and suicidal ideation. It is a prevalent comorbidity among patients undergoing total knee arthroplasty (TKA) and may affect recovery. While prior research has emphasized pain and functional outcomes, the broader impact of MDD on postoperative psychiatric, systemic, and mortality outcomes remains underexplored. This study investigated whether MDD is independently associated with increased risk of newly diagnosed psychiatric, somatic, and mortality complications following primary TKA.

METHODS: In a novel analysis of a large electronic health record network, we identified adult patients undergoing primary, elective TKA from 2015 to 2024, excluding those who had prior knee pathology or psychiatric diagnoses other than MDD. A propensity score matching analysis adjusted for demographics, comorbidities, laboratory values, and medication use was used to compare postoperative outcomes at 90 days and one year using odds ratios (ORs).

RESULTS: Compared to controls, MDD patients experienced significantly higher odds of developing generalized anxiety disorder (OR90day 13.66; OR1year 11.73), adjustment disorder (OR90day 11.11; OR1year 6.91), post-traumatic stress disor […]

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Delayed Administration of Apixaban, but Not Rivaroxaban, Reduces Transfusion Risk Without Increasing Thromboembolic Rates Following Revision Total Hip Arthroplasty.

Telang SS, Lim MA, Kumaran P, Culler MW, Palmer R, Telang S, Burdick G, Lieberman JR et al.

BACKGROUND: The optimal timing for initiating direct oral anticoagulants for venous thromboembolism prophylaxis following revision total hip arthroplasty (rTHA) is unknown. This study compared rates of thromboembolic and bleeding complications between patients beginning prophylactic courses of apixaban or rivaroxaban on postoperative day (POD) zero versus POD one.

METHODS: A national health care database including approximately 25% of all surgeries performed in the United States was used to identify patients who underwent aseptic both-component rTHA between 2016 and 2023. Patients receiving apixaban or rivaroxaban for chemoprophylaxis after rTHA on POD zero were compared to patients who received the same anticoagulant on POD one. Primary outcomes included 90-day rates of postoperative bleeding complications (acute anemia, hematoma, hemorrhage, and transfusion) and thromboembolic complications (deep vein thrombosis, pulmonary embolism, stroke, and myocardial infarction). Multivariable regression analysis was used to assess differences between cohorts. In total, 5,607 patients were identified. Of the 2,990 (53.33%) patients administered apixaban, 646 (21.61%) received anticoagulation beginning POD zero compared to 2,344 (78.39%) on POD one. Of the 2,617 (46.67%) patients given rivaroxaban, 598 (22.85%) received anticoagulation on POD zero compared to 2,019 (77.15%) on POD one. […]

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Recovery Benchmarks After Total Knee Arthroplasty: The Impact of Motion-Restoring Surgery, Contracture, and Comorbidities on Costs and Utilization.

Karim M, Beckley SJ, Stinton SK, Branch TP

BACKGROUND: This study aimed to establish a claims-based timeline for recovery following unilateral total knee arthroplasty (TKA), including the impact of comorbidities and postoperative complications such as motion-restoring surgery (MRS) and joint contracture.

METHODS: Health care claims data were analyzed to determine the costs and recovery timeline after unilateral TKA. Costs related to (1) index surgery, (2) complication surgery, (3) TKA revision, (4) nonoperative hospitalization, (5) MRS, and (6) outpatient surgery were reported. The effect of joint contracture and comorbidities, including diabetes, obesity, peripheral vascular disease, joint infection, and cardiovascular disease, was assessed. The effect of postoperative complications was determined using data from patients rehospitalized with or without additional surgery.

RESULTS: Index surgery median cost was $30,346 (interquartile range: 14,203 to 350,304). Median recovery time (from index surgery to last physical therapy claim) was 126 days (interquartile range: 30 to 975), based on physical therapy billing claims, which may not reflect full functional recovery. While 54% of patients completed their postoperative period in six months, 46% took over six months. Recovery periods for patients who required a complicated surgery were six times longer, and costs were 14 times higher than for those who did not have compl […]

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Fractures of the coronoid process: state of the art.

Marinelli A, Riva M, Coliva F, Minerba M, Carbone G, Guerra E

Coronoid fractures are rarely isolated and are much more frequently associated with other osseous or ligamentous structures injuries. On the basis of the coronoid fracture patterns, described by the O'Driscoll classification, it is possible to recognize three main patterns of injury that differ on traumatic mechanism and on associated lesions: posterolateral rotatory instability, posteromedial rotatory instability, and axial load injuries. The management of coronoid fractures is challenging and varies according to characteristics of the fracture, associated lesions, and amount of elbow instability. In general, operative treatment is indicated in every case the fracture is at least 50% of the whole coronoid, whether the sublime tubercle is involved, and whether the ulno-humeral joint is not perfectly reduced. In conclusion, the correct management of the coronoid, especially in the setting of complex elbow instability, represents a predictive factor for patient outcomes and functional results. The stability of the elbow, rather than the size of the coronoid fragment, is the main parameter for surgical indication, aimed to fix the coronoid and/or repair the associated lesions.

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Total hip arthroplasty in Italy: an observational, population-based study on surgical volume growth from 2001 to 2023 and forecasts until 2050 with six different statistical models.

Ciminello E, Cuccu A, Romanini E, Venosa M, Cazzato G, Tucci G, Boniforti F, Carpanese L et al.

BACKGROUND: The number of total hip arthroplasty (THA) procedures has been steadily increasing worldwide, driven by aging population, improvements in surgical techniques and implant design. This study aimed to analyze the temporal trends of elective THA in Italy since 2001-2023 and forecast THA volumes up to 2050 to provide insights for healthcare planning.

MATERIALS AND METHODS: International Classification of Diseases, 9th Revision, Clinical Modification (ICD9-CM) coding system was used to extract records of interest (elective THA) from the Italian National Hospital Discharge Record database. Six statistical models were applied to forecast future THA volumes: logistic regression; Poisson regression; logarithmic regression; inverse/power regression; Poisson log-normal regression; and hierarchical Poisson regression with temporal effects (HPTE). Model performances were assessed by using error metrics and internal validation on the basis of a rolling-origin approach. An out-of-sample validation was conducted to ensure a robust assessment of forecasting reliability. THA volume forecasts were provided with 95% prediction intervals.

RESULTS: A total of 1,318,400 records for primary elective THAs performed in Italy since 2001-2023 were analyzed. The number of THAs increased by approximately 80%, rising from 68.270 in 2001 to 122.777 in 2023. Among the tested models, HPTE generally […]

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Clinical relevance of patient-reported outcome measures in the surgical management of focal chondral defects of the knee: a systematic review.

Migliorini F, Maffulli N, Memminger MK, Hofmann UK

INTRODUCTION: To evaluate clinical outcome following surgical management of focal chondral defects in the knee, patient-reported outcome measures (PROMs) are used. To give these measures meaning, parameters such as the minimal clinically important difference (MCID), patient-acceptable symptom state (PASS), minimally detectable change (MDC), clinically important difference (CID) and substantial clinical benefit (SCB) have been introduced. This systematic review investigated the MCID, SCB, CID, PASS and MDC of the most commonly used PROMs for assessing patients following surgical repair of focal chondral defects of the knee.

METHODS: This systematic review was conducted in accordance with the 2020 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines and the recommendations of the Cochrane Handbook for Systematic Reviews of Interventions. All clinical studies investigating tools to assess the clinical relevance of PROMs in the surgical repair of focal chondral defects of the knee were reviewed. In April 2025, the following databases were accessed: PubMed, Web of Science and Embase. The PROMs of interest included: the International Knee Documentation Committee (IKDC) questionnaire, the Knee Injury and Osteoarthritis Outcome Score (KOOS) and its related subscales activities of daily living (ADL), pain, quality of life (QoL), sports and recreational […]

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Simultaneous versus staged bilateral total hip arthroplasty: a meta-analysis.

Soler F, Hernández J, Lamo-Espinosa JM, Benlloch M, Mariscal G

BACKGROUND: There is debate regarding the optimal timing of bilateral total hip arthroplasty (THA), with simultaneous or staged approaches considered. Cost-effectiveness is an important factor that influences resource allocation. The main aim of this study was to assess the cost implications of bilateral simultaneous (sim-THA) versus staged (st-THA) total hip arthroplasty. The secondary objective was to evaluate the efficacy and safety of sim-THA compared those with of st-THA.

METHODS: A systematic review and meta-analysis were conducted according to the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines. Studies comparing simultaneous and staged bilateral THA in terms of cost, complications, length of hospital stay, and patient outcomes were eligible. Odds ratios (ORs), mean differences (MD), and standard mean differences (SMD) with 95% confidence intervals (CIs) were calculated. The risk of bias was assessed using the Methodological Index for Non-randomised Studies (MINORS). Meta-analyses were performed using Review Manager version 5.4 (Cochrane, Oxford, United Kingdom).

RESULTS: A total of 20 observational studies including 13,984 patients were included. Sim-THA was associated with significantly lower total costs (SMD -0.54, 95% CI -0.92 to -0.16; p = 0.005) and shorter hospital stay (MD

-2.90, 95% CI -4.38 to -1.42; p = 0.0001) than tho […]

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Long-term outcomes of the modified Dunn procedure in moderate and severe slipped capital femoral epiphysis: a prospective case series with 7-year follow-up.

Fahmy M, Abdelazeem AH, Shawky MA

INTRODUCTION: Slipped capital femoral epiphysis (SCFE) is the most common hip disorder in adolescents and may lead to femoroacetabular impingement, early osteoarthritis, and long-term functional disability if inadequately treated. While in situ pinning remains the standard treatment for mild slips, it fails to correct the deformity in moderate and severe cases, potentially predisposing to degenerative changes. The modified Dunn procedure (MDP) was developed to restore proximal femoral anatomy through surgical hip dislocation while preserving vascular supply. The aim of the study is to evaluate the long-term radiological and functional outcomes of the MDP in patients with moderate (14 cases) and severe (10 cases) SCFE, and to assess the incidence of avascular necrosis (AVN), osteoarthritis, and other complications.

METHODS: A prospective case series was conducted between August 2015 and January 2019 at a single tertiary institution. A total of 24 hips with moderate-to-severe SCFE and open physis were treated using the MDP via surgical hip dislocation. Mild and acute-only slips were excluded. MDP was used as a primary procedure, performed early in severe slips and in selected moderate slips after clinical assessment. Patients were followed clinically and radiologically for a mean duration of 84 ± 2.6 months (range 80-88 months). Functional outcomes were assessed using the Harris […]

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The presence of a positive blood culture should be regarded a risk factor for DAIR failure in cases of haematogenous total knee arthroplasty infection.

Pedemonte-Parramón G, Reynaga E, Molinos S, de Los Ríos JD, Vivero A, García-Oltra E, López-Pérez V, Paredes R et al.

PURPOSE: Late acute haematogenous total knee arthroplasty (LAH TKA) infections represent approximately 20% of all prosthetic infections. Debridement, antibiotics, and implant retention (DAIR) is a widely accepted treatment; however, it carries a significant risk of failure. The study aims to identify risk factors for DAIR failure in LAH TKA infections, focusing on blood cultures (BC) as a potential contributor.

METHODS: A retrospective cohort study was conducted on 37 cases of LAH TKA infections from 2015 to 2023. Patients were divided into two groups: 20 in the success group (SG) and 17 in the failure group (FG). The study analyzes various factors, including demographics, comorbidities, serum and joint fluid biomarkers, blood and intraoperative cultures, prior infection or antibiotic use, and surgical and post-surgical variables.

RESULTS: Positive BC were more frequent in the FG compared with the SG (P = 0.03), and were also more common in patients with a history of prior infection (P = 0.03). Logistic regression identified positive BC as the only significant predictor of DAIR failure (odds ratio (OR) 12, 95% confidence interval (CI) 1.1-18, P = 0.04), even after adjusting for other variables. Positive intraoperative cultures were more frequent in the FG, particularly in those with a prior infection history (P = 0.08), even if they were receiving antibiotics (P = 0.05).

CON […]

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Iatrogenic femoral neck fractures or separation of the proximal femoral epiphysis during closed reduction of irreducible femoral head fracture-dislocations in children: a review of 12 cases.

Lu Y, Xue C, Canavese F, De Salvo S, Yao H, Huang D, Lin R, Zheng J et al.

OBJECTIVES: This study aimed to analyze the radiologic characteristics of irreducible pediatric femoral head fracture-dislocations to prevent potential iatrogenic femoral neck fractures (FNF) or separation of the proximal femoral epiphysis (SPFE).

METHODS: This is a retrospective review of patients who were skeletally immature and diagnosed with traumatic hip dislocations combined with femoral head fractures. The collected data included patient demographics, fracture classification, fragment ratio, combined injuries, urgent reductions, treatment strategies, complications, and final outcomes.

RESULTS: We treated 12 patients with femoral head fractures and dislocations; 11 out of 12 patients underwent urgent closed reduction (91.7%). Six of the patients failed closed reduction and experienced FNF (n = 2, 33.3%) or SPFE (n = 4, 66.7%). Five patients presented with avascular necrosis of the femoral head after open reduction with internal fixation via a surgical hip dislocation approach, and two patients required further surgical treatment. Analysis of radiographs and computed tomography (CT) scans of irreducible femoral head fracture-dislocations revealed that the fractured femoral head was perched on the sharp angle of the posterior wall of the acetabulum, with a fragment ratio of 18-27%. After recognizing the irreducibility, one case with a fragment ratio of 26% underwent immed […]

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Orthopaedic publications receive fewer citations than general medicine articles: a scientometric analysis of temporal trends.

Migliorini F, Vaishya R, Maffulli N, Eschweiler J, Kobbe P, Betsch M, Schäfer L, Oliva F et al.

BACKGROUND: Orthopaedics, as a surgical speciality with a structurally narrower citing audience and lower citation density than general medicine, is systematically disadvantaged in any cross-disciplinary comparison relying on absolute citation counts. Whether this disadvantage is stable or has changed over time has not been previously quantified.

METHODS: The 100 most-cited articles indexed in the Web of Science Core Collection were identified for medicine and orthopaedics in 2014 and 2024. Citation data were extracted from a single snapshot taken on 19 January 2026. Total citations and citation velocity, defined as citations per year since publication, were compared between fields and across time points using Mann-Whitney U tests. Ratio-based analyses and distributional characteristics were also assessed.

RESULTS: Median citation counts were substantially higher in medicine than in orthopaedics in both years, reaching 5055 versus 288 in 2014 and 720.5 versus 34.0 in 2024, corresponding to 17.6- and 21.2-fold differences, respectively. Citation velocity followed the same pattern, with medicine maintaining an approximately 17- to 21-fold advantage at both time points. Within-field temporal analysis showed that citation velocity remained broadly stable in medicine between cohorts, whilst orthopaedics demonstrated reduced early citation accrual in the more recent cohort. Citatio […]

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Sleep quality and nocturnal pain in patients with elbow disorders: a prospective multicenter study.

Schneider MM, Nietschke R, Migliorini F, Schoch C, Burkhart K, Hollinger B, Kimmeyer M, Lehmann L et al.

BACKGROUND: Mental health disorders have become a growing global concern. Sleep quality is known to have a crucial impact on mental well-being. Poor sleep contributes to increasing economic costs owing to reduced productivity as well as the development of mental disorders. Conversely, anxiety and depression often lead to sleep disturbances. Pain is a key factor affecting sleep quality. Studies have shown that musculoskeletal disorders, especially in the shoulder, may cause sleep disturbances. Research on the relationship between elbow pain and sleep quality remains limited. This study aims to evaluate the impact of common elbow pathologies on sleep quality.

METHODS: This prospective multicenter cohort study used self-reported outcome measures in patients with elbow disorders, including the Pittsburgh Sleep Quality Index (PSQI), to assess sleep quality. The PSQI was compared with other outcome measures.

RESULTS: A total of 664 patients (284 female, 380 male) were included, of whom 81% reported nocturnal pain. The mean age was 47.9 ± 14.2 years, body mass index (BMI) 26.9 ± 6.7 kg/m2 and symptom duration 15.8 ± 7.3 months. The most common diagnoses were lateral epicondylar pain (66%), medial epicondylar pain, arthritis, ulnar nerve entrapment and biceps/triceps tendinopathies. The overall PSQI was 7.3 ± 3.8, disabilities of arm, shoulder and hand questionnaire (DASH) 35.0 ± 19. […]

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A novel figure-of-eight polymeric cerclage wiring technique for greater trochanter fixation in revision arthroplasty and conversion of failed proximal femur fracture fixations to total hip arthroplasty.

Leggieri F, Martín Cocilova FN, Secci G, Stimolo D, Zanna L, Civinini R, Innocenti M

INTRODUCTION: Periprosthetic femoral fractures occur in 0.8-10% of primary total hip arthroplasties (THA) and up to 17.5% for revision THA (revTHA), with isolated greater trochanter (GT) fractures being the most predominant pattern. This study aimed to evaluate a novel figure-of-eight cerclage wiring technique for greater trochanter fixation in hip replacement cases.

METHODS: These retrospectively collected data included 36 hips (36 patients) undergoing revision THA or conversion of failed intertrochanteric fracture to THA with figure-of-eight cerclage wiring between April 2021 and June 2023. Patients under 18 years, with incomplete records, or lost to follow-up before 24 months were excluded. Primary outcomes included Trendelenburg sign, periprosthetic infections, and hip dislocation rates. Secondary outcomes evaluated THA stem subsidence and trochanteric migration using calibrated digital imaging. Statistical analysis included descriptive statistics, independent t-tests, linear regression, and correlation analyses with significance set at $p < 0.05$.

RESULTS: Mean follow-up was 404.76 ± 22.51 days. Patient demographics showed mean age 76.5 ± 12.3 years, body mass index (BMI) 25.8 ± 4.5 kg/m2, with 73.8% female patients. At final follow-up, one patient (2.8%) experienced post-surgical infection, one patient (2.8%) had dislocation, and three patients (8.3%) exhibited positive T […]

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Minimal clinically important difference (MCID), patient-acceptable state (PASS), minimally detectable change (MDC), and substantial clinical benefit (SCB) in patients who have undergone arthroscopic surgery for femoroacetabular impingement: a systematic review.

Migliorini F, Maffulli N, Schäfer L, Jeyaraman M, Betsch M, Mahmoud M

INTRODUCTION: Hip arthroscopy has been established as a successful management for femoroacetabular impingement (FAI) syndrome. Patient-reported outcome measures (PROMs) are frequently used to assess treatment results in patients. However, clinically important outcome values have recently been calculated to better understand and interpret PROMs and define successful treatment. This systematic review aimed to investigate the minimal clinically important difference (MCID), patient-acceptable state (PASS), substantial clinical benefit (SCB), and minimally detectable change (MIC) of the most commonly used PROMs for assessing patients who have undergone arthroscopic surgery for FAI.

MATERIAL AND METHODS: This study was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses: the 2020 PRISMA statement. In October 2025, the following databases were accessed: PubMed, Web of Science, and Embase. All clinical studies investigating tools to assess the clinical relevance of PROMs used to evaluate patients who have undergone arthroscopic surgery for FAI were accessed. Only studies that assessed the minimal MCID, PASS, SCB, and/or MIC were considered.

RESULTS: The methodological quality of the 21 included studies was assessed, and the overall risk of bias was found to be low-to-moderate. The main characteristics of the included studies were analyzed and […]

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Efficacy and safety of MIPO versus ORIF in distal tibial fractures: a systematic review and meta-analysis.

Hassaan AM, Hassan AA, Abdelkhalik WS, Saad AM, Hosny AS, Elfeky BAEH, Mandour I, Ragheb MG et al.

BACKGROUND: Distal tibial fractures are challenging to manage owing to limited soft-tissue coverage and compromised blood supply. Minimally invasive plate osteosynthesis (MIPO) and open reduction and internal fixation (ORIF) are among the widely used surgical options, but the optimal approach remains unclear. MIPO preserves periosteal blood flow with smaller incisions, while ORIF offers direct visualisation but increases soft-tissue damage. With growing comparative evidence, an updated evaluation of these techniques is necessary.

AIM: To compare the efficacy and safety of MIPO versus ORIF in adult patients with distal tibial fractures.

METHODS: Following PRISMA guidelines and the Cochrane Handbook, we searched in PubMed, Embase, Cochrane, Scopus, and Web of Science till November 2025. Randomised trials and cohort studies comparing MIPO and ORIF were included. Data extraction was performed independently by two reviewers. Risk of bias was assessed using RoB 2 for RCTs and the Newcastle-Ottawa Scale for observational studies. Random-effects models were used to estimate pooled mean differences and odds ratios, and heterogeneity was assessed using I2. Sensitivity and subgroup analyses were conducted by study design.

RESULTS: Nine studies involving 530 patients met our inclusion criteria. No significant differences were found between MIPO and ORIF in AOFAS scores, union time, oper […]

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Comparative efficacy and safety of aspirin, low-molecular-weight heparin, and rivaroxaban for deep venous thrombosis prophylaxis after hip arthroplasty for femoral neck fracture: a single-center randomized controlled trial.

Miao S, Yue Q, Zhang X, Lu J, Zhang Y, Lu N, Wei L

OBJECTIVE: To evaluate the efficacy and safety of aspirin, low-molecular-weight heparin (LMWH), and rivaroxaban for deep vein thrombosis (DVT) prophylaxis after total hip arthroplasty (THA) or bipolar hemiarthroplasty (BHA) for femoral neck fracture, with subgroup analysis stratified by surgical type.

METHODS: This single-center randomized controlled trial (RCT) was conducted between June 2024 and June 2025. A total of 217 consecutive patients with femoral neck fracture undergoing hip arthroplasty were initially screened, and 161 eligible patients were finally enrolled and randomly assigned to the aspirin group, LMWH group, or rivaroxaban group at a 1:1:1 ratio. All patients received standardized intermittent pneumatic compression (IPC) and graduated compression stockings for mechanical DVT prophylaxis, as well as a unified postoperative rehabilitation protocol. The primary outcome was the incidence of lower extremity DVT within 12 weeks postoperatively, confirmed by color Doppler ultrasound. The secondary outcome was the incidence of bleeding-related complications graded by the International Society on Thrombosis and Hemostasis (ISTH) criteria. All patients were followed up at 6, 8, and 12 weeks postoperatively, with routine bilateral lower extremity ultrasound performed at 12 weeks regardless of symptoms. Subgroup analysis was performed on the basis of surgical type (THA ver […]

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Does the surgical sequence of lumbar spine fusion and total hip arthroplasty affect the dislocation and revision risk? A meta■analysis and systematic review.

Liu F, Jia C, Wu X, Zhou Y

BACKGROUND: Patients who underwent primary total hip arthroplasty (THA) after lumbar spine fusion (LSF) had a higher incidence of complications compared with those who had not previously undergone LSF. This study aimed to determine whether the timing of THA before or after LSF impacts the incidence of THA complications.

METHODS: Following the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines, we searched PubMed, Web of science, Embase, and Cochrane Library from their inception until 1 October 2025. Data from all eligible published studies meeting the inclusion criteria were extracted and subsequently analyzed using a random-effects model to calculate the rates of all-cause revision, dislocation, periprosthetic joint infections, and aseptic loosening.

RESULTS: Our analysis included 10 studies involving 27,605 patients who underwent THA followed by LSF and 32,002 patients who received LSF prior to THA. There are no significant differences in the rates of dislocation (odds ratio [OR] 1.19, 95% confidence interval (CI) 0.73-1.86, p = 0.45), periprosthetic joint infections (OR 0.91, 95% CI 0.52-1.62, p = 0.76), and aseptic loosening (OR 0.89, 95% CI 0.74-1.08, p = 0.23) between the two surgical sequences, while for all-cause revision, the OR was (OR 1.01, 95% CI 0.70-1.47, p = 0.94), which reached statistical significance.

CONCLUSIONS: Compare […]

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Effects of anterior cruciate ligament resection and total knee arthroplasty on lower limb alignment and tibiofemoral rotation.

Fan Y, Chen S, Jiang L, Chen Z, Wu M, Wang C

BACKGROUND: The anterior cruciate ligament (ACL) is a key structure that restricts excessive anterior translation of the tibia and maintains rotational stability. This study aims to dynamically and quantitatively assess the direct effects of ACL resection and total knee arthroplasty (TKA) on lower limb alignment and tibiofemoral joint rotation under passive, unloaded conditions and in the pathological state of osteoarthritis.

METHODS: This retrospective study included 110 patients with varus knee osteoarthritis who underwent Mako for robotic-assisted TKA. Dynamic intraoperative measurements of the lower limb mechanical axis (flexion angle at maximum extension; varus angle) and tibiofemoral rotation at multiple flexion angles (min to max flexion) were recorded at three intervals: pre-ACL resection, post-ACL resection and post-TKA. The variation in rotation amplitude was calculated for successive flexion intervals.

RESULTS: Post-ACL resection, a significant decrease in the flexion angle at the maximum extension and varus angle (both $p < 0.001$) was observed, which was accompanied by increased tibial internal rotation at flexion angles of 60° and above ($p < 0.05$). Varus alignment was successfully corrected following TKA ($p < 0.001$). However, tibiofemoral rotational kinematics were significantly modified: the amplitude of internal rotation decreased in the initial flexion arc (min […]

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Comparison of Bankart repair with remplissage and Latarjet procedure for anterior shoulder instability: a systematic review and meta-analysis.

Nong K, Liu G, Jiang S, Li Q, Kwabena BR, Yuan D, Xiao W, Zhang K et al.

BACKGROUND: Anterior shoulder instability is the most common type of shoulder instability. For cases with subcritical glenoid bone loss, both Bankart repair with remplissage (BR) and the Latarjet procedure are viable surgical options. The aim of this study was to compare the postoperative outcomes in patients undergoing BR and Latarjet procedure.

METHODS: This review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guideline. The methodological index for nonrandomized studies (MINORS) was used to characterize the methodological quality. A literature search was conducted using PubMed, the Cochrane Library, Embase, Web of Science, and Scopus. The analyzed outcomes included postoperative Rowe score, visual analog scale (VAS) score, the Single Assessment Numeric Evaluation (SANE) score, range of motion (ROM), return-to-sport (RTS) rate, RTS time, recurrence rate, revision rate, and complications. Subgroup analysis was performed on the basis of follow-up duration.

RESULTS: Overall, 12 studies involving 1186 patients were included. The meta-analysis showed that BR was associated with a significantly lower complication rate than the Latarjet procedure. No significant differences were observed in functional outcomes, ROM, recurrence, revision, RTS, or RTS time. Subgroup analysis showed a significant interaction between f […]

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Risk of periprosthetic joint infection within 1 year following robotic-assisted versus conventional primary total knee arthroplasty: a propensity-score-matched cohort study.

Tian S, Li Y, Mu W, Ji B, Zhang X, Cao L, Zhao J

BACKGROUND: Robotic-assisted total knee arthroplasty (RA-TKA) is increasingly being adopted for its ability to enhance bone-resection accuracy and component alignment. However, whether these technical gains influence the risk of periprosthetic joint infection (PJI) remains uncertain, especially in the context of prolonged operative duration. This study aimed to compare the 1-year rate of PJI following conventional total knee arthroplasty (cTKA) and RA-TKA in a propensity-score-matched cohort.

METHODS: We retrospectively reviewed 1284 consecutive patients who underwent primary TKAs at a single centre between 2021 and 2023. The patients were stratified according to surgical technique (cTKA versus RA-TKA) and subsequently matched 1:1 using propensity score analysis (age, sex, body mass index [BMI], American Society of Anesthesiologists [ASA] score, Charlson Comorbidity Index [CCI] score, CCI components and smoking), resulting in 522 pairs (1044 patients) for the final comparative analysis. Operative time and 1-year PJI were assessed using multivariable logistic regression. Infections were stratified according to timing: ≤ 90 days and from 90 days to 1 year after surgery.

RESULTS: The 1-year rate of PJI was 0.77% (4/522) after RA-TKA and 0.96% (5/522) after cTKA (P = 1.000). All PJIs in patients who underwent RA-TKA occurred within 90 days, whereas PJIs in patients who underwent […]

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Measured resection, gap balancing, and force-sensor-guided total knee arthroplasty result in different femoral and tibial rotational alignment but similar combined knee rotation and clinical outcomes: a randomized controlled trial.

Hernández-Hermoso JA, Nescolarde L, Yañez-Siller F, Calle-García J, Pérez-Andrés R, García-Perdomo D

BACKGROUND: Different alignment philosophies and balancing methods may alter femoral and tibial component rotation in distinct ways within the same knee. This study aimed to (1) determine which of three surgical techniques-measured resection (MR), gap balancing with computer-assisted surgery (GB-CAS), or a force-sensor soft-tissue balancing device (FS-STB)-most closely reproduces native femoral, tibial, and combined rotational alignment in mechanically aligned total knee arthroplasty (TKA), and (2) assess whether differences in rotational alignment affect outcomes at 5 years postoperatively.

MATERIAL AND METHODS: A total of 60 patients undergoing primary mechanical alignment TKA were randomly assigned to one of three surgical approaches (n = 20 per group): MR, GB-CAS, or FS-STB. Blinded observers assessed the Knee Society Score (KSS), Western Ontario MacMaster Universities Osteoarthritis Index (WOMAC) score, and hip-knee-ankle angle preoperatively and at the 5-year follow-up visit. Pre- and postoperative two-dimensional (2D)-computed tomography scans were used to measure femoral rotation (BFA), tibial rotation, and combined femur-tibia rotation (TE_PTCA and BC_PTCA). Statistical analyses included paired t-tests, one-way analysis of variance, and effect size calculations.

RESULTS: Femoral rotation remained unchanged in the MR and GB-CAS groups, but decreased slightly (1° exter […]

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Risser growth plate injury in unstable paediatric pelvic fractures: a multicentre retrospective study.

Oransky M, Aulisa AG, Roncoroni A, Zoppi AR, Mata M, Rohayem MA, Falciglia F, Toniolo RM

INTRODUCTION: Unstable pelvic fractures in children are rare but severe injuries and are associated with high-energy trauma. Unlike adults, children have open growth plates, particularly the Risser growth plate (RGP), which is vulnerable to injury. The presence of growth plates, such as the triradiate cartilage and Risser's plate, represents point of vulnerability that determines distinctive fracture patterns, similar to those seen with Salter-Harris growth plate injuries. Lateral compression fracture is the most common mechanism and results in an irreducible crushing of the sacral alae, causing a rotational deformity. Both vertical and lateral compression displaced pelvic fractures require careful evaluation of the posterior iliac apophysis-associated injury. Failure to treat Risser growth plate injury (RGPI) can lead to severe long-term problems such as limb length discrepancy and scoliosis. This study aimed to determine how commonly RGPI occurs in unstable paediatric pelvic fractures, link it to specific fracture patterns and propose a subgroup type to improve treatment.

MATERIALS AND METHODS: This multicentre retrospective study included 40 children aged up to 12 years with unstable pelvic fractures (AO/OTA 61B and 61C). The patients were treated between 1987 and 2022 at trauma centres in Italy, Argentina and Venezuela. We reviewed patient demographics, injury mechanisms a […]

PubMed · DOI

Total femur replacement: function, quality of life, and gait: a multicentric EMSOS study.

Valentini M, Svehlik M, Leithner A, Bergovec M, EMSOS TFR Study Group, Smolle MA

BACKGROUND: Total femur replacements (TFRs) involve the replacement of the whole femoral bone, reconstruction of the hip and knee joints, and soft tissue reconstruction for function and joint stability. The aim of this multicentric European Musculo-Skeletal Oncology Society (EMSOS) study is to report on the functional outcome, quality of life, as well as gait analysis in patients with TFR.

MATERIALS AND METHODS: Retrospective data from 12 international participating centers were collected. In total, 65 patients [median (IQR) age at surgery 40 (18-56) years, 60% male] who received a TFR owing to oncologic or nononcologic indications between 1 January 1990 and 31 March 2024 were included. Patient-related outcome measures (PROMs), range of motion (ROM) of the hip and knee, and gait videos were collected.

RESULTS: At a median (IQR) follow-up of 28 (14-126) months, the median (IQR) percentage Musculoskeletal Tumor Society (MSTS) score was 63% (43-73%), 12-Item Short-Form Health Survey (SF-12) was 70% (55-79%), Harris Hip Score (HHS) was 63% (52-77%), and Oxford Knee Score (OKS) was 63% (52-79%); the median (IQR) extension-to-flexion ROM were 80° (70-97.5°, hip) and 70° (40-90°, knee). Gait videos were available for 24/65 patients (37%). The median (IQR) total Edinburgh Visual Gait Score of the operated side was 8.5 (6-13), with deviations from norm in all patients (24/24). Compens […]

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Vascularized versus nonvascularized free fibular grafts in reconstruction of post-traumatic critical long bone defects, a comparative study.

Fouaad AA, Massoud AH, Shaban M, Abdel-Naby M, Abulsoud MI

BACKGROUND: Reconstruction of critical-sized long bone defects is a complex orthopedic challenge. Free fibular grafts, vascularized (FVFG) and nonvascularized (NVFG), are established reconstructive options, but comparative clinical outcomes remain uncertain.

PURPOSE: To compare clinical and radiological outcomes of FVFG and NVFG in patients with post-traumatic critical bone defects more than 10 cm.

METHODS: A randomized controlled trial was conducted with 50 patients assigned equally to FVFG or NVFG groups. The primary outcome was time to union, while secondary outcomes included graft hypertrophy, functional scores (DASH and LEFS), complication rates, and donor site morbidity.

RESULTS: The mean time to union was 5.78 months in the FVFG group and 6.17 months in the NVFG group, showing no statistically significant difference ($p = 0.447$). Rates of graft hypertrophy, functional recovery, and complications were comparable between the groups.

CONCLUSIONS: Both FVFG and NVFG provide effective reconstruction for critical bone defects, with nearly similar healing times and functional outcomes. NVFG may represent a less complex alternative in selected cases.

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Five-year outcomes of medial open-wedge high tibial osteotomy with lateral supplemental lag screw in patients with obesity.

Tian YH, Lee KH, Huang CW, Lin SY, Yang JC

BACKGROUND: Obesity contributes to the accelerated progression of knee osteoarthritis. Medial open-wedge high tibial osteotomy (MOWHTO) is a joint-preserving surgical intervention for unicompartmental knee osteoarthritis; however, its efficacy in patients with obesity remains controversial. This study aimed to evaluate the 5-year clinical and radiographic outcomes of MOWHTO with lateral supplemental lag screw fixation in patients with obesity.

MATERIALS AND METHODS: This retrospective cohort study included patients with obesity who underwent MOWHTO between 2017 and 2020, with a minimum follow-up of 5 years. All procedures were performed by a single surgeon using three-dimensional printed, patient-specific instrumentation, locking plates, and a lateral supplemental lag screw. Clinical outcomes were assessed using the visual analog scale (VAS) and Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC). Radiographic parameters, including the weight-bearing line percentage (WBL%) and medial proximal tibial angle (MPTA), were evaluated.

RESULTS: Significant improvements in VAS and WOMAC scores were observed at all postoperative time points (p < 0.001) and were accompanied by improved radiographic alignment, with WBL% shifting toward the Fujisawa point and increased MPTA values. At 5 years, mild regression in alignment was noted; however, the overall correction was […]

PubMed · DOI

Outcomes of surgical fixation of osteochondral fracture: a systematic review.

Pang AS, Oviya R, Tan JJ, Lim KSA, Hui JHP, Tan SHS

BACKGROUND: Dealing with knee osteochondral defects presents a substantial clinical challenge due to the variable nature of repair outcomes and intricate biomechanical complexities inherent to the knee joint. Numerous surgical alternatives exist for addressing knee osteochondral lesions, yet there is limited evidence available for comparing their respective outcomes.

METHODS: According to the Preferred Reporting Items for Systematic Review and Meta-analyses (PRISMA) guidelines, a systematic literature search was conducted to identify publications from inception to September 2023 on PubMed, Cochrane, and Medline academic search engines with the MeSH terms "osteochondral" AND ("patella" OR "patellar" OR "patellofemoral" OR "femoral condyle lesions" OR "trochlear lesions"). Inclusion criteria were clinical human studies, English language, subjects who underwent surgical management (fixation, reconstruction, or debridement) for knee osteochondral lesions with reported treatment outcomes, and a minimum sample size of ten patients. The methodological index for non-randomized studies (MINORS) was used to evaluate the non-randomized studies' methodological quality. Main outcome measurements Pediatric International Knee Documentation Committee (Pedi IKDC) scale, Tegner-Lysholm Scoring Scale, Tegner Activity Score, Visual Analog Scale, Knee Society Score (KSS) Score, Kujala Score, and F […]

PubMed · DOI

Do modular tapered long stems differ in stability and function? A prospective comparison of two stems in complex femoral revision THA.

Fahmy M, Shawky MA

BACKGROUND: Revision total hip arthroplasty (rTHA) in the setting of substantial proximal femoral bone loss is technically challenging. Modular tapered fluted stems provide predictable diaphyseal fixation while allowing independent adjustment of version, offset, and limb length. Among these, two commonly used systems-modular tapered fluted stem Type A (Revitan™) and Type B (LIMA Modulus™)-have limited direct comparative evidence. This study aimed to prospectively compare radiographic stem subsidence (primary outcome), as well as functional outcomes, complications, survivorship, and secondary outcomes, between Type A and Type B modular long stems in femoral rTHA.

METHODS: In this single-center randomized prospective study, 110 patients undergoing femoral revision rTHA were randomly assigned to receive either Type A (n = 55) or Type B (n = 55) stems. All procedures were performed by experienced revision surgeons under standardized perioperative and rehabilitation protocols. Radiographs were analyzed for stem subsidence, osseointegration, and limb-length restoration. Functional outcomes were assessed using the Harris Hip Score (HHS), Oxford Hip Score (OHS), and European Quality of Life Visual Analogue Scale (EQ-VAS) at baseline and final follow-up (mean 61.4 months). Complications and stem survivorship were recorded prospectively. Statistical analysis included paired and unpaired […]

PubMed · DOI

Clinical comparison of single posterolateral plate with medial-cannulated-screw fixation and double-plate fixation for extra-articular distal humerus fractures.

Jung HS, Chu MS, Lee JS

BACKGROUND: Double-plate fixation is the gold standard for extra-articular distal humerus fractures, but it carries a substantial risk of postoperative ulnar neuropathy. Fixation using a single posterolateral plate with a medial cannulated screw may reduce ulnar neuropathy while maintaining fracture stability. This study aimed to compare the clinical outcomes of single-plate-with-medial-screw fixation versus double-plate fixation for extra-articular distal humerus fractures.

MATERIALS AND METHODS: Fifty-six patients who underwent surgery for extra-articular distal humerus fractures (Arbeitsgemeinschaft für Osteosynthesefragen/Orthopaedic Trauma Association [AO/OTA] classification A2 or A3) between January 2018 and August 2024 were divided into a double-plate group and a single-plate-with-medial-screw group. We conducted a retrospective, nonrandomized comparative study. The double-plate fixation was used in 30 patients from January 2018 to October 2021, while the single-plate fixation with a medial screw was used in 26 patients from November 2021 to August 2024. All surgeries were performed using a posterior paratricipital approach. Bony union, radiographic healing, and loss of reduction were evaluated. Postoperative pain scores (visual analog scale at 2 days after the operation), operative time (minutes), elbow range of motion, elbow function (Mayo Elbow Performance Score [MEP […]

PubMed · DOI

A study on the association between tibial plateau fractures and intra-articular soft-tissue injuries under valgus injury mechanisms.

Duan S, Wang S, Zhu T, Li Q, Zhang M

BACKGROUND: While recent investigations have focused on injury mechanism classifications of tibial plateau fractures (TPFs), the association between valgus TPFs and concomitant soft-tissue damage involving menisci and ligaments remains insufficiently elucidated. This study aimed to characterize intra-articular soft-tissue injuries associated with various valgus TPFs and assess the predictive value of lateral plateau depression (LPD) and widening (LPW). Additionally, the analysis extended to other injury mechanisms.

MATERIALS AND METHODS: This study included adult patients with acute tibial plateau fractures who had complete imaging data, excluding patients with open fractures and multiple fractures throughout the body. Imaging examinations were used to assess the fracture mechanism and intra-articular soft-tissue damage.

RESULTS: The study retrospectively analyzed the clinical data of 138 patients with valgus injury TPFs in our hospital. The study compared the incidence of TPFs with intra-articular soft-tissue damage under different injury mechanisms. The result demonstrated that the incidence of valgus hyperextension TPFs combined with medial collateral ligament injuries was relatively high (61.5%) compared with TPFs with other injury mechanism. Multivariable logistic regression and smooth curve fitting revealed significant dose-response relationships of LPD (odds ratio [OR] […]

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The coronoid as the key fragment of trans-ulnar fracture-dislocations of the elbow: Insights from a retrospective cohort comparison using the coronoid-centric Mayo classification system.

Henssler L, Kerschbaum M, Zanklmaier M, Wieczorek LA, Alt V, Klute L

BACKGROUND: Trans-ulnar fracture-dislocations of the elbow are rare injuries with complex fracture patterns and variable outcomes. Traditional classification systems offer limited prognostic value. A recently introduced coronoid-centric Mayo classification distinguishes injury subtypes based on coronoid attachment and identifies trans-ulnar basal coronoid (TUBC) fractures as a particularly challenging entity. This study evaluated outcomes across Mayo fracture types and explored factors associated with inferior results in TUBC injuries.

MATERIALS AND METHODS: In this retrospective cohort study, surgically treated trans-ulnar elbow fracture-dislocations managed at a level I trauma center between 2010 and 2022 were identified and classified according to the Mayo system. Demographic data, injury characteristics, surgical management, radiographic outcomes, and complications were recorded. Functional outcomes were assessed after a minimum follow-up of 12 months using the Mayo Elbow Performance Score (MEPS); Oxford Elbow Score (OES); Quick Disabilities of Arm, Shoulder and Hand Questionnaire (QuickDASH); European Quality of Life Five-Dimension, Five-Level Version (EQ-5D-5L); and range-of-motion measurements. Radiographs were analyzed for union, instability, heterotopic ossification, and post-traumatic osteoarthritis (OA).

RESULTS: A total of 52 patients were included (14 trans-olecr […]

PubMed · DOI

Feasibility study of a robot-assisted endoscopic technique for plate osteosyntheses of the acetabulum and anterior pelvic ring.

Hinz N, Althoff G, Thomaschewski M, Bonke LA, Eberenz A, Münch M, Behrends L, Männel G et al.

BACKGROUND: Minimally invasive stabilization of acetabular and pelvic ring fractures using endoscopic techniques has become increasingly important. The logical advancement of conventional endoscopic techniques is a robot-assisted approach, which benefits from the advantages of robot-assistance systems (e.g., more degrees of freedom for instruments and improved visualization). The aim was therefore to investigate the feasibility of a robot-assisted endoscopic technique for plate osteosyntheses of the acetabulum and anterior pelvic ring.

MATERIALS AND METHODS: In this two-part feasibility study, four different plate osteosyntheses were performed endoscopically using the Hugo™ robot-assisted surgery system, first on ten synthetic pelvic models and then on ten human cadavers. During robot-assisted dissection for the preperitoneal approach, identification of ten relevant anatomical landmarks was also assessed. In both parts, the success, number of drilling errors, and time for each plate were described as learning curves and analyzed using linear regression.

RESULTS: The infrapectineal plate could be successfully performed in 100% of synthetic models, the posterior column plate in 100%, the suprapectineal plate in 90%, and the superior pubic ramus plate in 80%. Learning curves could be observed for the number of drilling errors per plate (e.g., 0.67 to 0, $p = 0.009$) and the time r […]

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Favorable subjective clinical outcomes after revision hip arthroscopy for femoroacetabular impingement syndrome despite a high rate of capsular defects and increased capsular thickness.

Cao Z, Gao G, Lin W, Zhu Y, Zhou X, Wang J, Xu Y

BACKGROUND: The capsular healing status and capsular thickness changes in patients with femoroacetabular impingement syndrome (FAIS) following revision hip arthroscopy are poorly documented, and their relationship with subjective clinical outcomes remains unclear. The purpose of this work is to evaluate the incidence of capsular defects and changes in capsular thickness using magnetic resonance imaging (MRI) following hip arthroscopy in patients with FAIS, and to compare the subjective clinical outcomes between patients with and without capsular healing after revision hip arthroscopy.

PATIENTS AND METHODS: Consecutive patients with FAIS who underwent revision hip arthroscopy between 2013 and 2023 were included. Patients were categorized into two groups on the basis of capsular healing status after revision hip arthroscopy. Patient-reported outcomes (PROs) were collected preoperatively and at minimum 2-year follow-up, including the modified Harris Hip Score (mHHS), International Hip Outcome Tool-12 (iHOT-12), Hip Outcome Score-Activity of Daily Living Scale (HOS-ADL), Hip Outcome Score-Sport Specific Subscale (HOS-SSS), and visual analog scale (VAS). Achievements in minimal clinically important difference (MCID) and patient acceptable symptom state (PASS), patient satisfaction, and re-revision rates were evaluated and compared between groups. Capsular thickness was assessed by […]

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From bench to bedside: a novel suture-augmented prosthesis for greater trochanteric fixation in osteoporotic hip arthroplasty.

Yang M, Yang J, Wang X, Wu Z, Bian S, Sheng R, Liao F, Qin K et al.

BACKGROUND: Stable fixation of the greater trochanter during hip arthroplasty for unstable osteoporotic intertrochanteric fractures remains a significant challenge. Conventional techniques depend on securing the bone-implant interface, which is particularly compromised in osteoporotic bone. Here, we propose a novel conceptual approach that establishes a direct mechanical bridge from the abductor tendon to the prosthesis, thereby reducing reliance on the fragile bone-implant interface.

METHODS: In this two-stage translational study, we progressed from biomechanical validation to clinical application. First, using a decalcified caprine femoral model simulating osteoporosis, three fixation constructs were compared: locking plate (LP), suture-augmented locking plate (LPSA), and Kirschner-wire tension band (KWTB). Ultimate load and construct stability were evaluated. Biomechanical testing confirmed the principle of suture-mediated load sharing and highlighted the intrinsic weakness of screw fixation in osteoporotic bone. Guided by these results, we designed a novel femoral prosthesis that eliminates screw fixation, employs sutures as the primary load-bearing element, and incorporates integrated suture anchor tunnels. This prosthesis was then assessed in a retrospective series of 15 consecutive elderly patients with osteoporotic intertrochanteric fractures. Clinical outcomes were ev […]

PubMed · DOI

Updating radiographic parameters for the healthy growing hip: are current parameters still valid?

Hütter K, Smolle MA, Butter S, Schroedter R, Tschauner S, Kraus T

BACKGROUND: This study aimed to evaluate whether the acetabular angle (AA) by Tönnis and lateral center-edge angle (LCEA) by Wiberg-key radiographic parameters in pediatric hip assessment-have changed in a contemporary pediatric population compared with historical reference values, considering trends in earlier skeletal maturation and body composition.

MATERIALS AND METHODS: This retrospective diagnostic accuracy study assessed changes in AA and center-edge angle (CE) angles in a contemporary cohort. A total of 1774 anteroposterior pelvic radiographs (3548 hips) from patients aged 0-18 years without hip pathology were analyzed. Radiographs were obtained between 2006 and 2018 and measured using the Supervisely digital platform. Standardized anatomical landmarks were applied. Data were stratified by age and sex and compared with historical values using two-sample t-tests ($p < 0.05$). To minimize measurement bias, two independent observers conducted all assessments using standardized digital protocols.

RESULTS: A total of 1774 patients aged 0-18 years (mean age, 8.30 ± 5.20 years; 666 females, 1108 males) were evaluated. AA values did not show statistically significant differences in 11 of 16 age groups compared with historical data ($p > 0.05$). In contrast, LCEAs were significantly higher in the contemporary cohort, especially in the 5-8, 9-12, and 12-16 age groups (all $p < 0.01$) […]

PubMed · DOI

In-hospital deep vein thrombosis after tibial plateau fractures: incidence, laterality/anatomy, and risk factors in a multicenter retrospective cohort of 3366 patients.

Yang S, Li G, Li Y, Long Y, Zhang J, Liu L, Zhao Z, Sun C et al.

BACKGROUND: We aimed to determine the cumulative in-hospital incidence, anatomic distribution, and predictors of duplex ultrasonography-detected lower-extremity deep vein thrombosis (DVT) in patients hospitalized with tibial plateau fractures under routine thromboprophylaxis and protocolized surveillance.

METHODS: We retrospectively analyzed consecutive adults (≥ 18 years) admitted to two level-I trauma centers (November 2014-January 2024). Bilateral duplex ultrasonography was performed on admission/preoperatively (as soon as feasible after arrival and prior to surgery) and repeated serially during hospitalization (approximately every 5-7 days and when clinically suspected). The primary outcome was cumulative in-hospital DVT from admission to discharge; isolated intermuscular calf-muscle vein thrombosis (soleal/gastrocnemius) was recorded descriptively but excluded from the primary endpoint. Multiple injuries (polytrauma) were defined as ≥ 1 additional acute traumatic lesion beyond the index tibial plateau fracture documented on admission. Pulmonary embolism (PE) was assessed only when clinically suspected and confirmed by computed tomography pulmonary angiography (CTPA). Multivariable logistic regression identified independent predictors; receiver operating characteristic (ROC) analysis evaluated discrimination for key continuous predictors.

RESULTS: Among 3366 patients, 675 […]

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Retained articulating spacers (1.5 stage) in chronic knee periprosthetic joint infections: an analysis of associate factors and outcomes in a single center cohort of patients.

Balato G, Ascione T, Festa E, De Mauro D, Di Gennaro D, Rosa D, Mariconda M

BACKGROUND: Two-stage revision remains the gold standard for chronic periprosthetic joint infection (PJI), but it can significantly impact patients' quality of life and functional recovery. The 1.5-stage revision, involving indefinite spacer retention, has emerged as a potential alternative in selected high-risk patients. Our questions were: (1)

What are the early functional outcomes, including their infection-free survival? (2) Which pre- and perioperative factors are associated with spacer retention? and (3) What is the mechanical survivorship of retained spacers?

MATERIALS AND METHODS: This was a retrospective observational study including 108 consecutive patients with chronic PJI between 2016 and 2020 who were followed prospectively for clinical outcomes. Functional status and quality of life were assessed using the EuroQol-5-Dimension-5-Level (EQ-5D-5L), Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC), and Knee Society Score (KSS) before spacer placement and prior to scheduled reimplantation. Infection-free status was defined as the absence of clinical, radiographic, and laboratory signs of infection recurrence over a 24-month follow-up. Univariate and multivariate logistic regression were used to identify factors associated with spacer retention.

RESULTS: All patients initially underwent articulating spacer implantation as part of a planned two-st […]

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Moderate leg length discrepancy: a long-term risk factor for hip osteoarthritis?

Aprato A, Donis A, Via RG, Scandurra A, Tuè F, Audisio A, Fusini F, Massè A

BACKGROUND: Leg length discrepancy (LLD) has been implicated as a biomechanical factor contributing to hip osteoarthritis (OA), yet the extent of its influence remains unclear. This study examines the correlation between moderate LLD ($\geq$ 10 mm) and hip OA progression, focusing on asymmetrical OA distribution in patients over 65 years old.

MATERIALS AND METHODS: A retrospective analysis was conducted from a database of 1672 full-length standing X-rays. Patients under 65 years, with deformities, unilateral limb issues, or prosthetics, were excluded; therefore, the study group was composed of 220 patients. Tibial and femoral lengths were measured bilaterally, and hip OA severity was assessed using the Tönnis classification. Statistical analyses included Pearson's Chi-squared test and linear regression to explore correlations between LLD and OA distribution.

RESULTS: Among the sample, 18% showed an LLD $\geq$ 10 mm. A significant correlation was found between LLD and the asymmetrical distribution of hip OA ($p = 0.002$), with higher OA severity observed in the hypometric limb. Linear regression analysis suggested that each millimeter of LLD corresponded to a 0.74-unit change in OA severity difference between hips.

CONCLUSIONS: This study highlights a significant association between moderate LLD and contralateral hip OA in the elderly, emphasizing the biomechanical impact of asymmetrical […]

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Authorship, titles and open access as drivers of citation performance in orthopaedics: a scientometric analysis.

Migliorini F, Vaishya R, Rivera F, Eschweiler J, Kobbe P, Betsch M, Oliva F, Maffulli N

BACKGROUND: Bibliometric analyses are increasingly used to explore how scientific knowledge is created, disseminated, and perceived. In orthopaedics, research output has expanded rapidly over the past decade, yet the factors determining whether an article achieves wide visibility and scholarly impact remain poorly understood. Beyond the inherent quality of a study, elements such as authorship patterns, title construction, and open access (OA) availability may play an essential role in shaping citation performance. However, evidence in this field is still limited and sometimes contradictory, highlighting the need for large-scale, field-specific analyses.

METHODS: Orthopaedic publications from 2010 to 2020 were identified in Scopus using the keyword 'orthopaedic'. After duplicate removal, 97,806 unique articles were included with complete data on authorship, titles, citation counts, study design, and OA status. Citation rates were normalised per year since publication. Associations between bibliographic features and citation performance were assessed using multiple linear regression, while differences across title styles and study designs were evaluated with comparative statistical testing. Exploratory modelling was performed to identify combinations of authorship and title characteristics linked to the highest predicted citation rates.

RESULTS: Larger author teams were associa […]

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Interpreting nonsignificant findings: Reassessing the safety of self-locking standalone cages in octogenarian patients undergoing ACDF.

Abudayeh A, Fishchenko I

PubMed · DOI

A novel, simple, and affordable technique for arthroscopic soft-tissue tenodesis of the long head of the biceps tendon.

Fossati C, Ioppoli A, Ravaglia R, Vaja F, Menon A, Randelli PS

BACKGROUND: The purpose of this study was to describe a novel, simple, and implant-free arthroscopic soft-tissue tenodesis technique of the long head of the biceps tendon (LHBT) to the rotator cuff using an absorbable monofilament suture and to evaluate its clinical and functional outcomes at a minimum 1-year follow-up.

METHODS: A retrospective case series of 23 patients (mean age 58.0 ± 7.8 years) who underwent arthroscopic rotator cuff repair with concomitant LHBT soft-tissue tenodesis between June 2021 and June 2023 was analyzed. Functional outcomes were assessed using Constant-Murley Score (CMS), American Shoulder and Elbow Surgeons (ASES) score, Single Assessment Numeric Evaluation (SANE), visual analog scale (VAS) for pain, and the Long Head of Biceps (LHB) score. Supination strength was measured with a handheld dynamometer and compared with the contralateral side. Incidence of Popeye deformity and tenderness over the bicipital groove were recorded.

RESULTS: Objective Popeye deformity was observed in 13% of patients, with subjective concern reported by only 4.3%. Supination strength and LHB scores were similar to the contralateral side (means of 104.65 versus 104.64 N; LHB 93.7 versus 94.6 points). The mean CMS and ASES scores were 90.3 ± 12.4 and 89.6 ± 15.9 points, respectively. The SANE score averaged 87.4 ± 20.9, and the VAS for pain was low (1.65 ± 2.59 cm).

CONCL […]

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ORIF of three- and four-part proximal humeral fractures: a retrospective study of long-term clinical and complication outcomes of two-plate fixation methods.

Conteduca J, Gasbarra E, Rausa I, Casto A, Valentino L, Solarino G, Meccariello L, Rollo G

BACKGROUND: The aim of this study was to compare clinical outcomes and complication rates during long-term follow-up (≥ 6 years) after treatment for three- and four-part proximal humeral fractures using two different types of locking plates.

MATERIALS AND METHODS: A total of 113 patients with three- and four-part proximal humeral fractures who underwent surgery between September 2012 and January 2019 were enrolled retrospectively. Data for 49 patients [9 males, 40 females; mean age [standard deviation] (SD) 68.9 (5.8) years} treated with a PGR (intrauma) plate (group A) and 51 patients [10 males, 41 females; mean age (SD): 66.0 (7.5) years] treated with a PHILOS humeral plate (PHP, group B) were available at the last follow-up. The mean follow-up periods in groups A and B were 10.8 and 10.2 years, respectively, with a minimum follow-up of 6 years. At the final follow-up evaluation, functional outcomes were assessed using the Oxford Shoulder Scale (OSS), Simple Shoulder Test (SST), and Disabilities of the Arm, Shoulder and Hand (DASH) questionnaires. X-ray evaluation was also performed, and complications were recorded.

RESULTS: The mean OSS score in the PGR and PHP groups at follow-up was 42.9 and 39.1, respectively ($p > 0.05$). The SST score (PGR: 7.5 ± 2.0; PHP: 7.3 ± 4.0) and DASH score (PGR: 22.1 ± 5.5; PHP: 20.5 ± 4.0) were similar in both groups ($p > 0.05$). In our serie […]

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Incidence rates and treatment of the transcervical fracture of the neck of femur in Italy: is total hip arthroplasty an increasingly preferred approach? A population study on trends between 2001 and 2023 based on 1,120,770 hospital discharge records.

Ciminello E, Romanini E, Venosa M, Cazzato G, Tucci G, Boniforti F, Carpanese L, Cuccu A et al.

INTRODUCTION: Transcervical femoral neck fractures (TFNFs) are among the most devastating fragility fractures in the elderly. TFNF are associated with excess 1-year mortality rates ranging from 15% to 30%. Treatments include conservative methods, internal fixation, and arthroplasty (partial or total hip arthroplasty). This study aims to analyze the changes in incidences of TFNF in the Italian population between 2001 and 2023 and the evolution of the choices of treatment.

MATERIALS AND METHODS: Using hospital discharge record (HDR) data from 2001 to 2023, records with ICD9-CM codes for femoral neck fractures (820.0 and 820.1) among diagnoses were selected and categorized into four treatment groups: total arthroplasty, partial arthroplasty, fixation, and conservative. Time series were analyzed with stratification by sex and age.

RESULTS: The extracted data included 1,120,724 records of TFNFs, with 871,161 cases treated surgically (total or partial arthroplasty or internal fixation) and 249,563 treated conservatively; the average patient age was 79.1 years, with a higher proportion of women (72.8%). Partial hip arthroplasty was the preferred treatment overall. For younger patients, in the age classes < 45 and 45-54 years, fixation was the most chosen treatment. Over time, the use of the conservative treatment decreased from 27.5% in 2001 to 14.6% of cases in 2023. The use of par […]

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Does latissimus dorsi tendon transfer provide a normal scapular rhythm and external rotation strength in posterosuperior massive irreparable rotator cuff tears? A kinematic analysis in a retrospective cohort.

Calvo C, Fiumana G, Brigo A, Ruggiero ED, Bonfatti R, Donà A, Micheloni GM, Giorgini A et al.

BACKGROUND: Posterosuperior massive irreparable rotator cuff tears (PMIRT) are rare and disabling conditions. When conservative treatment fails, latissimus dorsi tendon transfer (LDTT) is a viable surgical option for symptom relief in carefully selected patients. However, its effectiveness in restoring glenohumeral function and its influence on scapulothoracic rhythm remain subjects of ongoing debate.

PURPOSE: The purpose of this study was to evaluate the clinical outcomes of LDTT and assess its impact on scapulothoracic rhythm using kinematic and electromyographic analysis.

MATERIAL AND METHODS: A total of 18 patients with PMIRT underwent LDTT. Functional scores consisted of Constant-Murley score (CMS), America Shoulder and Elbow Surgeons (ASES) score, and Quick Disabilities of the Arm, Shoulder, and Hand (QuickDASH) score. Electromyography (EMG) activity of the latissimus dorsi muscle was evaluated in conjunction with three-dimensional (3D) kinematic tracking system. Differences in external rotation strength were compared with and without active adduction.

RESULTS: The mean age was 55.9 years (range 40-69 years), with a mean follow-up of 20.4 months (range 6-39 months). At final follow-up, the mean CMS was 68.2 (95% CI 64.9-71.5), ASES was 76.9 (95% CI 66.1-87.6), and QuickDASH was 17.6 (95% CI 9.5-25.6). A significant difference in external rotation strength was observed […]

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Is there a need for exploration in pulseless supracondylar fractures of the humerus? A systematic review and individual patient data meta-analysis.

Haoyang C, Chen JG, Lai H, Lim AKS, Tan SHS, Hui JHP

BACKGROUND: Different approaches have been proposed to treat patients with pulseless supracondylar humeral fractures (SHF). We aim to analyze the current outcomes of patients who have undergone different treatment options for pulseless SHF, namely watchful waiting versus urgent surgical exploration.

METHODS: Electronic databases including PubMed, Embase, and Cochrane Library were explored. Our study included patients from all age groups but only included English-language articles. Single case reports and case studies with adequate description of patient population, injury, and outcomes were included. An individual patient data meta-analysis was done to evaluate key outcomes of surgical exploration versus no surgical exploration in pulseless patients with SHF.

RESULTS: Overall, the data for 1070 individual patients from a total of 48 studies were included. Patients with pulseless SHF with open fractures have a higher probability of requiring vascular intervention ($p < 0.001$) as they are more likely to have disrupted arteries ($p = 0.050$) and more likely to require vascular repairs ($p < 0.001$) than patients with closed fractures. Similarly, patients with pulseless SHF with pucker sign ($p = 0.003$) and ecchymosis ($p = 0.002$) were more likely to undergo surgical exploration. However, the neurological status had no relation to them undergoing surgical exploration ($p = 0.382$), and al […]

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Increasing complexity and plateauing outcomes in AO type C distal humerus fractures: a 50-year institutional case series.

Hönck K, Wenzelburger P, Eibinger N, Sadoghi P, Puchwein P

BACKGROUND: Distal humerus fractures in adults are rare but complex injuries, particularly AO type C3 fractures, often involving comminution, instability, and osteoporotic bone. Demographic trends indicate rising incidence, especially among elderly patients, with increasing functional demands challenging modern osteosynthesis. Open reduction and internal fixation (ORIF) with bicolumnar plating remains the preferred reconstructive approach, although complications remain substantial. This study aimed to evaluate trends in fracture complexity, patient demographics, and functional outcomes over five decades, focusing on the most recent cohort (2009-2018, series E) and comparing with historical cohorts (series A-D, 1969-2008).

METHODS: This retrospective analysis included five consecutive 10-year institutional cohorts (series A: $n = 43$; B: $n = 29$; C: $n = 47$; D: $n = 58$) of 233 patients with 235 supradiacondylar distal humerus fractures (AO type C1-3) treated with ORIF. Functional outcome parameters (Jupiter and Cassebaum Scores) and demographic data were available for all cohorts; additional parameters (range of motion (ROM), QuickDASH Score) were available for series D and E, Mayo Elbow Performance Score (MEPS), Short Form (SF)-36, and strength were assessed specifically in series E ($n = 56$; follow-up $n = 21$). Cross-cohort comparisons of categorical variables used chi-square tests; […]

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Magnetic resonance imaging-based fat infiltration grading improves reparability prediction in massive rotator cuff tears.

Hu Q, Zhang G, Wang D, Tung TH, Ding J, Zhou X

PURPOSE: This study aims to investigate the extent of fat infiltration (FI) in the rotator cuff (RC) muscles in more medial slices from the Y-shaped view (Y-view) on shoulder magnetic resonance imaging (MRI) and assess whether these slices provide a more accurate prediction of the repairability of massive RC tears (massive RCTs).

METHODS: The retrospective study included 57 patients with massive RCTs who successfully underwent arthroscopic surgery between 1 January 2023 and 31 December 2023. Patients were categorized into two groups: the irreparable group and the reparable group. All patients underwent shoulder MRI covering the region from the acromion to the medial border of the scapula in oblique coronal and oblique sagittal orientations. The FI stage of the RC was assessed across three different views to determine which view most effectively predicts the repairability of patients with massive RCTs.

RESULTS: The FI stage of the infraspinatus (IS) muscle across three distinct views demonstrated a significant correlation with the repairability of massive RCTs. The FI stage was more severe in the irreparable group. Receiver operating characteristic (ROC) analysis revealed that the area under the curve for the 1/2 scapula view (0.737) and the most severe view (0.745) exceeded that of the traditional Y-view (0.644). Paired-sample ROC curve analysis revealed a significant differe […]

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Mid-term clinical and functional outcomes after reverse shoulder arthroplasty with latissimus dorsi transfer.

Colombini AG, Rab P, Macken AA, Soares MN, Kimmeyer M, Shirinskiy IJ, Popescu IA, Lafosse L et al.

BACKGROUND: Although reverse total shoulder arthroplasty (rTSA) with concomitant latissimus dorsi transfer (LDT) has been shown to effectively treat external rotation (ER) deficits, there are limited data regarding its outcomes with modern implants and its impact on activities of daily living (ADLs) requiring ER. The purpose of this study was to assess the mid-term clinical and radiographic outcomes of rTSA with concomitant isolated LDT in patients with an ER lag sign and posterior rotator cuff deficiency.

METHODS: This retrospective cohort study with prospective follow-up included consecutive patients who underwent rTSA with concomitant isolated LDT between 2010 and 2022 with a minimum follow-up of 2 years. Primary outcomes included the resolution of ER lag sign, active ER, and the Activities of Daily Living and External Rotation (ADLER) score. Secondary outcomes included the Auto-Constant Score (CS), Subjective Shoulder Value (SSV), activities of daily living requiring internal rotation (ADLIR) score, visual analog scale (VAS) for pain, and radiographic analysis of standardized radiographs.

RESULTS: In total, 32 procedures in 32 patients were identified. Of these, 22 procedures in 22 patients (68% female, 72.9 ± 8.4 years at surgery) were available for follow-up at 4.8 ± 2.2 years postoperatively (response rate 73%). The ER lag sign resolved in 95.5% of patients, the active […]

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Partial versus full revision in cementless single-stage exchange for hip periprosthetic joint infection: a multicenter comparative study with 10-year survivorship analysis.

Mu W, Jarusriwanna A, Xu B, Brown SA, Zhang X, Dawes A, Parvizi J, Cao L

BACKGROUND: Periprosthetic joint infection (PJI) is one of the most challenging complications following total hip arthroplasty (THA). Traditional management typically involves full revision to ensure comprehensive infection eradication. However, for patients with well-fixed implants, partial revision in a single-stage cementless approach may provide a viable alternative, potentially preserving bone stock and reducing operative time. The relative efficacy of this approach compared with full revision remains to be fully explored. This multicenter study aims to determine whether partial revision in cementless single-stage exchange surgery offers comparable infection control outcomes to full revision for select patients with well-fixed implants.

METHODS: We conducted a retrospective, multicenter cohort study involving 226 patients who underwent cementless single-stage exchange hip arthroplasty for PJI between 1 October 2013 and 31 July 2022. Patients were divided into partial revision (n = 61) and full revision (n = 165) groups. The primary outcome was treatment success, defined as the absence of clinical symptoms and signs of infection at a minimum follow-up of 2 years.

RESULTS: The success rates were 77.0% for the partial revision group and 80.0% for the full revision group, with no significant difference (p = 0.629). Both groups showed comparable 10-year survival rates for ove […]

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Internal Rotation and Transthoracic Lateral Radiographs Frequently Disagree on Sagittal Angulation in Humeral Shaft Fractures (AO/OTA 12).

Zhang D, Brooks J, Schimizzi G, Erdman MK, Maassen N, Christiano AV

OBJECTIVES: The aim of this study was to compare humeral shaft fracture sagittal angulation between transthoracic and internal rotation radiographs at the time of injury, and to identify fracture patterns associated with discrepancies between views.

DESIGN: Retrospective cohort study.

SETTING: Single Level 1 academic trauma center.

PATIENT SELECTION CRITERIA: Adult patients presenting with closed humeral shaft fractures (AO/OTA 12A-C) between May 2018 and December 2024 were screened for inclusion. Patients with radiographs that included both internal rotation and transthoracic lateral views at the time of injury before bracing or splinting were included.

OUTCOME MEASURES AND COMPARISON: Demographic variables, injury mechanism, AO/OTA classification, and fracture location were collected. Sagittal angulation was measured on both views, and differences of greater or less than 10 degrees between views were noted. Acceptable humeral shaft fracture alignment was defined as less than 20 degrees of sagittal angulation. A paired sample t test was used to compare angulation between the 2 views. Fisher Exact Test was used to compare fracture characteristics between patients with less than 10 and more than 10 degrees of difference.

RESULTS: A total of 31 patients were included. The cohort was 54.8% female, with a mean age of 42.4 ± 20.4 years. The 2 views disagreed on acceptable align […]

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Results of a 2025 US Practice Survey of Orthopaedic Traumatologists.

Puckett HD, Borgida JS, Del Olmo M, Lundy D, Cunningham BP, Cannada LK

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Shifts in Medicare Reimbursement for Common Lower Extremity Orthopaedic Trauma Procedures, 2006-2024.

Hoveidaei AH, Mosalamiaghili S, Feng JE, Anoushiravani AA

OBJECTIVES: The aim of this study was to analyze Medicare reimbursement trends from 2006 to 2024 for open treatment of proximal femoral fractures [Current Procedural Terminology (CPT) 27236], femoral shaft fractures (CPT 27506), tibial shaft fractures (CPT 27759), and bimalleolar ankle fractures (CPT 27814), using data from the Centers for Medicare and Medicaid Services (CMS) Physician Fee Schedule (PFS).

DESIGN: This study analyzed Medicare reimbursement trends for the 4 most common lower extremity trauma procedures from 2006 to 2024.

SETTING: Reimbursement data were obtained from the Centers for Medicare and Medicaid Services (CMS) Physician Fee Schedule (PFS), and procedure frequency data were obtained from the M165Ortho dataset from PearlDiver Technologies.

PATIENT SELECTION CRITERIA: Procedures were selected based on frequency and included CPT codes 27236, 27506, 27759, and 27814.

OUTCOME MEASURES AND COMPARISONS: Reimbursement data were obtained from the CMS PFS using the corresponding Healthcare Common Procedure Coding System codes. Work reimbursement was calculated by multiplying work relative value units (RVUs) by the CMS conversion factor (CF), and total facility reimbursement was calculated by multiplying total facility RVUs by the CF. Values were adjusted to 2024 US dollars using the Consumer Price Index. Percentage changes from 2006 were calculated. A 5-year pr […]

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Infection Rate by Gustilo-Anderson Classification in Open Tibial Shaft Fractures Treated With an Intramedullary Nail-A Meta-Analysis.

Schermerhorn JT, McGlone PJ, Torrie AM, Carlini AR, Castillo RC, Abar B, Hendren S, Wesorick BR et al.

OBJECTIVES: To quantify deep surgical site infection (SSI) rates after intramedullary nailing of open tibial shaft fractures stratified by Gustilo-Anderson (G-A) classification.

DATA SOURCES: A systematic review of MEDLINE, Embase, and Scopus screened for English language studies published from January 2000 through May 2025 in accordance with PRISMA guidelines.

STUDY SELECTION: Studies included skeletally mature patients with open tibial shaft fractures (OTA/AO 42) treated with locked intramedullary nailing and reporting deep SSI rates by G-A type.

DATA EXTRACTION: Two independent reviewers screened studies and extracted data, with conflicts resolved through consensus or adjudication by a third independent reviewer.

DATA SYNTHESIS: A single-arm meta-analysis was performed to estimate pooled infection rates using fixed-effects and random-effects models, with between-study heterogeneity assessed using Cochran Q and the I2 statistic.

RESULTS: Seventeen studies composed of 2063 total patients met inclusion criteria. The overall pooled deep SSI rate was 13.2% [95% confidence interval (CI), 11.8%-14.8%]. Stratified pooled infection rates were 6.2% (95% CI, 3.7%-10.3%) for G-A type I, 7.6% (95% CI, 5.6%-10.1%) for type II, 11.9% (95% CI, 9.5%-14.6%) for type IIIA, 25.0% (95% CI, 18.6%-32.6%) for type IIIB, and 41.8% (95% CI, 21.3%-65.5%) for type IIIC fractures. Subgroup analysis […]

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Adverse Mental Health Outcomes After Modern External Ring Fixation Versus Internal Fixation for Severe Open Tibia Fractures: A Secondary Analysis of the FIXIT Study.

Simske NM, Solarczyk JK, Thompson AR, Reider L, O'Toole RV, Carroll EA, Karunakar MA, Obremskey W et al.

OBJECTIVES: To report the prevalence of adverse mental health outcomes 1 year after severe open tibial shaft fractures and their association with treatment and complication risk factors.

DESIGN: Randomized controlled trial.

SETTING: Multicenter.

PATIENT SELECTION CRITERIA: Eligible patients were 18-64 years of age with severe open tibial shaft fractures (AO/OTA 41-43) treated with modern external ring fixation or internal fixation. All Gustilo-Anderson type IIIB fractures were included and some type IIIA meeting specific severity criteria.

OUTCOME MEASURES AND COMPARISONS: Outcome measures included Patient Health Questionnaire scores for depression symptoms, posttraumatic stress disorder (PTSD) Checklist for DSM-IV (PCL-S), and the Brief Pain Inventory. Primary comparisons assessed depression, PTSD, and pain between treatment groups (modern external ring fixation vs. internal fixation).

RESULTS: A total of 254 patients were treated with external ring fixation (n = 121) or internal fixation (n = 133). The mean age was 39 years (SD = 13) and 16% were female. At 12 months after injury, 34% of patients had Patient Health Questionnaire-9 scores consistent with moderate to severe depression. Risk factors for depression were female sex (RR: 1.70 [1.01-2.86]), high school educational attainment or less (RR: 1.93 [1.27-3.04]), and any baseline mental health condition (RR: 1.91 [1.2 […]

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Metabolic Derangements Are Independent Additive Risk Factors for Tibial Diaphyseal Nonunion.

Kotzur T, Peterson B, McCoy L, Pereira DE, Hosseinzadeh P, Martin C

OBJECTIVES: The aim of this study was to evaluate whether altered metabolic factors (metabolic comorbidities) increase the risk of postoperative tibial diaphyseal nonunion after fracture fixation, and to determine whether additional metabolic risk factors contribute cumulatively.

DESIGN: Retrospective cohort analysis.

SETTING: The TriNetX Research Network.

PATIENT SELECTION CRITERIA: Patients with operatively treated tibial diaphyseal fractures (AO/OTA 42) between January 1, 2013 and December 31, 2023 were identified. A minimum of 1 year of follow-up was required. Patients with a previous diagnosis of tibial nonunion or underlying metabolic bone disease were excluded.

OUTCOMES MEASURES AND COMPARISONS: The primary outcome was tibial diaphyseal nonunion after the index fracture. Outcomes were compared between patients with and without each metabolic comorbidity of interest (obesity, hypertension, diabetes, and dyslipidemia), and across groups stratified by cumulative metabolic burden to evaluate additive risk.

RESULTS: A total of 36,474 patients (mean age 41.5 years, range 18-90; 62.7% male) were included in the analysis. Metabolic comorbidities were independently associated with a higher risk of nonunion, including diabetes (Hazard Ratio (HR) 2.82, P < 0.001), hypertension (HR 1.87, P < 0.001), obesity (HR 1.45, P < 0.001), and dyslipidemia (HR 1.25, P < 0.001). When strat […]

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Comparative Efficacy of Anterior MIPO With PHILOS Versus Posterior ORIF With EA-LCP in Distal-Third Humerus Fractures: A Randomized Controlled Trial.

Garika SS, Yadav P, Jyoti NJ, Bansal H, Mittal S, Trikha V, Sharma V

OBJECTIVES: To compare anterior minimally invasive plate osteosynthesis (MIPO) using the proximal humerus internal locking system in an upside-down reverse configuration with conventional posterior open reduction and internal fixation (ORIF) using extra-articular locking compression plate for the treatment of extra-articular distal-third humeral shaft fractures.

DESIGN: Prospective randomized controlled trial.

SETTING: Single Level 1 trauma center (New Delhi, India).

PATIENT SELECTION CRITERIA: Adult patients aged 18-55 years with extra-articular distal-third humeral shaft fractures (AO/OTA types 12A-C) presenting within 2 weeks of injury were randomized to undergo anterior MIPO or posterior ORIF.

OUTCOME MEASURES AND COMPARISONS: Radiological union, functional outcomes (Mayo Elbow Performance Score, University of California-Los Angeles shoulder rating scale), and elbow range-of-motion were assessed and compared at 2 weeks, 6 weeks, 3 months, and 6 months postoperatively. Operative time, intraoperative blood loss, radiation exposure, postoperative pain (visual analog scale scores) and complications were also compared.

RESULTS: Thirty patients were included (anterior MIPO, n = 14; posterior ORIF, n = 16). The mean age was 27.9 years (range 18-55) in the MIPO group and 29.8 years (range 19-48) in the ORIF group, P = 0.598. There were 11 men and 3 women in the MIPO group and […]

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Blade Versus Screw Complication Risk for Intertrochanteric Fracture Fixation.

Richey Levine A, Cross JL, Klug T, Salameh M, Riedel M, Leslie M

OBJECTIVE: To determine whether the use of a screw or blade for fixation in intertrochanteric fracture intramedullary nailing is associated with bony complications, specifically cut-through, cut-out, malunion, nonunion, proximal hardware backout, and reoperation.

DESIGN: Retrospective cohort study.

SETTING: Single Level I Trauma Center.

PATIENT SELECTION CRITERIA: Patients with intertrochanteric hip fractures (OTA/AO 31A) between 2014 and 2023 who underwent fixation with the same type of cephalomedullary nails (CMN) with either a blade or screw for proximal femur stabilization were identified based on Current Procedural Terminology Codes 27245.

OUTCOMES MEASURES AND COMPARISONS: Bony complications, specifically cut-through, cut-out, malunion, nonunion, proximal hardware backout, and/or reoperation, were compared across blade and screw implants for patients. This comparison was completed for the entire cohort, with patients stratified by fracture stability, and with the application of propensity matching across fracture and demographic characteristics.

RESULTS: In total, 1078 patients were included. The 615 patients treated with CMN blade had a mean age of 83.1 (range 65-106) years and 451 were women. The 463 patients treated with CMN screw had a mean age of 83.3 (range 65-103) years and 354 were women. No statistical significance was found between CMN blade versus screw co […]

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Interpretation Accuracy Between Orthopaedic Surgeons and Qualified Remote Spanish Interpreters in a Trauma Setting.

Jaeblon T, Monarrez R, Molestina C, DaConceicao L, Babecki MG, Bauer BJ, Talwar S, Demyanovich HK

OBJECTIVES: To assess the interpretation accuracy of remote qualified Spanish medical interpreters (QMI) in an orthopaedic trauma setting and determine whether interpretation method (telephonic or video), delivery speed, and subject complexity were associated with interpretation accuracy.

METHODS: Three clinical vignettes-simple, moderate, and complex-describing fictitious orthopaedic trauma injuries, treatments, complications, and outcomes were written by the investigators. An investigator read each vignette type to a simulated patient through 20 randomly selected QMIs (10 through telephone and 10 through video), all of whom were trained, proficiency-tested, and blinded to the study design, yielding 60 total interpretations. Data collected included interpretation method, English delivery speed (words per minute), and vignette complexity. One orthopaedic traumatologist and 1 bilingual physician assistant evaluated transcriptions of recorded QMI Spanish interpretations using a nonvalidated, criterion-referenced, expert-derived categorical assessment reflecting the degree to which an interpreter preserved the fidelity of the original vignette for patient ascertainment of diagnosis, treatment, risks, and prognosis, termed overall interpretation designation (OID). OID was classified, "acceptable," "borderline," or "unacceptable." A second orthopaedist, blinded to the first 2 inves […]

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Intramedullary Reaming Biopsy in Diagnosing Metastatic Long Bone Disease.

Stevens KN, Pan X, Smith KL, Blackburn CW, Jonard BW, Napora JK

OBJECTIVES: The aim of this study was to describe the diagnostic accuracy of intramedullary reaming biopsy in cancer patients with long bone lesions or suspected pathologic fractures in determining the presence of metastatic disease.

DESIGN: A retrospective review of electronic medical records.

SETTING: Single academic Level I Trauma Center.

PATIENT SELECTION CRITERIA: Patients with a known cancer diagnosis who underwent intramedullary nail fixation for a long bone lesion or pathologic fracture of the femur, humerus, and tibia for whom intramedullary reamings were sent to pathology between January 2013 and October 2021 were included. Patients for whom open biopsy was performed were excluded.

OUTCOME MEASURES AND COMPARISONS: The primary study measure was intramedullary reaming biopsy sensitivity.

RESULTS: Forty-six reamings from 44 patients were included, with a mean patient age of 69 years (range, 41-85 years) and 35 being female (76%). Approximately half of cases were therapeutic fixation of a pathologic fracture (52%) versus prophylactic (48%) in nature. Most reaming biopsies were from the femur (42/46, 91%), followed by humerus (3/46, 7%), and tibia (1/46, 2%). The most common diagnoses were breast (34%) and lung (20%) carcinoma and multiple myeloma (20%). Thirty-two reamings (70%) were able to establish a diagnosis of metastatic disease. Five samples (11%) were crushe […]

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Early surgery of trochanteric femoral fractures in patients on direct oral anticoagulants (DOACs) reduces the length of stay (LOS) and the 1-year mortality rates without increasing the risk for revision surgery: a matched-pair analysis.

Weihs V, Laggner R, Humenberger M, Duma A, Frossard M, Hajdu S

BACKGROUND: It remains unclear whether early surgical fixation of trochanteric femoral fractures in patients with direct oral anticoagulation (DOAC) therapy has an impact on the long-term outcome.

METHODS: A retrospective cohort study on patients with trochanteric femoral fractures and DOAC therapy between 2016 and 2024 was conducted. Patients were divided into two groups according to their time to surgery and matched with age and ASA score as covariates. The primary outcome of interest was the impact of early surgery on the long-term outcome.

RESULTS: Of 233 included patients, 102 patients were assigned in each group after propensity score matching. Patients with early surgery had lower in-hospital complication rates (16.0% vs. 26.3%; $p = 0.077$) and a significantly shorter median length of stay (LOS) (11.5 days (IQR 6) vs. 15 days (IQR 6); $p < 0.001$). No difference in the rates of revision surgery (9.8% vs. 5.9%; $p = 0.298$) was observed. Kaplan-Meier curves revealed a significant difference in the mortality rates during the follow-up period ($p = 0.0057$) with lower in-hospital (2.0% vs. 6.9%; $p = 0.088$), 30-day (3.9% vs. 9.8%; $p = 0.097$) and 1-year mortality rates (20.6% vs. 31.4%; $p = 0.079$).

CONCLUSION: Early surgical fixation of trochanteric femoral fractures in patients on DOAC therapy leads to a significant reduction in the LOS without increasing the risk for peri- and […]

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Platform-specific learning curve and early tibial radiographic agreement of ROSA®-assisted medial unicompartmental knee arthroplasty in an experienced high-volume surgeon: a consecutive case series.

Petrillo S, Ardiri D, Marullo M, Romagnoli S

PURPOSE: To evaluate the platform-specific learning curve and early tibial radiographic agreement of imageless ROSA®-assisted medial unicompartmental knee arthroplasty (UKA), with short-term clinical findings assessed exploratorily.

METHODS: This retrospective consecutive case series included 30 patients who underwent ROSA®-assisted medial UKA with the Persona® Partial Knee implant between January and April 2026 at a high-volume arthroplasty referral centre. A continuous learning-curve cumulative summation (LC-CUSUM) analysis of total surgical duration was performed against an independently defined surgeon-specific historical benchmark derived from institutional operative-register skin-to-skin times for manual UKA performed by the same surgeon (20 ± 4 min). Adequate performance was defined as a mean duration of 25 min, inadequate performance as 30 min, and the reference standard deviation as 4 min. Type I and type II error targets were 0.05 and 0.10, respectively; the decision boundary was calibrated by Monte Carlo simulation over a 30-case monitoring horizon. Robotic time and a supportive comparison of cases 1-5 versus 6-30 were secondary efficiency analyses. Postoperative radiographic measurements were performed by an observer blinded to the planned and robot-validated ROSA® values. Agreement was quantified using Bland-Altman bias and 95% limits of agreement.

RESULTS: Mean […]

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Cartilage disruption patterns associated with percutaneous intramedullary screw fixation for metatarsal fractures: a cadaveric study.

Keith B, Burke J, Hynes K, Baker J, Huh J

PURPOSE: Intramedullary screw fixation is an emerging technique for operative metatarsal fractures, yet the extent and location of articular cartilage disruption from retrograde screw insertion remain undefined. This study quantifies cartilage surface disruption and maps injury locations in the metatarsal head following screw placement.

METHODS: Fourteen fresh-frozen cadaveric feet (56 lesser metatarsals) underwent standardized retrograde intramedullary screw fixation using 3.6 mm screws. Following dissection, the total articular surface area and cartilage defect were measured to calculate the percentage of disruption. The cartilage surface was divided into quadrants (dorsal-to-plantar) and thirds (medial-to-lateral) to determine screw trajectory.

RESULTS: Mean cartilage disruption increased progressively from the 2nd to 5th metatarsals: 10.17%, 11.2%, 14.93%, and 17.43%, respectively. All screws were located within the middle third in the medial-to-lateral plane. In the dorsal-to-plantar plane, the majority of screws were positioned in the 2nd and 3rd quadrants. No screws were placed in the medial or lateral thirds.

CONCLUSION: Retrograde intramedullary screw fixation results in measurable cartilage disruption, with a tendency for screw placement in the middle third in the medial-to-lateral plane and 2nd and 3rd dorsal-to-plantar quadrants. While the relative cartilage surf […]

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Locking plate fixation for proximal humerus fractures: are four proximal screws adequate? A clinical and radiological analysis of 105 consecutive cases.

Michelitsch C, Stillhard PF, Zindel C, Sommer C, Acklin YP

INTRODUCTION: Proximal humerus fractures are common injuries, and achieving stable fixation is essential to prevent complications such as varus collapse and screw perforation. While optimal screw positioning has been emphasized biomechanically, clinical evidence regarding the required number of screws in the humeral head, particularly when minimally invasive aiming devices limit placement to four screws, remains limited. This study evaluated whether fixation with four cranial screws provides adequate clinical and radiological stability.

MATERIALS AND METHODS: A retrospective cohort study was conducted including 105 patients with displaced proximal humerus fractures treated with plate osteosynthesis. Patients were grouped according to the number of proximal screws used: four screws (n = 58) or more than four screws (n = 47). The primary outcome was the Constant-Murley Score. Secondary outcomes included secondary screw perforation, head-shaft angle maintenance, restoration of medial cortical support, and avascular necrosis. Mean follow-up was 18 ± 6 months.

RESULTS: Functional outcomes were comparable between groups, with Constant-Murley Scores of 78 ± 7 in the four-screw group and 73 ± 13 in the multi-screw group (p = 0.23). Rates of screw perforation (8.6% vs. 10.6%; p = 0.72) and postoperative head-shaft angles were likewise similar, and no significant alignment change occur […]

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Nighttime intramedullary nailing of femoral fractures is not associated with worse outcomes: a 13-year Level I trauma center cohort.

Abialevich A, Benkovich V, Osinsky V, Adiniaev Y, Tzaytlin I, Heinemann Y, Acker A

BACKGROUND: Whether nighttime orthopedic trauma surgery adversely affects outcomes remains controversial. Evidence specifically addressing intramedullary fixation across femoral fracture locations remains limited. This study evaluated whether nighttime surgery was associated with a higher rate of recorded postoperative complications than daytime surgery.

METHODS: This retrospective observational cohort included adult patients with nonperiprosthetic femoral fractures treated with intramedullary fixation between January 2010 and January 2023 at a single Level I trauma center. Of 311 screened records, 23 were excluded because the patient was younger than 18 years (n = 17) or the fracture was periprosthetic (n = 6), leaving 288 procedures (242 daytime and 46 nighttime). The primary outcome was any recorded postoperative complication during available follow-up. Secondary outcomes included radiographic abnormality, delayed union (> 24 weeks), nonunion, revision surgery, and mortality. Surgeon experience was categorized as senior surgeon or resident. Exploratory fracture-location subgroup analyses and multivariable logistic regression were performed.

RESULTS: Nighttime patients were younger (35.1 ± 19.1 vs. 46.6 ± 24.6 years, p < 0.001), more frequently male (82.6% vs. 64.0%, p = 0.014), more often injured by a high-energy mechanism (80.4% vs. 56.6%, p = 0.002), and underwent surger […]

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Patient-reported outcomes after lateral unicompartmental and total knee arthroplasty: a propensity score-matched cohort study.

Hariri M, Sorbi R, Moradi B, Weishorn J, Trefzer R, Reiner T, Walker T, Nees TA

INTRODUCTION: Comparative evidence on patient-reported outcomes (PROs) after lateral unicompartmental knee arthroplasty (UKA) and total knee arthroplasty (TKA) remains limited. We compared PROs between cohorts matched for demographic characteristics and baseline knee function.

MATERIALS AND METHODS: Prospectively collected single-center data were retrospectively analyzed. Lateral UKA was performed for isolated lateral-compartment osteoarthritis and TKA predominantly for multicompartmental disease. One-to-one propensity score matching incorporated age and body mass index, exact matching on sex, and restrictions on age and preoperative Oxford Knee Score (OKS), yielding 31 pairs. Postoperative OKS was the primary outcome. Secondary outcomes included OKS improvement, Forgotten Joint Score (FJS), University of California, Los Angeles (UCLA) activity score, OKS minimal clinically important difference (MCID) achievement, and FJS patient acceptable symptom state (PASS) attainment.

RESULTS: All matching covariates achieved acceptable balance (absolute standardized mean differences < 0.20). Postoperative OKS did not differ significantly (UKA, 38.9 ± 7.0; TKA, 39.8 ± 7.5; mean paired difference [TKA minus UKA], 0.94; 95% confidence interval [CI], - 2.57 to 4.44; p = 0.590). No significant differences were detected for OKS improvement (difference, 0.77; 95% CI, - 2.62 to 4.17; p = 0.644) […]

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Pre- and postoperative 4D CT analysis of thumb CMC joint kinematics after first metacarpal extension osteotomy.

Guebeli A, Honigmann P, Dobbe JGG, Keller M

INTRODUCTION: First metacarpal extension (Wilson) osteotomy combined with modified Brunelli ligamentoplasty is performed for early thumb carpometacarpal (CMC1) osteoarthritis to improve pain and function. The purpose of this study was to quantify in vivo changes in CMC1 joint contact distribution and kinematics after this combined procedure and to determine whether postoperative patterns resemble those of healthy volunteers. We hypothesized that the procedure would reduce cumulative palmar contact during opposition-retroposition.

MATERIALS AND METHODS: In a prospective single-center observational cohort study, six adults with early CMC1 osteoarthritis (Eaton-Littler I-II) underwent standardized preoperative and 6-month postoperative four-dimensional computed tomography (4D CT) of the CMC1 joint during active opposition-retroposition. Five healthy volunteers served as reference with a single 4D CT scan. First metacarpal-trapezium motion was registered across frames. Joint contact was estimated using a 2.0 mm proximity surrogate on the trapezial surface. The primary outcome was change in cumulative palmar contact area. Secondary outcomes included change in dorsoradial contact, temporal variability of contact curves, and contact-centroid path length and mean position. Within-subject pre- versus postoperative comparisons were performed; healthy scans provided reference patterns. […]

PubMed · DOI

Minimal kyphotic progression following pediatric vertebral fractures: implications for reduced radiographic surveillance.

Wegmann S, Lenz M, Babasiz T, Eysel P, Mueller LP, Weber M

INTRODUCTION: Pediatric spinal fractures are rare injuries (14.24 per 100,000 per year), representing only 0.6-3% of all spinal injuries. While most stable compression fractures are managed conservatively, concerns exist regarding progressive vertebral collapse and kyphotic deformity. Unlike extensive adult literature, pediatric-specific data on radiographic progression following conservative management remain limited, creating uncertainty about the clinical utility of routine follow-up imaging in this radiosensitive population.

MATERIALS AND METHODS: Retrospective single-centre cohort study of 54 pediatric patients (mean age 12.3 years) with traumatic vertebral body fractures diagnosed between 2012 and 2024. Kyphosis angles were measured using Cobb methodology at baseline and follow-up intervals (1, 3, and 6 weeks). The primary outcome was the change in kyphotic angle; secondary outcomes included the identification of predictors of progression. Paired t-tests and multivariable linear regression analysis were performed.

RESULTS: Kyphosis angles remained essentially unchanged across all follow-up intervals (ΔFU1 + 0.19°, ΔFU2 + 1.27°, ΔFU3 - 0.14°), with no statistically significant differences detected ($p > 0.05$ at all timepoints). No clinical or radiographic factors predicted progression of kyphosis. Only three patients (5.6%) required treatment escalation. Subgroup analyses […]

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Correction: Different sagittal cutting plane orientations do not affect posterior tibial slope in open wedge high tibial osteotomy: a Sawbone model study.

Gurbanov E, Cochard B, Bauer DE, Miozzari HH, Tscholl PM

PubMed · DOI

Resident autonomy in orthopedic trauma: independent femoral IMN by senior residents yields comparable outcomes to fellowship-trained surgeons.

Balziano S, Nesher G, Ben Nun N, Segev T, Freidlander A, Prat D

INTRODUCTION: Intramedullary nailing (IMN) of femoral shaft fractures is a fundamental cornerstone procedure in orthopedic trauma and an important benchmark of resident surgical competency. While independent resident operating is essential for skill development, its safety in complex trauma cases remains debated. This study evaluated whether senior residents can safely perform femoral IMN without direct supervision compared with fellowship-trained trauma surgeons.

MATERIALS AND METHODS: A retrospective cohort analysis was performed of adult patients undergoing IMN for isolated femoral shaft fractures between 2015 and 2023 at a level I trauma center. Cases were grouped by primary surgeon: senior residents (postgraduate year ≥ 4) versus attending trauma surgeons. Demographics, fracture characteristics, operative parameters, complications, and revision rates were compared. To minimize selection bias, 1:1 propensity score matching (PSM) was performed based on preoperative covariates including age, comorbidity, fracture type, injury mechanism, and severity.

RESULTS: A total of 207 patients met inclusion criteria; 150 (75 per group) were matched. Operative duration was longer in resident-led cases (median 120 vs. 91 min, $p = 0.003$). Postoperative hospital stay (5.0 vs. 6.0 days, $p = 0.674$) and weight-bearing protocols ($p = 0.218$) were similar. Overall complication rates did not dif […]

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Racial and ethnic disparities in perceived health and barriers to care among adults with hip osteoarthritis: a national analysis of 16,000 + patients from the all of us program.

Khan ST, Devadiga SA, Emara AK, Brahmbhatt S, Elmenawi KA, Delaney PG, Piuzzi NS

OBJECTIVE: To explore racial and ethnic disparities in perceived health, healthcare access, and perceived barriers among adults with hip osteoarthritis (OA) in a large, nationally representative U.S.

METHODS: Cross-sectional analysis of 16,743 adults with hip OA from the All of Us Research Program. Participants self-identified as Black, Hispanic, or White. Outcomes included self-reported general, mental, physical, and social health, provider communication, and structural barriers to care. Between-group comparisons used Chi-square tests.

RESULTS: Black and Hispanic participants were younger, had lower income and education, and relied more on Medicaid than White participants (all $p < 0.001$). They reported higher rates of unaffordable deductibles, difficulty accessing follow-up care, insurance coverage issues, transportation challenges, and medication cost-related nonadherence (all $p < 0.001$) Table 3. Minority groups also more often experienced poor provider respect, trust, and communication. Across all health domains, Black and Hispanic participants reported poorer outcomes and greater emotional distress (all $p < 0.001$).

CONCLUSION: Significant racial and ethnic disparities exist in structural and perceptual barriers to care among adults with hip OA. These findings identify potentially modifiable barriers that warrant further study and may inform future efforts to improve equi […]

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Postoperative periprosthetic fractures associated with compressive osseointegration: a systematic review of the Zimmer Biomet Compress® device.

Hashimoto K, Nishimura S, Goto K

The Zimmer Biomet Compress® Compliant Pre-Stress (CPS) device uses a compressive osseointegration technique for endoprosthetic fixation and provides an alternative to stemed implants. Despite having benefits, such as bone stock retention, periprosthetic fractures continue to remain a concern. Therefore, this systematic review presents the prevalence, risk factors, and outcomes of the CPS device and periprosthetic fractures after treatment. Medical databases (PubMed, Embase, and Google Scholar) were searched to identify studies reporting complications related to the CPS device. Peer-reviewed studies with at least 2 years of follow-up for fracture rates, revision results, and implant survivorship were included. The primary data collected were fracture incidence and timing of complications, and the CPS device was compared with the traditional cemented stem. This review highlights important studies, including a cohort of 221 patients and pediatric data from 36 patients. Adult populations showed a periprosthetic fracture rate of 2.7-3.9%. A distinct temporal pattern emerged: the frequency of fractures was most pronounced within the first 2 years after surgery. In the pediatric context, spindle survivorship was 86.3% at 5-10 years, despite higher rates of mechanical complications. Notably, for most fracture cases, the bone-implant interface was radiographically stable, providing easi […]

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Hip resurfacing arthroplasty at a minimum of 14 years: revision is concentrated among female patients with dysplasia and small components.

Lin YC, Chang CH, Hsieh PH, Huang YY, Chen DW, Lau NC

INTRODUCTION: Hip resurfacing arthroplasty (HRA) is a bone-preserving option for young, active patients. The influence of the underlying diagnosis on long-term outcome remains unclear, particularly the distinction between primary osteoarthritis (OA), developmental dysplasia of the hip (DDH) and non-OA conditions.

MATERIALS AND METHODS: We retrospectively reviewed 104 consecutive metal-on-metal HRAs in 101 patients (three bilateral) performed by one surgeon between 2006 and 2011, stratified into Primary OA (n = 50), DDH (n = 29) and Non-OA (n = 25: avascular necrosis [AVN] n = 15, inflammatory arthritis n = 5, post-traumatic arthritis n = 3, Legg-Calvé-Perthes disease n = 2). AVN was analysed additionally as an independent group, and exploratory analyses restricted to female patients and to cups ≤ 50 mm addressed the collinearity of dysplasia, sex and size. Survivorship used Kaplan-Meier estimation with revision as the endpoint.

RESULTS: Implant status was known for all 104 hips; the 91 unrevised assessed hips had a mean follow-up of 16.2 ± 1.5 years. Overall 15-year survivorship was 95.2%. Primary OA, Non-OA and AVN each showed 100%; DDH 82.8% (versus Primary OA p = 0.002, versus Non-OA p = 0.031). All five revisions occurred in female DDH patients with a cup ≤ 50 mm; DDH survivorship remained lower within female patients (p = 0.025) and within small cups (p = 0.006), with no […]

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Elective arthroplasty on "high-risk" calendar dates: no evidence for worse outcomes in 43,000 procedures.

Walter N, Lowenberg DW, Heiss C, Lau EC, Rupp M

BACKGROUND: Concerns regarding the timing of elective surgery remain common among patients and may influence surgical scheduling. Superstition-associated calendar dates such as Friday the 13th are frequently perceived as "high-risk" time points, yet evidence regarding their impact on arthroplasty outcomes is lacking. This study investigated whether primary total hip arthroplasty (THA) and total knee arthroplasty (TKA) performed on these dates are associated with differences in revision risk, mortality, or procedure frequency.

METHODS: A nationwide retrospective cohort study was conducted using Medicare physician service records (2009-2019). Primary THA and TKA procedures were identified in a 5% sample of Medicare beneficiaries aged ≥ 65 years. Outcomes included aseptic revision, septic revision, and mortality. Competing-risk survival analyses

(Fine-Gray) and multivariable Cox regression models adjusted for demographic, clinical, and socioeconomic covariates were applied. Procedure volumes were compared between superstition-associated and control dates.

RESULTS: A total of 945 arthroplasties were performed on Fridays falling on the 13th. No significant differences were observed in aseptic revision, septic revision, or mortality compared with control dates. For Friday the 13th, hazard ratios were 0.80 (95% CI 0.44-1.47) for aseptic revision, 0.83 (95% CI 0.31-2.25) for septic r […]

PubMed · DOI

Evaluation of a novel strap fixation method for biomechanical testing of sheep Achilles tendons.

Basci O, Aydemir S, Husemoglu RB, Çeltik M

BACKGROUND: Reliable fixation of tendon specimens is a major challenge in biomechanical testing, as slippage and tissue damage can reduce accuracy and reproducibility. Although various clamp- and suture-based fixation methods have been reported, there is still no widely accepted, practical, low-cost technique that preserves tendon integrity during testing. This study aimed to compare commonly used fixation techniques and to evaluate whether a novel strap-based fixation method provides a more stable and tissue-preserving alternative for biomechanical testing of sheep Achilles tendons.

METHODS: Forty fresh-frozen cadaveric sheep Achilles tendons obtained from a local butcher were used, and the specimens were stored at - 20 °C until testing. After controlled thawing at room temperature, all tendons were repaired using the Bunnell suture technique with No. 0 Vicryl. Four different proximal fixation methods were evaluated: (1) sutured fixation with strap, (2) direct clamp fixation, (3) Kocher clamp fixation, and (4) Krackow suture fixation. All specimens were mounted on a universal testing machine and subjected to a load-to-failure protocol at a constant loading rate of 10 mm/s. Maximum force (N), displacement at failure (mm), and mechanical failure mechanisms were recorded for each sample. Data were analyzed using the Kruskal-Wallis test with Bonferroni-corrected Mann-Whitney U te […]

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Implementation and early results of same-day hip and knee replacement in a public hospital without prior same-day surgical experience, a feasibility study.

Jørgensen CC, Sveen TM, Johansen B, Sander M, Steen-Hansen C, Sundstrup MS, Rothe C, Lange KH

BACKGROUND: Hip and knee replacements are increasingly performed as same-day surgery (SDS). However, the feasibility of SDS pathways in departments without prior experience is unknown. We report on the implementation process, feasibility and early results of implementing a SDS protocol in a department without prior experience with same-day hip and knee replacement.

METHODS: The study design was based on Bowens feasibility study framework on implementation processes. Data were collected prospectively through patient reported questionnaires and medical records during, and from the local Department of Digitalisation and Analytics after implementation. Outcome data included discharge on day of surgery, 30-days primary-care contacts, 90-days readmissions and patient satisfaction.

RESULTS: Implementation was initiated as a dedicated fellowship from September 2023 to October 2024. Patients were admitted and discharged through a dedicated day-surgical department with transfer to the regular ward in case of admission. The first SDS patient underwent surgery the 4th of September 2023. By October 2024, 150 of 713 operated patients (21.0% CI:18.2-24.2) had SDS. Of these, same-day discharge was achieved in 125 (83.3% CI:76.6-88.5) patients of which 6 (4.0% CI:0.9-7.1) were readmitted within 90-days. From October 2024 to October 2025, 178 of 800 patients were planned for SDS (22.3% CI:19.5 […]

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Relationship of foot anthropometry with Modified Pirani score during Ponseti correction of idiopathic congenital talipes equinovarus: a prospective observational study.

Suresh M, Pinto DJV, Rajasekharan P

BACKGROUND: The Ponseti method is the gold standard treatment for idiopathic congenital talipes equinovarus (CTEV). Clinical monitoring during correction primarily relies on the Modified Pirani score, which evaluates deformity severity based on clinical findings. However, objective quantitative assessment of morphological foot changes during treatment remains limited. This study aimed to evaluate the relationship between serial foot anthropometric measurements and Modified Pirani scores during Ponseti correction of idiopathic clubfoot.

METHODS: A prospective observational cohort study was conducted in 30 children with idiopathic unilateral CTEV undergoing Ponseti correction at a tertiary care center between June 2024 and February 2026. Five anthropometric parameters were measured weekly for eight weeks: great toe to midheel distance (GT-MH), little toe to midheel distance (LT-MH), second toe to midheel distance (ST-MH), medial malleolus to midheel distance (MM-MH), and lateral malleolus to midheel distance (LM-MH). Modified Pirani scores were assessed simultaneously by two independent observers. Pearson correlation analysis and inter-observer reliability testing were performed.

RESULTS: Significant improvement was observed in all anthropometric parameters and Pirani scores over the study period ($p < 0.001$). Mean GT-MH increased from 80.73 ± 3.36 mm to 87.56 ± 3.41 mm, LT-MH f […]

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Union outcomes and associated factors in adolescent scaphoid fractures.

Torun Ö, Girgin AB, Ka■■kç■ M, Acar A, Çevik HB

PURPOSE: Scaphoid fractures (SF) in adolescents differ from adult fractures in fracture pattern, healing capacity, and skeletal maturity. Limited data exist on factors affecting union and healing time in this population. This study evaluated clinical, radiological, and treatment-related factors associated with union and healing time in adolescent SF.

METHODS: This study included 77 adolescents (aged 10-18 years) treated for SF. Chronological age, bone age, fracture characteristics, treatment modality, union status, and healing time were recorded. Fractures were classified by anatomical location, displacement, and time to diagnosis (< 6 weeks or ≥ 6 weeks). CT-based morphometric measurements, including scaphoid height, lateral intrascaphoid angle, and dorsal cortical angle, were obtained from initial computed tomography images. Functional outcomes were assessed using the QuickDASH and patient-rated wrist evaluation (PRWE) scores.

RESULTS: Union was achieved in 63 patients (81.8%), with a median union time of 8 weeks (range, 6-10 weeks). Union rates were highest in distal pole fractures (100%) and lowest in proximal pole fractures (33.3%). Non-displaced fractures had higher union rates than displaced fractures (92.5% vs. 56.5%), and fractures diagnosed within 6 weeks demonstrated better union outcomes. Surgical treatment, performed in 16.9% of patients, was associated with lowe […]

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The effect of surgeon volume on outcomes after total hip arthroplasty: a systematic review and meta-analysis.

Meissner N, Ortwig K, Ramos-Pascual S, Agu C, Stoeve J, Schrednitzki D, Rolvien T, Halder AM

PURPOSE: To synthesize, critically appraise and systematically review studies comparing outcomes after primary total hip arthroplasty (THA) performed by low- vs. high-volume surgeons, and to propose evidence-based threshold definitions for these categories.

METHODS: This review followed the PRISMA guidelines and was registered in PROSPERO. Medline and Embase were searched on March 24, 2025. Prospective and retrospective clinical studies were included if they reported THA thresholds for low- and high-volume surgeons. Articles not written in English or German were excluded. Outcomes were pooled and presented in forest plots. Methodological quality was assessed.

RESULTS: Of 872 identified articles, 18 were included, reporting on 796,061 patients undergoing THA. Methodological quality was high in all studies (>6 of 7 points). Thresholds used to define high- vs. low-volume were determined using arbitrary values ($n = 11$), tertiles ($n = 1$), quartiles ($n = 3$), percentiles ($n = 2$), or receiver operating characteristic curves ($n = 1$). High-volume surgeons had significantly lower 1-month readmission rates [odds ratio (OR) = 1.6; $p = 0.033$] and 3-month mortality rates (OR = 1.8; $p = 0.008$). In contrast, there were no significant differences between high- and low-volume surgeons for 3-month revision rates (OR = 2.0; $p = 0.228$), 12-month revision rates (OR = 1.8; $p = 0.228$), and 3-month re […]

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Hexapod-assisted limb salvage after talectomy: outcomes of tibiocalcaneal arthrodesis using the TL-hex system.

Guerrero Serrano JA, Sánchez Morata E, García López JM, Mencía González S, Vilá Y Rico J

BACKGROUND: Severe talar destruction due to infection, trauma, avascular necrosis, or neuroarthropathy represents a major reconstructive challenge. Talectomy followed by tibiocalcaneal arthrodesis (TCA) is a limb-salvage option in selected high-risk patients, although optimal fixation remains controversial. Modern hexapod external fixators offer computer-assisted multiplanar control that may improve alignment and fusion after talectomy. The aim of this study was to evaluate the clinical and radiographic outcomes of talectomy followed by TCA using the TL-HEX system in a series of high-risk patients.

METHODS: A retrospective observational study was conducted between January 2020 and January 2024 including all patients who underwent total talectomy followed by TCA using a TL-HEX circular external fixator. Indications included talar destruction secondary to infection, avascular necrosis, post-traumatic collapse, Charcot neuroarthropathy, and aseptic loosening of total ankle arthroplasty. Primary outcomes were radiographic fusion and hindfoot alignment. Secondary outcomes included time to union, functional scores (AOFAS, EFAS), pain (VAS), complications, and patient satisfaction. Preoperative and final follow-up outcomes were compared using paired statistical tests, with significance set at $p < 0.05$.

RESULTS: Thirteen patients (mean age 57 years) were included with a mean follow-u […]

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Radiographic changes in midfoot alignment after hallux valgus correction with and without concomitant second tarsometatarsal arthrodesis: a comparative study.

Kim J, Kyung MG, An Y, Kim NY, Lee KM, Lee DY

INTRODUCTION: Second tarsometatarsal (TMT) joint osteoarthritis occurs in a subset of patients with hallux valgus (HV). While second TMT arthrodesis is performed to address symptomatic arthritis, its association with midfoot alignment changes during HV correction remains unclear. This study aimed to compare changes in radiographic arch parameters between patients undergoing HV correction with and without concomitant second TMT arthrodesis.

MATERIALS AND METHODS: This retrospective comparative study included 73 feet with HV and medial column malalignment (lateral Meary's angle > 10°). The TMT group (n = 37) underwent HV correction combined with second TMT arthrodesis for symptomatic osteoarthritis, while the control group (n = 36) underwent isolated HV correction. These patients were matched to achieve comparable baseline characteristics. Radiographic parameters of the transverse arch (M2-M5 angle, C1-M2 distance) and medial column alignment (lateral Meary's angle) were evaluated preoperatively and at a minimum 1-year follow-up. Functional outcomes were assessed as a secondary measure.

RESULTS: The TMT group demonstrated greater changes in transverse arch parameters compared with the control group. The M2-M5 angle decreased by 2.1° (95% CI, 1.4° to 2.7°) in the TMT group, whereas no significant change was observed in controls (-0.2°; 95% CI, -0.7° to 0.4°). The C1-M2 distance […]

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Coronal shear fractures of the distal humerus in children and adolescents: proposal for a modified classification.

Lieber J, Esser M, Fernandez FF, Schmittenbecher PP, Schneidmueller D, Strohm P, Kaiser MM, Kraus R

PURPOSE: Coronal shear fractures (CSF) of the capitulum and trochlea humeri are rare injuries during growth age. No systematic, prospective analysis of the classification, procedure, and outcome has yet been found in the literature.

MATERIALS AND METHODS: Based on the results of a multicentre study of CSF in growing patients, we propose a redesign of the AO Paediatric Comprehensive Classification of Long-Bone Fractures (PCCF) for children and adolescents. We also assessed inter-rater agreement consistency with the modified classification and calculated the intra-class correlation coefficient (ICC) and its 95% confident interval (CI).

RESULTS: A detailed classification has been proposed that refers to the location of the fragment, the extent to which the trochlea is involved, and the distinction between single and multiple fracture injuries. This modified AO classification distinguishes between five types of injury (13-E/3.1I - 13-E/3.1IV and 13-E/3.2), replacing the current code 13r-E/8.1. Inter-rater reliability using this modified classification was moderate (ICC 0.452 (95% CI 0,284; 0,661). This is more accurate than all previous classifications tested for CSF.

CONCLUSION: A revised classification of coronal shear fractures of the distal humerus in children and adolescents is required. Our proposed classification considers the various types of this injury and their implic […]

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Correction: Prevalence and risk factors for stem subsidence following reverse shoulder arthroplasty using on-lay press-fit short stems.

Nishiura R, Nakazawa K, Manaka T, Hirakawa Y, Ito Y, Terai H

PubMed · DOI

Reverse shoulder arthroplasty vs. hemiarthroplasty for proximal humerus fracture: in-hospital outcomes in a national trauma registry.

Raizah A

BACKGROUND: Reverse shoulder arthroplasty (RSA) has rapidly supplanted hemiarthroplasty for acute proximal humerus fracture, driven by superior functional outcomes and lower revision rates. Whether this shift has altered perioperative morbidity at the population level remains poorly characterized. This study compared in-hospital outcomes between RSA and hemiarthroplasty in a national trauma cohort.

METHODS: This retrospective cohort study used the National Trauma Data Bank (2017-2022). Adults (> = 18 years) admitted with proximal humerus fracture who underwent RSA (n = 2657) or hemiarthroplasty (n = 428) during the index hospitalization were included. The primary outcome was non-home discharge; secondary outcomes were any in-hospital complication, extended length of stay exceeding 7 days, and intensive care unit (ICU) admission. Multivariable logistic regression and inverse probability of treatment weighting (IPTW) were used to address confounding by indication.

RESULTS: Among 3085 patients (median age, 73 years; 2304 [74.7%] female), RSA patients were older (median, 74 vs. 64 years; P<.001) and more frequently female (77.3% vs. 58.6%; P<.001). After multivariable adjustment, RSA was not significantly associated with non-home discharge (adjusted odds ratio [aOR], 1.20; 95% confidence interval [CI], 0.93-1.55), any complication (aOR, 1.01; 95% CI 0.58-1.77), extended length o […]

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Impact of cognitive impairment on postoperative delirium and discharge disposition following knee arthroplasty: a Japanese nationwide propensity score-matched study.

Mori Y, Tarasawa K, Tanaka H, Kanabuchi R, Mori N, Fushimi K, Aizawa T, Fujimori K

PURPOSE: As populations age, the number of older patients undergoing knee arthroplasty continues to increase. Cognitive impairment is prevalent among older adults and has been associated with adverse perioperative outcomes; however, its impact on postoperative delirium and discharge disposition following knee arthroplasty remains insufficiently defined, particularly in large Asian populations.

METHODS: We conducted a nationwide retrospective cohort study using Japan's Diagnosis Procedure Combination database from April 2016 to March 2023. Older patients who underwent primary total knee arthroplasty (TKA) or unicompartmental knee arthroplasty (UKA) were identified. Cognitive impairment was defined using ICD-10 diagnostic codes at admission. The primary outcome was postoperative delirium, and secondary outcomes included discharge to home, length of hospital stay, and perioperative blood transfusion. Propensity score matching (1:1) was performed to adjust for demographic factors, comorbidities, and surgical characteristics. Multivariable logistic regression and sensitivity analyses were conducted.

RESULTS: Among 259,319 eligible patients, 3,934 matched pairs were identified after propensity score matching. Patients with cognitive impairment had a significantly higher risk of postoperative delirium compared with those without cognitive impairment (absolute risk difference, 3.2%; […]

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Association of primary tumor surgery with survival outcomes in patients with bone metastases across multiple cancer types.

Xu PH, Shi B, Li T, Tian M, Fang M, Cheng M, Huang W, Ren F et al.

BACKGROUND: Synchronous bone metastases (BM) have a notable impact on the survival rates of patients across various cancer types. While primary tumor surgery (PTS) has shown potential benefits, its impact across different cancers and patient subgroups needs further clarification. This study aimed to investigate the survival benefits of PTS across various cancer types.

METHODS: This retrospective cohort study analyzed data from the Surveillance, Epidemiology, and End Results (SEER) database, covering 53,015 patients diagnosed with BM from 2010 to 2019. Propensity score matching (PSM) was performed to reduce baseline imbalance. Overall survival (OS) and cancer-specific survival (CSS) were evaluated using Kaplan-Meier analysis and Cox proportional hazards regression.

RESULTS: PTS substantially improved overall and cancer-specific survival in patients with lung, kidney, breast and bladder cancers. All surgical modalities benefited breast and renal cancer survival. In prostate cancer, PTS produced no overall survival gain; survival varied by surgery type, being better after prostatectomy and worse with local resection. For liver cancer, PTS improved cancer-specific but not overall survival, and the small matched cohort necessitates circumspect interpretation.

CONCLUSION: The findings indicate that PTS can significantly enhance survival in selected patients with BM, particularly i […]

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Liquid nitrogen-treated autograft combined with megaprosthesis versus total femoral replacement for large-volume malignant femoral bone tumors: a comparative study.

Sang NTQ, Trung ND, Huy LD, Phuc LH, Thang VD, Anh LN, Hung LT, Hien TTT et al.

BACKGROUND: Limb-salvage surgery for extensive femoral osteosarcoma presents major reconstructive challenges. Conventional options such as total femoral replacement (TFR) and long-segment biological reconstructions are associated with significant complications. We evaluated the clinical outcomes of a novel hybrid approach combining a liquid nitrogen-treated autograft with a megaprosthesis (LN-MP) and compared them with those of TFR.

MATERIALS AND METHODS: This retrospective study included 28 patients (range 8-24 years old) with malignant femoral bone tumors treated between 2020 and 2024. Patients underwent either LN-MP ($n = 10$) or TFR ($n = 18$). Perioperative outcomes, functional results (Musculoskeletal Tumor Society score-MSTS), bone union, oncological control, and survival were analyzed.

RESULTS: LN-MP demonstrated better functional outcomes (MSTS 27.0 ± 2.1 vs. 24.4 ± 2.2, $p = 0.014$), shorter operative time, and lower blood loss compared with TFR. Length of stay and follow-up duration were comparable. Continuous disease-free survival was higher in the LN-MP group (90% vs. 44%, $p = 0.058$), with lower recurrence (10% vs. 28%). Bone union was achieved in 70% of LN-MP cases at a mean of 9 months, with longer graft length associated with delayed union.

CONCLUSIONS: LN-MP reconstruction may serve as a joint-preserving alternative to total femoral replacement for patients with e […]

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Tibial bone transport using external fixation in adults: a systematic review.

Giuli C, Di Pietro G, Farine F, Bocchi MB, Vitiello R, Palmacci O

INTRODUCTION: Tibial bone transport using external fixation is widely employed for the reconstruction of large and complex tibial bone defects; however, indications and surgical strategies remain highly heterogeneous. Tibial bone transport using external fixation is a demanding but reliable reconstructive technique, characterized by low failure rates and generally favorable outcomes despite frequent complications. Adequate soft-tissue management, selective docking-site grafting, and early functional rehabilitation were commonly reported. Further prospective studies are needed to standardize treatment strategies and improve clinical decision-making. MATERIALS AND METHODS: A systematic review was conducted in accordance with PRISMA guidelines in April 2025, searching PubMed/MEDLINE and the Cochrane Library. Extracted data included patient demographics, comorbidities, indications, defect length, surgical technique, frame type, external fixation time, healing index, complications, failures, and ASAMI bone and functional outcomes. Methodological quality was assessed using the MINORS score and results were synthesized descriptively.

RESULTS: A total of 3543 patients from 89 studies were included. Post-traumatic infection or osteomyelitis accounted for 58.5% of indications. The mean defect length was 7.04 cm. Bifocal bone transport was performed in 82.4% of cases, and circular exter […]

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Accuracy and risk factors for poor cup orientation in total hip arthroplasty: a comparison between supine and lateral registration using a portable navigation system.

Matsumoto S, Tetsunaga T, Yamada K, Okuda R, Masada Y, Yamamoto T, Okazaki Y, Ozaki T

INTRODUCTION: Accurate acetabular cup orientation is essential in total hip arthroplasty (THA). Portable inertial navigation may improve reproducibility, but accuracy in lateral decubitus THA may depend on the pelvic registration workflow. We compared cup placement accuracy between supine and lateral registration workflows using NAVBIT (Smith & Nephew PLC) and explored factors associated with large navigation errors.

MATERIALS AND METHODS: We retrospectively reviewed 110 consecutive primary cementless THAs performed in the lateral decubitus position using NAVBIT. The supine-registration workflow included 52 hips (supine pelvic registration followed by repositioning), and the lateral-registration workflow included 58 hips (registration and cup implantation in the lateral decubitus position). NAVBIT recorded radiographic inclination (RI) and radiographic anteversion (RA) intraoperatively; postoperative RI and RA were measured via computed tomography. Accuracy was defined as the absolute difference between navigation records and postoperative measurements. We compared navigation accuracy outcomes between propensity score-matched groups. Outliers were defined as an absolute error > 5° in RI or RA; factors associated with outliers were evaluated in the overall cohort.

RESULTS: After propensity score matching, 46 matched pairs (n = 92) were analyzed. Postoperative cup orientation w […]

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Differences in the coronal plane alignment of the knee (CPAK) measurements using long-leg radiographs and computed tomography.

Kawaguchi K, Tay ML, Young S

INTRODUCTION: Coronal plane alignment of the knee (CPAK) parameters are increasingly used to characterize native coronal alignment and guide alignment strategies in total knee arthroplasty (TKA) based on long-leg radiographs (LLR). Computed tomography (CT) based robotic TKA software enables calculation of CPAK parameters based on anatomical reference points. However, potential differences in CPAK measurements between LLR and CT remain incompletely defined. This study compared CPAK coronal alignment parameters and phenotype distributions between LLR and CT, and evaluated factors associated with measurement discrepancies.

METHODS: This was a radiographic subanalysis of a prospective randomised controlled trial including 241 patients undergoing robotic-assisted TKA. Medial proximal tibial angle (MPTA) and lateral distal femoral angle (LDFA) were measured and arithmetic hip-knee-ankle-angle (aHKA), joint line obliquity (JLO) and CPAK phenotype distribution were calculated and classified from preoperative LLR and CT. Discrepancy groups were defined as LLR = CT (≤ 2°), LLR < CT (> 2°) and LLR > CT (> 2°). Associations between discrepancy groups and preoperative deformity and flexion contracture were assessed.

RESULTS: Compared with LLR, CT demonstrated a more varus MPTA (mean 86.7° vs. 87.1°, $p = 0.04$), a more valgus LDFA (87.1° vs. 87.8°, $p < 0.01$) and a lower JLO (173.9° vs. 174. […]

PubMed · DOI

Conservative and surgical treatment of wrist osteoarthritis: a systematic review.

Boever J, Unglaub F, Wulf J, Spies CK, Langer MF, Kußmaul AC

INTRODUCTION: Wrist osteoarthritis leads to pain and functional impairment. Despite conservative treatment options, surgery remains a central therapeutic component. In Germany, the number of patients rose from over 472,000 cases in 2018 to more than 580,000 in 2024. This review examines the most common surgical treatment approaches, focusing on Proximal Row Carpectomy (PRC) and Four-Corner Fusion (4CF), discussing indications, contraindications and potential complications.

METHODS: A systematic and narrative literature review was conducted on treatment options for wrist osteoarthritis. The systematic review focusing on PRC and 4CF followed PRISMA guidelines and included searches in PubMed, Cochrane and Embase. Included were RCTs, systematic reviews and meta-analyses from January 2010 to May 2025 that compared both procedures in terms of pain, range of motion, grip strength, function, complication rates and reoperation rates.

RESULTS: Eight studies were included. Both procedures significantly reduced pain and improved function. PRC showed advantages in range of motion and lower complication rates, while 4CF was associated with higher grip strength. Partial arthrodesis (e.g., scaphotrapeziotrapezoid (STT) arthrodesis or radioscapholunate (RSL)) arthrodesis may be selectively used but are linked to higher nonunion rates. Total wrist arthrodesis or wrist arthroplasty may be consi […]

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Factors associated with selection of total hip arthroplasty versus hemiarthroplasty for femoral neck fracture in older adults: a nationwide analysis of 99,084 cases.

Maman D, Steinfield Y, Berkovich Y

BACKGROUND: Surgical management of displaced femoral neck fractures in older adults typically involves hemiarthroplasty or total hip arthroplasty (THA). Although clinical guidelines suggest that THA may be considered in selected healthier and cognitively intact patients, real-world procedure selection varies widely. This study evaluated patient factors associated with selection of THA versus hemiarthroplasty in a contemporary U.S.

METHODS: A retrospective cohort study was conducted using the 2022 Nationwide Readmissions Database (NRD). Patients ≥ 65 years hospitalized with femoral neck fracture were identified using ICD-10-CM codes. Those treated with internal fixation or non-arthroplasty procedures were excluded. Weighted analyses characterized demographics, comorbidities, and hospital utilization between THA and hemiarthroplasty groups. A multivariable logistic regression model identified factors independently associated with receiving THA. All analyses accounted for NRD survey design.

RESULTS: Among 142,013 operative cases, 99,084 met inclusion criteria (81.8% hemiarthroplasty; 18.2% THA). Patients receiving THA were younger (76.6 vs. 81.9 years), had shorter length of stay (6.06 vs. 7.24 days), and were more frequently discharged home (17.5% vs. 6.2%). Metabolic conditions were associated with increased odds of THA, including obesity (OR 1.17) and sleep apnea (OR 1.12). F […]

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Effect of distal screw number on axial deformation and fragment-specific stability in AO/OTA 2R3 C2.1 distal radius fractures.

Dietz L, Spiegel C, Kohler FC, Biedermann U, Zderic I, Feist A, Gueorguiev-Rüegg B, Lenz M et al.

PURPOSE: To investigate the biomechanical influence of distal screw number in palmar plate fixation of unstable intra-articular distal radius fractures with radiodorsal metaphyseal comminution.

METHODS: Nine matched pairs of cryopreserved human radii were prepared using a standardized AO/OTA 2R3 C2.1 fracture model with a defined radiodorsal metaphyseal defect zone. For each pair, specimens were randomized to palmar fixation using a 2.4-mm variable-angle locking plate with either four or six distal screws. After embedding, constructs were subjected to cyclic axial loading (150 N, 5,000 cycles) simulating early postoperative rehabilitation. Global axial deformation and fragment-specific three-dimensional interfragmentary motion were recorded using an optical motion-tracking system. Statistical analysis was performed using non-parametric tests ($p < 0.05$).

RESULTS: Two specimens from one donor pair were excluded, leaving 16 radii (8 pairs) for analysis. Bone mineral density and initial construct stiffness did not differ between groups. The six-screw configuration showed significantly lower axial deformation after cyclic loading ($p = 0.047$) and lower radial fragment-shaft translation at both time points (p = 0.028; p = 0.045). Radial fragment-shaft rotation was lower in the six-screw group at baseline (p = 0.018), but no rotational difference remained after cyclic loading. In the […]

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Crowe severity and femoral stem geometry are associated with lower-limb alignment after total hip arthroplasty for unilateral developmental dysplasia of the hip-related secondary osteoarthritis: valgus alignment trajectories.

Afacan MY, Gurcinar MG, Davulcu CD, Unlu MC, Kaynak G

BACKGROUND: Total hip arthroplasty (THA) for secondary osteoarthritis related to developmental dysplasia of the hip (DDH) restores hip mechanics. We aimed to quantify pre-to-postoperative changes in ipsilateral knee and global lower-limb alignment after unilateral THA to determine whether alignment changes differ by Crowe classification and femoral stem geometry.

METHODS: This retrospective cohort study included 74 patients who underwent unilateral primary THA for DDH-related secondary osteoarthritis, with a mean follow-up of 45.4 months. The predefined primary radiographic outcome set comprised preoperative-to-postoperative changes and final follow-up values in coronal lower-limb alignment parameters, including mechanical and anatomic lateral distal femoral angles (mLDFA and aLDFA), mechanical medial proximal tibial angle (mMPTA), joint line convergence angle (JLCA), global offset (GO), hip-knee-ankle angle (HKA), knee mechanical axis deviation ratio (KMADR), and femoral mechanical-anatomic axis angle (FMMA). The contralateral reference limb was used for side-to-side radiographic comparison. Patients were stratified by Crowe classification (I-II vs. III-IV), and femoral stem geometry analyses were performed according to conical, wedge, and rectangular stem designs. Clinical outcomes included hip/knee visual analog scale, Harris Hip Score, Knee Society Score, Oxford Knee Score […]

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Can ChatGPT pass the polish national medical specialization examination in orthopedics and traumatology?

Maci■g B, Bujak K, Jegierski D, Stolarczyk A

INTRODUCTION: Artificial intelligence (AI) has evolved rapidly in recent years and is becoming increasingly integrated into many areas of medicine. In the medical field, these systems have attracted considerable attention because of their potential applications in clinical decision support, medical education, scientific communication, and postgraduate training. The aim of the present study was to evaluate whether ChatGPT could achieve a passing score on the Polish National Medical Specialization Examination (Panstwowy Egzamin Specjalizacyjny, PES) in orthopedics and traumatology and to determine how different prompting strategies influenced its performance.

MATERIALS AND METHODS: Authors systematically assessed the performance of ChatGPT-4 across five consecutive official PES exams (from Autumn 2023 to Spring 2025) in orthopedics and traumatology. Each exam was administered using three distinct prompting strategies: Professor prompt, Specialist prompt, Resident prompt. For each exam session (e.g., Spring 2024, Autumn 2023), the prompts were submitted to ChatGPT sequentially. The model's responses were evaluated against the official answer key published by the Polish Center of Medical Exams (Centrum Egzaminów Medycznych, CEM).

RESULTS: The results were averaged across all prompts. The overall accuracy was 81% (range: 68.33%-85.83%). The highest score (85.83%) was recorded mult […]

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Two-year outcomes following concomitant ACL and extra-articular ligament reconstruction: a nationwide analysis of over 5,000 patients.

Abid R, J Burkhart R, J Moyal A, M Adelstein J, Kodsy M, G Calcei J, E Voos J, M Apostolakos J

PURPOSE: The anterior cruciate ligament (ACL) is the most commonly injured and reconstruction ligament in the United States. Associated injuries are common, oftentimes including components of the extra-articular knee. However, complications following concomitant ACL and extra-articular ligaments (EAL) has not been defined in a large-scale, population-based sample. We aimed to analyze outcomes following concomitant ACL and EAL reconstruction using the TriNetX nationwide database.ep METHODS: Deidentified data from the TriNetX Collaborative Network was used to identify all patients who received ACL and EAL reconstruction on the same day. Demographic information including age, sex, race, and geographic data were collected. Rates of medical and orthopaedic complications were analyzed at 30, 90, and 180 days and at 2 years. A propensity matched sub-analysis comparing outcomes patients with concomitant and staged repair was performed. Patients were identified using Current Procedural Terminology (CPT) coding.

RESULTS: We identified 5944 patients who received concomitant ACL and EAL reconstruction with postoperative outcomes available at 30, 90, and 180 days. 5040 patients had data available at 2 years. Serious medical complications like surgical site infection, deep vein thrombosis, acute kidney failure, and sepsis were rare at each point of follow-up. Rates of revision surgery were […]

PubMed · DOI

Recovery of gait after ankle fracture surgery following early mobilization and weight-bearing.

Bauer L, Schlegel AS, Scheuer A, Böcker W, Polzer H, Baumbach SF

BACKGROUND: Ankle fractures are among the most common fractures in adults. Despite significant advancements in surgical techniques, the postoperative treatment regimens remain conservative. This study aims to objectively evaluate gait recovery following early mobilization and weight-bearing after surgical ankle fracture treatment using instrumented gait analysis.

METHODS: This prospective, single-armed, longitudinal study enrolled adult patients following surgical treatment for isolated ankle fractures with early pain dependent weightbearing. Gait analysis, utilizing a treadmill integrated with a pressure plate and IMU-based motion capture systems, was conducted at 6 weeks, 3-, 6-, and 12-months post-surgery, assessing spatio-temporal and kinematic parameters. Results were compared longitudinally and against a healthy control group.

RESULTS: Longitudinal data were available for 44 patients. Significant ($p<.001$) improvements of self-paced walking velocity (SPV) were observed from 6 weeks (2.2 ± 1.1 km/h), to 3 (2.9 ± 0.9 km/h) and 6 months (3.4 ± 0.9 km/h). The spatio-temporal parameters showed significant changes between 6 weeks and all further time points only at SPV. Kinematic parameters showed significant longitudinal improvements at SPV for hip, knee, and ankle motion (including all three ankle planes). Patient gait parameters at 6 and 12 months were comparable to healthy […]

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A comparative analysis of three knee arthrodesis techniques to manage failed total knee arthroplasty due to infection: intramedullary nailing, LRS external fixator, and compression plating.

Rajasekaran RB, Ponniah HS, Palanisami DR, Sundaram VP, Palanivelayutham SS, Jayaramaraju D, Dhanasekaran S, Rajasekaran S

INTRODUCTION: This study compared external fixation (EF), intramedullary nailing (IMN), and compression plating (CP) for knee arthrodesis (KA) following failed total knee arthroplasty (TKA) due to periprosthetic joint infection, focusing on complications, radiological, and functional outcomes.

METHODS: Patients aged ≥ 18 who underwent knee arthrodesis following failed TKA due to infection from 2008 to 2023 were included. Four surgeons used three techniques: EF, IMN, and CP. Retrospective data on demographics, surgical techniques, complications, and outcomes were collected. Postoperative patient-reported outcome measures (PROMs), pain, and functional status were assessed. Radiographic data was reviewed for limb shortening and union status.

RESULTS: 46 patients (mean age 61.9 years, BMI 28.2), predominantly male (84.8%) were involved in this study. Twenty patients underwent EF, 16 underwent CP, and 10 underwent IMN, all for infected primary TKA. Pseudomonas was the most common infecting organism (15.2%). Average follow-up was 29.1 months. Bony fusion was achieved in 41 knees (89.1%), with five requiring additional surgery; four eventually required amputation due to non-fusion. EF group had the shortest time to union and no major complications or amputations. Time to union was significantly shorter in the CP group compared to the IMN group ($p < 0.001$), and EF group also had a si […]

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Accuracy of iliac crest reference pin placement in total hip arthroplasty: a CT-based analysis of risk factors for cortical perforation.

Masada Y, Tetsunaga T, Yamada K, Inoue T, Okuda R, Tetsunaga T, Okazaki Y, Ozaki T

INTRODUCTION: Accurate iliac crest reference pin placement is essential for safe navigation-assisted total hip arthroplasty (THA). Although complications are uncommon, inaccurate placement may cause neurovascular injury. This study aimed to evaluate pin insertion accuracy using postoperative computed tomography (CT), identify risk factors for cortical perforation, and determine whether CT-detected cortical perforation was associated with postoperative acetabular cup positioning accuracy.

MATERIALS AND METHODS: We retrospectively analyzed 200 primary THAs performed using portable or CT-based navigation between December 2022 and July 2025. Pin position was assessed on CT 1 week postoperatively and classified as complete intraosseous insertion or cortical perforation. Risk factors were examined using univariable and multivariable logistic regression analyses. Perforation direction was evaluated using Firth's penalized logistic regression. Postoperative cup positioning accuracy was assessed using CT-based radiographic inclination (RI), radiographic anteversion (RA), and absolute errors from target angles of 40° for RI and 15° for RA.

RESULTS: Cortical perforation occurred in 70 of 200 hips (35.0%): 32 medial and 38 lateral. On a pin-by-pin basis, 113 of 400 pins (28.3%) demonstrated perforation. In multivariable analysis, smaller ASIS width, lateral decubitus positioning, and 4-m […]

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Long-term outcomes of arthroscopic single-anchor fixation versus microfracture for osteochondral fractures of the knee weight-bearing area in adolescents and young adults.

Yang W, Pan Q, Wang H, Peng Y, Meng C, Huang W

PURPOSE: Osteochondral fractures involving the weight-bearing area of the knee in young patients require stable restoration of the native osteochondral fragment whenever possible. This study aimed to evaluate the long-term clinical and radiological outcomes of arthroscopic single-anchor fixation and to compare them with those of microfracture.

METHODS: This retrospective comparative study included 24 adolescents and young adults with osteochondral fractures of the femoral condyle or trochlear region treated between 2016 and 2017. Twelve patients underwent arthroscopic single-anchor in situ fixation, and 12 underwent microfracture. The fixation technique used one double-loaded suture anchor to obtain multi-point compression of the osteochondral fragment while preserving the subchondral plate. Clinical outcomes were assessed using the Lysholm, International Knee Documentation Committee, and Tegner scores. Magnetic resonance imaging was evaluated using the MOCART 2.0 score. The mean follow-up was 93.7 months.

RESULTS: The single-anchor group showed a significantly higher intraoperative contact area ratio than the microfracture group (96.5% ± 3.2% vs. 85.8% ± 5.6%, $P < 0.001$). At final follow-up, the single-anchor group had superior Lysholm scores (91 ± 5 vs. 83 ± 8, $P = 0.009$), IKDC scores (86 ± 6 vs. 78 ± 7, $P = 0.010$), and Tegner activity levels (5.0 ± 1.1 vs. 3.9 ± 0.9, P = 0 […]

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Isometric points in cerclage for patellar tendon repair: a 3-dimensional, weightbearing computed tomography kinematic simulation.

Zindel C, Hodel S, Luo Y, Meisterhans M, Fucentese SF, Vlachopoulos L

PURPOSE: The gold standard for the management of ruptured patellar tendon is the surgical repair with nonabsorbable sutures and additional reinforcement e.g. cerclage wires as described by McLaughlin in 1947. Despite the early description, the exact placement of such cerclage wires remains unclear. The aim of the study was (1) to analyze how the cerclage elongates during knee flexion and (2) to define the insertion points with the most isometric behavior at the patella and tibial tuberosity (TT).

METHODS: 3D simulations of 10 healthy knees in the range from 0° to 120° of flexion using weightbearing computed tomography scans were generated. 12 insertion points were defined at the patella in equal distances along the lateral and medial border from the proximal to the distal pole. 42 insertion points were defined at the tibial tuberosity (TT) in a 3.0 × 3.0 cm large grid around its center. An automatic string simulation algorithm was used to calculate the length for all possible combinations of the cerclage from 0° to 120° flexion. The most isometric combination of insertion points was found using the isometric score.

RESULTS: The most isometric insertion at the TT is distally ($p < 0.01$) and towards the center of the TT ($p < 0.01$). The most isometric points at the TT is located 61 mm distal to the joint line and 5 mm from the anterior border of the TT in a true lateral view fluo […]

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Patient mortality exceeds implant revision risk with equivalent survival following unicompartmental versus total knee arthroplasty in octogenarians.

Innocenti M, Leggieri F, Massenzi M, Stimolo D, Akkaya M, Matassi F, Civinini R

BACKGROUND: While traditional surgical decision-making emphasizes long-term implant survivorship for octogenarians knee reconstruction, this paradigm may be inappropriate when patient mortality risk substantially exceeds revision risk. This study compared mid-term mortality between unicompartmental knee arthroplasty (UKA) and total knee arthroplasty (TKA) in octogenarians, hypothesizing that implant type would not significantly influence patient survival.

METHODS: A retrospective cohort study was conducted on 238 consecutive octogenarian patients (age ≥ 80 years) who underwent primary knee arthroplasty at a single tertiary orthopaedic center between January 2018 and December 2020. Patients with revision arthroplasty, fracture, tumor, inflammatory arthropathy, or incomplete data were excluded. The primary outcome was all-cause mortality at minimum 5-year follow-up. Secondary outcomes included age-stratified mortality (80-84, 85-89, ≥ 90 years) and identification of independent mortality predictors. Statistical analyses included independent t-tests, chi-square/Fisher's exact tests, standardized mean differences, and multivariable logistic regression adjusting for age, follow-up duration, and registry year.

RESULTS: Of 238 patients, 185 (77.7%) underwent TKA and 53 (22.3%) underwent UKA. Overall mortality was 35.3% (84/238): 33.5% in TKA versus 41.5% in UKA (OR = 1.41, 95% CI: 0 […]

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Short- to mid-term clinical outcomes and return to sport and work after arthroscopic debridement of the extensor carpi radialis brevis in patients with chronic lateral epicondylopathy.

Siebenlist S, Doucas A, Hatt FL, Lappen S, Vieider RP, Lacheta L, Kadantsev P

BACKGROUND: This study aimed to evaluate the short- to mid-term clinical outcomes following arthroscopic extensor carpi radialis brevis (ECRB) debridement in patients with chronic epicondylopathy humeri radialis (cEHR). We hypothesized that arthroscopic ECRB debridement would result in favorable short- to mid-term patient-reported outcomes, improving pain levels, strength, range of motion (ROM) and patient satisfaction.

METHODS: A retrospective case series was conducted on patients who underwent arthroscopic ECRB debridement for cEHR with symptoms > 6 months. Patients were assessed at a minimum of 12 months postoperatively using the Mayo Elbow Performance Score (MEPS), Patient-Rated Tennis Elbow Evaluation (PRTEE), Disabilities of the Arm, Shoulder, and Hand (DASH) score, and Visual Analog Scale (VAS) for pain. Pre- and postoperative scores were compared using DASH and VAS. Maximum isometric wrist extension strength and postoperative ROM were measured bilaterally. Return to sport and work, patient satisfaction, and complications were documented.

RESULTS: Of the 35 eligible patients, 30 completed follow-up (86% follow-up rate; mean age 48.4 ± 6.6 years; range: 36-65 years; 47% male). The mean duration of symptoms before surgery was 12 months (IQR 8-20, range 6-62). The mean follow-up duration was 28.13 ± 10.9 months (range 12-45). At final follow-up, median outcome scores were […]

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Complications and outcomes after operative treatment of low-energy pelvic ring and acetabular fractures.

Salonpää KB, Nurkkala J, Lantto I, Kakko S, Kaakinen T, Aitavaara-Anttila M, Liisanantti J

BACKGROUND: Fragility fracture of the pelvis (FFP) with a low-energy mechanism of injury is relatively common among the growing geriatric population. The incidence of postoperative complications after surgically treated FFP is not well documented. The aim of this study was to report complications after operatively treated FFP and to determine whether these complications would have a detrimental effect on outcomes.

METHODS: Retrospective study of 200 consecutive patients who underwent operative treatment at Oulu University Hospital, Oulu, Finland, between 1.1.2010 and 15.4.2021 for FFP with a low-energy injury mechanism. Postoperative complications were recorded, and associations with outcomes were depicted.

RESULTS: Postoperative complications were recorded in 31 (15.5%) patients. Patients with complications had higher intraoperative bleeding (OR 2.8 (1.0-7.9), p = 0.048) and higher ASA classification 3.4 (1.3-8.4), P = 0.009). Patients with complications had longer hospital stays, but no clear association between complications and mortality or discharge location was observed.

CONCLUSIONS: The incidence of postoperative complications was comparable with that reported in previous literature. Patients who experienced complications had longer hospital stays; however, no definitive association between complications and outcomes was identified.

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Minced cartilage implantation for cartilage regeneration: a survey of current clinical practices.

Yilmaz T, Leutheuser S, Niemeyer P, Behrendt P, Roessler P, Gille J, Mumme M, Vogt S

INTRODUCTION: The objective of this study is to evaluate how minced cartilage implantation (MCI) is currently applied in clinical settings, focusing on the selection criteria, procedural execution, handling of concomitant joint pathologies, and postoperative management.

MATERIAL AND METHOD: An online survey was conducted among all 4,915 members of the German Society for Arthroscopy and Joint Surgery (AGA). Invitations with a survey link were emailed, and the questionnaire was accessible anonymously from January 10 to February 20, 2024. A total of 927 responses were received and analyzed, resulting in a 19% response rate. Demographic data were described descriptively; survey answers were reported as percentages. The 25-item questionnaire, developed by AGA and the Quality Circle for Cartilage Repair & Joint Preservation (QKG e.V.), included single- and multiple-choice questions on experience with cartilage therapy, minced cartilage procedure application and postoperative care, and the overall assessment of this technique within cartilage repair.

RESULTS: MCI has emerged as one of the three most commonly used cartilage repair techniques in the knee joint. Defect sizes up to 4 cm^2 represent the largest treated group. The application of the procedure is quite heterogeneous, with some practitioners using a shaver and others manually mincing with a scalpel. Fixation methods also var […]

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Variability in United States online rehabilitation protocols after open and endoscopic hip abductor repair.

Anwunah C, Rosenberg S, Hartwell M

PURPOSE: The purpose of this study was to evaluate readily accessible U.S. based online physical therapy protocols for open and endoscopic hip abductor repairs and to assess the variability in post-operative rehabilitation protocols.

METHODS: Online physical therapy protocols from U.S. based orthopaedic institutions for gluteus medius/minimus repair were collected using a web-based query utilizing search terms "hip abductor repair rehabilitation protocol" and "gluteus repair rehabilitation protocol." Published protocols from both Electronic Residency Application Service (ERAS) academic orthopaedic surgery programs and private groups were collected. All publicly available protocols were included. Protocols were analyzed using a standardized spreadsheet to assess rehabilitation components and timelines. Descriptive statistics (median values, percentages, ranges) were calculated.

RESULTS: 54 rehabilitation protocols were assessed, including 22 (41%) from ERAS-affiliated academic programs or associated physicians. Postoperative weight-bearing was addressed in 53/54 (98%) protocols, with 41 (77%) recommending toe-touch/touchdown/flat-foot/20-lb weightbearing. Median duration of restricted weight-bearing was 6 weeks (range, 2-8 weeks). Regarding range of motion restrictions, 37 (69%) protocols limited hip flexion to 90°, 42 (78%) advised against passive adduction past neutral, and […]

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The incidence and epidemiology of patellar fractures following medial patellofemoral ligament reconstruction: a systematic review and meta-analysis.

Abelleyra Lastoria DA, Hunt A, Ross CK, Yemane M, Keynes S, Ghosh A, Hing CB

INTRODUCTION: Patellar instability can arise from a traumatic event with anatomical predisposing factors increasing the risk of dislocation. Medial patellofemoral ligament reconstruction (MPFLR) is commonly performed to treat patellar instability. The aim of this systematic review was to determine the incidence of and risk factors for patellar fractures after MPFLR.

MATERIALS AND METHODS: The protocol for this review was registered on PROSPERO with registration number CRD420251066107. The PRISMA 2020 checklist was followed. Electronic databases and conference proceedings were searched to the 15th of May 2025. Studies were eligible if they reported incidence of patellar fractures after MPFLR. A random-effects meta-analysis was performed. Included studies were appraised using tools respective of study design.

RESULTS: A total of 4,972 records were screened in the initial search. A total of 110 studies satisfied the inclusion criteria, evaluating 115,496 patients. The evidence was moderate to low in quality. The incidence of patellar fracture ranged from 0 to 9.50%, with a pooled incidence of 1.17% (95% confidence interval [CI] 0.76- 1.68%). There was no difference in incidence of patellar fractures between patients who did and did not undergo bone tunnel fixation (0.94%, 95% CI 0.69-1.24 vs. 1.00%, 95% CI 0.15-2.44%, respectively). There was no difference in fracture incidence […]

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Predictive accuracy of a general-purpose artificial intelligence model for cut-out following proximal femoral nailing.

Ben Arie G, Warschawski Y, Amzallag N, Gan-Or H, Ben-Tov T, Khoury A, Horev I, Graif N et al.

INTRODUCTION: Cut-out is the most consequential mechanical complication after proximal femoral nailing and requires prompt recognition. General-purpose artificial intelligence (AI) models with image-interpretation capability are increasingly accessible, yet their diagnostic performance for this task is unknown. We evaluated ChatGPT's accuracy for predicting cut-out following proximal femoral nailing and compared TFNA and Gamma nail subgroups.

MATERIALS AND METHODS: In this retrospective predictive-accuracy study, 989 patients (683 TFNA, 306 Gamma nail; mean age 83.8 years; cut-out prevalence 2.5%) were analysed. For each case, three baseline radiographs - the injury anteroposterior and lateral views and the immediate postoperative control radiograph, all obtained before any complication could be radiographically present - together with the full clinical data set (excluding complication-related information) were submitted to ChatGPT, which returned a binary prediction (yes/no) and probability estimate (0-100%). The follow-up radiographs on which cut-out becomes apparent were not provided to the model. Ground truth was established on subsequent follow-up radiographs by a departmental review board requiring agreement of two senior surgeons. Accuracy metrics were calculated with Wilson 95% confidence intervals (CI); exact McNemar's and Mann-Whitney U tests assessed directional bia […]

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Staged management of a compound combined pelvi-acetabular injury with perineal degloving in a young female: a case report and literature review.

Dhankhar K, Kekatpure AL, Kekatpure AL

We present the case of a 30-year-old woman, the patient who sustained a compound combined pelvi-acetabular fracture with extensive perineal internal degloving after high-energy trauma. Computed tomography revealed an unstable pelvic ring injury (Tile type C1; anteroposterior compression type III) with right transverse acetabular fracture. The occurrence of extensive internal degloving was confirmed intraoperatively. The injury was treated using a graduated, principle-based approach incorporating early soft-tissue debridement, dead-space management and temporary pelvic stabilization, infection control, and delayed definitive internal fixation. An intercurrent pin-site infection was treated successfully without progression to deep infection. The patient demonstrated progressive radiological union at 4 months with a Harris Hip Score (HHS) of 80 and a Majeed pelvic score of 55. At one-year follow-up, complete union occurred and functional recovery was achieved, characterized by HHS of 90 and a Majeed score of 70. In this case, infection-guided staging as well as extremely aggressive soft tissue treatment may allow for safe delayed fixation even where pelvi-acetabular trauma has been compromised.

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Concomitant upper extremity fractures do not worsen outcomes in conservatively managed pelvic fractures: a cohort study of 1840 geriatric patients.

Graif N, Elbaz E, Kazum E, Rachevsky G, Gurel R, Factor S

INTRODUCTION: Pelvic fractures in elderly patients are associated with significant pain, functional decline, and prolonged rehabilitation, with the majority managed non-operatively. In the conservative management paradigm, intact upper extremity function is critical for mobilization with assistive devices; concurrent upper extremity (UE) fractures may therefore complicate recovery. The aim of this study was to evaluate the clinical outcomes associated with this injury combination in patients with conservatively managed pelvic fractures.

METHODS: A retrospective cohort study was conducted at a Level 1 trauma center, including all patients aged ≥ 65 years admitted with pelvic ring or acetabular fractures managed non-operatively between January 2010 and January 2024. Patients were stratified by the presence of a concomitant UE fracture sustained during the same traumatic event. Primary outcomes were hospital length of stay (LOS) and all-cause mortality at multiple time points. Secondary outcomes included readmission rates and systemic infection rates.

RESULTS: Of 1840 patients, 1693 (92.0%) sustained isolated pelvic fractures and 147 (8.0%) had concomitant UE fractures. Groups were comparable at baseline across all measured covariates. LOS did not differ significantly between groups (11.0 ± 14.2 vs. 11.8 ± 16.9 days; $p = 0.796$). Mortality was similar at all evaluated time points […]

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Single double-loaded anchor repair of isolated Lafosse III subscapularis tears in patients not involved in physically demanding activities provides satisfactory functional and radiologic outcomes at minimum 5-year follow-up.

Aral F, Oklaz EB, Ahmadov A, Asiret E, Delice C, Guler Oklaz EB, Akdulum I, Kanatli U

PURPOSE: To evaluate the outcomes of single-anchor repair for isolated Lafosse III subscapularis tears in patients not engaged in physically demanding activities.

METHODS: Patients with isolated subscapularis tears who underwent arthroscopic surgery between 2014 and 2020 were retrospectively evaluated. The inclusion criteria were patients who underwent single double-loaded anchor repair for Lafosse III tears and were not involved in physically demanding activities. Assessments included the American Shoulder and Elbow Surgeons (ASES) score, Constant-Murley Score (CMS), Visual Analog Scale (VAS), range of motion, specific clinical tests, and magnetic resonance imaging encompassing tendon integrity, cross-sectional area (CSA), vertical diameter, and transverse diameters.

RESULTS: 27 patients (mean age 52.4 ± 10.3 years; mean follow-up 80.8 ± 18.6 months) were included in the study. The final follow-up demonstrated significant improvements in the ASES (44.0 ± 12.4 to 82.4 ± 14.1), CMS (49.1 ± 15.3 to 85.3 ± 11.1), VAS (5.7 ± 2.1 to 1.3 ± 2.2), and internal rotation (L4 to T8) ($p < 0.001$ for all). The positivity of clinical tests decreased substantially: Bear-Hug, from 93% to 18%; Belly-Press, from 89% to 15%; and Lift-Off, from 89% to 18% ($p < 0.001$ for all). Radiological analyses indicated maintained tendon integrity in 93% of patients and increases in CSA (1.292 ± 453 to 1.544 […]

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Shoulder injuries in rugby union: a systematic review and meta-analysis.

Doyle J, Bellamy M, Fowler K, Keiller H, Gunasekera R, Funk L

Shoulder injuries are common in Rugby Union, resulting in prolonged absence and considerable burden. Reported incidence, severity, and mechanisms vary widely, reflecting the multiple high-impact actions in the sport. Various management approaches exist to treat a wide range of pathologies. This systematic review and meta-analysis aimed to synthesise evidence on the epidemiology, mechanisms, and treatment of shoulder injuries in rugby. PUBMED, SCOPUS, and WEB OF SCIENCE were searched from database inception until 1st June 2025. From 1,099 abstracts screened, 37 studies were included. The pooled match incidence was 11.02 injuries/1,000 player-hours (95% CI 7.23-16.82), though heterogeneity was very high ($I^2 = 97\%$) and the 95% prediction interval was wide (2.11-57.61 injuries/1,000 player-hours), indicating the pooled rate should be read as an average across diverse populations rather than a single representative value. Higher rates were seen in male elite (13.74/1,000 h) and high school/university players (12.28/1,000 h). Training incidence was considerably lower (0.20/1,000 h). Substantial between-study heterogeneity was observed across all incidence analyses, reflecting genuine variation across playing levels, sex, and surveillance methods. Tackling was the predominant injury mechanism, most often affecting the tackler. Bankart, Latarjet, and Bristow procedures all produced fav […]

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The pre-operative physical component subscale of the 12-item Short Form Health Survey was the only factor associated with 2-year clinical outcome and satisfaction after hip abductor tendon repair.

Ebert J, Edwards P, Klinken S, Janes G

INTRODUCTION: Surgical repair in the context of symptomatic hip abductor tears unresponsive to non-operative management has demonstrated encouraging outcomes. This study sought to investigate demographic, injury/management and post-operative factors associated with clinical outcome after hip abductor tendon repair.

METHODS: A total of 104 patients undergoing repair were included. Patients were assessed pre-surgery and 2-years post-operatively with the modified Harris Hip Score (mHHS), Oxford Hip Score (OHS), physical (PCS) and mental (MCS) component subscales of the 12-item Short Form Health Survey (SF-12), and satisfaction. Regression analysis assessed the contribution of pre-operative demographic (age, sex, body mass index, PCS, MCS), injury/treatment (symptom duration, prior corticosteroid injections, gluteus minimus/medius tear grade and fatty infiltration) and post-operative (hip abduction strength symmetry) variables, to the 2-year mHHS and being 'very satisfied'.

RESULTS: Mean improvement in the mHHS and OHS from baseline to 2 years was 36.0 points (95% CI, 31.9 to 40.2; $p < 0.001$) and 17.5 points (95% CI, 15.8 to 19.3; $p < 0.001$), respectively. Seven (6.7%) patients presented with symptomatic re-tearing. Of the 97 patients with 2-year follow-up, 78 (80.4%) were 'very satisfied'. The pre-operative SF-12 PCS was the only factor associated with the 2-year mHHS ($B = 0.46$; […]

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Temporising external fixation reduces loss of reduction compared with plaster splinting in ankle fracture-dislocations: a systematic review and meta-analysis of cohort studies.

Suravaram V, Solomon L, Bhat S, Wittauer M

PURPOSE: The optimal temporising strategy for ankle fracture-dislocations is a highly debated topic. These unstable injuries are commonly managed with staged fixation to allow for soft tissue recovery, using either plaster splinting or ankle-spanning external fixation. This study aimed to compare the efficacy and safety of these two approaches.

METHODS: A systematic search of MEDLINE, Embase, Scopus, Cochrane CENTRAL, and Google Scholar was conducted. Comparative studies evaluating temporary external fixation versus plaster splinting in skeletally mature patients with ankle fracture-dislocations were included. Meta-analyses were performed to assess differences in loss of reduction prior to definitive fixation, soft tissue complications, and time to definitive fixation. Risk of bias was assessed using ROBINS-I tool, and certainty of evidence was appraised using GRADE.

RESULTS: Nine retrospective cohort studies comprising 1638 patients were included (589 external fixation, 1049 splinting). Loss of reduction occurred in 3% of injuries in the external fixation cohort compared with 14.6% in those managed with plaster splinting (RR 0.22; 95% CI 0.09-0.52). Rates of overall soft-tissue complications were similar between cohorts. The data regarding time to definitive fixation were highly heterogeneous across studies. Overall certainty of evidence was moderate for LOR and very low for […]

PubMed · DOI

Comprehensive ultrasound assessment of thumb metacarpophalangeal collateral ligament injuries reveals frequent concomitant abnormalities.

Bain O, Farkash U, Atzmon R, Shemesh S, Ohana N

INTRODUCTION: Ulnar collateral ligament (UCL) tears of the thumb are well-recognized injuries, while radial collateral ligament (RCL) and dorsal extensor-related abnormalities are less frequently studied. The pattern of concomitant ultrasound abnormalities in patients after thumb trauma is not well documented. The aim of this study was to characterize concomitant UCL, RCL, and dorsal extensor-related abnormalities within a selected cohort referred to thumb ultrasound after trauma, and to explore associations with injury mechanism.

MATERIALS AND METHODS: A retrospective review was performed of 42 consecutive adult patients who underwent high-resolution ultrasound of the thumb metacarpophalangeal joint between April 2014 and February 2025 for suspected ligamentous injury after acute trauma. Patients were included if ultrasound demonstrated pathologic findings in at least one stabilizing structure, UCL or RCL, and the extensor mechanism was systematically assessed. Collateral ligament injuries were graded using standard sonographic criteria, while dorsal extensor-related findings were recorded descriptively. Associations between injury severity and mechanism of injury were tested using chi-square or Fisher's exact analysis, as appropriate.

RESULTS: UCL pathology was present in 35/42 (83.3%) patients, RCL pathology in 23/42 (54.8%), and dorsal extensor-related abnormalities in 17 […]

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Holding the line: predictors of coronal and sagittal loss of reduction after tibial plateau fracture fixation.

Hesmerg MK, Benner JL, Joosse P, Keijser LCM

INTRODUCTION: Tibial plateau fractures (TPFs) are challenging fractures that often suffer from secondary Loss of Reduction (LoR) after fracture fixation. In this study, we aimed to investigate the factors and characteristics associated with LoR after surgical fixation, using measurements of both coronal and sagittal displacement.

METHODS: A retrospective analysis of 213 TPF patients (79.3% female, mean age 57.5 ± 14.4 years) surgically treated between 2017 and 2024 in a level 1 trauma center was performed. Patients were included if they had a minimum of 100-days radiological follow-up. Loss of reduction was defined as late postoperative depression ≥ 3 mm, condylar widening ≥ 5 mm, varus/valgus or tibial slope collapse > 5 degrees, breakage, loosening, or conversion to total knee arthroplasty (TKA). Radiographic images were assessed for preoperative, direct postoperative, and late postoperative (up to 1 year) displacement. Univariable and multivariable logistic regression analyses were employed to study factors associated with LoR..

RESULTS: At a mean follow-up of 325 days (SD ± 161) 77 patients (36.2%) suffered from LoR. The adjusted multivariable logistic regression model revealed that patient age (OR 1.03, p = 0.028), involvement of the posterior column (OR 2.87, p = 0.014), preoperative step-off (OR 1.08, p < 0.001), postoperative step-off (OR 1.61, p = 0.007), and postope […]

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Using an App based algorithm to determine axial rotation and pelvic tilt at the time of an AP radiograph: a sawbone study.

Boettner F, Premkumar A, Sterneder M, Aboaf C, Haralambiev L

INTRODUCTION: Axial rotation of the pelvis impacts inclination and anteversion measurements of the acetabular component on plain radiographs. Currently the axial rotation is not taken into account when providing acetabular component measurements on plain radiographs. The current study evaluates a novel App based algorithm to determine pelvic axial rotation at the time of an AP pelvis radiograph.

MATERIALS AND METHODS: AP and lateral radiographs were taken of the sawbone pelvis to calibrate the App. The sawbone was then rotated 11 times to the left (average 5.3°, range 0.2°-10.5°) and 10 times to the right (average 5.2°, range 1.2°-11.3°). The sawbone was then tilted forward 14 times (average 5.6° range 0.1°-11.6°) and 13 times backward (average 5.9°, range 0.1°-11.2°). The amount of axial rotation and tilt of the sawbone pelvis was determined using a digital caliper. AP pelvis radiographs were taken for each position. Using a novel App based calculation algorithm the amount of axial rotation and pelvic tilt was determined from the AP Pelvis radiograph.

RESULTS: The algorithm determined the axial rotation of the pelvis at the time of the AP pelvis radiograph with an average difference of 0.6° (range 0°-1.8°) compared to the digital caliper measurement. For measuring pelvic tilt, the difference was 2.3° (range 0.4°-4.5°) compared to the digital caliper measurement.

CONCLUSIONS […]

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Risk factors for implant-associated infection after bone allograft reconstruction: a 20-year retrospective cohort study.

Scheurer F, Smolle M, Hauer G, Ruckenstuhl P, Valentin T, Leithner A

AIMS: Reconstruction of bone defects is a key challenge in orthopaedic surgery. While autologous bone grafts are considered the gold standard, osseous allografts are increasingly used as an alternative as they allow for treatment of large bone defects. However, they may be associated with an increased risk of postoperative infections. This study aims to analyze the frequency of and risk factors for implant-associated infections after bone allograft transplantation.

METHODS: All patients who underwent reconstruction with bone allografts at a single orthopaedic and trauma unit between 2003 and 2023 were retrospectively included. Data on demographics, disease-associated parameters, risk factors at index surgery, and development of implant-associated infections during follow-up were documented. Potential risk factors for development of implant-associated infections during follow-up were investigated with logistic regression models.

RESULTS: A total of 220 patients were included (46.8% males; median age at surgery 46.5 years [IQR: 22-67]; n = 23 with implant-associated infections during follow-up). Factors independently associated with subsequent implant-associated infections were smoking (p = 0.001), BMI (p = 0.015) and radiotherapy (p = 0.013). Gender, age, preoperative chemotherapy, allograft material, preoperative C-reactive protein, hemoglobin and reason for surgery (tumor, n […]

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Why it's better to believe in a larger definition of the diametaphyseal junction zone in pediatric distal radius fractures.

von Schrottenberg C, Beck SM, Schwerk P, Fitze G, Schultz J

INTRODUCTION: Diametaphyseal radius fractures (DMRF) in children are characterized by special biomechanical behavior. They present with a stereotypical loss of reduction (radial translation and ulnar tilt) of the distal fragment when stabilized with elastic stable intramedullary nailing (ESIN), the standard osteosynthesis for pediatric diaphyseal forearm fractures. No consensus has been found on the definition of the diametaphyseal junction zone (DMJZ). Some authors claim that diaphyseal fractures immediately proximal to the metaphysis, as defined by the AO Pediatric Comprehensive Classification of Long Bone Fractures (AO-PCCF), can safely be stabilized with ESIN. They confine the DMJZ to a small part within the metaphysis. Other definitions of the DMJZ extend slightly more proximally, including a small part of the distal diaphysis. The forearm fracture index (FFI) calculates the ratio of the fracture's distance to the radius' growth plate over its width and defines DMRF to have an FFI between 1 and 2. The aim of this case series was to demonstrate that these particular fractures, which can be considered diaphyseal according to the AO-PCCF, still exhibit biomechanical characteristics typical of DMRF.

MATERIALS AND METHODS: Radiologic and clinical outcomes of 11 DMRF, that would be considered diaphyseal by the AO-PCCF classification but fall into the DMJZ when using the FFI, ar […]

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Optimizing initial biomechanical strength ("Time Zero") of posterior repair in total hip arthroplasty: a biomechanical comparison of suture materials and knot configurations.

Chen M, Liu X, Chen D, Li X, Du H, Sun Y, Wang H

INTRODUCTION: Posterior capsule repair (PCR) is commonly performed during total hip arthroplasty (THA) via the posterolateral approach to improve postoperative stability. Because biological healing of the repaired posterior structures requires time, the repair construct must provide sufficient immediate mechanical support after surgery. This study aimed to identify the suture material and knot configuration with favorable initial mechanical performance in a standardized in vitro model.

MATERIALS AND METHODS: This in vitro biomechanical study was conducted in two sequential phases. Phase 1 evaluated the baseline tensile properties of three commonly used suture materials: #2 Ethibond, #0 Vicryl, and 3 - 0 PGLA. Phase 2 used a standardized porcine dermal surrogate model to compare eight suture configurations, including single- and double-strand techniques. The primary biomechanical outcomes included maximum failure load, load at 2 mm gap formation, and construct stiffness during uniaxial quasi-static load-to-failure testing.

RESULTS: In Phase 1, non-absorbable #2 Ethibond showed significantly greater ultimate tensile strength (92.44 ± 8.72 N) and higher stiffness than the absorbable alternatives ($P < 0.001$). In Phase 2, double-strand configurations outperformed all single-strand techniques. Among them, the double-strand Nice knot achieved the highest maximum failure load (104.04 […]

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Intravenous versus oral tranexamic acid in shoulder arthroplasty.

Parmar T, Adio A, Daher M, Khan A, Horneff J, Abboud J

PURPOSE: Tranexamic acid (TXA) is used in shoulder arthroplasty to reduce blood loss. While intravenous (IV) TXA is standard, oral TXA offers advantages in cost and administration. This study compares 90-day complications and postoperative laboratory values between IV and oral TXA in total shoulder arthroplasty (TSA).

METHODS: A retrospective cohort study using TriNetX identified adults undergoing primary TSA (2005-2025) receiving TXA on the day of surgery, stratified by route (IV vs. oral). One-to-one propensity score matching was performed. Ninety-day outcomes included transfusion, thromboembolic events, cardiac and renal complications, neurologic events, infection, wound dehiscence, readmissions, emergency department visits, opioid utilization, and mortality. Postoperative laboratory outcomes included hemoglobin, erythrocyte count, hematocrit, and platelet count. Outcomes were compared using odds ratios (ORs) with 95% confidence intervals (CIs) and t tests where appropriate.

RESULTS: After matching, 5722 patients were included in each cohort. No significant differences were observed between IV and oral TXA in 90-day transfusion rates ($P = 0.873$) or thromboembolic events, including deep vein thrombosis and pulmonary embolism. Rates of myocardial infarction, stroke, cardiac ischemia, acute renal failure, infection, readmission, emergency department visits, opioid use, and mo […]

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Robotic-arm assistance in revision total hip arthroplasty improves the accuracy of acetabular component positioning: a cadaver-based study.

Fitzgerald K, Borukhov I, Chandra V, Smitterberg C, Mont MA, LiArno S

INTRODUCTION: Acetabular component positioning is a key factor for stability, impingement, wear, and biomechanics in revision total hip arthroplasty (THA). However, anatomic distortion and previous implants may complicate positioning. Although evidence on computed-tomography (CT)-based robotic assistance in revision THA is limited, its planning features and precision may improve surgical plan execution. This study aimed to measure the precision and accuracy of acetabular component positioning of revision robotic-arm assisted total hip arthroplasty (RA-THA) compared with a published manual primary total hip arthroplasty (M-THA) benchmark for: (a) acetabular inclination; (b) acetabular version; and (c) center of rotation (COR), with comparison by surgeon case volume.

MATERIALS AND METHODS: There were seven cadavers (nine hips) that previously underwent primary THA revised with RA-THA by independent surgeons with varying primary RA-THA case volumes, surgical approaches, and acetabular component types. Due to surgical protocol deviations, two hips were excluded leaving seven hips from six cadavers. Pre- and postoperative CT scans were compared to calculate absolute error for inclination, version, and COR; normality was established with the Anderson-Darling test and two-variance tests assessed differences in standard deviations ($\alpha = 0.05$). Mean differences between cadaver-based rev […]

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Correction: Modified rotational wedge distal metatarsal osteotomy versus chevron osteotomy for hallux valgus: long-term radiographic and clinical outcomes.

Aybar A, Gokkus K, Özden E, Çetin MH, Çetin MÜ

PubMed · DOI

Isolated compartment arthrosis of the knee: contemporary concepts, persistent controversies, and future directions in unicompartmental knee arthroplasty.

Akkaya M, Innocenti M

PubMed · DOI

Preoperative lower radiographic osteoarthritis severity in primary total knee arthroplasty increases revision risk by tenfold compared to severe osteoarthritis.

Stroobant L, Vermue H, Droogsma T, Van Onsem S, Arnout N, van Hellemondt GG, Victor J, Heesterbeek PJC

BACKGROUND: Dissatisfaction following primary total knee arthroplasty (pTKA) is higher in patients with preoperative low-grade osteoarthritis (OA), but its impact on revision risk remains unclear.

QUESTIONS/PURPOSES: This study aimed to identify predictors for revision surgery within 10 years following pTKA, focusing on radiographic severity of OA using the Kellgren-Lawrence (KL) grading system, along with other demographic and radiographic factors.

PATIENTS AND METHODS: This retrospective case-control study was conducted at two European tertiary referral centers. The case group included 142 patients who underwent aseptic revision total knee arthroplasty (rTKA) between 2007 and 2023, within 10 years following pTKA. A 2:1 control group of 284 patients who had pTKA between 2011 and 2014, with no revision surgery within 10 years, was selected. Collected data included age, sex, body mass index (BMI), side of surgery, and American Society of Anesthesiologists (ASA) classification. Radiographic data were collected, requiring a preoperative radiograph to assess the KL grade. Univariate analyses identified potential predictors for rTKA, and multivariable logistic regression assessed their relationship with revision risk.

RESULTS: Patients requiring rTKA were significantly younger at the time of pTKA. For each additional year of age, the revision risk decreased by 6% (OR 0.94, 95% CI […]

PubMed · DOI

Correction: Non-tobacco nicotine dependence and postoperative complications following operative fixation of tibial shaft fractures: a retrospective cohort analysis.

Berk A, Bogart M, Eghtedari C, Good L, Florentino S, Moyal A, Burkhart R, Ochenjele G et al.

PubMed · DOI

Comparison of pediatric proximal femoral locking plates and dynamic hip screws in the fixation of pediatric intertrochanteric fractures, a biomechanical study.

Wang JH, Shih CA, Wang CH, Lee CC, Wu KW, Wang TM, Kuo KN, Hong CK

Treatment options for pediatric intertrochanteric fractures range from conservative casting in young children to surgical fixation in older patients, most commonly using dynamic hip screws (DHS) or pediatric proximal femoral locking plates (PF-LCP). As there is limited biomechanical evidence favoring one method over another, this study aimed to compare the biomechanical stability of DHS and PF-LCP for the treatment of pediatric intertrochanteric fractures using a synthetic bone model. Sixteen synthetic composite femurs were allocated into two groups stabilized with either a DHS or a PF-LCP. To simulate Delbet type IV intertrochanteric fractures, a vertical osteotomy was created in each specimen. Biomechanical evaluation was carried out using a universal testing machine, assessing axial stiffness, vertical, horizontal, and total displacement, as well as ultimate load to failure under standardized loading conditions. Comparative statistical analyses were then performed to determine differences in biomechanical performance between the two fixation methods. Group P demonstrated significantly smaller vertical and total displacements (0.15 ± 0.15 mm and 0.41 ± 0.31 mm) compared with the DHS group (0.59 ± 0.42 mm and 0.75 ± 0.33 mm) ($p = 0.018$ and $p = 0.045$), whereas horizontal displacement showed no difference ($p = 0.142$). Axial stiffness and ultimate failure load were comparable bet […]

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Impact of severe varus deformity on the outcome of mobile-bearing unicompartmental knee arthroplasty: a comparison of patients with varus deformity of less than and more than 15 degrees using matched-pair analysis.

Mullaji AB, Gupta AK, George GK, Kazi R

INTRODUCTION: The outcome of patients undergoing UKA with larger preoperative deformity (> 15º varus) and lesser deformity is sparsely compared in the literature. This matched-pair retrospective study attempts to fill this research gap by comparing the effect of preoperative and postoperative residual varus on outcome.

METHODS: Patients undergoing UKA with preoperative varus deformity > 15º (HKA > 15º group) were selected and matched for age and gender with those with < 15º of preoperative varus deformity (HKA < 15º group). There were 72 age and gender-matched pairs knees. HKA was measured and Forgotten Joint Score, OKS, VAS and WOMAC scores were obtained at a minimum follow up of 5 years. Postoperative residual varus was classified into 3 groups based on postoperative HKA: varus < 3º, varus 3 to 7º and varus > 7º.

RESULTS: Mean correction of deformity was significantly more ($p < 0.001$) in the HKA > 15º group (10.5º) when compared to HKA < 15º group (4.5º). There was no significant difference in outcome measures OKS ($p = 0.21$), FJS ($p = 0.35$) and VAS ($p = 0.6$) between patients in the two groups. There was no significant correlation between the preoperative and postoperative HKA of the patient with postoperative OKS, FJS and VAS in the two groups. There was no statistically significant difference between FJS ($p = 0.73$) of the three postoperative varus groups. Excellent to good […]

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A multimodal imaging approach to predict carpal tunnel syndrome after distal radius fractures.

Önüt M, Alkol N, Önalo■lu Y, Özer M, Apayd■n E, Talmaç MA

OBJECTIVE: Distal radius fractures are commonly associated with the development of carpal tunnel syndrome (CTS), yet current evidence primarily focuses on radiographic alignment parameters. The role of carpal tunnel morphology, particularly total carpal tunnel cross-sectional area (CSA) measured on computed tomography (CT), remains insufficiently investigated. This study aimed to evaluate the relationship between CSA, volar tilt, and the development of CTS in conservatively managed distal radius fractures, and to assess the predictive performance of these parameters when used in combination.

METHODS: This retrospective case-control study included 150 patients, comprising 50 patients who developed CTS and 100 controls without CTS. CSA was measured on CT scans obtained at initial presentation, while volar tilt was assessed using standard radiographs. Additional variables, including the presence of ulna fracture and fracture type, were analyzed. Group comparisons were performed using nonparametric tests. Multivariable logistic regression analysis was conducted to identify independent predictors of CTS. Diagnostic performance was evaluated using receiver operating characteristic (ROC) curve analysis.

RESULTS: Patients who developed CTS had significantly lower CSA and higher volar tilt values (both $p < 0.001$). The presence of an ulna fracture was inversely associated with CTS (p = […]

PubMed · DOI

Unicompartmental knee arthroplasty and total knee arthroplasty in patients with 15 or more degrees of varus deformity: age- and gender-matched comparative study of radiographic and clinical outcomes.

Mullaji A, Chopperla S, George GK, Kazi R

INTRODUCTION: To date, few studies have directly compared matched cohorts of patients with severe varus deformity undergoing UKA versus TKA. The current study seeks to address this gap. The hypothesis was that, in patients with deformity of > 15°, postoperative PROMs following UKA would be inferior to those achieved with TKA.

METHODS: Records of patients operated for UKA and TKA from January 2016 to December 2021 were evaluated, and the data for patients with deformity of ≥ 15° were selected. The UKA patients were then matched for age and gender with TKA patients. 52 matched pairs of patients were obtained. The HKA, mLDFA, MPTA were measured. OKS, SF12, and FJS were collected. The magnitude of change in the mLDFA and MPTA was assessed using the Estimated Marginal Means.

RESULTS: The mean follow-up times for UKA and TKA were 3.7 years and 5.5 years, respectively. The mean deformity correction (ΔHKA) was significantly greater in TKA (16.3° ± 4.5°) than UKA (10.6° ± 3.1°) ($p < 0.001$), with significant changes in mLDFA and MPTA ($p < 0.001$). In UKAs, 68% of patients demonstrated a postoperative HKA between 171.2° and 175.7°, whereas in TKAs, values clustered between 174.5° and 179.8°. Mann-Whitney U test demonstrated no statistically significant differences between the UKA and TKA cohorts across all assessed PROMs, including OKS, SF-12, and FJS ($p > 0.05$ in all). 98% of the UKA an […]

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Acetabular superior wall fractures: a distinct acetabular fracture entity.

Gänsslen A, Krappinger D, Liodakis E, Clausen JD, Omar-Pacha T, Lindtner RA, Sehmisch S, Osche DB

INTRODUCTION: A transitional "grey-zone" exists between typical locations of classical trapezoidal anterior wall and large-fragment posterior wall acetabular fractures. Part of this grey zone is the superior acetabular dome, which is commonly involved in acetabular fracture patterns, especially in geriatric patients. The aim of this study was to characterize isolated acetabular superior wall fractures and to assess their relevance as a distinct fracture pattern.

METHODS: A retrospective analysis of pelvic fracture databases from three tertiary trauma centers was performed. Acetabular fractures with superior dome involvement were screened, and cases with isolated superior articular involvement were included. Overall, seven patients were identified. Clinical and radiological data were analyzed regarding fracture morphology, associated fracture characteristics (including dislocation, comminution, marginal impaction, intra-articular fragments, and femoral head injury), concomitant pelvic ring injuries, treatment, and clinical outcomes.

RESULTS: Isolated superior wall acetabular fractures occurred following both high-energy and low-energy mechanisms, including simple falls. The mean age was 48 years and four of seven patients were male. Hip dislocation or subluxation was observed in four cases (anterior-superior or purely superior). Surgical treatment was performed in six cases, p […]

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A novel biomechanical model for reproducing and analysing the causal mechanisms of PFNA "cut-in" phenomenon.

Rasappan K, Joshua K, Chou SM, Shaw LKRM, Huang D, Yew A, Kwek EBK

INTRODUCTION: "Cut-in" is a rare but serious complication associated with cephalomedullary nail fixation for intertrochanteric hip fractures, particularly with Proximal Femoral Nail Antirotation (PFNA). This phenomenon involves paradoxical superomedial migration of the helical blade through the femoral head. Although increasingly recognised, the underlying mechanisms remain incompletely defined. This study aims to describe a new superolateral tensile loading set up in a bidirectional loading model, with the aim of incorporating possible mechanical deviations introduced by abductor weakness.

MATERIALS AND METHODS: This study was conducted in two stages. An initial stage with synthetic femurs (SYNBONE® 2420) was used to calibrate the set up. Within this stage, intramedullary canal diameter was compared (12 mm vs. 18 mm). In the final stage, 5 osteoporotic synthetic femurs (Sawbones® 3503) were used. In both stages, standardised AO/OTA 31-A1.1 intertrochanteric fracture were created in the synthetic femurs and fixed with PFNA-II implants. All femurs were subjected to cyclical bidirectional loading with a universal testing machine. Each cycle consisted of axial compression (720 N) and superolateral tensile loading (120 N) at 2 Hz. This configuration was designed to simulate altered hip biomechanics postulated with the trendelenburg gait. Testing was continued until mechanical fail […]

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Routine intraoperative infection testing in presumed aseptic hip and knee revision: a matched cohort retrospective study.

Loppini M, Rocchi C, Di Maio M, Chiappetta K, Bulgarelli A, Grappiolo G

INTRODUCTION: Periprosthetic joint infections (PJIs) complicate 1-2% of primary and up to 4% of revision arthroplasties, and their diagnosis remains challenging due to the lack of a diagnostic gold standard. Differentiating PJI from aseptic failure is particularly difficult in low-grade or biofilm-related infections. Routine intraoperative microbiological screening is not recommended in presumed aseptic revisions. This study aimed to evaluate the association between routine intraoperative microbiological screening and implant survival in presumed aseptic revision arthroplasty. A secondary objective was to identify preoperative predictors of unexpected PJIs.

METHODS: Retrospectively collected data of patients undergoing presumed aseptic THA and TKA revisions between 2013 and 2019 were included. Two matched cohorts were compared: a screened group (2016-2019) undergoing routine intraoperative microbiological sampling, and an unscreened one (2013-2015) without systematic screening. Kaplan-Meier survival analysis with log-rank testing assessed overall and infection-free survival. Univariate conditional logistic regression and receiver operating characteristic (ROC) analyses evaluated potential associations between preoperative serum markers and unexpected PJI.

RESULTS: A total of 559 patients were included, of whom 295 underwent screening and 264 did not. Unexpected infections occ […]

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Artificial intelligence advancements for orthopaedic clinical reasoning: longitudinal assessment of newer models (ChatGPT-5, Grok-3, Gemini 2.5 Flash) compared to clinicians.

Agharia S, Soroush S, Ameen D, Zhou Y

INTRODUCTION: This descriptive study aimed to longitudinally evaluate the performance of contemporary large language models - ChatGPT-5, Gemini 2.5 Flash, and Grok-3 - on orthopaedic clinical multiple-choice tasks, benchmarked against pooled clinician consensus. A secondary aim was to assess whether recent advances in generative AI translated into improved alignment with clinician consensus compared with previous AI models.

MATERIALS AND METHODS: A total of 97 multiple-choice clinical cases spanning eight orthopaedic subspecialties were sourced from OrthoBullets and previously benchmarked against aggregated responses from thousands of practising clinicians. Using identical methodology to our 2023 study of ChatGPT-3.5, ChatGPT-4, and Bard, each model was prompted with standardised case stems and response options. The primary outcome was the proportion of AI responses matching the most popular clinician response; secondary analyses assessed agreement within 10% and 20% of clinician consensus, performance on 'controversial' (< 25% margin) questions, and inter-model concordance using Cohen's kappa coefficients.

RESULTS: Gemini 2.5 Flash achieved the highest alignment with clinician consensus (69.1%), followed by Grok-3 (66.0%) and ChatGPT-5 (58.8%). None of the LLMs refused to respond to any prompts, representing a reduction from 7.2% from our 2023 study. Subspecialty analysis de […]

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Trajectory of urinary incontinence symptoms following total hip arthroplasty for hip osteoarthritis.

Kadowaki M, Fukutani S, Ushio K, Uchio Y

INTRODUCTION: Urinary incontinence (UI), a prevalent quality-of-life-limiting condition, frequently coexists with hip osteoarthritis, potentially through pelvic floor dysfunction related to obturator internus muscle involvement. Several reports suggest improvement after total hip arthroplasty (THA), although postoperative time-course changes remain unclear. This study aimed to clarify the temporal trends in UI after THA and examine association of postoperative UI trajectories with functional recovery.

MATERIALS AND METHODS: This retrospective study included 118 female patients who underwent THA for hip osteoarthritis between October 2021 and March 2024. UI, assessed using the International Consultation on Incontinence Questionnaire-Short Form (ICIQ-SF), was defined as ICIQ-SF ≥ 1 preoperatively and at 3, 6, 9, and 12 months postoperatively, and classified as stress (SUI), urge (UUI), or mixed (MUI). Hip-related patient-reported outcomes were evaluated using the Japanese Orthopaedic Association Hip Disease Evaluation Questionnaire (JHEQ). Associations with BMI, age, surgical approach (direct lateral/Bauer vs. modified Watson-Jones/OCM), and UI type were tested using Mann-Whitney U, chi-squared, and repeated-measures ANOVA tests.

RESULTS: Of 118 patients (mean age at surgery, 67 years; mean BMI, 24.3 kg/m2) preoperatively, 73 (61.9%) reported UI (mean ICIQ-SF 7.88), including S […]

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The grand piano ratio: an adjunctive intraoperative screening tool for detecting excessive femoral component external rotation in total knee arthroplasty.

Avci Ö, Öztürk A, Güler BO, Bayrak HÇ, Kaya AÖ, Do■an Y

BACKGROUND: Accurate femoral component rotation is essential for optimal outcomes in total knee arthroplasty (TKA), yet reliable intraoperative assessment remains challenging. The "grand piano sign" has been described as a qualitative visual cue, but its quantitative clinical value has not been clearly established. This study aimed to evaluate the diagnostic performance of the grand piano ratio as an intraoperative tool for detecting excessive femoral component external rotation using advanced postoperative imaging as a reference standard.

METHODS: A retrospective analysis was conducted on 170 knees undergoing primary TKA. The intraoperative grand piano ratio was measured after completion of all femoral resections and before trial component implantation using the anterior femoral resection surface. Postoperative femoral component rotation was assessed using MAVRIC-sequence magnetic resonance imaging referenced to the surgical transepicondylar axis (sTEA). Excessive femoral component external rotation was defined as postoperative external rotation greater than 3° relative to the sTEA reference. Receiver operating characteristic (ROC) analysis was performed to evaluate diagnostic performance of the grand piano ratio and determine the optimal cutoff value, while multivariable logistic regression analysis was used to identify independent predictors of excessive external rotation. […]

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The shoulder stiffness scale: a validated tool for monitoring stiffness in arthroscopic rotator cuff reconstructions.

Bouaicha S, Biner M, Selman F, Wieser K, Müller AM, Audigé L

BACKGROUND: Postoperative shoulder stiffness after arthroscopic rotator cuff repair is common and clinically heterogeneous. Established shoulder scores assess pain and function, but do not specifically isolate stiffness during follow-up. The aim of this study was to validate the newly developed Shoulder Stiffness Scale (SSS) as a pragmatic monitoring tool for postoperative shoulder stiffness after arthroscopic rotator cuff repair (ARCR).

METHODS: From a cohort of 973 primary arthroscopic rotator cuff reconstructions, patients without concomitant joint pathology of the same or contralateral arm were included. The new SSS was based on three items related to shoulder pain, subjective and objective ROM ranging from 0 to 10 points. It was collected before surgery, as well as 6 and 12 months postoperatively, along with other pain level parameters, Constant Score (CS), Subjective Shoulder Value (SSV), Oxford Shoulder Score (OSS). Changes of the SSS and its subscales over time, Cronbach's alpha for internal consistency, bottom and ceiling effects were reported. Validity was investigated through SSS associations with other parameters of similar dimension, and functional scores.

RESULTS: The selected cohort with 740 patients showed postoperative shoulder stiffness in 12.7 %. The SSS averaged 5.1 points (SD 2.5) preoperatively, and decreased significantly to mean 2.9 (SD 2.4) and 1.6 (S […]

PubMed · DOI

Isolated screw fixation of posterior wall fractures.

Clausen JD, Özbek H, Omar-Pacha T, Sehmisch S, Gänsslen A

INTRODUCTION: Posterior wall (PW) fractures are common acetabular fractures and are most often treated by open reduction and internal fixation. The gold-standard of surgical stabilization is screw fixation of large wall fragments with additional buttress plating. Some reports focussed on screw fixation alone presenting adequate results.

MATERIAL AND METHODS: From a total of 208 PW-fracture treated between 1972 and 2008, 57 patients were identified with open reduction and internal fixation (ORIF) using isolated screw fixation. These patients were analyzed regarding demographical data (patient age, sex), type of accident, injury mechanism, Injury Severity Score (ISS), concomitant pelvic ring injuries, associated injuries, perioperative parameters and long-term results.

RESULTS: 44 patients were male and 13 were female with a mean age of 36.8 years. More than 90% sustained high-energy trauma with 34 sustaining multiple injuries or polytrauma. The mean Injury Severity Score was 11.5 points. All patients had fracture dislocations, which were reduced within 24 h after admission except one patient with secondary transfer. 11 patients presented with an initial sciatic nerve deficit. 29 patients had a single PW-fragment, 16 presented with two PW-fragments and in 12 patients marginal impactions were present. Surgery was performed in average after 7 days using the Kocher-Langenbeck appr […]

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Research progress on debridement, antibiotics, and implant retention (DAIR) for the treatment of periprosthetic joint infection after artificial joint replacement.

Feng W, Zhang Z, Pu R, Wang H, Liu R, Sun Y, Zhang G

Periprosthetic joint infection (PJI) is one of the most serious complications following artificial joint replacement, posing significant treatment challenges and a heavy clinical burden. Debridement, antibiotics, and implant retention (DAIR) has become the preferred treatment strategy for acute PJI due to its advantages of minimal invasiveness, functional preservation, and cost-effectiveness. This article reviews the latest advances in optimizing the indications, timing of intervention, key technical details, and multimodal anti-infection strategies for DAIR. It focuses on discussing patient selection models (e.g., KLIC score), the impact of modular component exchange on biofilm eradication, the synergistic role of chemical debridement, and the central position of biofilm-active therapy in penetrating biofilms. Furthermore, this article proposes that future efforts should integrate molecular diagnostics, intelligent antimicrobial materials, and artificial intelligence prediction models to achieve individualized precision therapy for PJI, providing a theoretical basis for clinical decision-making.

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Tranexamic acid in high-risk shoulder arthroplasty patients: safety across thromboembolic, cardiac, renal, and neurologic risk profiles.

Parmar T, Adio A, Handy A, Daher M, Kellish A, Khan A, Abboud J

INTRODUCTION: Tranexamic acid (TXA) effectively reduces blood loss and transfusion requirements in shoulder arthroplasty. However, concerns regarding thromboembolic, neurologic, cardiac, and renal complications have limited its use in medically high-risk patients. This study evaluated temporal trends and the safety of TXA in high-risk patients undergoing total shoulder arthroplasty (TSA).

METHODS: A retrospective cohort study was conducted using TriNetX, identifying patients who underwent shoulder arthroplasty (2012-2025). Patients were stratified by preexisting high-risk conditions, including prior thromboembolism, renal failure, atrial fibrillation, seizure disorders, and visual disturbances. Within each subgroup, patients receiving TXA were propensity score-matched to those who did not. Perioperative TXA utilization trends were assessed. Ninety-day postoperative outcomes were compared, including transfusion requirements, thromboembolic events, cardiac complications, renal failure, neurologic events, infections, readmissions, emergency department visits, and mortality. Outcomes were reported as odds ratios (ORs) with 95% confidence intervals (CIs).

RESULTS: Reported perioperative TXA use began in 2012 and increased through 2025, with over 70% of patients in both standard- and high-risk cohorts receiving TXA. TXA use was associated with significantly lower transfusion rates […]

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Osteoporosis treatment gap prior to femoral fracture and prevalence of pharmacological risk factors: a prospective observational study.

Böck L, Kerschan-Schindl K, Frossard M, Hajdu S, Crevenna R, Anditsch M, Stemer G, Antoni A

INTRODUCTION: Hip fractures are a major complication of osteoporosis and are associated with high morbidity, mortality, and costs. Despite clear guideline recommendations, pharmacological osteoporosis treatment remains underutilized. In addition, medications that increase fall and fracture risk further contribute to fracture occurrence. The aim of this study was to assess guideline concordance of osteoporosis therapy and to quantify the prevalence of medications associated with increased fall and fracture risk prior to the fracture in patients admitted with femoral fractures.

MATERIALS AND METHODS: In this prospective observational study, 145 patients aged ≥ 55 years admitted with femoral neck, per- and subtrochanteric, femoral shaft and distal femur fractures to a tertiary academic trauma center between April and June 2025 were included. Pre-admission medication was assessed using structured medication reconciliation. Guideline concordance of pre-fracture osteoporosis therapy was evaluated according to the current Austrian osteoporosis guideline (Dimai et al., 2024). Fall-risk-increasing drugs (FRIDs) were identified using the STOPPFall criteria, and fracture-associated medications were recorded based on Austrian osteoporosis guideline-defined drug classes.

RESULTS: Among 145 included patients (mean age 80.4 years; 71% female), 117 (81%) met criteria for pharmacological oste […]

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Gait speed recovery after iliofemoral ligament-preserving hip arthroscopy for femoroacetabular impingement: associations with early postoperative gait kinematics.

Machida S, Tsutsumi M, Utsunomiya H, Kudo S

INTRODUCTION: Whether iliofemoral ligament-preserving hip arthroscopy facilitates recovery of gait speed-an indicator of functional recovery following hip arthroscopy in patients with femoroacetabular impingement-remains unclear. We investigated whether gait speed improved after iliofemoral ligament-preserving hip arthroscopy and examined whether pelvic jerk, defined as the time derivative of acceleration and assessed using inertial sensors at 1 month postoperatively (1M), was associated with gait speed at 6 months (6M).

MATERIALS AND METHODS: Eighteen patients who underwent primary hip arthroscopy for femoroacetabular impingement were included. Gait speed and pelvic jerk were assessed during a 10-m walk at a self-selected speed, using an inertial sensor positioned at the third lumbar level. Peak pelvic jerks during the 1st and 2nd halves of the stance phase (1st - and 2nd -peak pelvic jerks) were calculated. Measurements were obtained preoperatively and at 1M and 6M. Gait parameters were compared across time points using a linear mixed model. Bayesian regression was used to compare the observed change in gait speed from preoperatively to 6M with previously reported values. Associations between gait variables at 1M and gait speed at 6M were also examined.

RESULTS: Gait speed was significantly higher at 6M than that pre surgery (preoperative, 1.15 [95% confidence interval, 1.0 […]

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Computed tomography findings in 11,504 adult patients with traumatic brain injury: a large real-world cohort study with a S100B subgroup analysis.

Clar C, Puchwein P, Moshammer M, Sadoghi P, Kramer D, Leithner A, Reinbacher P

OBJECTIVE: The aim of this study was to characterize acute cranial CT findings in a large cohort of adult patients with traumatic brain injury and to explore the diagnostic performance of S100B in a selected subgroup. The hypothesis was that most cranial CT scans in adult patients with traumatic brain injury would not demonstrate acute traumatic abnormalities, and that, within a selected subgroup, negative S100B levels would be associated with the absence of such findings on CT.

METHODS: This retrospective cohort study included 11,504 adult patients presenting with traumatic brain injury to a level I trauma center between 04/2016 and 05/2024. All patients underwent cranial CT, while serum S100B measurements were available in a selected subset. CT findings were classified as normal or showing acute traumatic abnormalities. In the S100B subgroup, biomarker levels were analyzed in relation to CT findings to assess their diagnostic performance for excluding acute pathology.

RESULTS: A total of 11,504 adult patients with traumatic brain injury were included; 27.4% were inpatients and 72.6% outpatients. Overall, 81.9% of CT scans showed no acute pathology, while 10.7% demonstrated abnormalities. 7.4% of all cases were excluded. Acute TBI-related interventions were rare (3.7% of inpatients vs. 0.3% of outpatients). In patients with dementia and those receiving anticoagulation, CT sc […]

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Early post-operative CRP is a better predictor of DAIR failure than pre-operative CRP in total knee PJI.

Beadel H, Chaffey R, Kim K, Zhu M, Coleman B

INTRODUCTION: Debridement, antibiotics and implant retention (DAIR) surgery is often performed as a first-line treatment for periprosthetic joint infection (PJI), however, success rates are variable. This study aimed to determine whether post-operative C-reactive protein (CRP) levels predict DAIR failure.

MATERIALS AND METHODS: A multi-centre retrospective cohort study was performed over a 15-year period. All total knee arthroplasty (TKA) patients undergoing DAIR for first episode PJI were included. CRP was measured at admission and for 6 weeks post-operatively. Treatment success was defined as implant retention without the need for revision surgery or long-term suppressive antibiotics. Trends in CRP were compared between two groups: failed and successful DAIR. Receiver operating characteristic (ROC) curves were analysed.

RESULTS: 189 DAIR procedures were included, of which 49% (92) failed and 51% (97) were successful. Mean follow-up was 7.5 years. Overall, CRP trended down following surgery. Mean CRP was significantly higher in the failed DAIR group at all time points. The greatest difference was at week one post-operatively (mean 76 vs. 101, $P < 0.01$). This remained significant when patients experiencing DAIR failure within 90 days were excluded (mean 76 vs. 107, $P < 0.01$). CRP was most accurate as a predictor of failure at week one post-operatively, with an area under the […]

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Tuberosity refixation in reverse proximal humerus fracture arthroplasty: a prospective cohort study with two-year follow-up.

Oberrauter CH, Hess F, Welter J, Pape HC, Mayer S, Dullenkopf A, Mazzucchelli R, Marintschev I

INTRODUCTION: To evaluate the outcomes of a simplified tuberosity refixation technique using a two-suture construct in patients with complex proximal humerus fractures treated with reverse shoulder arthroplasty (RSA). The technique reduces the number of fixation sutures rather than the traditional four to five, aiming to decrease operative complexity while promoting greater tuberosity (GT) healing and improving the healing rate. A secondary exploratory analysis compared clinical and radiological outcomes between cemented and uncemented humeral stem fixation.

MATERIALS AND METHODS: This prospective study included 73 patients who underwent RSA (Global Unite Fracture Reverse System, DePuy Synthes, Warsaw, Indiana, USA). Clinical and radiological outcomes were assessed at two-year follow-up. Bone quality was evaluated using the Deltoid Tuberosity Index (DTI). Functional outcomes and GT healing were analyzed, and stem fixation type was assessed in a secondary subgroup analysis.

RESULTS: The mean patient age was 79 ± 8 years (range 61-95), and 65 (89%) were female. Osteoporotic bone quality (DTI < 1.4) was observed in 51 (70%) of cases. At two years, the median outcome scores were: Subjective Shoulder Value, 90 (IQR 80-95); Constant-Murley score, 76 (IQR 73-81); active forward flexion, 140° (IQR 120-160); abduction, 140° (IQR 120-160); external rotation, 8 (IQR 8-8); and internal r […]

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Is it time for a systematic and objective assessment of knee laxity? The impact of objective knee laxity after TKA on patient-reported outcomes: a comprehensive review.

Vermue H, Klincke V, Stroobant L, Tampere T, Victor J, Arnout N

PURPOSE: Over the past decade, robot assistance has emerged as an innovative and rapidly advancing tool in total knee arthroplasty (TKA). Despite technological advancements, the evaluation of the soft-tissue envelope during TKA often relies on subjective and rudimentary assessments by the surgeon. This literature review explores the relationship between objective intraoperative and postoperative laxity measurements in the coronal and sagittal planes and patient-reported outcome measures (PROMs) following TKA.

METHODS: A literature review was conducted using PubMed/MEDLINE and Embase to identify studies on intra- or postoperative laxity measurements in TKA and their correlation with PROMs. Inclusion criteria required studies on TKA patients with quantifiable coronal and/or sagittal laxity measurements and postoperative PROMs linked to these measurements.

RESULTS: Thirty-four studies met inclusion criteria: ten on sagittal laxity, twenty-seven on coronal laxity, and three on both. Follow-up ranged from 1 month to over 8 years. Sagittal laxity thresholds of 5-10 mm under applied loads of 89-133 N were associated with better functional outcomes, whereas excessive (> 10 mm) or overly tight (< 5 mm) laxity correlated with reduced satisfaction. Medial laxity in the coronal plane was linked to lower PROMs, while the impact of lateral laxity on PROMs was less conclusive.

CONCLUSIONS: […]

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Silver-impregnated occlusive dressing is associated with a lower rate of acute surgical site infection after direct anterior total hip arthroplasty.

Tsai YY, Shen PH, Wu CC, Pan RY, Lin LC, Wang SH

INTRODUCTION: The direct anterior approach (DAA) for total hip arthroplasty (THA) has gained increasing adoption in recent years; however, it has been associated with a higher risk of wound-related complications compared with other surgical approaches. Silver-impregnated occlusive dressings have demonstrated antimicrobial potential, though their role in DAA THA remains underexplored.

MATERIALS AND METHODS: We conducted a retrospective cohort study with a historical control group, reviewing 321 consecutive primary DAA THAs performed by a single surgeon (September 2020-December 2025). Standard sterile gauze was used before August 2021 (n = 76) and a silver-impregnated occlusive dressing (AQUACEL® Ag SURGICAL; ConvaTec, Deeside, UK) thereafter (n = 245). Despite the chronological difference between cohorts, baseline demographic and clinical characteristics did not differ significantly between groups (Table 1). The primary endpoint was acute surgical site infection(SSI) within 30 days (CDC criteria); all patients had ≥ 3 months follow-up to capture early wound-related reoperations as secondary outcomes.

RESULTS: Four acute SSI events occurred (1.25%). SSI incidence was 3.94% (3/76) with gauze versus 0.41% (1/245) with the silver-impregnated dressing (Fisher's exact p = 0.043). This corresponds to an absolute risk reduction (ARR) of 3.54% (95% CI, 0.34-10.57) and a number needed t […]

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Minimally invasive stabilisation of anterior pelvic ring injuries through anterior infix versus percutaneous anterior retrograde pubic screw: a prospective comparative cohort study.

Goda El-Hamalawy A, Abdelmoneim M, Al-Sayed M, Youness M, Sadek F, Kassem E

BACKGROUND: Management of unstable anterior pelvic ring fractures has shifted toward minimally invasive fixation techniques. This prospective randomized cohort study compares anterior subcutaneous internal fixator with percutaneous retrograde pubic ramus screw fixation, evaluating radiological outcomes, functional scores, complications, and peri-operative parameters.

METHODS: Fifty adult patients with unstable anterior pelvic ring injuries (Tile B & C; Young-Burgess lateral compression and vertical shear mechanisms) were enrolled between 2023 and 2025 and randomized into two equal groups: group 1 (internal fixator) and group 2 (retrograde pubic screw). Both groups underwent individualized posterior ring fixation as required. Outcomes assessed included operative time, blood loss, fluoroscopy time, union rate, Matta radiological score at 6 months, Majeed and Pelvic Outcome Scores at 12 months, and post-operative complications.

RESULTS: Management of anterior pelvic injury showed that retrograde screw fixation demonstrated significantly shorter operative time (32.0 ± 9.2 vs. 50.3 ± 8.4 min, p < 0.001) and less intraoperative blood loss (68.3 ± 16.2 vs. 122.3 ± 31.8 ml, p < 0.001). Radiological outcomes were comparable: excellent Matta score in 68% vs. 64% (p = 0.834). Union time showed no significant difference (14.1 ± 2.4 vs. 15.1 ± 1.5 weeks, p = 0.076). Functional outcomes at […]

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Prioritisation of assessments, diagnostic classifications, and outcome measures in Perthes disease: a Delphi survey of international health professionals.

Ball S, Davies LM, Gray K, Pacey V

INTRODUCTION: The aim of this study is to reach consensus and prioritise the most clinically important assessments, diagnostic classifications, and outcome measures by an international panel of experts for Legg-Calve-Perthes disease.

MATERIALS AND METHODS: A three-round modified e-Delphi survey was conducted between April and May 2025. Participants rated the clinical importance of 14 clinical assessments, 26 radiological assessments, eight diagnostic classifications, and 26 outcome measures identified by a scoping review. Consensus was defined as ≥ 75% agreement. In the final round, items in each category were prioritised for overall condition and by specific age groups (< 6 years, 6-8 years, > 8 years).

RESULTS: Thirty-eight health professionals, from nine countries participated in round 1. Thirty-four completed all three rounds, a retention rate of 89.5%. Consensus and prioritisation for clinical use were achieved for 10 clinical assessments, three radiological assessments, three diagnostic classifications, and three outcome measures.

CONCLUSION: Paediatric specialists have reached consensus and prioritised methods to evaluate and diagnose Legg-Calve-Perthes disease in clinical practice. These results represent an important step towards improving continuity of care and the foundation for future development of clinical practice guidelines.

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Compensatory activation of the deltoid and remaining rotator cuff muscles in patients with rotator cuff tears using real-time tissue elastography.

Hoshikawa K, Oishi R, Yuri T, Uno T, Nagai J, Giambini H, Mura N

INTRODUCTION: Rotator cuff (RC) tears are a prevalent condition leading to dysfunction and shoulder pain. Spatial changes in the behavior of the deltoid and remaining RC muscles during arm elevation in patients with RC tears have not been investigated. The purpose of this study was twofold: (1) to investigate factors related to the changes in the deltoid muscle behavior during scapular plane abduction (scaption) in patients with RC tears, and (2) to examine the functional relationship between the deltoid and remaining RC muscles, particularly remaining infraspinatus (ISP) and teres minor (TMin) muscles, in these patients using ultrasound elastography.

METHODS: Twenty-four patients (mean age, 66 ± 6 years) with RC tears (mean antero-posterior diameter, 17.4 ± 8.3 mm) were evaluated. Antero-posterior and medio-lateral diameters of the posterosuperior RC and Subscapularis (SSC) tears were measured intra-operatively. Muscle elasticity was evaluated at passive and active states, as a proxy for muscle activation, at 30°, 60°, and 90° during scaption for the deltoid, ISP, and TMin muscles using real-time tissue elastography. Based on the deltoid muscle activity ratio at 30° and 90° scaption, patients were divided into normal and abnormal activation groups.

RESULTS: Activity of the deltoid muscle in the abnormal activation group exhibited consistently high activity across various ele […]

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Pin orthosis extension block pinning versus conservative treatment for doyle type 4B mallet fractures.

Col KA, Demirsu O, Ayd■n M, Surucu S, Yilmaz M, Atlihan D

INTRODUCTION: Although numerous treatment options have been reported for mallet fractures, a universally accepted gold-standard approach has not yet been established. The purpose of this study was to compare the clinical and radiographic outcomes of the pin-orthosis extension-block pinning technique (PO-EBPT) with those of conventional conservative treatment in patients with Doyle type 4B mallet fractures.

MATERIALS AND METHODS: This study included 62 patients with Doyle type 4B mallet fractures involving 20-50% of the distal interphalangeal (DIP) joint articular surface, treated between March 2022 and April 2024. Patients were randomized into two groups: Group 1 (n = 33) underwent PO-EBPT, whereas Group 2 (n = 29) received conservative treatment with splint immobilization. Outcome measures included DIP joint extension lag, range of motion, fracture union, complication rates and functional outcomes according to the Crawford criteria.

Follow-up evaluations were performed at 2, 4 and 6 weeks and at 3, 6 and 12 months.

RESULTS: A total of 62 patients were analyzed (33 PO-EBPT; 29 conservative). No statistically significant differences were observed between the groups with respect to sex, affected side, injured finger, or complication rates ($p = 0.461$, $p = 0.658$, $p = 0.763$ and $p = 0.165$, respectively). However, the PO-EBPT group demonstrated significantly improved DIP joint exten […]

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Impact of cannabis use and tobacco smoking on outcomes of open carpal tunnel release surgery: a nationwide study in the United States.

Hoveidaei AH, Esmaeili S, Gilmor R, Chen Z, Conway JD, Ingari JV

PURPOSE: To evaluate the impact of cannabis use, tobacco smoking, and their concurrent use on postoperative complications following open carpal tunnel release (CTR) using a nationwide database.

METHODS: A retrospective cohort study was conducted using the PearlDiver database (2010-2022) to identify adult patients who underwent open CTR. Patients were categorized into four groups: cannabis-only users (n = 800), tobacco-only users (n = 167,803), concurrent users (n = 3529), and non-users (n = 167,803 matched controls). Postoperative complications, surgical site infection, wound disruption, joint stiffness, and complex regional pain syndrome, were assessed at 6 weeks and 3 months postoperatively.

RESULTS: At 6 weeks and 3 months, surgical site infection was more common among concurrent cannabis and tobacco users, cannabis-only users, and tobacco-only users. Wound disruption occurred more frequently in concurrent users and tobacco-only users, but not in cannabis-only users. Complex regional pain syndrome was associated only with tobacco use. No association was found between any substance use and joint stiffness.

CONCLUSION: Cannabis use, tobacco smoking, and especially concurrent use increase the risk of surgical site infection and wound disruption following open CTR. Tobacco use is independently associated with complex regional pain syndrome, while cannabis may have a protectiv […]

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Timing and microbiological profile influence long-term outcomes after debridement, antibiotics, and implant retention (DAIR) in acute hip periprosthetic joint infection.

Muñoz-Mahamud E, Perdomo-Lizarraga JC, Combalia A, Alías A, Serra A, Fernández-Valencia JÁ, Verdejo MÁ, Soriano Á

INTRODUCTION: The strategy of debridement, antibiotics, and implant retention (DAIR) represents one of the main therapeutic modalities for acute periprosthetic joint infection (PJI). However, reported outcomes remain highly variable, with success rates ranging from 16 to 92%. This study aimed to evaluate long-term treatment failure-free survival and identify risk factors for treatment failure in patients with acute hip PJI treated with DAIR using time-to-event analytical methods.

MATERIAL AND METHODS: This retrospective study evaluated the treatment failure rate in 115 patients treated with the DAIR strategy for acute PJI following primary or aseptic revision hip arthroplasty between 1999 and 2018. Potential predictors of treatment failure-free survival, including patient characteristics, infection microbiology, and surgical timing, were analyzed using univariate tests, Cox proportional hazards regression (hazard ratio [HR]), and Kaplan-Meier survival estimates.

RESULTS: Among the 115 patients included, 56 were women and 59 were men; 48 were aged 75 years or younger and 67 were older than 75 years. In 88 cases, the infection occurred after primary arthroplasty, whereas 27 followed aseptic revision surgery. The mean follow-up was 7.1 years (SD 4.4). At five years, treatment failure occurred in 30.4% of cases. Procedures performed within seven days of diagnosis showed lower fai […]

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Surgical treatment of varus unicompartmental knee osteoarthritis: indications, trends, and outcomes-a narrative review.

Burgio C, Bosco F, Cobisi C, Zepeda K, Giustra F, Capella M, Mayman D, Debbi E et al.

INTRODUCTION: Varus unicompartmental knee osteoarthritis (UC-KOA) is a frequent and heterogeneous clinical condition characterized by compartment-specific cartilage degeneration and frequently associated with lower-limb malalignment. The progressive understanding of knee biomechanics and alignment has led to the development of surgical strategies specifically tailored to address isolated compartment disease. The purpose of this narrative review is to evaluate indications, current trends, and clinical outcomes of the main surgical options for varus UC-KOA, with particular focus on realignment osteotomies and unicompartmental knee arthroplasty (UKA).

MATERIALS AND METHODS: A narrative review of the literature was conducted using major biomedical databases. Evidence related to biomechanics, alignment correction, osteotomy techniques, UKA indications, implant evolution, complication profiles, and comparative outcomes was analyzed and synthesized thematically.

RESULTS: Biomechanical and clinical evidence consistently identifies malalignment as a key driver of isolated compartment degeneration. Corrective osteotomies effectively restore physiological load distribution and provide durable outcomes in selected patients with extra-articular deformity. Advances in implant design and surgical techniques have substantially improved UKA survivorship, allowing broader patient selection wit […]

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Preoperative NarxCare overdose risk scores greater than 100 are associated with worse PROMs improvements and dissatisfaction after primary THA.

Khan ST, Emara AK, Patel SV, Elmenawi KA, Ibaseta A, Pasqualini I, Zhang C, Cleveland Clinic Adult Reconstruction Research Group et al.

INTRODUCTION: The NarxCare Overdose Risk Score (ORS) is a weighted measure of patient-specific prescription drug use. In patients undergoing primary total hip arthroplasty (THA), this study aimed to evaluate (1) the trend of the ORS from preoperative to 1-year postoperatively, and (2) the association of preoperative ORS with achievement of clinically meaningful improvements in patient reported outcome measures (PROMs), and satisfaction at 1-year.

METHODS: All patients who underwent primary unilateral elective THA at a USA tertiary healthcare system from January 2016-December 2022 were eligible. Patients with incomplete PROMs or missing baseline NarxCare ORS were excluded, leading to the inclusion of 5,424 patients. Multivariable regression models were used to assess the association between ORS and 1-year PROMs, including achievement of the minimum clinically important difference (MCID) and Patient Acceptable Symptom State (PASS) for the Hip Disability and Osteoarthritis Outcome Score (HOOS) subdomain scores, as well as satisfaction.

RESULTS: Preoperatively, 46.1% (n = 2501) had a NarxCare ORS of 0, 24.6% (n = 1336) had scores between 100 and 199, and 16.2% (n = 876) in the 200-299 range. After adjusting for confounding variables, a preoperative ORS of 100-199 was associated with failure to achieve MCID in HOOS Pain (OR 1.91;CI 1.32-2.77;p = 0.001), and HOOS JR (OR 1.54;1.10-2 […]

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Only a partial shield: local vancomycin postpones but does not prevent hip and knee prosthetic infections.

Darup A, Ettinger M, Savov P, Brand S, Seeber GH, Stauss R

INTRODUCTION: Periprosthetic joint infection (PJI) represents a severe complication following total hip arthroplasty (THA) and total knee arthroplasty (TKA). The intraarticular application of vancomycin powder is increasingly integrated into perioperative standards as an innovative strategy for PJI prevention. Yet the efficacy of this method remains a topic of debate. The aim of this retrospective cohort study was to evaluate whether topical vancomycin application reduces the incidence of PJI after primary total joint arthroplasty (TJA) and whether it is associated with an increased risk of wound complications.

MATERIALS AND METHODS: In this retrospective monocentric cohort study, all primary THA and TKA procedures performed between January 1, 2022, and December 31, 2023, were analysed. From January 1, 2023, intraarticular vancomycin powder was implemented as part of the standard perioperative protocol for PJI prevention. The PJI rates, postoperative surgery-related complications, and time to infection were compared between the vancomycin and control groups.

RESULTS: A total of 1,499 patients were included in the study. Patients were divided into two groups: a vancomycin group (VG, n = 818) and a control group (CG, n = 681). No statistically significant group differences were observed for wound healing disorders (p = 0.775) or PJI rates (1.0% VG vs. 0.9% CG, p = 0.846). Howev […]

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Validity and prognostic value of a novel intraoperative assessment tool for reduction in trimalleolar fractures: the SGSC checklist approach.

Gao H, Xu G, Ma J, Wang Y, Wang J, Zhou R, Liu Z, Wang B

BACKGROUND: Trimalleolar fractures present substantial surgical challenges, and the quality of fracture reduction significantly influences patient outcomes. Current intraoperative assessment methods, including fluoroscopy and arthroscopy, each possess inherent limitations when used independently. This study aimed to validate the effectiveness of a novel intraoperative evaluation tool, the SGSC (Step, Gap, Syndesmosis, Congruity) "traffic light" checklist, that integrates arthroscopic and fluoroscopic assessment, and to evaluate its predictive value for clinical prognosis.

METHODS: This single-center retrospective cohort study enrolled 36 patients who underwent surgical treatment for trimalleolar fractures. Validation proceeded in two stages: first, we calculated inter-rater reliability (Kappa statistic) and concordance with postoperative CT as the reference standard to establish the reliability and validity of the SGSC checklist (accuracy validation); second, patients were stratified into "All-Green" and "Non-All-Green" groups based on intraoperative SGSC assessment, and we compared AOFAS scores and early radiographic degenerative changes at final follow-up (prognostic validation).

RESULTS: For accuracy validation, inter-rater agreement between the two assessors demonstrated excellent reliability (Kappa = 0.827; 95% CI: 0.644-1.000). The overall concordance between SGSC check […]

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Radiographic assessment of femoral, acetabular and global offset following hip spacer implantation in staged total hip arthroplasty".

Cavagnaro L, Ferrari E, Providenti V, Marciante V, Carrega G, Formica M

BACKGROUND: The number of total hip arthroplasties (THAs) performed worldwide is steadily increasing, and, consequently, the demand for revision procedures is expected to rise as well. two-stage revision procedure continues to be considered the gold standard in many clinical scenarios. During the first step, failure to restore femoral offset has been implicated as a contributor to instability and mechanical failure, although a clear consensus is still lacking.

OBJECTIVES: The primary aim of this study is to evaluate the restoration of biomechanical parameters of the hip-specifically leg length discrepancy (LLD), femoral offset (FO), acetabular offset (AO), and global offset (GO)-following implantation of a specific type of articulating hip spacer during the first stage of a two-stage revision for PJI. A secondary objective is to assess the correlation between variations in offset parameters and the rate of interstage mechanical complications.

MATERIALS AND METHODS: We retrospectively reviewed all patients undergoing staged revision with a specific articulating spacer between 2020 and 2022 at a single institution. Harris Hip Scores, Oxford Hip Scores, and Visual Analogue Scales for pain were obtained at different time points. Radiographic analysis included LLD, FO, AO and GO measurements on the affected (spacer and post-reimplantation) and the contralateral side. Complications […]

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Simulated tears of antero-posterior rotator cuff force-couple and reduced glenoid concavity decrease anterior glenohumeral stability: a robot-assisted biomechanical analysis of a load and shift sequence.

Karlsfeld MJ, Wermers J, Tänzler S, Sußiek J, Herbst E, Katthagen JC

BACKROUND: The glenoid concavity and the compression applied by the rotator cuff (RC) are essential for glenohumeral stability (GHS). This study aimed to determine how different simulated rotator cuff tears (RCT) and the glenoid depth influence GHS.

METHODS: A Load and Shift sequence was performed with eight fresh-frozen cadaveric shoulders in a robotic-assisted setup. Differently configured static loading of the reinforced RC and deltoid muscle (DLT) simulated intact RC and anterior, superior, anterosuperior, posterosuperior, mass, and complete RC plus DLT tears. Anterior dislocation forces and their changes were determined as indicators of stability. The glenoid depth and Bony shoulder stability ratio (BSSR) were defined as indicators of concavity. To assess GHS, the maximal force (Favg), the maximal force increase (dFavg), and their mean deviations (ΔFmax, ΔdFmax) to the intact configuration during the sequence were evaluated.

RESULTS: Simulated tears of the subscapularis tendon (SCP) (ΔFmax = 7.90 N, ΔdFmax = 1.54 N/mm) and simulated tears of the infraspinatus (ISP) + teres minor (TM) tendons (ΔFmax = 7.19 N, ΔdFmax = 1.48 N/mm) resulted in greater differences to the intact shoulder than simulated tears of the supraspinatus tendon (ΔFmax = 3.82 N, ΔdFmax = 1.16 N/mm). High correlations were observed between maximal force concerning glenoid depth (r = 0.81) and BSSR (r = 0 […]

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Non-tobacco nicotine dependence is associated with increased complications following clavicle open reduction internal fixation.

Adio AA, Goyal RK, Skiena A, Daher M, Horneff JG, Kassam HF, Hill BW, Abboud JA

BACKGROUND: Tobacco dependence is well established to impair bone healing and increase postoperative complications. However, the independent effects of nicotine, separate from combustion-related factors, remain unclear. This study aimed to evaluate the impact of non-tobacco nicotine dependence (NTND) following open reduction and internal fixation (ORIF) of clavicle fractures, including midshaft and lateral fracture subtypes.

METHODS: A retrospective cohort study was performed using a large national database. Patients undergoing ORIF of clavicle fractures were identified using procedural and diagnostic codes. Patients with non-tobacco nicotine dependence (e.g., vaping, patches, or gum) were identified and propensity score matched to (1) patients without nicotine exposure and (2) patients with tobacco nicotine dependence. In a secondary analysis, patients were stratified by fracture type (midshaft and lateral), and outcomes were compared between nicotine exposure and no nicotine exposure within each subgroup. Outcomes were assessed at 90 days and 1 year and included medical complications, number of opioid prescriptions, and implant-related outcomes. Risk ratios (RRs) with 95% confidence intervals (CIs) were calculated, with $p < 0.05$ considered significant.

RESULTS: After matching, 936 non-tobacco nicotine users were identified. Non-tobacco nicotine dependence was associated wit […]

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Association between timing of surgery and postoperative outcomes in older adults with distal femur fractures: a systematic review and meta-analysis.

Hoshika S, Yamamoto N, Saka N, Tsuge T, Sato T, Isaji Y, Nakao S

INTRODUCTION: The association between surgical timing and clinical outcomes in older adults with distal femur fractures remains unclear. We conducted a systematic review and meta-analysis to evaluate whether earlier surgery is associated with improved outcomes in this population.

MATERIALS AND METHODS: In accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, two independent reviewers searched MEDLINE, Embase, the Cochrane Central Register of Controlled Trials, ClinicalTrials.gov, and the World Health Organization International Clinical Trials Registry Platform from inception to October 8, 2024. The primary outcome was 30-day mortality, and secondary outcomes included longer-term mortality and postoperative complications. Risk of bias was assessed using the Quality in Prognosis Studies tool, and the certainty of evidence was evaluated using the Grading of Recommendations Assessment, Development and Evaluation approach.

RESULTS: Of 6,101 records screened, 16 retrospective cohort studies comprising 31,213 patients met the inclusion criteria. Early surgery was not associated with reduced 30-day mortality (adjusted odds ratio [OR]: 0.77, 95% confidence interval [CI]: 0.56-1.05; low-certainty evidence) or with longer-term mortality at 90 days (crude OR: 0.91, 95% CI: 0.51-1.60), 180 days (crude OR: 0.46, 95% CI: 0.12-1.77), or 1 […]

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Is a central cavity necessary for bioactive glass-ceramic spacers in plated ACDF? A retrospective comparison of solid versus cavity designs.

Kang TH, Lee G, Kang B, Han J, Cho M, Lee JH

PURPOSE: Traditionally, interbody cages feature a central cavity for graft packing. However, for bioactive glass-ceramic (BGS-7) spacers achieving fusion via surface-mediated osseointegration, this cavity may be biologically redundant and structurally disadvantageous. We compared radiological and clinical outcomes of plated anterior cervical discectomy and fusion (ACDF) using BGS-7 spacers with or without a central cavity.

METHODS: We retrospectively reviewed patients undergoing plated ACDF using solid non-cavity (17 patients, 24 levels) or central-cavity (17 patients, 33 levels) BGS-7 spacers. Radiographic fusion was evaluated using dynamic interspinous distance (primary, ≤ 2 mm) and segmental angular motion (secondary, ≤ 4°), alongside CT-based osseointegration (direct contact ≥ 50%). Subsidence (≥ 2 mm) and clinical outcomes were evaluated at 12 months.

RESULTS: Both groups demonstrated comparable 12-month clinical improvements ($p > 0.05$). Radiographic fusion rates were equivalent between non-cavity and central-cavity groups based on the primary distance (91.7% vs. 84.8%, $p = 0.687$) and secondary angular criteria (79.2% vs. 63.6%, $p = 0.331$). The non-cavity group showed a trend toward superior segmental stability, with a smaller dynamic gap distance (0.82 ± 0.64 mm vs. 1.53 ± 1.99 mm, $p = 0.064$). CT-based fusion perfectly matched (90.5% vs. 90.9%, $p = 1.000$). Subsidence ra […]

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Clinical outcomes of internal iliosciatic fixation using a conventional reconstruction plate via an anterior approach for acetabular fractures involving the posterior column.

Xiong H, Chen H, Cao S, Chen X, Wang F

BACKGROUND: Acetabular fractures involving the posterior column remain surgically challenging because of the complex three-dimensional anatomy and restricted operative corridor. As an alternative to traditional posterior approaches, internal iliosciatic fixation through an anterior approach has gained increasing acceptance. Reconstruction plates, which offer the advantages of low cost and convenient intraoperative bending, are extensively used in clinical settings.

METHODS: A retrospective case series was conducted on 36 consecutive patients with acetabular fractures involving the posterior column who underwent internal iliosciatic fixation using an intraoperatively contoured reconstruction plate between July 2019 and November 2022. The modified Stoppa approach (n = 26) or the pararectus approach (n = 10) was employed. Fracture reduction was assessed using the modified Matta criteria, and hip function was evaluated using the modified Merle d'Aubigné-Postel score at one-year follow-up. Operative time, blood loss, fracture union, and complications were recorded.

RESULTS: The mean age of the patients was 42.6 ± 14.5 years (range: 21-71 years), with 25 males and 11 females. Mechanisms of injury included traffic accidents (n = 21), falls from height (n = 10), skiing injuries (n = 3), and others (n = 2). According to the Letournel-Judet classification, there were 13 both-column fra […]

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Novel UHMWPE sleeve for stabilizing loose suture anchors in rotator cuff repair.

Lin CK, Jaw FS, Sam SS, Siow TF, Yii SC, Chang YJ, Young TH

Suture anchors are frequently used for repairing torn rotator cuffs. However, post-operative anchor loosening is a common complication often caused by low bone density. The purpose of this study was to evaluate if a novel UHMWPE cortex locking sleeve could improve the anchor pullout strength after revision in synthetic bone. The initial failed anchor fixation was simulated by inserting a 4.5 mm anchor and pulling it out of the bone, which created an enlarged insertion hole for the revision anchor. After revision, two loosening conditions were simulated; (i) intra-operative anchor loosening where the revision anchor was directly pulled out from the block after insertion, and (ii) post-operative anchor loosening where the revision anchor was first subjected to cyclic loading and then pulled out. For both loosening conditions, the pullout strength and energy absorbed was recorded for the following configurations: (i) Group A, same-diameter anchor (SDA; 4.5 mm × 10 mm); (ii) Group B, larger-diameter anchor (LDA; 5.5 mm × 10 mm;); and (iii) Group C, same-diameter anchor combined with the novel sleeve (SDA + NS). For intra-operative anchor loosening, the highest pullout strength was recorded in the SDA + NS group, with a mean value of 370.077 ± 32.699 N, which was significantly greater than both SDA and LDA groups ($p < 0.05$).

Similarly with post-operative anchor loosening, the SDAP […]

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Impact of rheumatoid arthritis on complications and hospital outcomes following reverse shoulder arthroplasty: evidence from 389,135 hospitalizations in the national inpatient sample.

Mahamid A, Elgabsi M, Khatib M, Murad H, Qawasmi F, Lavon E, Zahalka S, Yassin A et al.

OBJECTIVE: To evaluate the impact of rheumatoid arthritis (RA) on perioperative complications and hospital outcomes following reverse total shoulder arthroplasty (RTSA) using the National Inpatient Sample (2016-2021).

METHODS: Adult patients undergoing RTSA were identified using ICD-10 procedure codes. Outcomes included prolonged length of stay (LOS), high hospital charges, inpatient complications, and mortality. Multivariable logistic regression adjusted for demographic, clinical, and hospital characteristics. Survey weights were applied to generate nationally representative estimates.

RESULTS: A total of 389,135 patients underwent RTSA, including 18,140 (4.66%) with RA. In unadjusted analyses, RA patients had higher rates of prolonged LOS (17.8% vs. 14.8%) and acute blood loss anemia (10.7% vs. 8.6%). After multivariable adjustment, RA remained an independent predictor of prolonged LOS (adjusted OR 1.18, 95% CI 1.14-1.23, $p < 0.001$) and acute blood loss anemia (adjusted OR 1.25, 95% CI 1.18-1.33, $p < 0.001$), but not urinary tract infection. In-hospital mortality was rare and could not be modeled due to zero events in the RA group. RA patients also had slightly longer hospitalizations and higher hospital charges.

CONCLUSION: RA is independently associated with prolonged hospitalization and increased risk of perioperative blood loss anemia following RTSA. These findings high […]

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Long-term results of thumb carpometacarpal joint arthrodesis.

Vogler P, Benedikt S, Bode S, Keiler A, Stock K, Arora R

INTRODUCTION: Thumb carpometacarpal (CMC 1) arthrodesis remains an established treatment option for patients with high functional demands and strength requirements. Long-term data on mobility, complication rates and clinical outcomes remain limited. This study aimed to analyze long-term clinical and radiological results after CMC 1 arthrodesis.

MATERIALS AND METHODS: A single-center, retrospective study included patients who underwent CMC 1 arthrodesis between 2004 and 2024. Following medical record analysis, eligible patients were invited to a standardized clinical and radiological follow-up examination. Outcomes included DASH, PRWHE and MHQ scores, pain intensity, mobility, strength and patient satisfaction. Radiological assessment covered fusion signs, implant position and adjacent joint arthrosis. The median follow-up was 9 years.

RESULTS: 34 patients were included, 16 participated in follow-up. The median DASH score was 14 points, PRWHE 4 points and MHQ 80 (operated) vs. 84 (contralateral). Grip strength reached 85% of the contralateral side, with preserved pinch strength. Compensatory adaptation of adjacent joints was observed regarding range of motion. Patients reported no relevant resting pain and only minimal load-related discomfort. 15 of 16 patients were satisfied with the outcome. Non-union occurred in 4 of 34 cases (11.8%), associated with absent cancellous bone […]

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Does the tendon retract in flexor hallucis longus muscle laceration? A cadaveric study.

Ayd■n M, ■ahin Y, Ya■ar H, Çiçek F, Cerano■lu FG, Mert A, Ç■naro■lu S

INTRODUCTION: Flexor hallucis longus (FHL) tendon injuries are common, particularly in open injuries caused by sharp objects. However, data regarding FHL tendon retraction following hallux-level FHL tendon lacerations remain limited. This study investigated the retraction and shortening of the FHL tendon using cadaveric models.

MATERIALS AND METHODS: Six fresh-frozen below-knee amputated cadaveric feet (mean age: 65 years, range: 62-88) were utilized. FHL tendons were dissected, and lacerations were simulated at the hallux level. Tendons were subjected to various tensile forces (20-120 N) using a mechanical device. Force was measured in Newtons via a loadcell, and retraction distance was recorded in millimeters using an encoder. Data were summarized as mean ± standard deviation (SD), median, and quartiles.

RESULTS: The FHL tendon retracted gradually as applied force increased. An average retraction of 7 mm was measured at 20 N, reaching 21 mm at 60 N. Beyond 60 N (> 21 mm retraction), the other four toes also flexed. This occurs due to the anatomical connection between the FHL and flexor digitorum longus (FDL) tendons.

CONCLUSIONS: Only slight retraction was observed in hallux-level FHL tendon lacerations. At most, the findings suggest that distal FHL lacerations may exhibit limited proximal retraction under the tested conditions, which may have implications for surgical pla […]

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Early-onset rice body synovitis of the knee in children under three years: a case series and review of the literature.

Zhu L, Liu C, Liu Y, Mao B, Huang Y, Wen J, Liu Y, Qian T

OBJECTIVE: Rice body synovitis reports have been rarely reported in children under three years of age. We aimed to explore the clinical characteristics and treatment experience of rice body-like synovitis in children's knees.

METHODS: We retrospectively analyzed the clinical and imaging data of 4 cases (5 knees) of rice body-like synovitis in children treated at our hospital from January 2014 to December 2021. Among the patients, 1 was male and 3 were female, with an average age of 22.5 months. The average length of symptoms before admission was 1.86 months. Clinical presentations included swelling and limping, with no fever or rash.

RESULTS: Three patients with unilateral onset and one patient with bilateral onset underwent unilateral arthroscopic excision of synovial lesions. All 4 patients had pathological findings consistent with chronic inflammation, leading to a postoperative diagnosis of juvenile idiopathic arthritis (JIA). Postoperatively, all patients received standardized pharmacological treatment and were followed for 24 to 48 months in collaboration with the rheumatology department. None of the 4 knees undergoing surgery experienced recurrence, and the symptoms of the knee treated conservatively resolved spontaneously. In the last follow-up session, MRI and ultrasound showed only mild synovial thickening, with effusion and loose bodies disappearing.

CONCLUSION: R […]

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Extravertebral temperature exposure during spinal radiofrequency ablation: an experimental surrogate assessment of neural injury risk.

Weigert A, Hoffmann RT, Büttner A, Zaccaria R, Wegener V, Holzapfel BM, Wegener B

BACKGROUND: Radiofrequency ablation (RFA) has become an important minimally invasive option for the treatment of painful spinal tumors and benign bone lesions. While its effectiveness within the vertebral body is well established, there is ongoing concern about unintended heat exposure of nearby neural structures. Experimental data describing how heat propagates beyond the vertebra during spinal RFA are still limited. This experimental in vitro study aims to assess the extravertebral temperature exposure during spinal radiofrequency ablation as a surrogate parameter for potential neural injury risk under controlled conditions.

METHODS: This experimental in vitro study was performed on four fresh-frozen human lumbar spines. Radiofrequency ablation was carried out using a commercially available system and standard clinical protocols. Two anatomical configurations were examined: ablation within the vertebral body and ablation within the pedicle. Temperatures were measured at predefined locations outside the vertebra, including the spinal canal and cortical boundaries, using multiple thermocouples. To aid interpretation of radiofrequency-related measurement artifacts, an additional control setup using an electrothermal heating system without radiofrequency energy was included. Temperature data were evaluated descriptively.

RESULTS: In the control experiments, temperature profiles […]

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Giant cell tumor of bone: exploring HtrA1 as a potential predictor of recurrence.

Mirioglu A, Dalkir KA, Gokmen MY, Bagir M, Ates KE, Deveci MA, Gönlü■en G, Ozbarlas HS

INTRODUCTION: Giant cell tumor of bone (GCTB) is a locally aggressive tumor with a considerable risk of recurrence, but reliable predictive markers are limited. High-temperature requirement A serine peptidase 1 (HtrA1) has been implicated in the progression of several cancers and may also play a role in GCTB recurrence.

MATERIALS AND METHODS: This retrospective study included 34 patients with GCTB treated with curettage between 2002 and 2016. HtrA1 expression was evaluated separately in giant cells (GC) and mononuclear cells (MNC) using immunohistochemistry. Based on high or low expression in each cell type, patients were categorized into four expression patterns: High GC / High MNC, High GC / Low MNC, Low GC / High MNC, and Low GC / Low MNC. Clinical data, including recurrence status and time to recurrence, were analyzed.

RESULTS: The High GC / Low MNC group showed the highest recurrence rate at 71.4%, compared with 28.6% in the High GC / High MNC group and 31.6% in the Low GC / Low MNC group. No recurrence was observed in the Low GC / High MNC group, which included only one patient. The High GC / Low MNC group showed a non-significant trend toward higher recurrence compared with all other groups combined, with a similar trend toward shorter recurrence-free survival.

CONCLUSION: HtrA1 expression patterns in GCTB may provide preliminary insight into recurrence risk. Although […]

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Coronal hamate body fractures with dorsal fourth and fifth carpometacarpal joint instability: fracture patterns, injury mechanisms, surgical strategies, and outcomes.

Lee SH, Chung HJ, Cha SM

INTRODUCTION: Coronal hamate body fractures are rare injuries, and studies regarding their morphology and management are limited. This study analyzed fracture patterns, injury mechanisms, associated injuries, surgical strategy, and clinical outcomes in patients with coronal hamate body fractures involving more than one-third of the articular surface and concomitant carpometacarpal (CMC) joint instability.

METHODS: Fifty-eight male patients were retrospectively reviewed. Fractures were classified as dorsal oblique or coronal splitting types based on computed tomography findings. Surgical treatment involved open reduction and multiple screw fixation of the hamate using a K-wire-guided technique, with supplementary Kirschner wire stabilization for associated metacarpal base fractures or CMC joint instability. Radiographic and functional outcomes were evaluated after a minimum follow-up of 6 months.

RESULTS: Dorsal oblique fractures (63.8%) occurred more frequently than coronal splitting fractures (36.2%) and were most commonly associated with punching (OR, 4.9; $p = 0.01$), whereas falls from height were more common in coronal splitting types. Concomitant fractures occurred in 82.8% of cases, most frequently involving the fourth metacarpal base. Among 35 patients with a minimum 6-month follow-up, 34 achieved successful healing. In the 33 patients with functional outcome data, the […]

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Effect of spinal fusion length on hip and knee osteoarthritis: a comparative cohort study of no fusion, short-segment fusion, and long-segment fusion.

Sülek Y, Demirkale ■, Erto■rul R, Özdemir HM

PURPOSE: Lumbar fusion alters spinal-pelvic biomechanics and may influence degenerative changes in the lower extremities. This study investigated whether fusion length is associated with radiographic progression of hip and knee osteoarthritis.

METHODS: In this retrospective, single-center cohort study, patients aged 40-65 years who underwent lumbar surgery between 2010 and 2021 were grouped as no fusion (N), short fusion (S; ≤3 levels), or long fusion (L; ≥4 levels). Hip joint space width was assessed using minimum joint space (MJS) and superointermediate joint space (SJS), including femoral head-standardized measures (sMJS and sSJS). Knee medial joint space width (KMJS) was measured when available. Spinopelvic parameters (LL, PI, PT, SS, PI-LL) and functional outcomes (ODI, HHS, WOMAC) were recorded. Multivariable linear and logistic regression analyses were performed to identify independent predictors of joint space narrowing and osteoarthritis risk.

RESULTS: Hip joint space narrowing differed significantly among groups, with the highest annual narrowing rates in the long-fusion group, followed by the short-fusion group, and the lowest in the no-fusion group ($p < 0.001$). In contrast, knee KL grade change and annual KMJS narrowing did not differ significantly between groups ($p > 0.05$). In logistic regression, each additional fused level increased the odds of postoperative MJ […]

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Biomechanical comparison of inverted triangle and L-shaped screw configurations with medial buttress and anteromedial support plate in Pauwels type III femoral neck fractures.

Erem M, Demirta■ U, Y■ld■r■m S, Selçuk E, Çopuro■lu C

INTRODUCTION: Surgical stabilization of Pauwels Type III femoral neck fractures remains a significant challenge due to high vertical shear forces. While the medial buttress plate is a recognized solution, it requires extensive deep dissection. This study aims to compare the biomechanical performance of various screw configurations combined with either a medial buttress or anteromedial support plate.

MATERIALS AND METHODS: Twenty-five third-generation synthetic femurs were used to create a standardized 70-degree (Pauwels III) fracture model. Specimens were divided into five groups ($n = 5$): (A) Inverted triangle (IT) screws with a Pauwels screw, (B) Inverted triangle (IT) with medial buttress plate (MBP), (C) Inverted triangle with anteromedial support plate (ASP), (D) L-configuration with medial buttress plate (MBP), and (E) L-configuration with anteromedial support plate (ASP). Axial loading was applied at 2 mm/min until construct failure, defined objectively by the real-time force-distance curve.

RESULTS: Although no statistically significant difference was found between groups ($p = 0.102$), a large effect size was observed ($\eta^2 = 0.309$). Group C (IT + ASP) demonstrated the highest mean failure load (1695 ± 494.6 N). Conversely, Group E (L-configuration + ASP) exhibited the lowest stability (977.2 ± 195.4 N) with a remarkably narrow standard deviation. The majority of failures […]

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Reverse total shoulder arthroplasty is a surgical option for patients over 60 years of age with locked shoulder dislocation.

Sebastia-Forcada E, Miralles-Muñoz FA, Martinez-Mendez D, Albero-Catala L, Climent-Peris V, Lizaur-Utrilla A

BACKGROUND: in locked dislocation of the shoulder, instead of reducing back to the glenoid, the humeral head remains incarcerated on the glenoid in a locked fashion. This clinical situation is fairly uncommon. It is essential to conduct an individual evaluation of each patient to determine the appropriate treatment.

OBJECTIVE: the aim of this study was to evaluate the functional outcomes of reverse total shoulder arthroplasty (rTSA) in the treatment of locked shoulder dislocation.

METHODS: patients with locked shoulder dislocation who underwent reverse shoulder prosthesis surgery and were admitted to our center between 2007 and 2023 were reviewed. The primary outcome was the Constant score. Secondary outcomes included the adjusted Constant, UCLA and DASH scores. Additionally, any signs of radiologic loosening were also documented.

RESULTS: the series consisted of 10 patients, six men and four women, with a mean age of 68.0 years. The average time from the traumatic injury to surgery was 7.5 months. All patients showed improved Constant, Adapted Constant, UCLA, and DASH scores compared to their preoperative values. When comparing the outcomes of chronic posterior and anterior dislocations, no differences in functional outcomes or shoulder motion were observed after rTSA implantation. There were no complications during or after surgery.

CONCLUSION: The results of the present […]

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Does the order matter? Comparing the order of stem placement and fracture reduction in revision total hip arthroplasty for Vancouver B2 and B3 periprosthetic femur fractures.

Antonioli SS, Khury F, Kennedy MF, Haider MA, Duke A, Schwarzkopf R, Aggarwal VK

BACKGROUND: Vancouver B2 and B3 periprosthetic femur fractures (PPFF) have posed significant treatment challenges due to stem instability and lack of adequate femoral bone stock. This study investigated subsidence, survivorship, and outcomes of Vancouver B2 and B3 fractures, based on the order in which revision stem placement and fracture reduction occurred during revision total hip arthroplasty (rTHA).

METHODS: This retrospective, cohort study included 46 rTHAs between June 2011 and April 2023. Included patients underwent rTHA for Vancouver B2 or B3 PPFF with minimum one-year radiographic and two-year clinical follow-up. All patients were treated with diaphyseal-engaging tapered fluted titanium stems and stem modularity decisions were based on surgeon preference. Cohorts were separated based on if stem placement (SF, n = 19), or fracture reduction (RF, n = 27) occurred first. Patient demographics, intraoperative information, and clinical and radiographic outcomes were collected.

RESULTS: The SF and RF cohort showed no statistically significant differences in rate of subsidence ≥5 mm [26.3%[SF], 22.2%[RF], P = 0.749), rate of subsidence ≥ 10 mm (15.8%[SF], 14.8%[RF], P = 0.928), nor average subsidence (4.1 mm[SF], 4.4 mm[RF], P = 0.861). We found no statistically significant differences in surgery-related clinical outcomes or all-cause revision rates within a two-year follow- […]

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Offset reconstruction in stemless and stemmed shoulder hemiarthroplasty: influence on long-term outcomes.

Trefzer R, Gurda A, Deisenhofer J, Weishorn J, Hariri M, Koch KA, Wolf M, Bülhoff M

BACKGROUND: Shoulder hemiarthroplasty (HA) is a well-established treatment for various humeral-side pathologies. Overstuffing should be avoided because it is associated with glenoid wear; however, specific radiological reconstruction targets remain unclear. This study assessed the association between radiological joint restoration in stemmed and stemless HA and long-term functional outcomes.

METHODS: Patients who underwent stemmed or stemless HA between 2001 and 2011 were included in a long-term follow-up. Cases with incomplete or non-calibratable digital radiographs or concomitant glenoid procedures were excluded. Clinical outcomes were evaluated using the Constant-Murley Score (CMS) and satisfaction questionnaires. Radiographic parameters-lateral glenohumeral offset (LGHO), lateral humeral offset (LHO), and center of rotation (COR)-were measured on standardized anteroposterior radiographs pre- and postoperatively by two independent observers. Differences from baseline to final follow-up (ΔLHO, ΔLGHO, ΔCOR) were calculated. Interobserver reliability was determined using the intraclass correlation coefficient (ICC). Associations between radiological measures and clinical outcomes were analyzed using linear regression; intergroup comparisons used unpaired t-tests ($p < 0.05$).

RESULTS: Forty-eight patients (12 stemmed, 36 stemless HA) were examined after a mean follow-up of 16.6 […]

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Emergency access to the subclavian vessels by non-thoracic surgeons: a cadaver-based learning model for orthopedic trauma surgery.

Grechenig P, Gänsslen A, Dauwe J, Wittig U, Sagmeister M, Koutp A, Puchwein P, Sadoghi P et al.

PURPOSE: The aim of this study was to evaluate the procedural time and safety of an infraclavicular approach to the subclavian vessels for damage control of major upper extremity vascular injuries. The study specifically focused on the performance of non-thoracic surgeons using a cadaveric model.

METHODS: A sample of 100 orthopedic trauma surgeons (residents, specialists, and attendings) was recruited from an AO cadaveric dissection course. Emergency infraclavicular access was performed on 50 cadavers over two consecutive days. The procedural time, successful vessel clamping, participant self-assessment, and the resulting learning curve were analyzed.

RESULTS: On day one, 27% (9/33) of participants who successfully achieved correct clamping of both subclavian vessels reported feeling confident. By day two, this proportion increased to 71% (55/77). Comparing day 1 to day 2 we found an improvement of 35% (subclavian artery) and 37% (subclavian vein) in the correct identification of subclavian vessels in our sample. The evaluations of this study show that there is no correlation between surgical experience and successful emergency access.

CONCLUSION: Anatomic dissection is of paramount importance for teaching rare and demanding surgical techniques. This study demonstrates that anatomical workshops significantly improve procedural safety and self-assessment when accessing subcla […]

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Operative timing and surgical complexity in hand trauma: a multicenter analysis of replantation and non-replantation centers in Germany.

Nicklas A, Will Marks P, Dragu A, Schmidt H, Rotärmel A, HandTraumaRegister DGH

BACKGROUND: Hand injuries are common traumatic conditions that often require specialized surgical care. Differences in hospital structures and levels of specialization may influence the timeliness and complexity of treatment. This study evaluates quality of care across different hospital types, focusing on time-to-skin-incision and surgical management of finger and hand injuries.

MATERIALS AND METHODS: This retrospective multicenter study is based on data from the HandTraumaRegister of the German Society for Hand Surgery (HTR DGH). A total of 16,726 surgically treated cases with documented finger or hand injuries recorded between 2018 and 2023 were analyzed. Hospitals were categorized as non-replantation centers, replantation centers, or FESSH-accredited replantation centers. Demographic characteristics, injury-patterns, treatment modalities, and time-related parameters were assessed. Statistical analysis included descriptive statistics, group comparisons, and multivariate regression models to evaluate factors associated with time-to-skin-incision.

RESULTS: The study population consisted predominantly of male patients (75.79%), and 91.6% were right-handed. Soft tissue injuries without fractures or amputations accounted for 46.57% of cases. Non-replantation centers demonstrated the longest time-to-skin-incision (median 13 h), whereas replantation centers (median 3.75 h) and FE […]

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Neglected acromioclavicular joint injury in coracoid process fractures: a retrospective cohort study.

Lee MS, Lee CH, Jo YH, Cho WT, Seo YW, Choi WS

INTRODUCTION: Coracoid process fractures are frequently associated with injuries to the superior suspensory shoulder complex, including the acromioclavicular (AC) joint. However, concomitant AC joint injuries may be overlooked in the acute phase, potentially limiting treatment options. This study aimed to investigate the association between coracoid process fractures and SSSC injuries and to identify imaging-based predictors of concomitant AC joint injury.

METHODS: A retrospective single-center cohort study including 60 patients with coracoid process fractures was conducted between February 2016 and April 2023. Coracoid fractures were classified based on anatomical location, and associated SSSC injuries were categorized according to the involved structures. These injuries were evaluated using 3D shoulder CT. Morphological deformity of the SSSC was assessed by measuring the glenoclavicular distance (GCD)-a longitudinal parameter-on final follow-up clavicle AP radiographs. In base fractures, the medial displacement gap was measured on sagittal CT. Statistical analyses included logistic regression and ROC curve analysis to identify predictors of AC joint injury.

RESULTS: Among the 60 patients, fractures were located anteriorly in 27 cases, in the middle region in 5 cases, and at the base in 28 cases. AC joint injury was identified in 15 patients, all of whom had base-type fractu […]

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Subclinical median nerve alterations after distal radius fractures: a comparative analysis of volar plate fixation and cast immobilization using dynamic ultrasonography and electrophysiology.

Karapinar SE, Arslan E, Oguzlar FC, Hatip AY, Akgun VO, Dincer R, Asci S, Kizilkaya V

BACKGROUND: Distal radius fractures may affect the median nerve through trauma-related and treatment-related mechanisms. While overt neurological deficits are uncommon, subclinical median nerve alterations may occur following both conservative and surgical treatment.

PURPOSE: To determine whether treatment modality influences subclinical median nerve behavior using combined dynamic ultrasonographic and electrophysiological assessment.

METHODS: This retrospective observational study included 42 patients with unilateral distal radius fractures. 18 patients were treated conservatively with circular casting, and 24 patients underwent volar plate fixation.

Ultrasonographic assessment of the median nerve was performed at the distal radius level in neutral wrist position, as well as during wrist flexion and extension, using a dynamic ultrasonographic approach, and was combined with electrophysiological evaluation including sensory and motor nerve conduction studies. In the surgically treated group, only patients with Soong type 1 or type 2 plate positioning were included.

RESULTS: Clinical functional outcome scores were comparable between groups, despite a higher frequency of mild clinical signs such as positive Tinel test and night pain in the plate group. However, ultrasonographic evaluation demonstrated significantly greater position-dependent median nerve enlargement and flatte […]

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Pectoralis major reference technique improves humeral height restoration and tuberosity healing after hemiarthroplasty for complex proximal humeral fractures: A comparative study.

Park BS, Kim DH, Sohn HJ, Cho CH

BACKGROUND: Accurate humeral stem positioning is critical for achieving favorable outcomes after hemiarthroplasty for complex proximal humeral fractures. This study aimed to compare the clinical and radiological outcomes of hemiarthroplasty performed using the pectoralis major reference technique (PMRT) with those performed using a conventional technique (CT) METHODS: Thirty-nine patients who underwent hemiarthroplasty for complex proximal humeral fractures were included in this study. The conventional technique was used in the first 24 cases (CT group) and PMRT was applied in the subsequent 15 cases (PMRT group). The mean follow-up period was 54.5 months (range, 12-96 months) in the PMRT group and 56.1 months (range, 24-132 months) in the CT group. Radiographic outcomes included postoperative head-tuberosity distance and tuberosity healing status, assessed using serial plain radiographs. Clinical outcomes were evaluated using the visual analog scale (VAS) pain score, the University of California, Los Angeles (UCLA) score, the American Shoulder and Elbow Surgeons (ASES) score, the subjective shoulder value (SSV), and active range of motions.

RESULTS: The rate of complete tuberosity healing was significantly higher in the PMRT group than in the CT group (80.0% vs 58.3%). The mean postoperative head-tuberosity distance was significantly shorter in the PMRT group compared with th […]

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Clinical outcomes of interventional radiology versus operative management in hemodynamically stable and unstable moderate-to-severe high-grade blunt renal trauma.

Hus A, Indorewala Y, Nasef H, Prashar S, Rogers L, Nasef Y, Elkbuli A

BACKGROUND: Blunt renal trauma remains a major source of morbidity and mortality despite advances in non-operative management. This study evaluates clinical outcomes in adults with moderate-to-severe high-grade blunt renal trauma, comparing interventional radiology (IR) and operative management.

METHODS: This retrospective cohort study utilized the ACS-TQIP-PUF dataset (2017-2023) to evaluate outcomes in adult (≥18 years) trauma patients with moderate-to-severe (AIS abdomen ≥ 2, ISS ≥ 9) high-grade (AAST grade ≥4) blunt renal trauma who underwent operative management or IR approach. Patients were stratified by hemodynamic stability, initial hypertension, and trauma center level. Primary outcomes included mortality and nephrectomy rates. Secondary outcomes included transfusion requirements, nonoperative management failure, and complications.

RESULTS: In hemodynamically stable adult trauma patients with moderate-to-severe blunt renal injuries, IR was associated with lower 24-hour (aOR 0.477, 95% CI 0.274-0.83, $p = 0.009$) and in-hospital mortality (aOR 0.210, 95% CI 0.116-0.378, $p < 0.001$). Among unstable patients, IR was associated with reduced in-hospital mortality (aOR 0.076, CI: 0.007-0.847, $p = 0.036$). IR was also associated with reduced in-hospital mortality when stratifying by presence of initial hypertension (aOR 0.317, 95% CI:0.103-0.973, $p = 0.045$) and trauma center le […]

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Validity of fluoroscopic methods along the transsyndesmotic axis: A cadaveric study.

Nakajima H, Kimura S, Yamaguchi S, Watanabe S, Sasho T, Ohtori S

PURPOSE: Alignment along the transsyndesmotic (TS) axis is necessary to fix syndesmotic diastasis associated with ankle fractures. Fluoroscopic clamp techniques for the tibiofibular joint along the TS axis include the center-center (CC), talar dome lateral (TL), and incisura tangent (IT) methods. However, whether the clamp placement along the TS axis is feasible under fluoroscopy and the effects of metal artefacts from fracture fixation remain unclear. The purpose of this study was to clarify the differences from the TS axis using three methods under fluoroscopy on a cadaver and to compare the differences from the TS axis before and after implant placement for each method.

MATERIALS AND METHODS: Ten fresh-frozen cadaveric feet were used in this study. We used three methods and determined the interaxis angle and distance as the differences from the TS axis. Furthermore, implant placement was made in cadaveric feet, and these differences were evaluated using the three methods. The differences obtained by the three methods were compared using one-way repeated measures analysis of variance. Paired t-tests were used to compare the differences before and after implant placement.

RESULTS: Both the interaxis angle and distance showed significant differences among the three fluoroscopic methods, irrespective of implant placement ($p < 0.001$). The CC method demonstrated pronounced inter […]

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Correlation of functional and radiological outcome after surgical treatment of acetabular fractures with quadrilateral plate involvement. A meta-analysis.

Solou K, Galanopoulos G, Lakoumentas J, Zacharakis E, Kyprianou A, Papachristos IV, Nikolopoulos F

PURPOSE: Quadrilateral plate fractures are complex injuries often requiring surgical management. Nevertheless, postoperative reduction quality has not been consistently correlated with functional outcome. We conducted a meta-analysis evaluating the association between functional and radiological outcomes of patients with acetabular fracture involving the quadrilateral plate who were treated surgically with open reduction and internal fixation.

METHODS: A comprehensive search of PubMed, Cochrane Library and Web of Science databases was performed to identify prospective and retrospective studies reporting the functional and radiological outcomes after internal fixation of acetabular fractures with quadrilateral plate involvement. Literature screening was based on eligibility criteria, extracted relevant data and on the quality of included studies. Meta-analysis evaluated functional (Harris Hip Score (HHS) and the Merle d'Aubigné and Postel (MAP)) outcomes as well as radiological (MATTA) outcomes, pooling agreement proportions, assessing heterogeneity, and performing meta-regression using R.

RESULTS: We included 12 studies with 336 patients treated for acetabular fracture involving the quadrilateral plate. Seven studies (177 patients) reported MAP scores: excellent 41%, good 47%, poor 12%. Seven studies (217 patients) reported HHS scores: excellent 47%, good 31%, poor 22%. Overa […]

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Surgical rib fixation in pediatric patients: Following the adult trend or a different story?

Hamdan HS, Halabi M, Nouiri AE, Gardner C, Djuric M, Hill L, Johnson JL

INTRODUCTION: For many years, non-operative management has been an established standard of care for rib fractures. In adults, surgical stabilization of rib fractures (SSRF) has become an increasingly accepted alternative, with growing evidence supporting its benefits. In contrast, the role of SSRF in pediatric trauma remains poorly defined. This study presents the largest known cohort of pediatric SSRF patients to describe their characteristics and assess whether pediatric practice mirrors adult trends.

METHOD: A retrospective review of pediatric patients (<18 years) with rib fractures was conducted using the TQIP database (2017-2023). Patients were stratified by treatment (SSRF vs. non-operative). Demographic, injury, surgical, and outcome characteristics were compared. Adult SSRF patients during the same period were included for secondary analysis.

RESULTS: Among 28,465 pediatric rib fractures, 87 (0.31%) underwent SSRF. Pediatric SSRF rates remained stable over time (0.16%-0.39%; $p = 0.17$), while adult SSRF use increased (1.5%-2.37%; $p = 0.0027$). Pediatric SSRF patients were older (14.6 vs. 13.0 years; $p < 0.001$), more likely to present with flail chest (25.3% vs. 1.2%, $p < 0.001$), and had higher ISS (25.9 vs. 20.3; $p < 0.001$) compared to non-SSRF pediatric patients. Compared to adults, pediatric SSRF patients underwent less extensive fixation but had more open thoracic in […]

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Delivery outcomes following previous pelvic ring injury: A systematic review, meta-analysis and evidence informed management pathway.

Gupta VK, Phua SN, Rai SK, Thakker A, Qayum K, Sanghera RS, Varma J, Vallier HA

Pelvic ring injuries (PRI) are hypothesised to cause mechanical obstruction to the pelvic outlet thus affecting delivery mode in pregnancy. Research has shown a higher caesarean section rate following PRI as a result. Our objective was to synthesise the current evidence for delivery outcomes following PRI and appraise the influence of fracture morphology, operative management and retained implants to derive an evidence informed management pathway. A systematic review with meta-analysis of proportions was conducted in line with PRISMA guidelines. Women of reproductive age with any previous pelvic ring injury were included. Maternal and neonatal outcomes were recorded as secondary outcomes. Ten observational studies (1983-2024) contributed 1253 women and 1719 deliveries. The pooled caesarean section (CS) rate was 40% (95% CI 27-53) with high heterogeneity ($I2=94\%$) and a wide prediction interval (8-82%). The pooled vaginal delivery rate was 60%. Trial of labour succeeded in 86% (95% CI 78-91), with intrapartum CS occurring in approximately 13% of attempted labours. Accounting for one delivery per woman and CS-naive cohorts did not change the pooled estimate. Operatively managed fractures (51% CS versus 32% non-operative, OR 2.24, 95% CI 0.92-5.44) and retained implants showed a non-significant trend towards higher CS rate. The data is inconclusive as to whether fracture morphology […]

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Beyond damage control: Functional preservation and avoidance of iatrogenic harm in combat vascular injury and fasciotomy.

Dmytro M, Ihor R, Iurii M, Serhii M

BACKGROUND: Damage-control interventions are essential in combat vascular trauma, but their benefits must be balanced against iatrogenic injury and long-term functional impairment. Extensive vascular exposure, temporary shunting, vessel ligation, and prophylactic fasciotomy may be lifesaving in selected patients, yet their indiscriminate use can increase operative burden and complicate definitive reconstruction.

METHODS: A focused narrative review of PubMed/MEDLINE and Google Scholar was performed through July 2026 using predefined combinations of terms related to vascular trauma, temporary shunting, ligation, endovascular management, fasciotomy, compartment syndrome, limb ischemia, and damage-control surgery. Civilian and military guidelines, reviews, comparative and observational studies, case series, and technical reports were included, with priority given to combat casualty care and staged evacuation. Owing to heterogeneity and the narrative design, findings were synthesized qualitatively without meta-analysis or formal certainty grading.

RESULTS: Vascular injury alone does not invariably mandate immediate extensive exploration. In stable patients, imaging, observation, and endovascular or hybrid techniques may avoid non-therapeutic dissection. Temporary shunts have an established damage-control role in reducing ischemic time during resuscitation, treatment of higher-prio […]

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Innovative intramedullary nail system with internally locking: A safe alternative fixation technique for humerus shaft fracture that reduces the risk of the iatrogenic radial nerve injury and surgical time.

Onder Y, Tuncez M, Kurtulus T, Bulut T

PURPOSE: Humeral shaft fractures may be managed conservatively or operatively; however, increasing non-union and complications after non-operative treatment have renewed interest in intramedullary fixation. Innovative intramedullary nails with distal internally locking, requiring neither fluoroscopy nor external guidance, aim to reduce surgical time, simplify the procedure, and minimise the risk of iatrogenic radial nerve injury. This study evaluates the clinical and radiological outcomes of a humeral nail utilising distal internally locking.

MATERIAL AND METHOD: A retrospective review included 75 patients treated with antegrade intramedullary nailing for humeral shaft fractures between 2020 and 2025. Patients with pathological fractures, preoperative radial nerve palsy, concomitant upper-limb injuries, or insufficient follow-up were excluded. Operative parameters, complications, union time, hospital stay, and return to activity were recorded. Functional outcomes were assessed at final follow-up using the Constant-Murley score, QDASH, VAS, and shoulder range of motion.

RESULTS: Operative time was 46.4 ± 11.1 min, and hospital stay was 3.2 ± 0.8 days. Radiological union occurred at 13.0 ± 2.9 weeks, and patients returned to daily activities at 12.6 ± 2.5 weeks. No postoperative radial nerve palsy or infections were observed. One non-union required exchange nailing, and three c […]

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Splenic artery embolization versus surgery for blunt splenic injuries: a systematic review, meta-analysis and trial sequential analysis stratified by hemodynamic status.

Bravin S, Menegat ALRS, Menegat BLRS, Mirón AM, Wiederkehr H, Garza EG, Melo SL

BACKGROUND: Surgical management of blunt splenic injuries is associated with long hospital stays and high risk of postoperative infections and thromboembolism. Splenic artery embolization (SAE) has emerged as a valuable alternative to preserve the spleen and limit complications in hemodynamically stable patients with moderate and severe injuries. However, how SAE compares to surgery in both hemodynamically stable and unstable populations is still debated, as international trauma guidelines advocate for primary open surgery in unstable cohorts. Therefore, we conducted a systematic review and meta-analysis to evaluate the role of endovascular management across hemodynamic statuses.

METHODS: We searched PubMed, Embase and Cochrane for randomized and nonrandomized studies comparing surgical management to SAE in adult patients with blunt splenic injuries. Risk ratios and mean differences with 95% confidence intervals were calculated for categorical and continuous outcomes, respectively. We conducted a trial sequential analysis to confirm the robustness of our findings.

RESULTS: We analyzed 33 148 patients, of which 9593 were initially treated with SAE. Compared to surgery, SAE was associated with significantly lower mortality (RR: 0.43; 95% CI: 0.25-0.72; p = 0.004). SAE also reduced the length of stay in the hospital and in the intensive care unit (MD: -3.1; p = 0.015 and MD: -4; […]

PubMed · DOI

Early mortality prediction of prognosis in cardiac arrest patients using machine learning: Development, external validation, and explainability with SHAP.

Sun Y, Wang L, You H, Liu X, Ren J, Wu Y, He Z

BACKGROUND: Cardiac arrest (CA) is a major global health challenge, accounting for a significant proportion of deaths and healthcare resource utilization. However, due to the lack of interpretability, most predictive models for CA mortality have not been applied in clinical practice. We aimed to construct an interpretable model for predicting in-hospital mortality for CA patients in the intensive care units (ICU).

METHODS: Data from the MIMIC-IV database were split, with 70% allocated to the training set and 30% to the internal validation set. The eICU database served as an external validation set. Using the variables selected by least absolute shrinkage and selection operator (LASSO), we constructed and evaluated seven machine-learning (ML) models. The optimal model was rendered explainable through the Shapley additive explanations (SHAP) approach. And, a web-based calculator was developed based on the optimal performance model for easy use.

RESULTS: A total of 1749 patients with CA were finally enrolled from MIMIC-IV in this study. By LASSO regression, 21 variables were selected to construct the ML models. Among seven constructed models, the eXtreme Gradient Boosting (XGBoost) model emerged as the best-performing model in internal validation set (AUC = 0.8486) and external validation set (AUC=0.8471). In addition■XGBoost model showed higher net benefit and wider threshold p […]

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Pediatric digit replantation: A mixed methods outcomes study.

Saoud K, Bruce JG, Bettlach C, Tung TH, Fox IK

BACKGROUND: Indications for pediatric digital replantation/revascularization (replant/revasc) are broader than those for adults. Pediatric patients suffer many of the same challenges seen in adult replant/revasc patients with similar rates of digit salvage. The purpose of this study was to review the incidence of and long-term outcomes after pediatric replant/revasc.

METHODS: Data was collected from medical records of patients presenting to our Level 1 Trauma Pediatric Hospital Emergency Room between January 2011 and March 2017. All charts with ICD-9 codes for upper extremity trauma were queried and reviewed. Additional data was collected on patients undergoing digit replant/revasc or revision amputation. Patients whose replant/revasc digits survived to the first post-operative visit underwent further outcomes assessment.

RESULTS: Of the 1364 patients who presented with upper extremity trauma over the study period, 19 underwent replant/revasc and 12 digits survived to the first post-operative visit. 10 of the patients and their families participated in interviews (mean follow up 89 months (range 74-119) post-procedure). Children and parents had different perspectives on their treatment experience. Six patients reported excellent function with no limitations in schoolwork or sports. Nine patients and seven parents stated they would choose to undergo replant/revasc surgery agai […]

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Anomalous occluded vessels and the "Corona Mortis": A reappraisal of nomenclature origins, anatomical variations, and surgical risks.

Song X, Wang F, Zhang Z, Zhang Q, Dong J, Wang J, Wang J, Shi G et al.

BACKGROUND: The "corona mortis" (CMOR) typically refers to arterial or venous anastomotic branches located posterior to the superior ramus of the pubis within the Retzius space, connecting the obturator vascular system to the extrailiac or subabdominal vascular system. This anatomical variation is closely associated with inguinal hernia repair, pelvic and acetabular surgery, pelvic tumor lymph node dissection, and pelvic trauma hemorrhage. However, its nomenclature origin, terminological boundaries, and actual bleeding risk have long been controversial across disciplines.

OBJECTIVE: To establish standardized terminology, trace the nomenclature origins of CMOR, review its anatomical variations and incidence, summarize key imaging identification features, and propose actionable surgical risk stratification and management recommendations.

MATERIALS AND METHODS: We conducted a narrative review by retrieving publicly published anatomical, imaging, and clinical studies and systematic reviews from 1996 to 2025, integrating data from representative autopsies, CTA/enhanced CT, and intraoperative observations.

RESULTS: Systematic reviews indicate that CMOR is present in approximately half of hemipelves, with venous variants significantly outnumbering arterial ones. Variations in CMOR definitions and the scope of "abnormal/accessory obturator vessels" included across studies account fo […]

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External fixation for displaced intra-articular calcaneal fractures: A case series.

Morelli F, Caperna L, Pepino M, Iorio R, Maffulli N

Intra-articular calcaneus fractures (CF) remain challenging. Surgery aims to restore subtalar joint congruence and calcaneus width, height, shape, and alignment, thereby avoiding medial and lateral impingement and enabling patients to resume a normal lifestyle. We retrospectively evaluated 32 patients (35 fractures) with a minimum displacement of 2 mm of the posterior facet and disruption of the calcaneus length, width, and height. Treatment involved a minimally invasive sinus tarsi approach and definitive stabilization with an external fixator (EF). The mean time to partial weight-bearing was 32 days, and the EF was removed at an average of 92 days. At final follow-up (mean 22.2 months), the mean American Orthopaedic Foot and Ankle Society (AOFAS) score was 75.4. Two patients developed superficial pin-tract infections (5.7%), both resolved without further sequelae. No patient experienced deep infection or wound-healing complications. Based on these findings, EF appears to be a valid alternative to open reduction and internal fixation, allowing adequate reduction, early mobilization, and acceptable functional outcomes with minimal complications.

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Expanding the indications of horizontal plate fixation to the proximal humerus: Technical note and clinical series.

Machado JKS, da Silva Cordeiro R, Dos Santos DR

INTRODUCTION: Isolated greater tuberosity fractures of the proximal humerus are relatively uncommon injuries, and there is no consensus regarding the optimal fixation method for displaced fractures. This study aimed to describe a horizontal plate fixation technique and to demonstrate its clinical applicability and feasibility in the treatment of isolated greater tuberosity fractures.

METHODS: This study consists of a technical description of the horizontal plate fixation technique, followed by its retrospective clinical application. A series of 11 patients with isolated greater tuberosity fractures treated with this method between 2009 and 2021 was included. All patients had a minimum follow-up of 24 months. Functional outcomes were assessed using the University of California at Los Angeles (UCLA) Shoulder Rating Scale.

RESULTS: All patients achieved fracture union. The mean UCLA score was 30, with predominantly good functional outcomes. Most patients reported mild or occasional pain and minimal functional limitation.

CONCLUSION: Horizontal plate osteosynthesis is a reproducible and accessible technique that demonstrated satisfactory clinical applicability in this series of patients with isolated greater tuberosity fractures. Further studies are needed to confirm its effectiveness.

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An ethnographic study of mobilization, basic mobility, physical activity, and exercise during acute hospitalization in patients following hip fracture surgery.

Hansen MS, Høite MES, Bandholm T, Lindholm ST, Skibdal KM, Pedersen MM, Kirk JW

INTRODUCTION: Existing evidence highlight the need for a standardized approach in treating patients hospitalized following a hip fracture surgery. Clinical guidelines highlight rapid recovery of basic mobility post-surgery and the importance of frequent mobilization, physical activity, and exercise in rehabilitation. However, findings from the first HIP-ME-UP study show increasing physical inactivity among these patients, indicating issues with guideline implementation in clinical practice. Therefore, this study investigated clinical practice in relation to mobilization, basic mobility, physical activity and exercise during acute hospitalization in patients following hip fracture surgery.

METHODS: An ethnographic field study was conducted at the orthopedic ward of a publicly funded hospital in the Capital Region of Denmark. Researchers carried out 22 observations from November 2023 to February 2024. A total of 230 pages of field notes were produced. An initial inductive qualitative content analysis was applied to find preliminary themes and subthemes. To nuance and validate these findings, 23 interviews with healthcare professionals and management were conducted and analyzed deductively. This deductive analysis was guided by the preliminary themes and subthemes that were revised in the final analysis if data provided nuances and new insights to the observations.

RESULT: Three […]

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Modified Delphi consensus recommendations for the use of negative-pressure wound therapy in orthopedic trauma and complex wounds in low- and middle-income countries: anAn African consensus study.

Manegabe JDT, Englebert A, Kabakuli AN, Buzisa Mbuku R, AFRICA-NPWT Group, Tissingh E, Schubert T, Munguakonkwa PB et al.

BACKGROUND: Open fractures and complex traumatic wounds remain a major cause of morbidity in low- and middle-income countries, where delayed presentation, limited reconstructive capacity, and restricted access to specialized wound care complicate management. Although negative-pressure wound therapy is increasingly used as an adjunct in complex wound management, evidence specific to low- and middle-income countries remains limited and practical consensus recommendations are lacking.

PURPOSE: To develop evidence-informed consensus recommendations for negative-pressure wound therapy use in orthopedic trauma and complex wounds in low- and middle-income countries and validate them through a multinational African modified Delphi process.

METHODS: A narrative literature review identified evidence regarding negative-pressure wound therapy in orthopedic trauma and complex wounds, with particular emphasis on low- and middle-income countries settings. Draft recommendations were developed by the writing committee and externally validated through a modified Delphi process involving orthopedic-trauma and plastic surgeons from across Africa. Recommendations were initially classified as strong, moderate, or conditional and evaluated by the expert panel. Consensus was predefined as ≥ 80% agreement with the proposed recommendation category, while ≥ 90% agreement was considered very strong cons […]

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An epidemiological evaluation of fatal and nonfatal assault related firearm injuries among youth aged 0-17 years.

Gilani N, Hunter AA, Lazzarini Z, DiVietro S

BACKGROUND: Firearm-related injuries are a leading cause of death and disability among youth in the United States (US). Children are at risk for assault-related firearm injuries based on their race, ethnicity, age, and gender. The present study quantifies assault-related firearm morbidity and mortality among US children aged 0-17 years.

METHODS: We utilized data from 41 states reporting to both the Nationwide Emergency Sample (NEDS) and the National Violent Death Reporting System (NVDRS) from 2019 to 2020 to calculate rates of nonfatal and fatal firearm-related assaults, respectively. We also calculated disproportionality and disparity, using the Disproportionality Representation Index (DRI) and Disparity Index (DI).

RESULTS: An estimated 6820 (weighted) children experienced a nonfatal firearm assault and over 2300 children died within the two-year study timeframe. Disproportionality and disparity calculations suggest that non-Hispanic Black, male, and older youth bore the greatest burden of morbidity and mortality.

CONCLUSION: Identifying root causes of fatal and nonfatal pediatric firearm assaults in future studies can support preventive healthcare and firearm policy reform in the US. Decreasing the burden of firearm-related death and disability will require additional funding and implementation of community violence reduction programs and councils.

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Transformation into chronic subdural hematoma following acute subdural hematoma: Incidence, outcomes, and predictors.

Chen H, McIntyre MK, Santdasani S, Lakhani DA, Malhotra A, Kan P, Gandhi D, Colasurdo M

IMPORTANCE: Acute subdural hematoma (aSDH) and chronic subdural hematoma (cSDH) have traditionally been considered distinct clinical entities. However, a subset of patients with aSDH may subsequently develop cSDH, representing a poorly characterized clinical phenomenon with unclear incidence, outcomes, and risk factors.

OBJECTIVE: To quantify the incidence and outcomes of cSDH-related readmissions following aSDH, identify independent predictors of transformation, and develop and validate a prediction score for risk stratification.

DESIGN: Retrospective cohort study SETTING: Nationwide Readmissions Database (2016-2022).

PARTICIPANTS: Adult patients (aged ≥18 years) hospitalized with primary aSDH were included. Exclusions included pre-existing subacute or chronic SDH, in-hospital mortality, prolonged hospitalization (≥14 days), comfort care status, middle meningeal artery embolization during index hospitalization, or insufficient follow-up (<30 days).

MAIN OUTCOMES AND MEASURES: The primary outcome was cSDH-related readmission within 180 days. Secondary outcomes included functional decline, mortality, surgical intervention, and hospitalization costs. Predictors were identified using least absolute shrinkage and selection operator (LASSO) regularization in a Cox proportional hazards framework, and the Post-acute Evolution Risk Score to Identify Subdural Transformation (PERSIST […]

PubMed · DOI

Association of CT-derived adipose measurements with prolonged intensive care unit stay in moderate and severe trauma patients.

Zander C, Mühlenfeld N, Erdle B, Scheef T, Rau A, Reisert M, Kalbhenn J, Klingler F et al.

BACKGROUND: Obesity as measured by body mass index (BMI) has been associated with worse clinical outcomes in moderate and severe trauma patients. However, while BMI does not differentiate between different fat compartments, routine CT-derived body composition provides detailed assessment and may thus provide more accurate risk assessment.

METHODS: 200 eligible patients with moderate or severe trauma (injury severity score (ISS)≥ 9 from a level-1 trauma center were retrospectively identified and assembled into a 1:1 matched cohort (obese, BMI ≥ 30 kg/m vs. non-obese, BMI < 30 kg/m²), with exact matching on sex and close balancing on age and ISS. Patients with low quality CT imaging (n = 10), without intensive care unit (ICU) stay (n = 27) and death while hospital stay (n = 21) were excluded. CT-derived body composition measures including visceral adipose tissue (VAT), subcutaneous adipose tissue (SAT), their respective indices (VATI, SATI) and a combined abdominal fat tissue index (AATI) were assessed from a single axial slice at the L3 vertebral level using a previously developed and validated deep learning segmentation model.

RESULTS: A total of 142 ICU survivors formed the final study cohort (median age: 56 years (IQR 47-65), 16% female, 52% obese). VATI (ρ = 0.25 and 0.27, p = 0.003 and 0.001) and SATI (ρ = 0.19 and 0.24, p = 0.024 and 0.003) were significantly correlated […]

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Chloral hydrate outperforms oral midazolam in facilitating venipuncture in children: A double-blind randomized trial.

Khakshour A, Akhondian J, Saeidinia A, Nirooman S

BACKGROUND: Venipuncture for intravenous access is a frequent and distressing procedure in pediatric hospitals and trauma centers. Achieving immobility in young, anxious children is essential yet difficult. A direct comparison of commonly used sedatives for this purpose was lacking.

OBJECTIVE: To directly compare the efficacy and safety of oral chloral hydrate versus midazolam for facilitating successful venipuncture in hospitalized children aged 1-5 years in emergency and trauma care.

METHODS: Prospective, double-blind, randomized trial in a tertiary hospital emergency department. One hundred children received either oral chloral hydrate (50 mg/kg) or oral midazolam (0.3 mg/kg) 30 min before venipuncture.

PRIMARY OUTCOME: number of attempts for successful venipuncture. Secondary outcomes included sedation time, venipuncture duration, total procedural time, recovery time, satisfaction, and adverse events. Analyses were adjusted for baseline age and weight differences.

RESULTS: Chloral hydrate significantly reduced the number of venipuncture attempts (1.30 vs. 1.96; mean difference -0.66, 95% CI: -0.92 to -0.40, p < 0.001). Time to adequate sedation (16.30 vs. 25.54 min), IV access time (6.80 vs. 16.74 min), and total procedural time (42.12 vs. 67.06 min) were all significantly shorter with chloral hydrate (all p < 0.001). Recovery time did not differ (p = 0.819). After adju […]

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Algorithm for the hemodynamically unstable patient with pelvic fracture - The Parkland experience.

Halvachizadeh S, Carron CJ, Katzir A, Sun J, Zhang A, Grewal I, Lamus D, Park C et al.

OBJECTIVE: To evaluate the outcomes of an institutional management algorithm for hemodynamically unstable pelvic ring injuries at a Level I trauma center with rapid access to interventional radiology.

METHODS: This retrospective cohort study included adult patients (≥18 years) who underwent surgical treatment for an unstable pelvic ring injury between 2024 and 2026. Patients who were hemodynamically unstable on admission were compared with hemodynamically stable patients managed according to the same institutional protocol. The primary outcome was time to definitive pelvic ring fixation. Hemorrhage-control interventions, fixation strategy, complications, and length of hospital stay were also assessed.

RESULTS: Eighteen patients (52.9%) presented with hemodynamic shock. These patients had greater physiological derangement and higher injury severity scores than hemodynamically stable patients. Angioembolization was used more frequently in the shock group (60.0% vs 10.5%), and no patient underwent preperitoneal pelvic packing. Despite their greater initial injury severity, patients presenting with shock underwent definitive pelvic ring fixation at a similar time and using similar fixation methods to stable patients. The choice and timing of fixation were determined mainly by fracture morphology rather than initial hemodynamic status. Patients in the shock group had longer hospit […]

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Impact of altered consciousness on the risk of fasciotomy for acute compartment syndrome following traumatic tibial fracture.

Bouklouch Y, Poirier P, Parent-Harvey S, Esfahani SJ, McBeth K, Harvey EJ

BACKGROUND: Acute compartment syndrome (ACS) following tibial fracture is a limb-threatening emergency that relies heavily on clinical assessment for diagnosis. In patients with altered consciousness (hereafter referred to as obtunded status), key symptoms such as pain out of proportion cannot be reliably assessed, potentially influencing surgical decision-making. Whether obtunded status independently affects fasciotomy rates remains poorly understood.

OBJECTIVE: To determine whether obtunded status is independently associated with increased likelihood of fasciotomy for suspected ACS in patients with traumatic tibial fractures.

METHODS: A retrospective cohort study was performed using the National Trauma Data Bank (NTDB) from 2016 to 2018. Patients with tibial plateau (AO/OTA 41) and tibial shaft (AO/OTA 42) fractures were included. Obtunded status was defined as sedation/intubation, severe alcohol intoxication, or Glasgow Coma Scale score < 9. The primary outcome was fasciotomy during hospitalization. Multivariable logistic regression was used to evaluate the association between obtunded status and fasciotomy while adjusting for age, sex, fracture location, fracture complexity, open fracture status, and skull fracture.

RESULTS: Among 119,393 patients with tibial fractures, 6854 (5.7%) underwent fasciotomy. On univariable analysis, obtunded patients experienced higher fascio […]

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A pragmatic clinical definition of traumatic shock using systolic blood pressure and shock index: A national registry study.

Noonan M, O'Reilly G, Balogh ZJ, Mathew J, Ellis DY, Groombridge C, Cameron P, Fitzgerald M

BACKGROUND: Traumatic shock (TS) is implicated in up to 40% of early trauma deaths; however, no consensus definition exists, and its epidemiology and outcomes have not been characterised at a population level in Australia.

AIMS: To apply a pragmatic clinical definition of TS, based on systolic blood pressure (SBP) and shock index (SI), to a national trauma registry, and to describe the epidemiology and outcomes of major trauma (MT) patients with and without clinical signs of shock.

METHODS: A retrospective cohort study was conducted using Australia New Zealand Trauma Registry (ANZTR) data (2017-2022). TS was defined as pre-hospital SBP ≤ 90 mmHg or SI > 1, or hospital arrival SBP ≤ 90 mmHg or SI > 1. Epidemiology and outcomes were compared with patients without these features. Outcomes were reported as absolute risk differences and risk ratios alongside odds ratios; analyses were unadjusted.

RESULTS: Among 47,730 adult MT patients in the hospital arrival analysis, 16,463 (34.5%) were excluded from the pre-hospital analysis due to incomplete centre-level data. Patients with pre-hospital or hospital arrival signs of TS were younger, more severely injured, and more likely to sustain penetrating trauma. Clinical signs of shock were associated with substantially higher absolute in-hospital mortality: h-SBP ≤ 90 mmHg, 30.9% versus 7.4% (ARD 23.5 %age points, 95% CI 21.7-25.3; RR 4 […]

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Raising the standard of fracture care in low- and middle-income countries.

Martin C, Mwanga J, Lekina FA, Adewole L, Itungu S, Harrison WJ

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Comparative performance of CRAMS, RTS, NISS, and ESI scores in predicting 24-hour mortality in adult multi-trauma patients: A prospective study.

Saznaghi FH, Hakimi H, Alizadeh A, Hosseinigolafshani S

BACKGROUND: Accurate mortality prediction in trauma patients is essential for guiding clinical decisions and optimizing resource allocation. This study compared the predictive performance of four commonly used trauma assessment tools-CRAMS (Circulation, Respiration, Abdomen, Motor, Speech), Revised Trauma Score (RTS), New Injury Severity Score (NISS), and Emergency Severity Index (ESI)-in predicting 24-hour mortality among adult multi-trauma patients.

METHODS: This prospective analytical study was conducted at the Emergency Department of Qazvin Trauma Hospital in Qazvin, Iran, from January to June 2025. A total of 265 adult multi-trauma patients were enrolled. Participants were selected via convenience sampling and were required to meet specific inclusion and exclusion criteria. Upon arrival at the hospital, patients were triaged and assessed using four scoring systems: CRAMS, RTS, NISS, and ESI. The primary outcome was 24-hour mortality. To compare the predictive performance of the scoring systems, Receiver Operating Characteristic (ROC) curves and DeLong's test were used.

RESULTS: Among 265 patients (mean age: 38.29 ± 12.99 years; 78.87% male), 16 (6.03%) died within 24 h. All four scoring systems demonstrated significant predictive ability ($p < 0.001$). The New Injury Severity Score (NISS) showed superior performance, with the highest area under the receiver operating chara […]

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Pre-hospital paediatric preventable injuries: Who, what, when and where?

Kocierz L, Bird F, Henry CL, Strutt M, Dobbie A, Lockey DJ

INTRODUCTION: Trauma remains the leading cause of mortality among children over one year of age in the United Kingdom (UK). Accidental injuries and violence represent some of the most preventable contributors to childhood morbidity and mortality. This study aimed to analyse data from a physician-led, urban pre-hospital service to identify paediatric populations at risk of preventable injury, characterise common mechanisms of injury, and explore temporal and geographic trends to inform targeted prevention strategies.

STUDY DESIGN: A retrospective review was conducted of paediatric patients (aged ≤16 years) attended by a physician-led pre-hospital trauma service between January 2017 and June 2022. Data collected included age, sex, time of day, location of incident and mechanism of injury. Cases were divided into three subgroups representing potentially preventable injuries: Falls from height (>2 m), penetrating trauma, and road traffic collisions (RTCs). Regression analysis was performed to assess for any association between injury and index of deprivation.

RESULTS: 782 patients met inclusion criteria. 607 (78%) patients sustained injuries classified as potentially preventable, the majority of which involved male patients (n = 483, 79.6%). Adolescents aged 12-16 years accounted for the highest proportion of penetrating trauma (n = 279, 96%) and RTCs (n = 92, 54%), whist falls f […]

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Prophylactic antibiotic use and infectious complications in open pelvic fractures with concomitant visceral injury.

Gochi AM, Patel S, Mathur T, Perez HO, Perez AH, Heximer A, Barrientos K, Gallo KG et al.

BACKGROUND: The objective of this study was to evaluate antibiotic treatment patterns administered with prophylactic intent and infectious outcomes in open pelvic fractures (OPFs). We hypothesized that antibiotic regimen and duration would demonstrate significant variability, particularly in the presence of visceral injuries, and that prolonged or broader antibiotic use would reflect clinical complexity rather than a standardized prophylactic approach.

METHODS: A retrospective cohort study of adult trauma patients (>16 years) presenting with an OPF to a Level I trauma center between 2014 and 2024, was performed. Antibiotic timing, spectrum, duration, and regimen variability were assessed for antibiotics administered at the time of OPF diagnosis and management, reflecting prophylactic intent. Prolonged antibiotic therapy was defined as > 72 h. Infectious outcomes included superficial, deep, and organ space surgical site infections (SSI), and osteomyelitis. Multivariable logistic regression was used to identify predictors of infectious complications and prolonged antibiotic use.

RESULTS: Among 160 patients (median age 30 years; 85% male), antibiotic management demonstrated substantial variability, with 16 unique antibiotic regimens used. Patients with concomitant visceral (gastrointestinal and/or genitourinary) injuries received significantly more antibiotic combinations ($p < 0$ […]

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Evaluating the efficacy of disinfecting agents against Staphylococcus aureus on external fixator devices.

Horowitz ME, Talsania A, Grando K, Walters T, Albicoro FJ, Tükel Ç, Rehman S

OBJECTIVE: The objective of this study was to evaluate and compare the efficacy of multiple disinfectant agents commonly used to remove bacteria from external fixators infected with Staphylococcus aureus.

METHODS: A sterile external fixator was colonized with Staphylococcus aureus for 48 h to allow biofilm formation on the device. Following incubation, the device was treated with one of the following: Isopropyl Alcohol (IPA), Chlorhexidine Gluconate Scrub (CHG) + IPA, Betadine (PI) + IPA, or ChloraPrep (2% CHG and 70% IPA). Swabs were taken from locations on each device and plated on Tryptic Soy Agar (TSA) plates. Colony forming units (CFUs) were counted and analyzed using Kruskal-Wallis tests along with Dunn's multiple comparisons. The frequency of zero vs. detectable (non-zero) CFU data was also recorded.

RESULTS: The median in the control group was 7.8 × 103 CFUs (interquartile range (IQR): 1.5 × 103-2.2 × 104 CFUs), compared to a median of 0 CFUs in each treatment group (IPA IQR: 0-0; CHG+IPA IQR: 0-0; ChloraPrep IQR: 0-7.5; Betadine+IPA IQR: 0-3.8 × 102). Three non-zero data were recorded in the IPA group ($n = 16$), two non-zero data were recorded in the CHG + IPA group ($n = 16$), four non-zero data were recorded in the ChloraPrep group ($n = 16$), and seven non-zero data were recorded in the Betadine + IPA group ($n = 16$). Kruskal-Wallis analysis indicated that all disinfect […]

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Validity of the non-union scoring system in management of long bone fracture non-union. A prospective study.

Wafa TA, Al Kersh MA, Ghaly NAM, Ahmed AR, Rizk MH, Mahran MA

PURPOSE: To validate the Non-union Scoring System by applying the score, prospectively, on long bone non-union cases and evaluate the results of the treatment according to the score.

METHODS: In order to assess treatment efficacy on non-union treatment, the NUSS was used in a series of patients diagnosed with femoral and tibial non-union. We included both septic and aseptic non-union and excluded skeletally immature patients, patients with autoimmune diseases, neoplasia and patients under immunosuppressive therapy. The patients were followed up for maximum of 30 months.

RESULTS: The NUSS was applied prospectively to 60 cases of fracture non-union. The outcomes demonstrated that most patients across different NUSS treatment groups achieved successful healing, with group 1 reporting 88.9%, group 2 achieving 80%, and group 3 achieving 81.5% union rates. The mean time for clinical and radiographic healing varied by group, with longer recovery times seen in more severe cases. Three patients who underwent amputation were not included in the final union assessment. Six patients were excluded from the study due to what was termed a "score-clinical mismatch," where intraoperative findings indicated a need for more aggressive treatment than the NUSS had suggested.

CONCLUSION: The NUSS is a reliable tool for managing fracture non-union, but it may underestimate treatment needs in some […]

PubMed · DOI

Epidemiological characteristics and temporal trends of adult acute compartment syndrome From 2015 to 2025.

Ruppert NF, Sanghvi PA, Good LM, Florentino SA, Berk AN, Adelstein JM, Farrell ND, Ochenjele G et al.

INTRODUCTION: Acute compartment syndrome (ACS) is a rare surgical emergency that carries limb- and life-threatening risk. Existing literature demonstrates limited patient cohorts, leaving knowledge gaps regarding the epidemiology, clinical characteristics, and outcomes of ACS. Using a large national dataset, this study characterized demographic variables, national trends, and epidemiological characteristics among patients undergoing fasciotomies for ACS.

METHODS: The TriNetX US Collaborative Network database was queried to identify patients aged 18 and older who underwent fasciotomy for ACS from 2015 to 2025 using relevant ICD-10 and CPT codes. The primary outcome was incidence proportion of ACS over time. Cohort characteristics were collected, including age, sex, race, body mass index (BMI), comorbidities, associated orthopedic injuries, anatomic location of compartment syndrome, and geographic distribution of patients. A secondary descriptive subgrouping of non-traumatic and traumatic ACS was also reported.

RESULTS: A total of 117,592,405 patients were identified from initial query within TriNetX, of which 10,045 patients were diagnosed with ACS. The mean age of the cohort was 44.8 ± 19.4 years with a mean BMI of 28.2 ± 6.9 kg/m2. Most patients were male (67%) and White (68%). Compartment syndrome was most often associated with open injuries (22%) and tibial shaft fractures […]

PubMed · DOI

Epidemiology, associated injuries, and surgical intervention of 255,081 clavicle fractures in the United States from 2015 to 2024.

Rodio LA, Adelstein JM, Burkhart RJ, Good LM, Moyal AJ, Napora JK

BACKGROUND: Clavicle fractures are a relatively common injury and are often associated with other injuries, especially those of the thorax. While they have traditionally been treated conservatively, rates of surgical intervention have been reportedly increasing. This study investigates recent epidemiological trends and demographic distributions of clavicle fractures in the US. Additionally, it reports on rates of injuries associated with clavicle fractures and on rates and trends of surgical intervention for clavicle fractures.

METHODS: The TriNetX US Collaborative Network database was used to identify patients who experienced a clavicle fracture in the US from 2015 to 2024. The primary outcome was incidence proportion (IP), stratified by both age and sex. Rates of associated injuries and surgical treatment were also assessed.

RESULTS: A total of 255,081 patients were identified as having a clavicle fracture in the US TriNetX database from 2015 through 2024. The average yearly IP was 52.11 cases per 100,000 patients. From 2015-2024, the IP increased by 68.2%. Males made up 64% of the cohort and had a larger IP than females every year. The mean age was 35 ± 27 and the age-stratified IP followed a bimodal distribution, with the highest values in those under 19 and over 80. Rib fractures (14%) were the most common associated injury assessed. Overall, 7.5% were treated with ORIF. […]

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Operative and non-operative management of people with hip fracture: A modified-Delphi study to develop an international consensus.

Mitchell RJ, Johansen A, Willems HC, Tabu I, Seymour H, Moppett I

OBJECTIVE: To develop a series of consensus statements regarding the operative and non-operative management of hip fracture patients.

METHOD: A two-round modified-Delphi study was used to obtain consensus on statements for the the operative and non-operative management of hip fracture patients. Panel members rated the importance of each statement to guide the decision to operatively or non-operatively treat hip fracture patients. High consensus was defined as agreement of ≥ 70% and moderate consensus 50-69% between participants.

RESULTS: There were 67 round 1 and 55 round 2 panellists. Panellists were working across 17 countries and commonly were anaesthetists (30%), orthopaedic or trauma surgeons (28%), or geriatricians (27%). Just less than one-quarter of panellists (24%) indicated the facility they predominately worked in had a protocol for the non-operative management of hip fracture patients. At the conclusion of the modified-Delphi study, 14 (93%) statements achieved high consensus and one (7%) statement moderate consensus.

CONCLUSION: The consensus statements are intended to support shared decision-making regarding non-operative hip fracture care and to promote a more consistent approach to the management of older adults with hip fracture at the end of life. Ultimately, the decision for operative or non-operative management should be made based on each patient's indiv […]

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Surgical stabilization of rib fractures in adult trauma patients with flail chest with or without severe traumatic brain injury: A national analysis.

Nasef H, Chin B, Nunes N, Werling A, Amin Q, Wright DD, Smith C, Zito T et al.

INTRODUCTION: We aim to compare clinical outcomes of surgical stabilization of rib fractures (SSRF) versus non-operative management (NOM) in adult non-TBI patients with isolated flail chest and severe TBI patients with flail chest.

METHODS: A retrospective cohort analysis was performed utilizing the American College of Surgeons Trauma Quality Improvement Program Participant Use File (ACS-TQIP PUF) dataset from 2017 to 2021. Adult non-TBI trauma patients with isolated flail chests and severe TBI patients with flail chests were included. Outcomes included in-hospital mortality, intensive care unit length of stay (ICU-LOS), ventilator-free days (VFDs), in-hospital complications, and discharge disposition.

RESULTS: Among non-TBI patients, there was no significant difference in in-hospital mortality between SSRF and NOM patients (OR 0.565, 95% CI 0.261-1.222, p = 0.147). Among severe TBI patients, SSRF was associated with significantly lower odds of in-hospital mortality (OR 0.275, 95% CI 0.088-0.859, p = 0.026). In both non-TBI and TBI patients, SSRF was associated with significantly longer ICU-LOS (B=2.434, 95% CI 1.796-3.098, p < 0.001; B=5.009, 95% CI 2.866-7.151, p < 0.001), increased odds of ventilator-associated pneumonia (VAP) (OR 4.051, 95% CI 1.573-10.435, p = 0.004; OR 4.066, 95% CI 1.501-11.019, p = 0.006), tracheostomy (OR 2.061, 95% CI 1.143-3.716, p = 0.016;OR 3.797 […]

PubMed · DOI

Cytokine release after geriatric coxal monotrauma.

Pass B, Boerner A, Bousch JF, Schoeps D, Maus U, Kupschus J, Eschbach D, Ruchholtz S et al.

PURPOSE: Interleukin-6 (IL-6), interleukin-10 (IL-10), and monocyte chemoattractant protein-1 (MCP-1), also known as chemokine (C-C motif) ligand 2 (CCL2), are pivotal mediators of the immune response following trauma. While IL-6 predominantly promotes inflammation, IL-10 serves a regulatory function, and MCP-1/CCL2 facilitates monocyte recruitment and differentiation. Dysregulation of these cytokines has been linked to adverse postoperative outcomes in geriatric patients. An investigation was conducted to analyze the cytokine release profile in geriatric patients with proximal femur fractures and to compare it with that of a geriatric non-trauma group undergoing a comparable operative procedure.

METHODS: Patients older than 70 with proximal femur fractures were enrolled alongside geriatric patients undergoing elective hip arthroplasty and a young (<50 years) healthy control group. Blood samples were collected preoperatively, immediately postoperatively, and on postoperative days 1, 4, and 7 for cytokine quantification.

RESULTS: Preoperative IL-6 and IL-6/IL-10 ratios were significantly higher in trauma patients. IL-6 remained higher immediately postoperatively but was surpassed by THA patients. MCP-1/CCL2 and IL-10 did not differ between groups.

CONCLUSION: Geriatric fracture patients exhibited a pronounced preoperative pro-inflammatory state, whereas elective THA was assoc […]

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High-dose melatonin for early tibial fracture healing: A prospective study of safety, feasibility, and preliminary biological effects.

Ninomiya AF, Bertolucci V, Nonose N, Messias LO, Beck WR, Pejon TMM, de Almeida Guerreiro F, Domingues VM et al.

BACKGROUND: Tibial shaft fractures represent a prevalent orthopedic challenge, particularly in public health systems, due to delayed healing and limited access to biological adjuvants. Melatonin, a pleiotropic indoleamine with antioxidant and osteogenic properties, has demonstrated consistent efficacy in preclinical bone repair models; however, its clinical application, especially at pharmacological doses, remains poorly explored. This prospective controlled study evaluated the effects of high-dose melatonin (30 mg/day) on early bone healing, safety, and tolerability following reamed intramedullary nailing for diaphyseal tibial fractures.

METHODS: Twenty-five adults (18-60 years) with closed tibial shaft fractures (AO/OTA 42-A/B/C) were sequentially allocated to a melatonin group (n = 12) or a standard care control group (n = 13), with nightly oral melatonin administered for 49 consecutive postoperative days. Primary outcomes included serum alkaline phosphatase (ALP) levels and the Radiographic Union Score for Tibial fractures (RUST) assessed at 21 and 49 days, while secondary outcomes encompassed adverse events, treatment adherence, and sleep quality.

RESULTS: Both groups showed significant increases in ALP and RUST scores over time ($p < 0.001$), reflecting expected fracture healing progression; however, no significant between-group differences were observed for ALP ($p = 0.32$ […]

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Prehospital national early warning score outperforms shock index in predicting mortality and resource utilisation in adult trauma: A systematic review.

Tolich A, Dicker B, Howie G, Todd VF

INTRODUCTION: Early recognition of severely injured trauma patients is essential in the prehospital setting. Emergency Medical Services (EMS) frequently rely on simple physiological measures to guide triage, yet validated prognostic tools for prehospital trauma remain limited. The Shock Index (SI) and National Early Warning Score (NEWS) are physiological scoring systems that may support early risk stratification. This systematic review evaluates the prognostic accuracy of prehospital SI and NEWS for mortality and resource utilisation in adult trauma.

METHODS: A systematic search of CINAHL, Medline and Cochrane databases (2015-2025) identified studies reporting associations between prehospital SI or NEWS and overall mortality in adult trauma patients. Paediatric studies, Emergency Department only cohorts, isolated body-region trauma, and modified scoring systems were excluded. Due to substantial heterogeneity in study design, populations and outcome measures, a narrative synthesis was undertaken.

RESULTS: Twenty-two studies met inclusion criteria, with numbers ranging from 184 patients to 1439,221 patients across diverse EMS systems. Across individual studies, prehospital SI showed variable discrimination for mortality, with AUROC values ranging from poor to moderate (0.48-0.79), although elevated SI (>0.9) consistently correlated with increased mortality, transfusion, and cri […]

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Comparative efficacy of the femoral neck system, cannulated cancellous screws combined with medial buttress plate, and cannulated cancellous screws in the treatment of femoral neck fractures in young and middle-aged adults.

Huang Y, Lin J, Chen X, Li R, Jiang X, Lin D, Lin F

OBJECTIVE: This study aimed to evaluate the long-term efficacy of the Femoral Neck System (FNS), cannulated cancellous screws (CCS) combined with a medial buttress plate (MBP), and CCS alone in treating femoral neck fractures in young and middle-aged adults, and to identify independent risk factors for postoperative complications.

METHODS: This retrospective single-center study enrolled 355 patients with femoral neck fractures. The inclusion criteria were patients aged 18-65 years with complete baseline data and a mean follow-up duration of 18.22 ± 6.31 months. Exclusion criteria included pathological/bilateral fractures and significant comorbidities. Based on the surgical technique employed, patients were assigned to the FNS group (n = 165), CCS+MBP group (n = 62), or CCS group (n = 128). Baseline characteristics, surgical parameters, radiographic outcomes (Garden index), complication rates, and functional outcomes were compared among the groups. Statistical analyses included one-way ANOVA, χ^2 tests, and multivariate logistic regression.

RESULTS: Baseline characteristics were comparable across the three groups. Mean fracture healing time was 13.30±1.34 weeks for CCS, 11.42±1.02 weeks for CCS+MBP, and 11.62±1.23 weeks for FNS ($P = 0.224$). At the 3-month postoperative assessment, the CCS+MBP group demonstrated significantly different functional outcomes compared to both the FN […]

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Dorsal subchondral arc line: A novel radiographic marker to avoid dorsal screw prominence during distal radius fixation with volar locking plates.

Worhacz K, Castaneda P, Hui C, Walker JB

OBJECTIVES: The aim of this study was to evaluate a novel radiographic marker for measuring screw length during fixation of distal radius fractures with volar locking plates. The Dorsal Subchondral Arc Line (DSAL) is visualized on the lateral wrist view and screw length is determined using the intersection of a line along the longitudinal axis of the radius and the most dorsal extent of the subchondral arc. The hypothesis was that use of this marker would yield screws that did not penetrate the dorsal cortex and would provide similar fixation in the distal segment to bicortical screws.

METHODS: This was a study performed on 10 matched pairs of cadaveric forearms. Each specimen in a matched pair was randomized into two groups; one group received distal locking screws measured using the DSAL and the other group received distal locking screws that were fully bicortical. Specimens were all instrumented using volar locking plates and subsequently denuded of all soft tissue to determine presence of dorsal cortex penetration. Osteotomies were performed in order to simulate and extra-articular distal radius fracture with significant comminution (OTA 23-A3). All specimens were then tested under cyclical axial compression utilizing a servoelectric load frame and then loaded under axial compression to failure.

RESULTS: All 10 matched pairs were included in final analysis. No distal scre […]

PubMed · DOI

Complications after operatively treated fractures in patients with solid organ transplantation - A systematic review of the literature.

Hübner CT, Anranter A, Kalbas Y, Klingebiel FK, Shehu A, Halvachizadeh S, Pfeifer R, Rössler F et al.

INTRODUCTION: Solid organ transplantation is associated with multiple risk factors, reduced bone quality and relevant comorbidities, which may negatively affect postoperative outcomes after fracture treatment. Furthermore, these patients usually require lifelong immunosuppressive therapy. A systematic review was performed to evaluate the incidence of postoperative complications and to assess whether specific considerations are required in this special patient group.

METHODS: A systematic review of the literature according to the PRISMA guidelines was performed. A total of 3743 studies were identified and subsequently assessed for eligibility. Studies investigating elective orthopedic procedures were excluded. After full-text screening, six studies were included in the final analysis.

RESULTS: Overall, the six studies investigated 2317 patients. Cohort sizes ranged from 12 to 2163 patients, with a median cohort size of 33.5 patients. Mean age of patients was 67.9 years. Follow-up durations varied between 18 weeks and 3 years. The most frequently investigated fracture type was proximal femoral fractures with three studies, followed by lower extremity fractures (two studies). One study examined distal radius fractures. Infectious complications were reported in four studies, with a pooled infection rate of 10%. The pooled 90-day readmission rate was 20%. Mortality was reported he […]

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Investigation of the biomechanically optimal length of intramedullary nails in unstable intertrochanteric femur fractures 31A2.2: A finite element analysis study.

Den S, Ueda K

INTRODUCTION: Achieving sufficient stability in unstable intertrochanteric femur fractures (AO/OTA 31A2.2) is important for bone healing. Although implant-related factors such as nail diameter, proximal fixation configuration, and nail length contribute to mechanical stability at the fracture site, the appropriate nail length for specific fracture patterns remains unclear. This study evaluated the effect of nail length on fixation stability using finite element analysis.

METHODS: Finite element models of AO/OTA 31A2.2 fractures, with and without a gap, were developed using CT data from the femur of an average-sized Japanese woman. ASULOCK® intramedullary nails ranging in length from 170 to 280 mm were virtually implanted, and stair climbing was simulated to assess interfragmentary motion and implant stress.

RESULTS: Interfragmentary motion significantly decreased with increasing nail length under both gap conditions and reached a plateau at nail lengths of 260 mm or greater. Nail stress tended to be higher in the gap model, but neither nail stress nor lag screw stress showed a significant association with nail length.

CONCLUSION: In an AO/OTA 31A2.2 fracture model based on an average-sized Japanese female femur, this study demonstrated that interfragmentary motion reached a plateau at nail lengths of ≥ 260 mm, suggesting that sufficient control of interfragmentary motion may […]

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Minimally invasive plate osteosynthesis, open reduction and plate fixation, and intramedullary nailing for humeral shaft fractures: A network meta-analysis of randomized controlled trials.

Protásio Netto J, Giordano V, Elias N, Belangero WD, Cavalcante B

BACKGROUND: The optimal surgical technique for humeral shaft fractures remains debated, and earlier syntheses often pooled all plate constructs, disregarding the distinction between open reduction and plate fixation (ORPF) and minimally invasive plate osteosynthesis (MIPO). This network meta-analysis compared ORPF, MIPO, and intramedullary nailing (IMN), focusing on iatrogenic radial nerve injury.

METHODS: We searched PubMed/MEDLINE, Embase, Cochrane CENTRAL, and Scopus from inception through December 2024, with the PubMed search updated through January 2026, following the PRISMA network meta-analysis extension (PROSPERO CRD420261321547). Randomized controlled trials comparing at least two of the three techniques in adults with closed diaphyseal humeral fractures were eligible. The primary outcome was iatrogenic radial nerve injury; secondary outcomes were nonunion, deep infection, and reoperation. A frequentist random-effects network model estimated odds ratios (OR) with 95% confidence intervals. Treatments were ranked using the surface under the cumulative ranking curve (SUCRA), which orders treatments by estimated effect and is not an absolute event rate.

RESULTS: Twelve RCTs (647 patients) met the criteria for the primary network; three non-randomized or non-comparable studies were reserved for sensitivity analysis. MIPO was associated with lower odds of radial nerve inju […]

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The penetrating injury risk of non-home discharge (PIRD) score: Development and validation of a predictive tool using the National Trauma Data Bank.

Phadke R, Salman S, Ilyas MH, Sontam T, Rana A, Leucht P

BACKGROUND: Penetrating trauma disproportionately affects young patients, yet many survivors cannot be discharged directly home and instead require post-acute care. Identifying these patients early could expedite rehabilitation referral and discharge planning, but no validated tool predicts non-home discharge in penetrating trauma. We developed and internally validated the Penetrating Injury Risk of Non-home Discharge (PIRD) score.

METHODS: Using the National Trauma Data Bank (NTDB) Trauma Quality Improvement Program Participant Use Files (2019-2024), we selected adult patients (≥18 years) with penetrating trauma who survived to discharge. The outcome was non-home discharge to post-acute care (e.g., skilled nursing or inpatient rehabilitation) versus home. The cohort was split 2:1 into derivation and validation sets, stratified by outcome. Candidate predictors available early in the hospital course (demographics, insurance, mechanism, emergency-department physiology, the Injury Severity Score [ISS], body-region injury, transport, and comorbidities) underwent univariate screening, collinearity assessment, and backward-stepwise multivariable logistic regression. Unlike the orthopedic FORD score, ISS was retained because penetrating trauma is a high-severity, mortality-driven population. Coefficients were converted to integer points, scaled 0-10.

RESULTS: Among 302,399 patients, […]

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A double-hit analysis: Evaluating the nephrotoxicity of dual intravenous and vancomycin impregnated cement exposure.

Vu NH, Esper GW, Egol KA, Ganta A, Konda SR

PURPOSE: The concomitant use of postoperative intravenous vancomycin (IVV) in patients treated with vancomycin-impregnated cement (VIC) may represent a "double hit" contributing to nephrotoxicity, yet its impact on acute kidney injury (AKI) remains unclear. This study evaluated whether postoperative IVV in patients treated with VIC is associated with increased AKI incidence (primary outcome) and greater changes in serum creatinine (secondary outcome) compared with VIC alone.

METHODS: A retrospective cohort study was conducted at a multi-hospital academic health system from 2012 to 2024. Adults undergoing VIC implantation for fracture-related infection were grouped by postoperative IVV exposure (IVV+VIC versus VIC alone). AKI was defined using Kidney Disease: Improving Global Outcomes (KDIGO) creatinine-based criteria. Secondary outcomes included mean absolute and percent serum creatinine change. A subgroup analysis evaluated 81 patients with chronic kidney disease (CKD). Multivariable regression adjusted for baseline creatinine, tobramycin dose, CKD, ASA class, and VIC dose.

RESULTS: A total of 589 patients were included: 326 (55.4%) received IVV + VIC and 263 (44.6%) received VIC alone, with similar baseline characteristics. Among IVV + VIC patients, mean IV vancomycin dose was 1160 mg (SD 354) over a mean of 6.6 doses (SD 7.6), with a mean VIC dose of 4.9 g (SD 3.3). The VI […]

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Higher Rate of Treatment Success of Infected Femoral Shaft Non-unions with Ultrathin Silver-Polysiloxane-Coated Implants.

Mueller MM, Niclassen CM, Kowald B, Klinger CE, Gerlach UJ, Grimme C, Schulz AP, Frosch KH et al.

BACKGROUND: Infected non-union represents a severe complication after femoral shaft fractures, often associated with high recurrence rates and the need for multiple surgical revisions. Antimicrobial implant coatings may improve treatment success; however, evidence on the use of silver as an implant coating remains limited. This study aimed to (1) report short-term outcomes following treatment of infected femoral shaft non-union and (2) investigate if silver-polysiloxane-coating of implants leads to a significantly higher rate of treatment success at one-year follow-up.

METHODS: This prospective cohort study (REFECT) with a historical control group included 70 patients treated for infected femoral shaft non-unions from 2013 to 2024 using either conventional, uncoated (UC, n = 62) or silver-polysiloxan-coated (SC, n = 8) implants with minimum of one-year follow-up. The primary outcome was successful treatment at one-year follow-up, defined as bony consolidation, and full weight-bearing and without recurrent infection. Secondary outcomes included clinical outcomes, blood serum analysis (including silver-ion concentration), radiographic outcomes, and patient reported outcome measures (SC-cohort only) including Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC), Lower Extremity Functional Scale (LEFS), and EuroQol-Visual Analogue Scale (EQ-VAS).

RESULTS: Mean f […]

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The evolution of primary subtalar arthrodesis for displaced intra-articular calcaneal fractures: A historical review.

Schepers T

BACKGROUND: Primary subtalar arthrodesis has been advocated for the treatment of severe displaced intra-articular calcaneal fractures for more than a century. Although the procedure remains a valuable option in carefully selected patients, its historical evolution has never been comprehensively reviewed. This study aimed to describe the development of primary subtalar arthrodesis from its introduction in 1911 to contemporary practice and to evaluate how indications, surgical techniques, and reported outcomes have evolved over time.

METHODS: A comprehensive search of PubMed and Embase was performed without date restrictions, supplemented by backward and forward citation tracking. Publications describing acute primary subtalar or triple arthrodesis for calcaneal fractures were included. Studies were grouped into four historical eras based on publication volume, treatment philosophy, and technical developments. Data regarding indications, operative techniques, timing, and clinical outcomes were extracted descriptively.

RESULTS: The concept of primary subtalar arthrodesis was first introduced by Van Stockum in 1911, who recognized the subtalar joint as the principal source of persistent pain after severe calcaneal fractures. Early reports regarded arthrodesis primarily as a salvage procedure when anatomical reconstruction was considered impossible. During the 1950s-1970s, the con […]

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Do all Gartland type II supracondylar distal humeral fractures need surgery and pinning?

Qiu R, Ferrie L, Buckley R

PubMed · DOI

Composite graft survival in allen zone II fingertip amputations: Independent predictors of graft failure.

Gürhan U, Kahve Y, Sar■ E, Umur FL

BACKGROUND: Composite grafting is a commonly used treatment option for Allen Zone II fingertip amputations; however, reported survival rates vary considerably across studies. Although several clinical and injury-related factors have been suggested to influence graft survival, independent predictors of failure and clinically meaningful ischemia duration thresholds remain insufficiently defined. This study aimed to identify independent clinical predictors of composite graft survival and to explore, secondarily, whether ischemia duration is associated with graft failure.

METHODS: Forty-two patients who underwent composite grafting due to Allen Zone II fingertip amputation between December 2021 and June 2023 were retrospectively evaluated. Baseline variables were collected from hospital records and surgical notes. Ischemia duration was calculated as the number of hours between the injury and the start of surgery. Diabetes mellitus (DM) and other comorbidities were verified using diagnostic codes and patient histories. The primary outcome measure was graft survival at 6 weeks postoperatively; necrosis affecting more than 25% of the graft surface area was defined as failure. Group comparisons, univariate and multivariate logistic regression, and receiver operating characteristic (ROC) curve analysis were performed. Because of the limited number of outcome events and quasi-complete s […]

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Collagen nanofibers enhanced by plasma rich growth factors (PRGF) for the burn wound healing.

Abedini-Esfahlani M, Khodabakhshi MR, Eslami-Mahmoudabadi M, Nourani MR

BACKGROUND: Severe burns impair the body's natural healing capacity, often resulting in delayed tissue repair. Platelets facilitate hemostasis and initiate wound healing through growth factor release. This study investigates the therapeutic potential of collagen nanofibers enriched with Plasma Rich Growth Factors (PRGF) to enhance cell proliferation, migration, and wound healing in burn injuries.

METHODS: PRGF powder was prepared from whole blood using a two-step centrifugation protocol followed by freeze-thaw activation and incorporated into collagen/polyethylene oxide (COL/PEO) electrospun nanofibers. The physicochemical properties and growth factor release profile of the nanofibers were characterized. In vitro assays evaluated the effects of PRGF-loaded nanofibers on fibroblast viability and migration. For in vivo analysis, a standardized full-thickness burn wound model was induced in male Wistar rats following Orio's rules. Wound healing was assessed over a 21-day period through macroscopic wound closure measurements and histological analyses, including H&E staining, Masson's trichrome staining, and VEGF immunohistochemistry.

RESULTS: PRGF-loaded collagen nanofibers showed an initial burst release of growth factors followed by sustained release. In vitro, PRGF incorporation significantly enhanced fibroblast viability and migration. In vivo, the PRGF + COL/PEO group demons […]

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Professional obstacles: A perception-based survey of women members of AO trauma - Latin America.

Medina C, Arroyo-Berezowsky C, Oberlohr V, Amaral A, Calcado I, Gelvez AG, Tobia R, Rodarte P et al.

PURPOSE: Orthopaedic surgery remains one of the most homogenous fields in medicine, with men accounting for over 95% of all orthopaedic surgeons worldwide. Regardless of age, experience, or academic achievement, women orthopedists have historically been underrepresented in leadership positions, academic medicine, and specialty societies and associations. While these trends are recognized globally, limited studies examine gender-based dynamics in Latin America. This perception-based survey aims to bring awareness to the experiences and challenges of women members of AO Trauma Latin America (AO T Lat).

METHODS: A questionnaire evaluating the experiences and challenges of women orthopedic surgeons was designed in collaboration with AO T Lat and several national Latin American orthopaedic societies. The questionnaire was divided into sections: demographics, motivations and opportunities, challenges and barriers, and AO-faculty-specific questions utilizing a 5-point Likert scale to rank items. The survey was distributed to women orthopaedic surgeons identified through AO T Lat, national orthopaedic societies, and professional networks throughout Latin America.

RESULTS: 116 women responded to the survey, representing current or former AO T Lat members from 16 Latin American countries. Intellectual stimulation and personal gratification and desire to work "hands-on" were the stronge […]

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Histologic and physiologic effects of PRP and HBO on muscle healing after contusion injury in an animal model.

Chen P, Chiu JC, Lei KF, Chen SC, Chan YS

BACKGROUND: Muscle contusion injuries frequently cause inflammation, hypoxia, and impaired regeneration. Platelet-rich plasma (PRP) and hyperbaric oxygen (HBO) therapy have been proposed as potential treatments for enhancing muscle recovery. This study aimed to investigate the effects of PRP and HBO therapy on muscle regeneration following contusion injury and assess their potential synergistic effect.

METHODS: A controlled laboratory study was conducted using a muscle contusion model in sixty C57BL6J mice, which were divided into five groups: control, sham, HBO, PRP, and PRP+HBO. HBO therapy was administered at 2.5 atm for 25 min, three cycles per day over 10 days, while PRP was injected twice, on days 1 and 6. Muscle regeneration was evaluated using histological staining, immunofluorescence, muscle wet weight, and contractile function tests, including assessments of fast-twitch and tetanic strength.

RESULTS: The results demonstrated that HBO treatment alone showed statistically significant improvements in the number of regenerating myofibers and tetanus strength, compared to the sham group. However, the magnitude of the HBO treatment effects might not have been as substantial as those observed in the PRP and PRP+HBO groups. PRP and PRP+HBO significantly increased regenerating myofiber counts compared to the sham group ($P < 0.001$) and prevented muscle atrophy, as shown by hi […]

PubMed · DOI

Reliability of patient and proxy reported pre-injury functional status in orthopaedic trauma: A prospective observational study.

Kelly R, Hong I, Mucharraz C, Taylor L, Bajwa S, Joyce C, Cohen J, Summers H et al.

BACKGROUND: Patient-reported outcome measures (PROMs) are essential for evaluating functional recovery after orthopaedic trauma. However, obtaining accurate pre-injury baseline PROM scores is often infeasible due to the emergent nature of traumatic injuries. This study aimed to assess the reliability of retrospective patient- and proxy-reported pre-injury functional status over time.

METHODS: This prospective observational study was conducted at a level I trauma center and included 173 adult patients presenting with acute upper or lower extremity fractures. Participants completed Patient-Reported Outcomes Measurement Information System Computer Adaptive Test (PROMIS CAT) surveys at week-0 (within 7 days of injury), week-2 and week-6 post-injury, all retrospectively recalling their pre-injury functional status. A subset of 91 patients also had designated proxies complete the same PROMIS surveys at the week-0 time point. Paired t-tests and Wilcoxon signed-rank tests evaluated score differences over time and between patients and proxies. Reliability was assessed using intraclass correlation coefficients (ICC), Bland-Altman plots and linear mixed-effects models adjusted for age, sex, comorbidities and baseline ambulatory status.

RESULTS: PROMIS scores remained consistent across the three time points, with minimal mean differences (e.g., lower extremity: week 0-6 difference = 0.2 […]

PubMed · DOI

The evolution of a novel technique for treatment of intercostal hernias and restoration of costal margin rupture.

Sharma R, Rafael A, Pailthorpe C, Bhattarai K, Leong JY, Sayed S

Costal Margin Rupture (CMR), with or without Intercostal Hernia (IH), is often a missed diagnosis. This has led to recent recommendations for a comprehensive and standardised classification, aiming to generate a heightened index of suspicion for these injuries. [1] While some surgical techniques have been published, surgical failure rates remain high and optimal surgical management remains undefined. We present three patients with IH, where the diagnosis of CMR played a pivotal role in the surgical management. In the first case, the CMR was identified post-operatively after a first surgical attempt to treat the IH had failed. In contrast, increased clinical suspicion in the second and third cases led to early diagnosis of the CMR. All three patients underwent a successful repair of the CMR using an innovative technique with a titanium bar crimped to rotatable connectors mounted on titanium rib clips. These cases highlight the importance of early recognition and comprehensive imaging and digital reconstruction techniques in detecting CMR. The surgical technique might be a useful addition to the surgical armamentarium/repertoire of the chest wall surgeon.

PubMed · DOI

A comprehensive review of trauma transfers to a rural pediatric trauma center.

Koenen M, Sang HI, Gorra A, Ryckman J

PURPOSE: The purpose of this study is to delineate transfer patterns to a rural Level 2 pediatric trauma center by identifying common diagnoses associated with potentially avoidable transfers (PAT) to a higher level of care with the end goal of optimizing resource utilization in the center's catchment area.

METHODS: Using trauma registry data, every pediatric trauma patient (< 15 years old) transferred to our Level 2 trauma center from January 2017 to December 2023, was included. "Potentially avoidable transfer" was characterized as a patient who, upon arriving at the trauma center, did not undergo any testing or intervention and was discharged after 0 or 1 midnights. Statistical analysis was performed to better characterize features associated with PAT.

RESULTS: 890 patients met inclusion criteria, of which 93 were identified as PAT (10%). The mechanism of injury most likely to result in a PAT was a fall. PATs were more likely to have had CT imaging performed prior to transfer. No injury, as well as head injuries confined to concussions, or skull fracture without hemorrhage were associated with PAT, while chest, long bone fractures, and abdominal injuries were associated with unavoidable transfer (UAT).

CONCLUSION: Analysis of transfer patterns in this study reveals consistent diagnoses associated with potentially avoidable transfers and highlights areas for enhanced educat […]

PubMed · DOI

Combined C5 and CD14 inhibition balances immunological miRNA responses after multiple trauma in pigs.

Groven RVM, Mert Ü, Greven J, He Z, Zhou N, Poeze M, Mollnes TE, Horst K et al.

BACKGROUND: Combined inhibition of complement-component C5 and Toll-Like Receptor CD14 is a promising therapeutic strategy to attenuate post-traumatic inflammatory-complications. The effect of this inhibition on miRNA expression at the fracture site and in systemic circulation was assessed in a porcine multiple-trauma model. Expression of the mRNA targets of deregulated miRNAs was determined at the fracture site and in lung and liver.

METHODS: Two experimental groups, intramedullary nailing (control; n = 8) and intramedullary nailing with combined C5/CD14 inhibition (n = 4), were compared using an established porcine multiple-trauma model. Fracture hematoma, fractured bone, unfractured control bone, lung, and liver samples were collected. Extracellular-vesicles (EVs) were isolated from plasma at 1.5, 2.5, 24, and 72 h post-trauma. MiRNA qPCR array analyses were performed on pooled fxH, fX, CB, and EV samples. In silico mRNA target prediction and mRNA qPCR analyses were conducted on all samples.

RESULTS: The fracture site miRNA signature in the C5/CD14 inhibition group was anti-inflammatory compared to intramedullary nailing alone, based on miRNA and mRNA qPCR analyses, with downregulation of anti-osteogenic miRNAs. Furthermore, C5/CD14 inhibition reduced pro-inflammatory and pro-fibrotic EV-carried miRNA expression in systemic circulation and their downstream mRNA targets in […]

PubMed · DOI

Prioritizing harm reduction in instant delivery crashes: Predictors of pedestrian injury severity and targeted interventions.

Yan X, Zhang X, Ye X, Chen J, Wang T, Du M, Bai H

Instant delivery crashes (IDCs) pose a growing public health challenge. Standard crash databases rarely capture the fine-grained variables necessary to model severe pedestrian disability. To address this gap, we analyzed a sample of 732 adjudicated court verdicts involving pedestrian-IDC collisions. This specific data source captures a highly selected subset of litigated, severe events rather than the broader crash population. We applied a Hierarchical Generalized Ordered Probit (HGOP) model. This framework accommodates the ordinal nature of disability grades and unobserved age-related heterogeneity. To achieve model convergence, we consolidated fatalities and severe disabilities into a single highest-severity category. Our modeling reveals that older pedestrian age (65-74 years) and motorbike involvement strongly predict higher-grade disability. Conversely, female pedestrians and winter conditions correlate with lower injury severity. We also identified a negative monotonic association between rider liability and pedestrian injury grade. As legal rider responsibility increased, the predicted severity of pedestrian injuries systematically decreased. Lacking direct kinematic data, we hypothesize this liability association reflects distinct conflict typologies rather than direct physical causation. Translating these findings into effective harm reduction requires a two-tiered app […]

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Navigated percutaneous column screw fixation combined with total hip arthroplasty for non-reconstructable acetabular fractures in geriatric patients.

Pozzi L, Wahl P, Kuttner H, Nehls F, Schläppi M, Meier C

BACKGROUND: Periprosthetic acetabular fractures and acetabular fractures are becoming increasingly common in the elderly population. The ultimate treatment goal in these patients should include immediate unrestricted mobilisation. Additionally, the burden of the operative procedure should be minimized. We therefore developed a technique of single stage navigated percutaneous screw fixation of the anterior and posterior columns combined with total hip arthroplasty for the treatment of non-reconstructable acetabular fractures in geriatric patients. The aim of this study is to retrospectively evaluate the clinical and radiological outcomes thereafter.

MATERIALS AND METHODS: This is a single-centre, prospective cohort study. All patients aged ≥ 60 years treated with the above-mentioned procedure at our institution between January 2020 and June 2025 were eligible for inclusion. Patient and surgery related parameters as well as level of independency of living and walking abilities, EQ-5D-5L and Forgotten Joint Score at latest follow-up were recorded.

RESULTS: Fifty-two hips in 51 patients (median age 85 years, range 64-95) with 30 (58%) native and 22 (42%) periprosthetic fractures were included. Median intraoperative blood loss was 600 mL (0-2'700). Median operating time was 255 min (range 169-430). Thirty-day mortality was 10% and revision rate 12%. Thirty-six patients (69%) retur […]

PubMed · DOI

Navigating the antithrombotic challenge in hip-fracture care: A framework to avoid delays and optimise patient outcomes.

Marta M, Costa L, Paulo Moreira J, Araújo F

INTRODUCTION: Hip fractures are a critical surgical emergency in older adults, frequently complicated by chronic antithrombotic therapy. Balancing expedited surgery with haemostatic and anaesthetic safety remains a major multidisciplinary challenge, with direct implications for survival and functional recovery.

OBJECTIVE: To propose an evidence-based decision-making framework for the perioperative management of anticoagulated patients with hip fractures, prioritising early surgery through an active clinical approach.

FRAMEWORK: This review proposes a pragmatic multidisciplinary framework, derived from practice in a Level 1 Trauma Centre and informed by current international guidelines (NICE, AAOS, Association of Anaesthetists, ESAIC/ESRA, and ASRA). It integrates recent evidence on antithrombotic management, regional anaesthesia safety, reversal strategies, and Patient Blood Management.

MAIN RECOMMENDATIONS: Early surgery-ideally within 24 h and no later than 48 h-is the overriding objective. Decision-making is guided by active haemostatic assessment, including drug-specific assays where available. A foundational principle is the clinical decoupling of safety thresholds for surgery and anaesthesia, recognising that the haemostatic requirements for surgery are distinct from the more stringent criteria for neuraxial or deep regional techniques. When neuraxial safety thresholds […]

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Artificial intelligence in the evaluation and treatment of osteoporotic and fragility fractures: A systematic review.

Liu X, Chui ECS, Wu X, Wang Y, Li T, Xie T, Cheung WH, Sang H et al.

BACKGROUND: Osteoporotic and fragility fractures impose a significant global health burden, especially with the aging population. Despite advancements in imaging, risk assessment, and surgical techniques, underdiagnosis and undertreatment persists. Artificial intelligence (AI), particularly machine learning (ML) and deep learning (DL), offers promise for enhancing fracture risk prediction, imaging-based diagnosis, clinical decision support, and postoperative outcome monitoring.

OBJECTIVE: To systematically review AI applications in the evaluation and treatment of osteoporotic and fragility fractures, summarizing performance, limitations, evidence gaps, and future directions for clinical translation.

METHODS: A PRISMA-compliant search was conducted in PubMed, IEEE Xplore, Google Scholar, and Web of Science from inception to October 2025. Inclusion criteria targeted original English-language studies in adults using AI/ML/DL for fracture risk prediction, diagnosis, treatment planning, intraoperative guidance, or postoperative management. Data extraction focused on study design, population, AI methods, performance metrics, and validation.

RESULTS: Of 1286 records, 21 studies were included, clustering into three domains: fracture risk prediction (n = 10) using clinical, biochemical, and imaging data, often outperforming tools like FRAX; bone mineral density (BMD) estimation and o […]

PubMed · DOI

Evaluation of erectile dysfunction following pelvic fractures: A retrospective analysis.

Tabassum A, Kruijning M, Wong J, Jenkins L, Langcake M

BACKGROUND: Pelvic fractures, typically resulting from high-energy trauma, are associated with substantial mortality and morbidity. Erectile dysfunction (ED) is a recognized yet underreported complication that significantly impairs the sexual, psychological, and overall quality of life in predominantly young male patients. While reported risk factors include specific fracture patterns and pelvic embolization, existing evidence remains limited and inconclusive.

OBJECTIVES: To determine the incidence of ED following traumatic pelvic fractures and identify significant demographic and clinical risk factors.

METHODS: This retrospective cohort study was conducted at St George Hospital, a Level 1 Trauma Centre in Sydney, Australia. Male patients with traumatic pelvic fractures admitted between June 1, 2020, and June 30, 2024, were identified from the institutional trauma registry and medical records. Incidence and risk factors for ED were analysed using univariate and multivariate logistic regression.

RESULTS: Of the 164 patients included, eight (4.9%) were diagnosed with ED. On univariate analysis, significant risk factors included Injury Severity Score (ISS), hospital length of stay, operative management of the fracture, and pelvic embolization. Following multivariate logistic regression, only embolization remained an independent predictor of ED (p = 0.030, OR 7.84, 95% CI 1.22-5 […]

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Diagnostic evaluation of delayed flexion-extension radiographs after blunt cervical spine trauma: A prospective study.

Emin L, Lafont C, Ravaillault S, Gardeton L, Lehuede A, Geay C, Montassier E, Moles A et al.

BACKGROUND: The value of delayed cervical flexion-extension radiographs after blunt cervical spine trauma remains debated in alert patients with persistent neck pain, normal neurological examination, and normal initial static imaging. At Nantes University Hospital, a dedicated delayed dynamic radiography pathway has been used for more than 20 years. This study reassessed its diagnostic relevance.

METHODS: Prospective, observational, single-centre study conducted in routine clinical practice between May 2022 and May 2024. Two populations were analyzed: patients referred to the delayed dynamic radiography pathway after blunt cervical trauma (n = 303), and patients who underwent surgery for unstable post-traumatic cervical ligamentous injury diagnosed during the acute emergency department assessment (n = 24). The objectives were to determine whether delayed flexion-extension radiographs identify surgically relevant unstable cervical injuries in alert patients without neurological/radiological abnormalities at initial emergency department assessment, and to describe the clinical and radiological features of surgically treated unstable ligamentous injuries.

RESULTS: Among 303 patients referred to the delayed dynamic radiography pathway, 4 unstable cervical ligamentous injuries were identified. Retrospective review showed that all 4 patients had abnormal initial imaging and/or neur […]

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Fracture orthopedic risk of non-home discharge score II (FORD-II).

Salman SG, Phadke R, Carlin T, Ramirez J, Mendez-Reyes JE, Hoffman MK, Leonard J, Davis RW et al.

INTRODUCTION: Post-acute placement after fracture trauma is a major driver of hospital length of stay and cost, but discharge needs are often identified late. We developed and validated the Fracture Orthopedic Risk of Non-Home Discharge score, version II (FORD-II), as an admission-time tool to identify adult fracture-trauma patients likely to require post-acute placement.

METHODS: We performed a retrospective development and held-out validation study using the National Trauma Data Bank from 2019 to 2024. Adults with International Classification of Diseases, Tenth Revision fracture diagnoses and classifiable discharge dispositions were included. The primary outcome was non-home discharge to inpatient rehabilitation, skilled nursing facility, long-term acute care hospital, or intermediate care. FORD-II was derived from 41 admission-time predictors using least absolute shrinkage and selection operator regression, converted to a 0-to-10 integer score, locked after derivation, and validated without recalibration. Discrimination and calibration were compared with the original Fracture Orthopedic Risk of Non-Home Discharge score (FORD-I), Geriatric Trauma Outcome Score II, and Trauma Risk Adjusted In-hospital Geriatric Evaluation Score. A hospital-perspective budget-impact analysis explored the potential cost implications of using FORD-II to trigger early discharge planning.

RESULTS […]

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Risk factors and hospital harm: A study of hip fracture patients in the community using population-based data.

Rashidian L, Ali A, Senthinathan A, Bai YQ, Wodchis WP, Bronskill SE, Backman C, Thavorn K et al.

BACKGROUND: Hip fractures are a leading cause of morbidity, mortality, and hospital-related complications among older adults. While many in-hospital complications are preventable, little is known about how individual and neighborhood-level factors contribute to preventable hospital harm in this population. This study examines associations between individual, neighborhood, and organizational-level characteristics and the likelihood of experiencing hospital harm during acute care for hip fracture among community-dwelling older adults in Ontario, Canada.

METHODS: We created a retrospective cohort using linked administrative databases to capture community-dwelling individuals (50+) hospitalized with hip fracture in Ontario, Canada from April 1st, 2008 to March 31st, 2022. We measured associations between preventable hospital harm and demographic, neighborhood-level and organizational risk factors. We included hospital harm as a binary variable (overall harm/no harm) and used multivariable logistic regression, stratified by length of stay (≤ 6 versus > 6 days), to examine associations between risk factors and hospital harm.

RESULTS: Our cohort consisted of 124,570 community-dwelling individuals with a mean age of 80.4 ± 10.6 years. 19.2% of the cohort experienced at least one incident of harm during their acute hospital stay. We observed that regardless of length of stay, individu […]

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Survival and quality of life after treatment of geriatric odontoid fractures.

Oberthür S, Singer AK, Roch PJ, Weiser L, Lehmann W, Sehmisch S

BACKGROUND: Odontoid fractures are the most common cervical spine injuries in geriatric patients and are associated with substantial morbidity and mortality. Their incidence is increasing with population aging. Optimal management remains controversial, particularly due to poor bone quality and comorbidities in this patient group. While surgical treatment is associated with lower nonunion rates, its impact on survival and patient-reported outcomes is unclear.

METHODS: This retrospective single-center study included patients with odontoid fractures who were treated conservatively or surgically between 2017 and 2020. Demographic and clinical data were obtained from medical records. Survival was analyzed using Kaplan-Meier estimates and Cox regression. Health-related quality of life was evaluated in surviving patients using Visual Analog Scale (VAS), Neck Disability Index (NDI), and Short Form-36 (SF-36) questionnaires. Comorbidity burden was determined using the age-adjusted Charlson Comorbidity Index (CACI).

RESULTS: A total of 93 patients were included. The mean age was 78.9 ± 13.7 years, and 75% of fractures were Type II. Surgical treatment was performed in 59% of patients, whereas 41% received conservative management, primarily due to comorbidities. The 1-year mortality rate was 26.9% and increased with age and comorbidity burden. Higher CACI scores and conservative treatmen […]

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External Iliac artery injury in severe pelvic ring trauma: Limb salvage or early hemipelvectomy? A case series.

Yang YJ, Lai CY, Tsai PJ, Chen IJ, Liu CH, Yu YH

PURPOSE: External iliac artery (EIA) injury associated with pelvic ring injury (PRI) is rare but represents a particularly devastating pattern of pelvic trauma, combining life-threatening hemorrhage and critical limb ischemia. Although vascular reconstruction is generally recommended, limb salvage may be unrealistic when severe soft-tissue destruction and contamination coexist. Guidance on abandoning salvage attempts and proceeding to early amputation or hemipelvectomy remains limited.

METHODS: We reviewed our institutional experience with patients sustaining combined PRI and EIA injury between 2013 and 2026. The injury characteristics, treatment strategies, and outcomes were analyzed. The severity of vascular injury, limb ischemia status, and pelvic fracture patterns were classified using established systems. Management decisions were examined within a staged surgical strategy focusing on damage control, stabilization, and reconstruction.

RESULTS: Five patients with high-energy trauma were identified. All patients had severe EIA injuries associated with major soft-tissue damage. Revascularization was attempted in two patients, but limb salvage was unsuccessful in all 5 cases. Despite repeated debridement, four patients ultimately required hemipelvectomy following progressive necrosis and infection. All patients survived after prolonged multidisciplinary treatment.

CONCLUSIO […]

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The biomechanical performance of the both-column screw in posterior acetabular column fractures.

Öztürk V, Hürmeydan ÖM, Çelik T, Bayrak A, Kural C, Bilgili MG

BACKGROUND: The both column screw (BCS) fixation technique has been described as a fixation method for acetabular posterior column fractures. This study aimed to biomechanically evaluate and compare different fixation strategies for acetabular posterior column fractures, including BCS-based, plate-based, and hybrid (BCS + P) constructs, using finite element analysis (FEA) METHODS: Within the scope of FEA, five different internal fixation models were constructed: fixation using a single BCS (sBCS), double BCS (dBCS), a single plate (sP), double plates (dP), and a combination of one BCS and one plate (BCS+P). During the modeling process, mesh generation was performed, material properties were assigned, and boundary and contact conditions were defined. All models were subjected to three physiological loading conditions: level walking, stairs up, and stairs down. Outcomes were evaluated in terms of stress on the screw, stress on the bone, total deformation, fracture gap, and sliding distance in fracture surface.

RESULTS: Biomechanical differences were observed among the five models. The sP model demonstrated the highest deformation and fracture gap values under all loading conditions, indicating inferior biomechanical performance. In contrast, the sBCS and dP models showed lower deformation values and generally more favorable fracture stability. The BCS + P model exhibited variabl […]

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Moral injury in orthopaedic surgery: Distinguishing burnout from system-level ethical conflict.

Saad J, Aboukar M, Quick JC, Saleh A, Saleh KJ

Orthopaedic surgeons experience some of the highest rates of burnout, depression, and suicidal ideation among all surgical specialties, yet existing wellness frameworks may not fully account for the specific nature of their distress. When surgeons are prevented from providing timely, indicated care by system constraints, pressured to participate in clinical decisions that conflict with their professional judgment, or repeatedly exposed to preventable disability and suffering, the resulting harm is not simply exhaustion. It is moral injury-a deeper disruption of professional identity, trust, and conscience that burnout frameworks alone do not capture. This manuscript introduces a three-domain model of moral injury in orthopaedic surgery, linking constraint-based, permission-based, and exposure-based mechanisms to measurable surgeon-level and system-level outcomes. We examine how this framework differs from burnout and moral distress, why orthopaedic surgery is uniquely vulnerable, and what individual, departmental, and institutional interventions are most applicable. A structured research agenda and GRADE-informed recommendations are provided to guide future investigation and departmental implementation.

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Riding the shockwave: Escalating spine injuries from electric micromobility devices in the United States.

Tannoury T, El-Hajj N, Dhunna DP, Khan RR, Ryali R, Zhou OT, Kim M, Tao BS et al.

BACKGROUND: Electric micromobility vehicle (electric scooter, bicycle, hoverboard, skateboard) usage has risen significantly since the mid-2010s, and is associated with rising orthopaedic trauma. Having a narrow base, upright posture, and allowing rapid changes in velocity, these vehicles cause disproportionately more head and spinal injuries than traditional bicycles. Understanding longitudinal trends in emergency department presentations of electric micromobility vehicle related spine injuries provides clinicians with tools to anticipate cases and develop spine-injury prevention strategies as micromobility ridership grows.

METHODS: A retrospective longitudinal study was conducted, using publicly-accessible National Electronic Injury Surveillance System (NEISS) data between 2015-2024. Study inclusion necessitated the documented diagnoses of: fracture, dislocation, sprain or strain, nerve injury, or contusion of the spine, localized as cervical, thoracic, lumbar, sacral, coccygeal, or general back, and caused by electronic scooters, hoverboards, skateboards, and bicycles. Variables included patient demographics of age, sex, and race, contextual factors such as substance and alcohol use, injury location, as well as case narratives, discharge status, causative product, and diagnosis.

RESULTS: A total of 2699 patients who were diagnosed with electric micromobility-related back i […]

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Subtrochanteric varisation explains age-related changes in proximal femoral morphology and has implications for hip arthroplasty and trauma surgery.

Ecker N, Simon M, Lisitano L, Fenwick A, Betsch M, Donhauser M, Perl M, Mayr E et al.

BACKGROUND: Age-related varus remodeling of the proximal femur is well recognized, but its anatomical center of rotation remains unclear. Varus proximal femoral morphology plays a relevant role in fracture patterns, implant positioning, and surgical exposure in geriatric trauma surgery. We investigated whether correlations between the caput-collum-diaphyseal angle (CCD) and trochanteric rotation/offset parameters indicate a subtrochanteric origin of femoral varisation.

METHODS: 100 CT angiographies of the pelvis-leg axis from patients of different ages were analyzed. CCD and the critical trochanter angle (CTA) were assessed in their conventional and modified forms (mCCD/mCTA). Additional parameters included greater (OGT/OGTI) and lesser trochanter offsets (OST/OSTI), femoral antetorsion, and trochanteric anteversion/retroversion. Bone quality was classified according to Dorr.

RESULTS: mCCD showed a significant negative correlation with age, confirming increasing varus alignment in older individuals. CTA and mCTA positively correlated significantly with both CCD and mCCD (CTA/CCD: $p = 0.005$ and CTA/mCCD $p < 0.001$; mCTA: with both CCD & mCCD $p < 0.001$), indicating that increased medial trochanteric overhang is strongly associated with varus morphology. OGT and OGTI demonstrated strong positive correlations with CCD and mCCD (OGT: with both $p < 0.001$, OGTI: with both $p = 0.001$), […]

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Today's perspectives and challenges in moving management of patients with open tibial fractures forward in the Netherlands.

Dibbits MHJ, Rakhorst HA, Stirler V, de Groot R, van Egmond PW, Krijgh DD, Verkerk EW, de Jong T

BACKGROUND & OBJECTIVE: Management of open crural fractures is challenging and requires close interdisciplinary collaboration. Dutch guidelines recommend soft tissue coverage, ideally within 72 h, no later than 1-week post-injury. The guideline does not recommend centralization or volume per center. This poses significant challenges in clinical practice. This study assessed current practice in the Netherlands regarding case volume, organization of care and timing of procedures, while also exploring barriers and facilitators for fracture fixation and soft tissue closure (fix and flap) within 72 h.

METHODS: A national survey was conducted among plastic and orthopedic trauma surgeons, covering January-December 2024. The 54-item questionnaire addressed caseload, logistics, timing of coverage, and perceived challenges. All Dutch level 1 trauma centers were represented.

RESULTS: All 11 of the level 1 centers in the Netherlands responded, in addition we got a response from 13 specialists working in level 2 centers. Eighteen Orthopedic trauma surgeons and 24 plastic surgeons responded. In 63% of centers, at least 10 open tibial fractures were managed annually; 28% reported treating over 20 open tibial fractures per year. Soft tissue coverage was performed within 72 h in only 6% of cases, and within one-week post-injury in 59% of cases. Although 59% of respondents regarded early soft […]

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AI-assisted holosight robotic percutaneous minimally invasive surgery for pelvic fractures: A retrospective cohort study.

Zhu L, Wang H, Qi X, Zhang T, Wang L, Jiao Y, Wang G, Zhao G et al.

BACKGROUND: Pelvic fractures pose significant surgical challenges due to complex anatomy and proximity to neurovascular structures. This study evaluated the clinical efficacy of artificial intelligence (AI)-assisted Holosight robotic percutaneous screw fixation compared to conventional fluoroscopy-guided internal fixation for pelvic fractures.

METHODS: A retrospective cohort analysis was conducted on 89 patients with pelvic fractures treated between September 2016 and December 2024. Patients were divided into two groups: those who underwent robot-assisted percutaneous screw fixation (Robot group, $n = 68$) and those who received conventional fluoroscopy-guided screw fixation (Control group, $n = 21$). In selected cases within the robot group, AI was utilized for preoperative screw trajectory planning. Clinical outcomes including operative time, intraoperative blood loss, fluoroscopy frequency, guidewire insertion attempts, Majeed functional scores, and fracture healing time were compared between groups.

RESULTS: The robot-assisted group demonstrated significantly shorter operative time ($P < 0.01$), reduced intraoperative blood loss ($P < 0.01$), fewer fluoroscopic exposures ($P < 0.01$), and fewer guidewire insertion attempts ($P < 0.01$) compared to the control group. No statistically significant differences were observed in Majeed functional scores between the two groups ($P > 0.05$). […]

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Prehospital pelvic binder use and transfusion requirements in patients with pelvic fractures: A nationwide propensity-matched cohort study.

Matsumoto S, Aoki M, Shimizu M

PURPOSE: Pelvic binders are recommended for suspected pelvic fractures to provide early stabilization and hemorrhage control, but robust comparative evidence regarding their real-world impact on transfusion requirements and subsequent hemorrhage-control strategies remains limited. We evaluated the association between prehospital pelvic binder use, early transfusion requirements, and pelvic angioembolization in a nationwide trauma registry.

METHODS: We conducted a retrospective cohort study using the Japan Trauma Data Bank from 2019 to 2023. Adult patients with blunt pelvic fractures were included. One-to-three propensity score matching was used to adjust for confounding by indication. The primary outcome was 24-hour red blood cell transfusion volume. Secondary outcomes included in-hospital mortality and hemorrhage-control interventions, particularly pelvic angioembolization. Multivariable regression analyses and mixed-effects models were used to assess associations between prehospital pelvic binder use and outcomes.

RESULTS: Among 10,332 eligible patients, 182 patients (1.8%) received a prehospital pelvic binder. After matching, 182 patients with prehospital pelvic binder use and 546 controls were analyzed. Median 24-hour red blood cell transfusion volume did not differ significantly between groups (2.0 [interquartile range, 0.0-10.0] vs. 0.0 [0.0-8.0] units; $P = 0.486$), and […]

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The application of allgöwer-donati suture in obese patients with closed Schatzker type V and VI tibial plateau fractures.

Feng J, Hou J, Wang Y, Lu H, Wang X

OBJECTIVE: To evaluate the efficacy of the Allgöwer-Donati suture in obese patients with closed Schatzker type V and VI tibial plateau fractures.

METHODS: A total of 83 eligible patients treated between October 2024 and February 2026 were enrolled. Patients were stratified by body mass index (BMI) and suture technique into four groups: Obese-Experimental,$n = 17$,Obese-Control,$n = 18$,Normal-Experimental,$n = 24$, and Normal-Control, $n = 24$. Outcomes included a modified ASEPSIS score (defined as the highest daily score during postoperative days 1-7■AP1-7■ and days 7-14■AP7-14■), which has not been formally validated and should be interpreted as exploratory, VAS pain scores on days 3, 7, and 14, and SF-36 physical health domain scores at 12 weeks.

RESULTS: Baseline characteristics were comparable across the four groups. All surgeries were completed uneventfully. For AP1-7 scores, the obese group had significantly higher scores than the normal-weight group (mean difference 1.63, 95% CI: 0.39-2.88, $P = 0.011$), while the experimental group had lower scores than the control group ($P = 0.055$). Two-way ANOVA revealed a significant main effect of BMI group ($P = 0.006$). The number needed to treat (NNT) was 5 in the obese group and 12 in the normal-weight group. For AP7-14 scores, the obese group also had significantly higher scores than the normal-weight group ($P = 0.011$), with no differen […]

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Posterior-only versus combined anterior-posterior stabilization in fragility fractures of the pelvis: Reoperations, perioperative burden, and radiographic outcomes.

Gewiess J, Karampitianis S, Bastian JD, Lustenberger T, Deml MC, Tinner C

BACKGROUND: Fragility fractures of the pelvis (FFP) are increasingly encountered in older adults, but the value of additional anterior ring fixation in surgically treated fractures involving both the anterior and posterior pelvic ring remains uncertain. We compared posterior-only with combined anterior-posterior stabilization in FFP.

METHODS: In this retrospective single-center cohort study, we included patients aged ≥ 65 years who underwent operative stabilization of FFP between 2013 and 2024. Eligible patients had osteoporosis- or low-energy-related fractures with preoperative computed tomography confirmation of combined anterior and posterior pelvic ring involvement. Patients underwent combined anterior-posterior stabilization (group 1, n = 31) or posterior-only stabilization (group 2, n = 24). The primary outcome was reoperation within 2 years. Secondary outcomes were blood loss, operative time, and length of stay. Radiographic outcomes were analyzed longitudinally with linear mixed-effects models.

RESULTS: Posterior-only fixation was not associated with a higher reoperation rate; reoperation occurred in 1/24 patients (4%) in group 2 and 5/31 (16%) in group 1 (odds ratio, 0.23; 95% confidence interval, 0.03-2.02; $p = 0.216$). Posterior-only stabilization was associated with substantially lower blood loss (median, 20 vs 200 mL; $p < 0.0001$) and shorter operative time (46 vs […]

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Regional versus general anesthesia in volar plate fixation of distal radius fractures: An AO/OTA-stratified matched-pair analysis of radiographic reduction quality.

Pieringer A, Hinkelmann MA, Zumsteg J, Pape HC, Allemann F, Barthel C

BACKGROUND: Regional anesthesia is increasingly used for volar plate fixation of distal radius fractures and may offer advantages over general anesthesia. However, its association with radiographic reduction quality remains unclear. This study compared regional versus general anesthesia with respect to radiographic reduction quality and early clinical outcomes using an AO/OTA-stratified matched-pair design.

METHODS: This single-center, retrospective study included operatively treated distal radius fractures from January 2016 to December 2024. A strict 1:1 matched-pair cohort compared regional anesthesia (RA; axillary brachial plexus block) with general anesthesia (GA), using exact matching on AO/OTA main type (A/B/C) and nearest-neighbor matching on a propensity score including age and sex. The primary outcome was radiographic reduction quality assessed across five radiographic parameters and summarized using a composite radiographic reduction quality score (0-5). Secondary outcomes were operative time, complications, Soong grade, and wrist flexion/extension at the last documented follow-up.

RESULTS: Of 797 screened cases, 660 met inclusion criteria. The matched cohort comprised 112 pairs (A=48, B=13, C=51). In AO/OTA type A and B fractures, radiographic reduction quality did not differ between anesthesia groups. In AO/OTA type C fractures, reduction quality score was 3.96±1. […]

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Comparing the epidemiology and associations of single- versus multi-level traumatic spinal fractures at a tertiary referral hospital in Ethiopia.

Mengistu MG, Yazdanpanah S, Damenu H, Baz S, Hanks HE, Dai J, Chandekar A, Cowman AW et al.

INTRODUCTION: Spinal fractures cause major morbidity and mortality globally, especially in low- and middle-income countries (LMICs); however, differences between single- and multi-level injuries remain underexplored. This study describes the epidemiology and associated factors of level-based traumatic spinal fractures in Ethiopia to inform care and policy.

METHODS: A prospectively maintained registry from a tertiary referral hospital in Ethiopia (2023-2026) was retrospectively queried, stratifying spinal fracture patients into single- and multi-level cohorts. Extracted variable analyses included independent t-tests, Pearson's chi-squares, Fisher's exact tests, and multivariable logistic regression.

RESULTS: A total of 223 patients were included: 85 single-level (mean age = 31.9±11.6 years; 83.5% male) and 138 multi-level (33.5±13.7 years; 81.2% male). Baseline demographics and socioeconomic characteristics were largely similar. Multi-level fractures were more frequently due to blunt trauma (23.9% versus 10.6%) and less due to road-traffic accidents (21.7% versus 34.1%; both p = 0.04). Neurological injury was more severe in the multi-level cohort, with higher American Spinal Injury Association (ASIA) A rates (49.3% versus 16.5%), and lower D (4.4% versus 20.0%) and E rates (19.6% versus 40.0%; all p < 0.001). Thirty-day mortality was higher in the multi-level cohort (27.5% ver […]

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The influence of pelvic tilt on regional bone mineral density of the sacrum: A quantitative CT analysis.

Klockner FS, von Stosch D, Jäckle K, Malik HA, Roch J, Meier MP, Lehmann W, Weiser L

BACKGROUND: Spinopelvic alignment plays a crucial role in the biomechanical load distribution of the lumbosacral junction. Pelvic tilt (PT), as a key sagittal alignment parameter, reflects compensatory pelvic rotation and increases with age and degenerative spinal changes. While sacral insufficiency fractures represent a growing clinical problem in elderly populations, the relationship between pelvic orientation and regional sacral bone mineral density remains poorly understood.

PURPOSE: To investigate the association between pelvic tilt and regional trabecular bone mineral density within the S1 vertebra using quantitative computed tomography (QCT).

METHODS: In this retrospective study, 204 patients who underwent CT imaging of the spine and standing radiographs of the lumbar spine between 2017 and 2024 were analyzed. Trabecular volumetric bone mineral density (vBMD) was measured using QCT at five predefined anatomical regions within S1: right lateral mass of S1 (MLR), right sacral ala (ALR), vertebral body S1 (CS1), left sacral ala (ALL), and left lateral mass of S1 (MLL). Pelvic tilt was measured on standing lateral radiographs. Associations between pelvic tilt, bone mineral density, age, and sex were assessed using correlation analysis and multivariable linear regression.

RESULTS: Pelvic tilt increased significantly with age and was higher in women than in men. Trabecular […]

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What is the relative importance of anterior ring intramedullary implant diameter and cortical purchase on outcomes following percutaneous fixation of lateral compression pelvic ring injuries?

Hoskins W, Parola R, Gusho C, Bravin D, Crist B, Wenzel A, Bellamy J, Della Rocca G et al.

OBJECTIVE: Fixation of the anterior pelvic ring improves outcomes in operatively treated lateral compression (LC) fractures (AO/OTA 61-B1/B2). Despite a high variability in the choice of intramedullary implants used for anterior ring fixation, the relative importance of implant diameter and cortical purchase (length) remains unclear. This study compared reoperation rates and radiographic outcomes between implants of various diameters and lengths in LC pelvic ring injuries.

METHODS: This retrospective multicenter study included patients from three international centers with acute (<3 weeks) LC injuries treated with percutaneous anterior and posterior ring fixation (January 2019-December 2025). Anterior ring implants (screws, Fasteners, or CurvaFix) were classified by outer diameter (small: 3.5-4.0 mm; medium: 4.5-5.0 mm; large: ≥ 6.5 mm) and purchase (unicortical vs. bicortical purchase), creating six possible groups for comparison. Variables collected included LC subtype, modified Nakatani zone, comminution, displacement, angle, bilaterality, and Mizzou Pelvic Displacement Predictor (MiPDiP) score. The primary outcome was unplanned reoperation (for loss of fixation, deformity correction, nonunion, or symptomatic implant). The secondary outcome was radiographic loss of reduction (>1 cm medial displacement on final healed radiographs vs. immediate postoperative). A sub-analysis […]

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Fall height is inversely associated with mortality in geriatric trauma patients undergoing trauma team activation and whole-body CT: A retrospective cohort analysis.

Erdle B, Pietsch C, Holzfurtner L, Siegel M, Mühlenfeld M, Frodl A, Klingler F, Wagner FC et al.

BACKGROUND: Fall height is commonly used as a surrogate marker of trauma severity, yet its prognostic relevance in geriatric major trauma remains unclear. We evaluated the association between fall height, injury severity, injury patterns, and mortality in elderly trauma patients undergoing trauma team activation and whole-body computed tomography (WBCT).

METHODS: This retrospective single-center cohort study included consecutive patients aged ≥ 65 years undergoing trauma team activation and WBCT after fall-related trauma between 2018 and 2024. Falls were categorized as ground-level (<1 m), intermediate (1-3 m), and high (≥3 m). Injury severity (ISS) and characteristics, clinical outcomes, prognostic geriatric trauma outcome (GTOS) and in-hospital mortality were analyzed. Multivariable logistic regression was performed to identify independent predictors of mortality.

RESULTS: A total of 349 patients were included (mean age 79.2 ± 8.2 years; 62.5% male). Ground-level falls accounted for 35.2% of cases and were associated with older age, a higher comorbidity burden, and more frequent antithrombotic therapy than higher falls. Injury patterns differed substantially between groups: severe head/cervical spine injuries predominated after ground-level falls, whereas thoracic and extremity injuries were more common after higher falls. However, ISS and GTOS did not differ significantly […]

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Injury mechanism patterns, mortality predictors, and resource utilization in pediatric trauma in Karachi, Pakistan: A seven-year registry study.

Lahoti GA, Ismail H, Kewalramani D, Dalal N, Benton J, Choron RL, Englert Z, DeAnnuntis J et al.

BACKGROUND: Pakistan faces a particularly severe pediatric trauma burden due to its large youth population, ubiquitous motorcycle use, and limited traffic safety enforcement. Low- and middle-income countries bear the vast majority of the global injury burden, yet data derived from their trauma registries remain sparse.

METHODS: We conducted a retrospective analysis of a prospectively maintained pediatric trauma registry in Karachi, Pakistan (2018-2024; n = 4576; outcomes cohort n = 3406). Injury patterns and resource utilization were compared across five age groups. Severity-adjusted mortality was benchmarked using observed-to-expected (O:E) ratios derived from the Revised Trauma Score. Multivariable logistic regression identified independent predictors of in-hospital mortality.

RESULTS: Of 4576 patients in the full cohort, motorcycle crashes were the most common mechanism (34%), followed by falls (33%) and pedestrian injuries (12%); falls predominated in children under 10 and motorcycle injuries in adolescents aged 11-18. Males comprised 82% overall and 93% of adolescents aged 15-18. Among 3406 patients with known disposition, crude mortality was 15.1% and varied by age group (χ^2=44.1, $p < 0.001$), ranging from 10.6% in children aged 11-14-34.9% in infants. Resource utilization included blood transfusion (33.4%), operative intervention (20.2%), ICU admission (11.9%), and intu […]

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Residual instability of the radiocapitellar joint after treatment of acute pediatric Monteggia fractures: A single-center experience.

Polat M, Akgün H, Çat G, Ayd■n A

PURPOSE: Pediatric Monteggia fracture-dislocations are clinically challenging injuries due to the risk of radiocapitellar joint (RCJ) instability and subluxation despite treatment. This study aims to evaluate the mid- to long-term clinical and radiological outcomes of acute pediatric Monteggia fractures and to identify factors associated with residual RCJ instability.

METHODS: Thirty-nine patients who met the inclusion criteria and were treated for acute pediatric Monteggia fracture-dislocation between January 2015 and December 2023, with a minimum follow-up of 2 yr, were retrospectively evaluated. Patient demographic data, fracture characteristics, treatment methods, and clinical and radiological follow-up findings were reviewed. Radiological evaluation included measurement of initial ulnar angulation, RCJ reduction, radial head displacement index (RHDI), and elbow carrying angle (CA). Clinical outcomes were assessed using the Mayo Elbow Performance Score (MEPS). Radiological parameters associated with treatment success were analyzed, and cutoff values for initial ulnar angulation were determined using receiver operating characteristic (ROC) analysis.

RESULTS: Successful outcomes were achieved in all patients who underwent surgical treatment, whereas the success rate of conservative treatment was 68.4% (P = 0.008). Initial ulnar angulation was significantly higher in the uns […]

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Post-operative weight-bearing practices following lower limb fracture surgery: A National multidisciplinary survey of concordance with AO foundation guidelines.

Tsohar TN, Khoury A, Barzel O, Dudkiewicz I

BACKGROUND: Post-operative weight-bearing (WB) prescription is the principal tool for regulating fracture mechanical environment. Low adherence to AO Foundation guidelines is documented across settings, yet the occupational, age-related, and semantic determinants of non-concordance in a multidisciplinary trauma workforce remain poorly characterised.

METHODS: A national cross-sectional survey was administered electronically to 79 Israeli clinicians (orthopaedic specialists n = 44, trauma specialists n = 16, residents n = 13, rehabilitation team n = 6) between December 2022 and March 2023. Four standardised clinical vignettes (AO 43A2, 43C1, 43B1 distal tibia; B-type pelvic ring fracture) were used to assess AO-concordant WB responses. Multivariable binary logistic regression, chi-square and Fisher's exact tests were performed.

RESULTS: Per-case AO concordance ranged from 25.3% (AO 43C1, complete articular) to 73.4% (B-type pelvic fracture); only 10.1% were concordant across all four cases. Trauma specialists demonstrated a statistically significant bimodal pattern: lowest concordance on the simple extra-articular C1 vignette (12.5% vs 44.4% non-trauma; Fisher p = 0.022), driven by paradoxical over-restriction (50.0% chose NWB-6 weeks vs 22.2%; p = 0.057), yet highest concordance on the complex pelvic fracture C4 (81.2%). WB escalation philosophy differed significantly by occup […]

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Pediatric firework-related upper extremity injuries: A 10-year review of cases presented to United States emergency departments, 2015-2024.

Zediker CR, Crowl KE, Krul JS, Megafu M, Li X, Parisien RL, Cusano A

CONTEXT: Fireworks remain a prominent feature of U.S. holiday festivities but continue to pose significant risks, particularly to children and teenagers. Although earlier studies have examined patterns of pediatric injuries linked to fireworks before 2015, recent nationwide data are sparse. This investigation seeks to fill that void by evaluating nonfatal upper extremity injuries related to fireworks among pediatric patients seen in American emergency departments from 2015 through 2024. The goal is to provide a current, detailed summary of injury rates, demographic trends, anatomical injury patterns, and age-related vulnerability.

STUDY DESIGN: This descriptive retrospective epidemiological study utilized data from the National Electronic Injury Surveillance System (NEISS), a database that generates nationally representative estimates of emergency department-treated injuries in the U.S. All pediatric firework-related injuries to the upper extremity reported between January 1, 2015, and December 31, 2024, were included in the analysis.

RESULTS: Between 2015 and 2024, 434 pediatric firework-related upper extremity injuries were identified (NE = 16,872). The majority occurred in males (77.9%) and adolescents aged 12-17 years (48.4%). Injuries most commonly involved the hand (48.8%) and fingers (32.7%). Thermal burns were the predominant diagnosis (74.0%), followed by amputations […]

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Impact of MRI on the management of CT-defined AO spine A/B thoracolumbar fractures.

Quilichini JB, Dien E, Ranc PA, Kilani V, Pavan LJ, Desalos K, Filippigh M, Bronsard N et al.

OBJECTIVES: To assess the added value of emergency spine MRI compared with CT alone in patients with AO Spine A/B thoracolumbar trauma without neurological deficit, focusing on clinical management.

MATERIALS & METHODS: This retrospective single-center study included consecutive trauma patients who underwent both CT and spine MRI between November 2023 and October 2025. CT served as the index test and

MRI as the adjunct modality for fracture characterization and posterior ligamentous complex (PLC) assessment. MRI was performed according to an indication-driven institutional protocol. Changes in treatment strategy (conservative, minimally invasive, invasive) before versus after MRI were analyzed. Paired Wilcoxon signed-rank, McNemar, and symmetry tests were used as appropriate. Agreement was assessed using Cohen's κ. Predictors of MRI-driven management change were explored using Fisher's exact test and multivariable logistic regression.

RESULTS: A total of 158 patients were included (114 men; mean age, 35.8 ± 12.4 years; range 18-59). MRI detected a higher number of fractured vertebrae than CT (2.2 ± 1.8 vs. 1.5 ± 1.0; p < 0.001). MRI led to a change in treatment strategy in 19 patients (12%), predominantly reflecting escalation of care (p < 0.001). Management change was driven by MRI-detected PLC injury in 17 patients (partial in 8, complete in 9), by occult spinal cord injury […]

PubMed · DOI

Percutaneous fixation as definitive care for AO spine type B thoracolumbar fractures in polytrauma patients.

Giorgi PD, Ramponi A, Legrenzi S, Villa F, Boeris D, Picano M, Colistra D, Bove F et al.

INTRODUCTION: Percutaneous pedicle screw fixation is considered the standard treatment for thoracolumbar compression fractures. Its role in unstable injuries with anterior or posterior tension band lesion (AO Spine Type B) remains controversial, as open posterior fusion is traditionally recommended. In polytrauma patients, associated injuries and hemodynamic instability may increase the morbidity of open surgery and delay definitive treatment. In this setting, percutaneous stabilization offers a minimally invasive damage-control strategy, allowing restoration of spinal stability and early mobilization of the patient. This study evaluated the safety and mid-term clinical and radiological outcomes of percutaneous posterior stabilization in polytrauma patients with AO Spine Type B thoracolumbar fractures.

METHODS: A retrospective analysis was performed on 43 consecutive polytrauma patients who underwent percutaneous pedicle screw fixation for a total of 48 thoracolumbar AO Spine type B fractures. Inclusion criteria required a minimum follow-up of 12 months. Patients with associated vertebral body fractures scoring ≥7 on the Load Sharing Classification were excluded. Operative time, blood loss and perioperative complications were recorded. Radiological follow-up assessed alignment maintenance, implant failure and hardware-related complications.

RESULTS: Thirty-two patients were m […]

PubMed · DOI

Deep-learning CT reconstruction improves image quality but not diagnostic accuracy for sacral fragility fractures: MRI remains essential.

Naisan M, Brockstedt LA, Grebe AW, Joumah M, Kurz E, Hartung P, Othman AE, Sanner A et al.

PURPOSE: To determine whether modern deep-learning CT reconstruction improves trauma-relevant diagnostic performance for sacral fragility fractures (SFF) and whether optimized CT can reduce the need for MRI in routine clinical practice.

METHODS: In this retrospective paired-reader study, 171 pelvic CT examinations with corresponding MRI (reference standard) were analyzed (fracture prevalence 63.2%). Two blinded readers independently assessed each dataset reconstructed using standard iterative reconstruction and deep-learning reconstruction (DLR). Binary fracture detection and qualitative image quality were recorded. Diagnostic performance was compared using paired statistical testing.

RESULTS: DLR significantly improved perceived image quality and diagnostic confidence across all evaluated domains (p < 0.001). However, no significant differences were observed in sensitivity, specificity, or overall diagnostic accuracy between reconstruction techniques for either reader (all p > 0.65). Diagnostic accuracy exceeded 87% across all conditions. Missed fractures on CT were primarily associated with nondisplaced or marrow-based injuries detectable on MRI.

CONCLUSION: Although advanced CT reconstruction improves image quality and reader confidence, it does not significantly enhance diagnostic accuracy for sacral fragility fractures. Even with optimized reconstruction techniques, CT […]

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Implementation of a thoracolumbar compression fracture remote review protocol safely reduces in-person spine consultations.

Nie J, Di Gerolamo J, Truong N, Grigorian A, Nguyen PD, Chan A, Finney N, Hashmi S et al.

BACKGROUND: Thoracolumbar compression fractures (TLCF) are often managed non-operatively with Thoracic-Lumbar-Sacral Orthosis (TLSO) bracing. This study evaluated the safety of a novel remote review protocol for non-operative TLCF management.

METHODS: A single-center retrospective analysis was performed on adult trauma patients with acute TLCF. The PRE cohort (5/2022-4/2023) received formal bedside consultation, while the POST cohort (6/2023-6/2024) underwent remote review by an on-call spine surgery resident. The primary outcome was the incidence of TLCF-related complications.

RESULTS: Of 120 patients, 79 (65.8%) were in the POST cohort. Demographics, injury mechanisms, vitals, and injury severity score were comparable between cohorts (all $p > 0.05$). Median time to discharge (PRE 28 h vs. POST 26 h, $p = 0.61$) was similar. No patients developed TLCF-related complications including neurological deficits or TLSO brace-related complications in either group.

CONCLUSIONS: Implementation of a remote review protocol for low-risk acute TLCF appears safe with similar outcomes and time to discharge compared to traditional bedside consultation.

LEVEL OF EVIDENCE: Level III.

PubMed · DOI

Epidemiology and injury characteristics of pelvic ring injuries at a level 1 trauma centre: A 10-year retrospective cohort study.

Kumar PP, Ramesh P, Imran A, Devendra A, Dheenadhayalan J, Rajasekaran S

PURPOSE: Contemporary epidemiological data describing the spectrum of pelvic ring injuries in developing trauma systems remain limited. This study evaluated the epidemiology, injury characteristics, management patterns, and early outcomes of pelvic ring injuries treated at a high-volume Level 1 trauma centre.

METHODS: A retrospective cohort study of consecutive patients with pelvic ring injuries treated at a Level 1 trauma centre between January 2014 and December 2024 was conducted. Demographic characteristics, injury patterns, associated injuries, physiological status, complications, mortality, length of hospital stay, and readmission were evaluated.

RESULTS: A total of 1451 patients with pelvic ring injuries were included, of whom 68.6% were male and 80.7% were aged ≤ 60 years. Road traffic accidents were the predominant mechanism of injury (67.5%), followed by falls from height (21.6%). Among classifiable adult injuries, lateral compression fractures were the most common pattern (79.1%). Combined pelvic ring-acetabular injuries were identified in 21.6% of patients, and 23.8% of patients met the predefined criteria for polytrauma. Lower limb fractures (33.6%), upper limb fractures (29.1%), and head injuries (28.9%) were the most frequent associated injuries. Most patients were managed non-operatively (81.9%), while 18.1% underwent surgical treatment. The overall mortality r […]

PubMed · DOI

Should we plate or IM nail simple displaced transverse proximal humeral fractures?

Gibbon S, Rangel ET, Buckley R

PubMed · DOI

Collagen sponge combined with recombinant human basic fibroblast growth factor for refractory venous leg ulcers: A prospective controlled study evaluating healing, skin graft avoidance, and costs.

Zhu Y, Lai Q, Yang S, Kang R, Wei Z, Cheng J

BACKGROUND: Refractory venous leg ulcers (VLUs) represent a significant clinical challenge, often requiring surgical intervention with skin grafting. This study evaluated whether a collagen sponge combined with recombinant human basic fibroblast growth factor (rh-bFGF) could accelerate healing, reduce the need for skin grafting, lower costs, and improve scar quality compared with standard alginate dressings.

METHODS: This prospective controlled study enrolled 120 patients with CEAP C6 VLUs refractory to standard compression therapy. Patients were allocated to treatment with either a collagen sponge saturated with rh-bFGF (n = 60) or standard alginate dressing (n = 60), both under compression therapy. The primary endpoint was complete wound healing at 4 weeks. Secondary endpoints included time-to-healing (analyzed by Kaplan-Meier with log-rank test), need for skin grafting, total hospitalization costs, and scar quality assessed by the Vancouver Scar Scale (VSS) at 3 months. Multivariable Cox regression was used to adjust for potential confounders.

RESULTS: The CS + rh-bFGF group achieved a significantly higher 4-week healing rate (85.0% vs. 55.0%; risk difference: 30.0%, 95% CI: 15.2-44.8; p < 0.001). Median healing time was shorter in the treatment group (26 days vs. 44 days; log-rank p < 0.001), with an adjusted hazard ratio of 2.54 (95% CI: 1.71-3.78). No patient in the tre […]

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Carpal tunnel release during distal radius fracture fixation is not associated with reduced subsequent carpal tunnel release: A propensity-matched cohort study.

Dinesh A, Poursalehian M, Marjoua Y, Billow D, Padubidri A, Cereijo C, Peirish RW, Styron J et al.

PURPOSE: Concomitant carpal tunnel release (CTR) during distal radius fracture (DRF) fixation is performed with the goal of preventing symptomatic carpal tunnel syndrome (CTS). However, evidence supporting this practice remains limited. This study evaluated whether CTR at the time of DRF open reduction and internal fixation reduces subsequent CTR procedures and other associated surgical complications.

METHODS: A retrospective cohort study using the TriNetX database identified adult patients who underwent DRF open reduction and internal fixation between 2010 and 2025. Patients were categorized based on whether they received CTR at the time of DRF fixation. Propensity score matching controlled for demographic and clinical variables, including fracture complexity. The primary outcome was subsequent CTR procedure within 2 years. Secondary outcomes included 90-day complications (emergency department visits, readmission, hematoma) and 1-year surgical complications (infection, malunion/nonunion surgery, revision surgery).

RESULTS: Among 53,565 patients identified, 5090 matched pairs were analyzed. Patients who received CTR at the time of DRF fixation did not have significantly different rates of subsequent CTR at 1 year (1.75% vs 1.61%; HR 1.112, 95% CI 0.824-1.501) or 2 years (2.24% vs 1.79%; HR 1.306, 95% CI 0.992-1.721). CTR at the time of DRF fixation was associated with higher […]

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Arterial hepatic perfusion is preserved during early resuscitation after hemorrhagic shock in a large animal polytrauma model.

Hübner CT, Klingebiel FK, Kalbas Y, Ricklin J, Bästlein C, Stoeck CT, Weisskopf M, Cinelli P et al.

BACKGROUND: Hemorrhagic shock is a major contributor to morbidity and mortality after polytrauma and plays a central role in the development of multiple organ dysfunction syndrome (MODS). The liver is particularly vulnerable due to its high metabolic demand and dependence on adequate oxygen delivery, and hepatic dysfunction is associated with poor outcomes after trauma. This study aimed to assess hepatic perfusion and biochemical liver function following hemorrhagic shock without direct hepatic trauma in a standardized porcine polytrauma model.

METHODS: 32 landrace pigs were used in the current study and randomized into four groups (n = 8): Tissue trauma group received a blunt chest injury and bilateral femur fractures. In the shock group, hemorrhagic shock was induced and maintained at a mean arterial pressure of 35 mmHg for 1.5 h by controlled blood withdrawal. The polytrauma group underwent both tissue trauma and hemorrhagic shock. 8 sham animals served as control. At the beginning of resuscitation, hepatic perfusion was assessed using spectral CT-based iodine density imaging. Regions of interest were placed in homogeneous liver parenchyma (segments III, IV, and VII) by a blinded investigator. Serum transaminases (AST, ALT, GGT), bilirubin, and lactate were measured at defined time points.

RESULTS: Arterial hepatic perfusion during resuscitation did not differ significantl […]

PubMed · DOI

Impact of osteoporosis on outcomes following ankle open reduction and internal fixation.

Guruprasad P, Sivaram P, O'Neill C, de Cesar Netto C, Adams S, Anastasio AT

OBJECTIVES: To determine whether osteoporosis is an independent risk factor for adverse postoperative outcomes following open reduction and internal fixation (ORIF) of ankle fractures in adults aged ≥ 50 years.

METHODS: Design: Retrospective cohort study using the TriNetX Research Network with 1:1 propensity score matching.

SETTING: Multicenter, international research setting using the TriNetX federated electronic health records database, which includes data from > 100 healthcare organizations across academic and non-academic institutions. Patient Selection Criteria: Adults aged ≥ 50 years who underwent ankle ORIF between January 1, 2004, and December 31, 2024, were identified using International Classification of Diseases, Tenth Revision (ICD-10) and Current Procedural Terminology (CPT) codes for ankle fractures and low-energy falls. Patients with pathologic fractures or additional non-ankle lower-extremity fractures were excluded. Two cohorts were created for comparison: patients with a prior diagnosis of osteoporosis and patients without prior diagnosis of osteoporosis. Outcome Measures and Comparisons: Postoperative complications, including malunion/nonunion/delayed healing, hardware failure or revision, readmission, infections, wound dehiscence, and pulmonary embolism, were assessed at 90 days, 1 year, and 3 years.

RESULTS: Across all time points, no statistically signi […]

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Outcomes of conservative management of CT-diagnosed hemothorax in chest trauma: A systematic review and meta-analysis.

Alomar Z, El-Khatib A, Khreibeh M, Refai A, Alomari M, Alomar A, El-Menyar A, Rizoli S et al.

BACKGROUND: Hemothorax is one of the most common findings in traumatic chest injuries. Emerging evidence suggests that small-volume hemothoraces (less than 300 cc) diagnosed by CT scan can often be managed successfully with conservative (non-interventional) treatment. This choice is clinically important, as chest tube drainage may itself lead to complications. The aim was to evaluate the outcomes and complications of conservative management in patients with small-volume hemothorax.

METHODS: This systematic review and meta-analysis followed PRISMA guidelines. We searched PubMed, Scopus, and the Cochrane Library for studies on CT-diagnosed traumatic hemothorax in stable patients managed conservatively. Articles meeting the inclusion criteria were reviewed. The main outcome measured was the failure rate of conservative management.

RESULTS: Of 3089 articles screened, 10 studies met inclusion criteria, reporting on 3030 patients with CT-diagnosed hemothorax. Among these, 1679 patients (55.4%) were managed conservatively, achieving an 81% success rate. Most cases (94.4%) involved a hemothorax volume under 300 cc, and 409 patients had an associated occult pneumothorax. Delayed tube thoracostomy was needed in 19% (319/1679) of conservatively managed patients; only 1.8% required further surgical intervention. The primary indications for delayed tube thoracostomy were progression of he […]

PubMed · DOI

Healing indices are poorly standardized in distraction osteogenesis literature.

Huang AY, Bernstein M, Blair JA, Rosslenbroich S, Scolaro J, Marecek GS

BACKGROUND: Distraction osteogenesis is the process by which tissue is created through gradual distraction at a regular rate and rhythm. It is commonly used for bone lengthening due to congenital limb differences, malunion, and bone defects. As the timing and complexity of this procedure are necessarily dependent on the length gained or transported, surgeons have utilized various healing indices to describe their results including bone healing index (BHI), lengthening index (LI), consolidation index (CI), and external fixation index (EFI). Accurate and consistent definitions are therefore needed to compare outcomes across studies to eliminate the variability in the reporting of these important indices. This review seeks to evaluate the current literature on the accuracy and consistency of how these indices are defined.

METHODS: Studies using the terms BHI, LI, CI, or EFI published prior to December 31, 2025 were searched and screened using the Pubmed/Medline and Web of Science databases. Separate searches were performed for each healing index; 62 articles reporting BHI, 41 for LI, 63 for CI, and 145 for EFI, met inclusion criteria.

RESULTS: Across the studies reporting on BHI, 29.0% did not explicitly define BHI and another 22.6% defined the parameter incorrectly, mostly confusing the term with EFI (78.6%). In 29.3% of the articles that included LI, a definition was not provi […]

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ABCB5■ mesenchymal stem cells enable high-ratio cellular augmentation of bone grafts without loss of osteogenic function.

Hofmann J, Bewersdorf TN, Sommer U, Borcherding K, Thiel K, Ganss C, Gerstner M, Frank MH et al.

Autologous bone grafting remains the standard for reconstruction of bone defects but is constrained by limited graft volume and loss of osteogenic capacity when expanded with acellular materials. We hypothesized that allogeneic ABCB5■ mesenchymal stem cells (MSCs) can serve as a biologic cell source to augment bone marrow-derived MSC (BM-MSC) populations without impairing osteogenesis. Primary human BM-MSCs from five donors were combined with GMP-manufactured ABCB5■ MSCs at graded replacement ratios (0-95%) and cultured under osteogenic conditions for 21 days. Osteogenic potential was evaluated using complementary assays of mineralization, biochemical activity, and matrix composition. Robust osteogenesis was maintained across all groups, with cultures containing 50% ABCB5■ MSCs demonstrating comparable functional osteogenic activity to BM-MSC controls, including similar 99mTc-HDP uptake (2.96 vs. 2.68 MBq) and calcium-phosphate ratios (1.2 vs. 1.1). Higher substitution levels showed a progressive decline in osteogenic output, consistent with dilution of BM-MSC-driven effects. These findings establish a quantitative threshold for biologic augmentation and demonstrate that substantial replacement of BM-MSCs is feasible without loss of function. This approach may enable cell-based expansion of bone grafts and supports further evaluation in translational in vivo models.

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Surgical outcomes in an improvised underground operating room during missile attacks.

Mitchnik IY, Essa A, Heinig O, Silberzweig JG, Ben Sira Y, Rabau O

PURPOSE: Armed conflict disrupts healthcare when hospital infrastructure is vulnerable to missile attacks. During a 12-day missile strike period, our hospital established an improvised operating room (OR) in an underground parking lot. This study evaluated perioperative and short-term outcomes among patients treated in the improvised OR compared with those treated in the hospital's fortified standard OR.

METHODS: This retrospective cohort study included patients undergoing general or orthopaedic surgery during the 12-day conflict. Procedures identified via surgical logs were categorized by OR location (improvised or standard). Demographic, perioperative, and post-discharge data were collected from electronic records. Primary outcomes included operative time, estimated blood loss, transfusions, complications, hospital and ICU length of stay, ED visits, readmissions, and 3-month mortality.

RESULTS: Overall, 74 surgeries took place in the improvised OR and 77 in the standard OR. Patients in each group were comparable in age, sex, comorbidities, and anesthesia. Improvised OR surgery time was longer (mean 68.1 vs. 47.7 minutes, $p < 0.001$). There were no significant differences between groups in terms of blood loss, transfusions, postoperative complications, or hospital length of stay. After 3-month follow-up, there were no significant differences in readmissions or mortality rates […]

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Repeated scoring with the adult appendicitis score improves the sensitivity and the specificity of appendicitis diagnosis in patients with early equivocal signs of appendicitis: a secondary analysis.

Lastunen KS, Leppäniemi AK, Mentula PJ

PURPOSE: The utilization of computed tomography in the early stage of acute appendicitis may result in overdiagnosis and unnecessarily expose patients to ionising radiation. The Adult Appendicitis Score (AAS) can be used to select patients for imaging. Observation and re-scoring in the DIAMOND trial reduced the need for imaging. Now, we wanted to determine if the change in AAS (ΔAAS) can serve as a diagnostic tool to select patients for imaging even more precisely.

METHODS: Eighty-eight patients with early equivocal appendicitis participated in the observation arm of the DIAMOND trial. The data for these patients were reanalysed, and ΔAAS during the observation was calculated. The baseline AAS, final AAS, and the change in C-reactive protein (ΔCRP) were selected as reference standards.

RESULTS: Eighty-three patients with complete data were included in the analysis. The AUROC (Area Under the Receiver Operating Characteristic) values are as follows: ΔAAS, 0.932 (95% CI 0.868-0.996); baseline AAS, 0.629 (95% CI 0.498-0.760); final AAS, 0.936 (95% CI 0.886-0.987); and ΔCRP, 0.796 (95% CI 0.696-0.897). Using receiver operating characteristic curves, we established the thresholds for low (AAS ≤ -2), intermediate (AAS -1 to 0), and high (AAS ≥ 1) probability of appendicitis. The negative predictive value for the low-probability group and the positive predictive value for the high-pr […]

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Long-term patient-reported outcomes and quality of life after tibiofemoral knee dislocations in patients with open growth plates: a retrospective case series from a Level 1 trauma center.

Hepp P, Youssef Y, Henkelmann R

BACKGROUND: Tibiofemoral dislocations in children and adolescents are rare, and treatment guidelines are primarily based on adult data and algorithms. This case series of four pediatric and adolescent patients aims to add to the limited literature and highlights key aspects of diagnosis, treatment, and patient-reported outcomes.

METHODS: We conducted a retrospective-prospective single-center case series of pediatric and adolescent patients (< 18 years, open growth plates) treated for tibiofemoral knee dislocations at a Level 1 trauma center between 2017 and 2019. Clinical, radiographic, and intraoperative data were reviewed. In addition, long-term patient-reported outcomes covering both functional recovery and health-related quality of life were assessed using KOOS, IKDC, and EQ-55 D-L.

RESULTS: Four patients (2 male, 2 female), aged between 12 and 16 years, were included in this case series. The mean follow-up was 7.3 years (range 6 to 8 years). All patients sustained Schenck Type III-L dislocations. Early MRI and vascular imaging identified frequent associated injuries such as meniscal and chondral lesions, avulsion fractures, in one case, peroneal nerve injury. A staged surgical approach was used in two of the four cases. No growth disturbances or deformities were observed in the patients based on the available clinical and radiological follow-up. Functional outcomes range […]

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The need for, feasibility of, and barriers to implementation of a trauma traveling fellowship: a survey of ESTES members.

Wohlgemut JM, Gupta S, Cioffi SPB, Llaquet-Bayo H, Wohlgemut HS, Hobby J, Mohseni S, Mariani D et al.

PURPOSE: Many European surgeons lack access to high-volume trauma training. A Trauma Traveling Fellowship, supported by the European Society of Trauma and Emergency Surgery (ESTES), could broaden educational and clinical exposure to trauma care, but feasibility and implementation factors are unclear. The aims of this survey are to assess the aspirations, expectations, and perceived challenges of prospective fellows and host centres for an ESTES Trauma Traveling Fellowship.

METHODS: A cross-sectional, web-based survey was distributed to all ESTES members, capturing demographics, fellowship aspirations, perceived barriers, and host centre capacities. The survey incorporated Likert scales, free-text responses, and was guided by the Consolidated Framework for Implementation Research (CFIR) and the Reach, Effectiveness, Adoption, Implementation, and Maintenance / Practical Robust Implementation and Sustainability Model (RE-AIM/PRISM).

RESULTS: Of 903 invited, 210 responses (23.3%) were analysed. The median age was 38 years; 71% were consultants, and 29% were residents. Most (71%) would consider applying for a Fellowship, and 47% stated their centre could host Fellows. Only 23% currently access funded fellowships. Key aspirations included hands-on participation in trauma care, although hosts could more often only guarantee observational roles. Major barriers included funding and in […]

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Antegrade transcystic balloon dilatation in emergency laparoscopic cholecystectomy with MRCP validation.

Lipping E, Saar S, Kirss J, Aaspõllu H, Bahhir A, Gapejeva O, Kangur T, Rauk M et al.

PURPOSE: Endoscopic retrograde cholangiopancreatography (ERCP) remains the predominant treatment of common bile duct stones but single-stage laparoscopic management via antegrade transcystic balloon dilatation (ATBD) offers a compelling alternative. All prior studies validate success solely by intraoperative fluoroscopic cholangiography, with no prospective trial employing postoperative magnetic resonance cholangiopancreatography (MRCP).

METHODS: This single-centre prospective study enrolled 100 consecutive patients with confirmed CBD stones requiring emergency laparoscopic cholecystectomy (LC). ATBD was attempted in all patients. Primary outcome was MRCP-confirmed CBD stone clearance. Secondary outcomes included operative time and postoperative complications.

RESULTS: One hundred consecutive patients were enrolled (median age 69 years; 38% male; 56% ASA III-IV). ATBD was performed in 88 patients (88%). Postoperative MRCP was performed in 86 of 88 patients (98%), with residual stones identified in 24 (28%; 95% CI 19.5-38.2%), corresponding to an MRCP-confirmed CBD clearance rate of 72% (95% CI 61.8-80.5%). The median operative time was 90 min (IQR 75-111.5). Among the 88 patients who underwent ATBD, postoperative complications occurred in 12 (14%), most commonly pancreatitis (4.5%) and deep surgical site infection (4.5%); one patient died following severe post-procedural panc […]

PubMed · DOI

Closed reduction and percutaneous fixation versus open reduction and internal fixation for displaced medial malleolar fractures: a randomized controlled trial.

Elkohail A, Elkasaby A, Shaheen AMZ, Teama M, Raza ML, Ibrahim IA

OBJECTIVE: To evaluate the radiographic and functional outcomes of closed reduction and percutaneous fixation (CRPF) compared to open reduction and internal fixation (ORIF) in the treatment of isolated displaced medial malleolar fractures.

METHODS: This randomized controlled trial included 50 adult patients with isolated displaced medial malleolar fractures, Herscovici classification type B and C, across two hospitals. Participants were randomly assigned (1:1) to either CRPF (Group A, n = 25) or ORIF (Group B, n = 25). The principal objectives were radiological union and clinical function evaluated by the American Orthopedic Foot and Ankle Society (AOFAS) score at a 6-month follow-up. Secondary outcomes included the incidence of postoperative complications and correlations with patient demographics and mechanism of injury.

RESULTS: At 6 months, there was no significant difference between groups (p > 0.05); in Group A, 13 patients rated as "Good," 11 as "Fair," and 1 as "Poor," while in Group B, 15 were "Good," 9 "Fair," and 1 "Poor." AOFAS outcomes showed 13 excellent, 7 good, 4 fair, and 1 poor in Group A, versus 12 excellent, 9 good, 3 fair, and 1 poor in Group B, with no significant difference (p > 0.05). Complication rates were similar: wound infection (CRPF 4% vs. ORIF 12%), malunion/non-union (8% vs. 4%), hardware discomfort (12% vs. 20%), and deep venous thrombosis (4% […]

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Long-term radiological and clinical outcomes after surgical treatment of non-union: a ten-year follow-up study.

Böpple JC, Berger J, Ferbert T, Bewersdorf TN, Schmidmaier G, Findeisen S

PURPOSE: Non-unions are a major challenge in trauma surgery. They are known to affect patients' lives for years after the initial trauma. Nevertheless, most studies only examine the first two years after treatment. The question what the long-term results look like ten years after surgery was posed in light of observations made in everyday clinical practice. To answer this, our study was designed to examine all patients who underwent non-union surgery in our department ten years ago.

METHODS: All patients who underwent surgery between July 2014 and October 2015 were invited to participate in the study. During a one-site appointment, an X-ray image was taken, which was then used to assess consolidation. The patients underwent a physical examination and answered a questionnaire, covering topics such as pain on a visual analogue scale (VAS) and overall quality of life with the short-form 12 (SF-12).

RESULTS: A total of 54 patients were included in the study. The median follow-up period was 122 months (IQR: 118-124). After 10 years 88.9% patients showed osseus consolidation with a median Lane-Sandhu-Score (LSS) of 4 (IQR: 3-4). The Wilcoxon signed-rank test revealed a significant difference compared to 24 months postoperatively ($Z = 4.716$, $p < 0.001$). The difference in the consolidation showed a marginally significant result in the McNemar test with $p = 0.063$. A total of 51 patien […]

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Artificial intelligence for predicting 30-day mortality after emergency laparotomy.

Bedrikovetski S, Murshed I, Seow W, Bunjo Z, De Fontgalland JS, Traeger L, Vather R

PURPOSE: Accurate pre-operative risk prediction in emergency laparotomy (EL) is essential for appropriate resource allocation and improve patient outcomes. This study aimed to establish the accuracy of an artificial intelligence (AI) model in predicting 30-day mortality after EL using a national dataset.

METHODS: Data was extracted from the Australian and New Zealand Emergency Laparotomy Audit-Quality Improvement (ANZELA-QI) database from 1 July 2018 to 24 July 2023. AI models were created employing multilayer perceptron (MLP) and radial basis function (RBF) architectures. Extended AI models were developed using all available pre-operative variables and simplified AI models were developed using statistically significant variables identified through univariate analysis. We assessed the performance of AI models with that of a multivariate logistic regression (LR) and the UK-based National Emergency Laparotomy Audit (NELA) models using the area under the receiver operator characteristic curve (AUROC).

RESULTS: Data from 8293 ELs were included in the model and were randomly divided into a training dataset of 6619 (80%) and a testing dataset of 1674 (20%). There were 537 (6.5%) deaths 30 days after EL. In the testing dataset, the NELA model performed the best (AUROC 0.836), followed by the multivariate LR model (0.824), MLP simplified (0.817). MLP extended (0.802), RBF extended (0 […]

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Predicting postoperative course in Fournier's gangrene surgery: evaluation of the E-PASS score in a single-center retrospective study.

Gercek O, Senkol M, Ulucak MA, Yazar VM

BACKGROUND: The E-PASS scoring system, which incorporates preoperative data and surgical stress factors, may provide a reliable risk classification for predicting mortality risk in patients with Fournier's gangrene. This study aimed to investigate the value of the E-PASS score in predicting mortality in patients with Fournier's gangrene.

METHODS: The study included 41 patients who were diagnosed with FG clinically and underwent surgery. Preoperative demographic and clinical data, laboratory values, FGSI, and E-PASS scores were recorded. The patients were divided into two groups: 8 (19.5%) patients who developed mortality and 33 (80.5%) patients who were discharged without mortality. Demographic, clinical, and laboratory data, as well as FGSI and E-PASS scores, were compared between the groups.

RESULTS: In the mortality group, CRP, serum creatinine levels, FGSI, E-PASS PRS, and E-PASS CRS scores were observed to be significantly higher, while serum albumin and hematocrit levels were lower compared to the other group (respectively, $p = 0.020$, $p = 0.001$, $p < 0.001$, $p < 0.001$, $p < 0.001$, $p = 0.001$, $p < 0.001$). FGSI, E-PASS CRS, CRP, and serum albumin values were found to have diagnostic value in predicting mortality. The cut-off values were determined as 7.5 for FGSI, 0.55 for E-PASS CRS, 186.25 for CRP, and 2.92 for albumin.

CONCLUSION: Our findings show that E-PASS can provide […]

PubMed · DOI

Thoracoscopic internal fixation, open surgical internal fixation, and conservative treatment of rib fractures: differences in postoperative rehabilitation and chest volume (a randomized controlled trial).

Li T, Zhao P, Li W, Liu B

BACKGROUND: Currently, the standard treatment for multiple rib fractures is still under debate. This study explores the efficacy of three treatments (thoracoscopic internal fixation, open surgical internal fixation and conservative treatment) for multiple rib fractures and their influence on thoracic cavity volume and postoperative rehabilitation.

METHODS: Patients with multiple rib fractures admitted to Weifang People's Hospital of Shandong Province from March 2019 to March 2021 were enrolled in this study. The thoracic volume changes before and after treatment were measured by three-dimensional computed tomography (CT) reconstruction. Twenty-seven patients in the open operation group underwent internal fixation with the rib cage, and 24 patients in the non-internal fixation group underwent external fixation with the chest band. The recovery of multiple rib fractures was evaluated by comparing the difference in thoracic volume before and after treatment, the incidence of related complications and the Comprehensive Assessment of Generic Quality of Life Inventory-74 (GQOLI-74) score one month after the operation.

RESULTS: After 12 months of follow-up, thoracic volume was significantly different between the thoracoscopic and open fixation groups ($P < 0.001$). One month after the thoracoscopic internal fixation of rib fractures, the scores of physical function, psychological func […]

PubMed · DOI

What is the rate and degree of deformity in femoral diaphyseal nonunion? A CT-based retrospective cohort study.

Nabben J, Wagner RK, Groepenhoff F, Verwest TA, Bont Y, Kloen P, Janssen SJ

BACKGROUND: This study investigated the rate and degree of deformity in femoral diaphyseal nonunion using preoperative bilateral CT-scans.

METHODS: A retrospective cohort study was conducted at a single Level 1 trauma center. Patients aged > 18 years who underwent operative treatment for femoral diaphyseal nonunion between 2015 and 2025 and had preoperative bilateral femur CT-scans were included. Axial, sagittal, coronal, and length differences between the femoral nonunion and the contralateral healthy femur were measured. Torsional deformity ≥ 15° and length discrepancy ≥ 20 mm were considered clinically relevant. Thresholds of ≥ 5° and ≥ 10° were used to identify outliers in coronal and sagittal plane deformity, respectively.

RESULTS: Thirty-two patients met the inclusion criteria (median age 57 years, 63% male). Torsional deformity ≥ 15° was present in 10 patients (31%), including internal rotation in 4 (13%) and external rotation in 6 (19%). Sagittal plane deformity ≥ 10° was observed in 5 patients (16%), with apex anterior angulation in 1 (3%) and apex posterior angulation in 4 (13%). Mean sagittal plane angulation was 9.7° ± 6.7° in the affected limb and 12.5° ± 2.6° in the unaffected limb ($p = 0.017$). Coronal plane deformity ≥ 5° was present in 10 patients (31%), including varus in 8 (25%) and valgus in 2 (6%). Mean coronal angulation was 5.3° ± 6.0° in the affected li […]

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The impact of pre- versus postoperative geriatric consultation in elderly patients with hip fractures.

Aydin FB, Laane DWPM, van Rossum du Chattel A, DiStefano MT, King AH, Wignakumar T, Nwagbata A, van der Velde D et al.

PURPOSE: To determine whether preoperative geriatric consultation was independently associated with lower incidence of delirium for elderly patients with hip fractures compared with postoperative geriatric consultation.

METHODS: This retrospective cohort study was conducted at two urban Level 1 trauma centers in the United States. Patients aged ≥ 70 years with isolated proximal femur fractures treated surgically between 2018 and 2024 were included. Patients with pathological or periprosthetic fractures, non-orthopedic admissions, and those without geriatric consultation were excluded. Patients were categorized based on timing of initial geriatric consultation: preoperative versus postoperative. Delirium was selected as the primary outcome because geriatric consultation most directly addresses modifiable delirium risk factors. Secondary outcomes included the incidence of one or more in-hospital complications, length of stay, 30-day all-cause readmissions, and mortality. Multivariable logistic and linear regression analyses were used to determine the independent association between timing of initial geriatric consultation and primary and secondary outcomes.

RESULTS: Of 1776 patients included, 1382 (78%) received initial geriatric consultation preoperatively and 394 (22%) postoperatively. Median age was 84 years (IQR 78-90), 70% were female, 73% ASA III, and 87% lived at home pr […]

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Volume-dependent detection of free intraperitoneal fluid by FAST: a prospective in-vivo study with sequential fluid instillation.

Güsgen C, Nada A, Kornek MT, Willms A, Schaaf S

PURPOSE: To assess the volume-dependent sensitivity of the Focused Assessment with Sonography for Trauma (FAST) and to analyse regional detection performance under controlled in-vivo conditions.

METHODS: In this prospective single-centre study, sequential intraperitoneal instillation of 0.9% NaCl (250-1500 mL) was performed during elective laparoscopic procedures in 35 adults. Standardised FAST examinations were conducted by two experienced sonographers using transverse and longitudinal views of Morison's pouch, the Douglas recess, and Koller's space. Sensitivity was calculated for each instilled volume and anatomical region. Specificity could not be assessed due to the experimental design with mandatory fluid presence. Regional differences were tested using one-way ANOVA, and a Monte Carlo simulation (1000 resamples) evaluated detection stability across body-mass-index (BMI) groups.

RESULTS: FAST sensitivity increased with volume: 14.3% (95% CI 4.8-30.3) at 250 mL, 37.1% (21.5-55.1) at 500 mL, 74.3% (56.7-87.5) at 1000 mL, and 91.4% (76.9-98.2) at 1500 mL. The highest detection rate occurred in Morison's pouch, followed by the Douglas and Koller recesses. Longitudinal scanning of the Douglas pouch improved sensitivity by > 20% compared with the transverse view. ANOVA confirmed significant regional variability ($F(5, 153) = 4.37$, $p = 0.001$, $\eta^2 = 0.125$). Detection rates decline […]

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When demography challenges policy: the rising burden of proximal femur fractures as a role model and driver for geriatric trauma care in Germany.

Klug A, Münzberg M, Sibbel R

PURPOSE: Germany's hospital reforms will substantially reshape service delivery and capacity planning. Proximal femur fractures (PFFs), the most frequent geriatric fractures, are highly sensitive to demographic aging and therefore serve as a model condition to assess future healthcare system requirements.

METHODS: Nationwide inpatient data from the German Federal Statistical Office (2000-2023) were analyzed for proximal femur fractures (ICD-10-GM S72.0-S72.2). Demographic projections up to 2050 were combined with Poisson regression, exponential smoothing, and autoregressive integrated moving average models to forecast future incidence and case numbers. Model performance was evaluated in a hold-out period (2016-2023) using mean absolute percentage error and root mean square error. Direct costs were estimated using the official German cost-of-illness framework. Staffing and hospital bed requirements were derived from case-based activity times, profession-specific full-time equivalent coefficients, and observed department-specific occupancy rates.

RESULTS: Between 2000 and 2023, annual proximal femur fracture cases in Germany increased from 118,243 to 182,447 (+ 54.3%). Forecasts project a rise to 317,877 cases by 2050 (95% CI 262,436-373,317), corresponding to an increase of approximately 75% compared with 2023. Patients aged ≥ 80 years are projected to account for 74% of all c […]

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Orbital floor fractures: clinical characteristics, mechanisms of injury, and surgical outcomes in a 25-year single-center study.

Kasap ■, Ayd■n MS, Özler G, Ni■anc■ M, Özvurmaz O

PURPOSE: Orbital floor fractures are common injuries following blunt midfacial trauma, presenting as isolated defects or as part of complex craniofacial fractures. This study aimed to evaluate clinical characteristics, fracture patterns, management strategies, and postoperative outcomes in surgically treated patients.

METHODS: This retrospective single-center study included 56 patients who underwent surgical treatment for orbital floor fractures between 2000 and 2025. Demographic data, injury mechanisms, imaging findings, associated fractures, clinical presentation, surgical techniques, and outcomes were analyzed. Minimum follow-up was 3 months. Statistical analysis was performed using IBM SPSS Statistics version 25.0, with $p < 0.05$ considered significant.

RESULTS: Among 56 patients, 80.4% were male, with a mean age of 38.6 years (10-70). Assault (44.6%) was the most common cause, followed by in-vehicle accidents (21.4%) and falls (17.9%). Inferior wall fractures were most frequent (55.4%), followed by medial (28.6%) and lateral (16.1%) involvement; no superior fractures were observed. Common findings included periorbital edema/ecchymosis (75.0%), diplopia (26.8%), and gaze limitation (19.6%); infraorbital hypoesthesia occurred in 12.5%. Associated fractures involved the maxilla (60.7%) and zygoma (46.4%).Surgical treatment included exploration/elevation (46.4%), titanium mes […]

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Epidemiology and healthcare burden of implant-associated complications in Germany: a nationwide population-based analysis.

Anees H, Heiss C, El Khassawna T

PURPOSE: Implant-associated complications remain a major challenge in orthopaedic and trauma surgery, particularly among elderly patients requiring revision procedures or infection management. This study analysed nationwide trends and healthcare burden of implant-associated complications in Germany.

METHODS: Nationwide German hospital discharge data from 2000 to 2023 were analysed and complemented by Diagnosis-Related Group (DRG) data from 2024. Cases were identified using ICD-10-GM code T84 and its subcategories. Outcomes included temporal trends, age-standardised incidence, demographic characteristics, in-hospital mortality, length of stay (LOS), and DRG-based healthcare resource utilisation.

RESULTS: Hospitalisations for implant-associated complications increased from 45,121 in 2000 to a peak of 95,154 in 2011, followed by a decline to 72,846 in 2023. Age-standardised incidence decreased by approximately 31% between 2010 and 2023. Segmented regression identified a temporal breakpoint in 2010.6, with annual percentage changes of + 6.21% before and - 3.55% after the breakpoint. Annual hospitalisations for periprosthetic joint infection increased from 5,367 to 11,215, while osteosynthesis-associated infections increased from 2,469 to 7,059. Mean LOS decreased from 20.8 to 11.8 days, whereas in-hospital mortality remained stable. Compared with low-complexity cases, infection-r […]

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Effect of periprosthetic greater trochanter fractures (Vancouver AG) on primary stability of the femoral stem in total hip arthroplasty: a biomechanical micromotion analysis.

Bruder J, Goronzi M, Cavalcanti Kußmaul A, Wulf J, Pachmann F, Holzapfel BM, Böcker W, Kistler M

PURPOSE: Primary stability is a crucial factor for the long-term success of cementless total hip arthroplasty (THA). Periprosthetic fractures represent an important cause of loss of primary implant stability. While Vancouver AG fractures rarely lead to stem loosening, their effect on the metaphyseal anchoring of the stems has not yet been investigated. The present study aimed to evaluate the influence of a Vancouver AG fracture on the primary stability of the CLS Spotorno stem.

METHODS: Six fourth-generation composite femora were implanted with cementless CLS Spotorno stems (Zimmer, Warsaw, USA). Primary stability was determined by three-dimensional digital image correlation under dynamic loading conditions (300-1700 N, 1 Hz) simulating physiological gait. Micromotion was recorded at six standardized measurement points (proximal, mid, distal, medial and ventral) before and after a simulated Vancouver AG fracture. Data was statistically analyzed using paired one-sided t-tests ($\alpha = 0.05$).

RESULTS: The results revealed no significant increase in micromotion at the primary anchorage zone of the stem. However, a non-significant increase in micromotion was observed at the prosthesis tip (DGventral and DGmedial), where the micromotion values increased from 164.4 µm (± 59.6 µm) and 140.4 µm (± 75.6 µm) to 193.1 µm (± 48.8 µm) and 186.9 µm (± 60.5 µm), respectively.

CONCLUSION: Vanco […]

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Injury patterns, management and outcomes of hemodynamically unstable pelvic fractures at an Australian level 1 trauma centre: a 10-year registry analysis.

Louw L, Raubenheimer K, Blakeney WG, Young S, Balogh ZJ, Weber DG

INTRODUCTION/BACKGROUND: Pelvic ring injuries (PRI) in traumatic shock are associated with over 30% mortality even in high income countries' mature trauma systems, but recent Australian data showed no haemorrhage related mortality in this group. We aimed to describe the epidemiology, management and outcomes of PRI with haemodynamic instability managed in a state's only Level 1 adult trauma centre.

DESIGN: We conducted a retrospective cohort study of all trauma patients with major PRI (Abbreviated Injury Scale (AIS) > 3) and hemodynamic instability (defined as systolic blood pressure (SBP) <90 mmHg or a shock index (heart rate (HR)/SBP) > 0.7) presenting to the state adult trauma centre between 2013 and 2022. Data on demographics, injury patterns, management strategies, and outcomes were collected from the State Trauma Registry and hospital records.

RESULTS: Of 1369 patients with pelvic or acetabular fractures during the 10-year study period, 152 (11%) had major PRI with hemodynamic instability at presentation. Most were male (72%) with a median age of 35 years. Motor vehicle and motorbike crashes accounted for 52% of injuries. The median Injury Severity Score was 33. Massive transfusion protocols were activated in 45% of cases, while angioembolisation was utilised in 5.9%. Overall in-hospital mortality was 7.2% and mortality due to haemorrhage was 2%.

CONCLUSIONS: In our mat […]

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Time to surgery and its impact on length of stay and mortality for neck of femur fractures at a major trauma centre.

Tang LY, Ho SYS, Zhou AK, Geetala R, Krkovic M

PURPOSE: Neck of femur (NOF) fractures are a major public health concern. Timely surgical intervention (current guidelines is within 36 h) is associated with improved survival, shorter hospital stays, and better functional outcomes. However, surgical delays remain common in hospitals, especially in major trauma centres, and their impact on length of stay (LOS) and mortality is not well defined across different NOF fracture subtypes.

METHODS: We conducted a retrospective cohort study of 4415 patients with NOF fractures, classified into intracapsular (n = 2507) and extracapsular (n = 1908). The primary outcome data gathered were LOS (from surgery to discharge) and in-hospital mortality. Regression analyses were performed to assess associations between LOS, and mortality, adjusting for age, sex, Charlson Comorbidity Index (CCI), American Society of Anesthesiologists (ASA) score, intra-operative blood transfusion, and surgical methods.

RESULTS: For intracapsular fractures, delays in time to surgery were significantly associated with increased LOS in patients expected to have short to median lengths of stay, after adjusting for other variables ($p < 0.05$). Delayed surgery was associated with increased mortality for both intracapsular and extracapsular fractures ($p < 0.0001$). Survival curves showed significant differences for the surgical delayed group. After controlling for time to […]

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Pattern of blast injuries. A Systematic Review: Part 1-Improvised, multiple and unspecified explosive devices.

Neubert A, Seidelmann M, Gaeth C, Bieler D

PURPOSE: This systematic review aims to comprehensively identify and categorize the specific injury patterns associated with IEDs and other explosive devices across military and civilian populations to improve medical preparedness, resource planning strategies, and treatment approaches.

METHODS: A search on PubMed und Scopus was conducted on 26. March 2025 for observational studies reporting blast injuries in civilian and military populations of all ages. Studies were clustered by explosion type - Part 1 IED and multiple/unspecified explosives. A narrative data synthesis was performed. Reporting was assessed with the JBI tool for case series.

PROSPERO: CRD42023391736.

RESULTS: 33 studies with 4,373 victims (mainly military personnel) from 2002 - 2022 reported injuries caused by IEDs. The average age was 25.1 years. 97.2% were male. Injuries to the upper and lower extremities were predominant in all study populations followed by head injuries among military personnel, and thorax and abdomen injuries among civilians. 37 studies with 24,600 casualties (mean age 25.7, 85.8% males) were included in multiple/unspecified explosives. Extremities were predominant followed by facial injuries among military personnel, and thorax injuries among civilians.

CONCLUSION: The first part of this systematic review on the pattern of blast injuries shows that injuries to the extremities, includ […]

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Clinical and laboratory predictors of mortality in Fournier's gangrene: prognostic role of the HALP score.

Yavuz A, Ersöz V, Çamöz MS, Yurdakul ZN, ■■deci Tut G, Demirci BF, Koçsoy NS, Dönmez M

OBJECTIVE: Fournier's gangrene (FG) is a rapidly progressive necrotizing soft-tissue infection associated with substantial mortality. This study evaluated the prognostic performance of the Hemoglobin-Albumin-Lymphocyte-Platelet (HALP) score and routinely available clinical and laboratory parameters in patients with FG.

METHODS: This retrospective cohort study included 236 consecutive patients who underwent surgical treatment for FG between February 2019 and May 2025. The primary outcome was in-hospital mortality. Prognostic factors were evaluated using univariate and multivariable logistic regression analyses, and discriminative performance was assessed by receiver operating characteristic (ROC) curve analysis.

RESULTS: The overall in-hospital mortality rate was 12.3%. Non-survivors had significantly lower HALP scores, hemoglobin, hematocrit, albumin, lymphocyte, and monocyte levels (all $p < 0.05$). In univariate analysis, younger age, disease confined to the perianal region, and higher albumin, monocyte, hemoglobin, hematocrit, and HALP values were associated with survival. In the primary multivariable model, advanced age (OR = 0.966, 95% CI: 0.936-0.997; $p = 0.030$) and monocyte count (OR = 5.243, 95% CI: 1.055-26.049; $p = 0.043$) remained independent predictors of survival. Hemoglobin showed the highest discriminative performance (AUC = 0.781), followed by albumin (0.767), he […]

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Surgical stabilization of rib fractures is associated with lower adjusted odds of adverse pulmonary outcomes across RibScore-defined injury severity: a single-center cohort study.

Muldiiarov V, Liu JL, Campbell A, Bouldoukian S, Daro A, Kemp SE, Matos M, Cantrell E et al.

PURPOSE: Surgical stabilization of rib fractures (SSRF) is increasingly used for selected patients with severe chest wall injury, but its relationship with pulmonary outcomes after accounting for RibScore-defined injury severity remains unclear. This study evaluated adverse pulmonary outcomes after SSRF versus nonoperative management (NONOP) among adults with traumatic rib fractures.

METHODS: Single-center retrospective cohort study at an ACS-verified Level I trauma center from January 2016 to April 2023. Adults with CT-confirmed blunt traumatic rib fractures were categorized as SSRF or nonoperative management. RibScore was calculated from initial CT imaging. The primary outcome was a composite of pneumonia, invasive mechanical ventilation for more than 48 h, or tracheostomy. Multivariable logistic regression adjusted for demographics, RibScore, thoracic and extra-thoracic injury severity, calendar year, COPD, and current smoking status.

RESULTS: Among 3,066 patients, 444 underwent SSRF and 2,622 were managed nonoperatively. SSRF patients had greater radiographic chest wall injury severity, including a higher median RibScore than NONOP patients (2 vs. 0; $p < 0.001$). Composite adverse pulmonary outcome occurred in 14.6% of SSRF patients and 16.6% of

NONOP patients. In adjusted analysis, SSRF was associated with lower odds of composite adverse pulmonary outcome (aOR 0.57, 95% C […]

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Physician-staffed prehospital units for severely injured patients in the Stockholm trauma system - is there a survival benefit?

Helmersson K, Lapidus O, Alm T, Bäckström D, Rubensson Wahlin R

BACKGROUND: The role of prehospital physicians for patients with major trauma has previously been examined with mixed results. No studies have assessed the survival benefit of physician-staffed prehospital units in the context of the Stockholm trauma system.

AIM: To investigate the association between exposure to prehospital physicians and mortality for severely injured trauma patients in the Stockholm trauma system.

METHODS: A total of 11,063 trauma patients were obtained from the Swedish Trauma Registry and stratified according to physician exposure in the prehospital setting. Logistic regression was used to evaluate the association between prehospital physician exposure and 30-day mortality and Glasgow Outcome Score (GOS).

RESULTS: No significant reduction in 30-day mortality was observed in the P-EMS group compared to the EMS group (OR 1.06, $p = 0.71$). However, transport with helicopter emergency medical services (HEMS) was independently associated with greater odds of survival. The P-EMS group had significantly lower GOS at discharge compared to EMS (OR 1.72, $p < 0.001$).

CONCLUSION: The potential benefit of prehospital physician exposure for major trauma patients remains uncertain. However, HEMS transport was associated with improved survival. Further research is needed to determine which patients benefit from prehospital physicians and to evaluate the optimal allocati […]

PubMed · DOI

Clinical impact and economic burden of electric scooter-related facial fractures in Barcelona.

Lugilde-Guerbek A, López-Masramon B, García Sánchez V, Andreu-Sola V, Collado-Delfa JM, Pueyo Morillo C, Arteaga A, Grabosky-Elbaile A et al.

PURPOSE: The landscape of facial trauma has evolved with the integration of micromobility devices into urban environments. This study aimed to characterize the clinical profile of electric scooter (e-scooter) accidents compared with traditional etiologies (traffic, assaults, falls) and evaluate the economic burden imposed on the public health system.

METHODS: A cohort of 119 patients who underwent surgery for facial fractures at a Spanish tertiary hospital between 2020 and 2024 was analyzed. To accurately assess economic impact, a specific subgroup of 51 "pure facial fractures"-those with isolated injuries-was utilized for cost analysis.

RESULTS: E-scooter users (10.1% of the total cohort) were significantly younger (mean age 27.9 years, $p = 0.044$), had shorter hospital stays (7.7 days, $p = 0.007$), and lower ICU requirements (16.7%) than the traditional group. Adverse outcomes accounted for 29.4% of cases overall, with a significantly higher rate in the e-scooter group compared to other etiologies (58.3% vs. 26.2%; $p = 0.039$). The median cost for an e-scooter injury was €8,092; though higher than other causes, this was not statistically significant ($p = 0.33$).

CONCLUSION: E-scooters generate complex facial injuries with clinical severity and complication rates comparable to high-energy trauma. The high economic cost and reconstructive complexity, suggest a need for stricter […]

PubMed · DOI

Significant other reported outcome measures within European cohorts of severely injured patients: A systematic review.

Van Ditshuizen JC, Berg LVD, Verhofstad MHJ, Leliefeld PHC, Jongh MAC, Hartog DD

PURPOSE: Major trauma can have a long-term impact, leading to diminished health-related quality of life (HRQoL) for patients and their significant others (SOs). Patient reported outcome measures (PROMs) make this impact visible, providing expectation- and self-management. European studies assessing Significant Other Reported Outcome Measures (SOROMs) in cohorts of severely injured patients were reviewed.

METHODS: A systematic literature search was conducted. European studies from 2000 onwards assessing SOROMs in severely injured patient cohorts were included. Two reviewers independently screened all records, consensus was reached by a third reviewer.

RESULTS: 2,909 studies were screened, resulting in 30 inclusions. Studies originated from Western and Northern Europe, overrepresented by traumatic brain injury (n = 27, 90%). Forty-three different SOROMs were identified. Twenty (67%) studies reported subacute and long-term reduction in HRQoL, and high prevalence (28-56%) of depression, anxiety, and stress within SOs. Outcomes of SOs and patients run parallel and are associated with coping strategies. Sixteen (53%) studies reported around 50% of unmet family needs (information and emotional support), and a disproportionate burden of care associated with loneliness and life satisfaction.

CONCLUSION: SOs of severely injured patients reported reduced HRQoL, with high levels of depr […]

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Early treatment goal uncertainty as a barrier to regional analgesia and respiratory deterioration in traumatic rib fractures: a propensity-weighted analysis.

Yi SB, Wu JH, Zhang XN, Lin DC, Zheng YL

PURPOSE: Patients with multiple rib fractures (≥ 3) admitted to the intensive care unit (ICU) are at high risk of early respiratory deterioration, for which timely and effective analgesia is essential. Although multimodal analgesic strategies are widely recommended, their real-world implementation may be influenced by variability in early clinical decision-making, including treatment goal uncertainty (TGU). We aimed to investigate the association between early TGU, analgesic strategy selection, and respiratory outcomes in critically injured patients with rib fractures.

METHODS: This single-center observational cohort study included 236 adult ICU patients with ≥ 3 radiologically confirmed rib fractures who did not require invasive mechanical ventilation at admission (2020-2024). TGU within the first 48 h was assessed using blinded electronic medical record review as a process-level marker of variability in early care prioritization. Analgesic strategies were categorized as regional-based analgesia (RA), systemic-dominant analgesia (SA), or multimodal conservative management. The primary outcome was respiratory deterioration, defined as a composite of escalation of respiratory support, endotracheal intubation, or pulmonary complications requiring intervention. Associations were evaluated using propensity score overlap weighting and adjusted regression models.

RESULTS: High TGU […]

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Multifocal necrotizing fasciitis: what makes it different? A German trauma center experience.

Keßler F, Frank J, Störmann P, Wagner N, Saman AE, Marzi I, Leiblein M

BACKGROUND: Multifocal necrotizing fasciitis (MulNF) is a rare presentation of necrotizing fasciitis (NF) characterized by involvement of more than one non-contiguous anatomical site during the same disease episode. Comparative data between monofocal and multifocal NF remain limited.

METHODS: We performed a retrospective single-center cohort study of patients treated for NF between 2015 and 2026. Patients were classified as MNF (single site) or MulNF (multiple non-contiguous sites) based on chart review. Demographics, comorbidities, microbiology, laboratory values, organ failure, ICU course, surgical burden, and outcomes were compared.

RESULTS: Forty-seven patients were included, of whom five (10.6%) had multifocal NF. Four of these showed clearly synchronous non-contiguous involvement, whereas one developed a delayed second non-contiguous focus within the first four days of the same disease episode. Compared with MNF, MulNF was associated with higher ICU burden, more frequent mechanical ventilation, and a greater number of debridements and total operations. Higher rates of septic shock, hemodynamic failure, and renal replacement therapy were observed in MulNF. No deaths occurred in the small multifocal cohort, whereas 11/42 patients with monofocal NF died during hospitalization. Given the small multifocal sample size and multiple comparisons performed, all findings should be […]

PubMed · DOI

Trauma-related retrobulbar hematoma: predictive risk factors and clinical outcome in 52 cases.

Bicsák Á, ■ahin H, Brandt L, Hassfeld S, Bonitz L

PURPOSE: Retrobulbar hematoma (RBH) is a vision-threatening emergency. This large-scale study integrates clinical data and 3D analysis to identify prognostic factors, refine diagnosis, and support guideline-based emergency management.

METHODS: This study standardizes terminology, applies AO fracture classification, and uses CT-based 3D segmentation to analyze anatomical patterns, treatment protocols, providing evidence-based guidance for diagnosis, surgical management, and postoperative care.

RESULTS: The Dortmund Maxillofacial Trauma Registry (2018-2025) recorded 52 RBHs among 7,671 trauma patients (0.7%). RBH occurred predominantly in middle-aged males and elderly females, with overproportionate incidence of anticoagulant therapy. Most fractures clustered around the orbital floor, medial wall, zygomatic surface, often near the Frankfurt horizontal plane. RBHs were mainly extraconal; poor outcome was predicted with intraconal location, optic nerve contact, orbital floor fracture, high RBH/orbital volume ratio, and aspirin use. Surgical decompression was performed in 15 cases (28.9%); 65,4% (34 of 52) retained vision. Neural network analysis confirmed fracture site and hematoma position as key prognostic factors over demographic or comorbidity variables.

CONCLUSION: In 7,671 midface fracture cases, RBHs occurred in 0.7% (n = 52), mainly after trauma. Falls predominated in ae […]

PubMed · DOI

Patterns of negative trauma laparotomy worldwide: a sub-analysis of a global dataset.

Laird WLA, Bath MF, Amoako J, Wohlgemut J, Nuño-Guzmán CM, Khajanchi M, Smith BG, Hobbs L et al.

PURPOSE: Trauma remains a major contributor to global disability. While the trauma laparotomy can be a life-saving procedure in abdominal injury, a select proportion are performed with no pathology identified intra-operatively, potentially exposing patients to unnecessary morbidity. We aimed to assess the global prevalence of negative laparotomy and identify any influencing factors at both the patient- and system-level.

METHODS: This was a secondary analysis of the GOAL-Trauma study, a multicentre prospective international observational study on trauma laparotomy patients, conducted from April to December 2024. We defined a negative laparotomy as where no intra-abdominal injuries were identified during the index operation. A multivariable logistic regression was performed to identify predictive factors for a negative laparotomy.

RESULTS: Of the 1769 patients included, 128 patients (7.2%) underwent a negative laparotomy. Regression analysis demonstrated that a penetrating mechanism of injury (OR 2.37, CI:1.53-3.69, $p < 0.001$) and a lower grade of surgeon (surgical registrar relative to consultant - OR 2.91, CI:1.90-4.47, $p < 0.001$) were independent predictors for negative laparotomy. Neither human development index nor hospital resource level impacted negative laparotomy rates.

CONCLUSION: One in every fourteen trauma laparotomies performed in our global cohort was negative, […]

PubMed · DOI

Preoperative stay under 48 h after admission for proximal femur fractures is linked to fewer complications and better outcomes.

Moskal M, Feulner S, Grimm A, Müller-Heck R, Heep H, Godolias P

PURPOSE: To evaluate the association between admission-to-surgery time (with a focus on < 48 h) and postoperative complications and discharge mobility in patients with proximal femur fractures (PFFs), using a large administrative dataset from routine inpatient care in Germany.

METHODS: In this retrospective database analysis, 128,092 patient records from external inpatient quality assurance (North Rhine-Westphalia, Germany, 2016-2020) were included. All patients underwent surgical treatment for PFFs and were stratified by time to surgery as early surgery (Group A; within 48 h of admission) or delayed surgery (Group B; more than 48 h after admission). Two cohorts were analysed separately: hip arthroplasty (HA: early 57,635; delayed 7,504) and osteosynthesis (OST: early 59,466; delayed 3,487). Associations between admission-to-surgery time, postoperative complications, pre-admission anticoagulant therapy and inability to walk at discharge were assessed using descriptive statistics and binary logistic regression. Patients unable to walk before the fracture were excluded from the mobility analysis. Reporting was guided by the STROBE statement.

RESULTS: In-hospital mortality was 4.7% after early versus 9.7% after delayed surgery (OST) and 5.8% versus 9.3% (HA). Delayed surgery was associated with higher odds of postoperative complications (OST: OR 1.21, 95% CI 1.11-1.34; HA: OR 1. […]

PubMed · DOI

Impact of routine use of the GFAP and UCH-L1 TBI blood test for management of mild TBI on patient care in a European emergency department.

Brüning-Wolter F, Wolff N, Schrader M, Musaelyan K, Chandran R, Hoffmann M, Marino JA, McQuiston B et al.

OBJECTIVES: Blood biomarkers glial fibrillary acidic protein (GFAP) and ubiquitin C-terminal hydrolase L1 (UCH-L1) were recently introduced into clinical practice for mild traumatic brain injury (mTBI) assessment. However, real world data demonstrating TBI biomarkers' impact on the emergency department (ED) practice is lacking. This study aimed to describe the experience of a single tertiary hospital ED that introduced TBI biomarkers GFAP and UCH-L1 measurements using Alinity i® TBI test into routine mTBI assessment.

METHODS: Retrospective data were extracted from hospital records of patients who had TBI test ordered in the ED. Test results and subsequent clinical care pathway data (CT order, CT scan result, admission or discharge) were analysed.

RESULTS: Data was available for 460 mTBI patients (median age 43, 54% male). Of the 216 patients (47%) with a negative TBI test, 198 did not have a CT scan, and 18 had a CT scan with no abnormalities detected. Among 244 patients with a positive TBI test, 162 had CT scans, with 12 (6.6%) showing intracranial lesions. Overall, among all 180 scanned patients every patient with intracranial lesion findings had a positive TBI test result. The test could rule out CT scans in 47% of mild TBI patients. In practice, 42% of patients avoided a CT scan based on the TBI test results with additional 12% of patients ruled out based on clinical asse […]

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Electric bicycle versus conventional bicycle crashes: comparative analysis of injury patterns, severity, and the modifying effect of body mass index.

Bauer D, Maier JP, Holzfurtner L, Pietsch C, Wagner FC, Schmal H, Erdle B, Mühlenfeld N

PURPOSE: E-bike trauma is often presumed to result in greater injury severity than conventional cycling, yet evidence remains inconsistent. Furthermore, the impact of factors such as Body Mass Index (BMI) on injury patterns is poorly understood. This study therefore evaluated whether e-bike trauma is associated with increased overall severity or distinct regional injury patterns.

METHODS: This retrospective single-center cohort study analyzed 261 adult trauma-bay patients (e-bike n = 37; conventional bicycle n = 224). Primary endpoints included Injury Severity Score (ISS) and regional maximum Abbreviated Injury Scale (AIS) scores. Secondary endpoints included physiological burden (SAPS II) and resource utilization (LOS, TISS-10). Multivariate logistic regression was used to identify independent predictors of intracranial hemorrhage, e-bike riding and extremity fractures. Event counts and events-per-variable (EPV) ratios are reported for each model.

RESULTS: E-bike riders had a significantly higher BMI than conventional cyclists (26.1 vs. 24.6 kg/m²; p = 0.029). However, the absolute difference of approximately 1.5 kg/m² is modest, and both group medians fall within the normal-to-overweight range. No statistically significant differences were observed in ISS, physiological burden, or resource utilization. Conventional cyclists had a higher proportion of patients with AIS Pelvi […]

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Effect of management in intermediate care unit versus conventional hospitalization on long-term mortality and autonomy in patients with rib fractures aged 75 years and older: a retrospective study.

Vallod E, Debord-Peguet S, Mikail N, Termoz A, Tripoz S, Abrard S, Gueret-du-Manoir M

PURPOSE: Rib fractures in older adults are associated with substantial morbidity and mortality. Management in an intermediate care unit (IMCU) may improve outcomes, but its long-term benefit remains uncertain. We compared IMCU versus conventional management regarding 12-month mortality and autonomy in patients aged ≥ 75 years with rib fractures.

METHODS: We conducted a retrospective observational cohort study in two Lyon teaching hospitals (Hôpital Edouard Herriot and Hôpital Lyon Sud, Hospices Civils de Lyon, France) between January 2015 and December 2019. The primary outcome was a composite of death or institutionalization within 12 months of hospital admission. Secondary outcomes included discharge modality, opioid prescription at discharge, hospital length of stay, unplanned readmission within 6 months, and ADL/IADL scores. Logistic regression was adjusted for hospitalization unit, age, sex, pre-existing institutionalization, and associated injuries.

RESULTS: A total of 196 patients were included: 126 (64%) in the conventional group and 70 (36%) in the IMCU group. Death or institutionalization within 12 months occurred in 31 patients (26%) in the conventional group and 8 (12%) in the IMCU group (p = 0.026). However, after adjustment, conventional hospitalization was not significantly associated with the primary outcome (OR 2.25, 95% CI 0.90-6.09; p = 0.092). Institutional […]

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The road to the Pearl-String Technique: lessons learned from early IMT modifications using RIA autograft and cancellous allograft granules.

Hückstädt M, Scholl C, Langwald S, Klauke F, Mendel T, Kobbe P, Fischer C

BACKGROUND: The induced membrane technique (IMT) is widely used for reconstruction of large bone defects; however, reported outcomes vary considerably among technical modifications. One potential mechanism impairing consolidation is graft sedimentation, particularly when low-density cancellous autograft harvested using the reamer-irrigator-aspirator (RIA) is applied. This study evaluates a standardized modification of the IMT and explores mechanical factors potentially affecting graft consolidation.

METHODS: In this retrospective single-center study, 28 patients were treated following eradication of osteomyelitis using a defined modification of the IMT. Mean follow-up was 7.4 years. Defect length ranged from 3.2 to 15.7 cm. During the second stage, RIA-harvested autologous cancellous bone was combined with lyophilized cancellous bone allograft granules. Definitive fixation was achieved using intramedullary nailing or locking plate fixation; one patient was treated with a ring fixator.

RESULTS: Primary consolidation was achieved in 57.1% of cases. Aseptic nonunion occurred in 10.7%, with secondary consolidation achieved after revision in all affected patients. Recurrent infection was observed in 35.7%; infection control was achieved in 90% of these cases following additional procedures, and reconstruction was ultimately completed using alternative strategies. Median time to fu […]

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Non-tourniquet strategy with high-position low-perfusion in distal humeral fracture surgery: reduces swelling without increasing blood loss.

Wang Z, Ai X, Ren T, Wu B, Zhang K, Li Z, Xue H, Li M et al.

INTRODUCTION: To evaluate the feasibility of non-tourniquet strategy combined with "High-position Low-Perfusion" positioning in distal humeral fractures ORIF (DHF-ORIF), and to investigate its effects on perioperative blood loss, postoperative rehabilitation, and functional outcomes.

METHODS: Prospective inclusion of patients aged 18-45 with distal humeral fractures, randomly divided into an experimental group without tourniquets (Group A) and a control group with traditional tourniquets (Group B). The Group A adopted a lateral position with high and low perfusion of the affected limb, and received intravenous injection of TXA (15 mg/kg) combined with continuous flushing with 4 ■ physiological saline during the operation; The control group received routine use of tourniquets.

PRIMARY OUTCOMES: perioperative blood loss, elbow ROM and degree of limb swelling.

SECONDARY OUTCOMES: Postoperative VAS score, Mayo elbow joint score and DASH score, and incidence of complications.

RESULTS: No significant difference was observed in total perioperative blood loss between the Group A and the Group B (240.78 mL vs. 228.53 mL, $p > 0.05$). However, the Group A demonstrated significantly reduced postoperative drainage volume and hidden blood loss ($p < 0.05$). The Group A exhibited significantly superior outcomes in terms of VAS scores and limb swelling, compared with the control group. Additi […]

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Impacts of intracranial pressure monitoring on mortality and length of stay in severe traumatic brain injury: a systematic review and meta-analysis.

Merakis M, Kim N, Velli G, Balogh ZJ

PURPOSE: This systematic review and meta-analysis examined the association between intracranial pressure monitoring (ICPm) and mortality in patients with severe traumatic brain injury (sTBI), with additional focus on functional neurological outcomes and intensive care unit (ICU) and hospital length of stay (LOS).

METHODS: This review followed Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines and was registered with PROSPERO (CRD42025643607). MEDLINE, Embase, and the Cochrane Central Register of Controlled Trials were searched through August 1, 2025, for studies comparing ICP-monitored and non-ICP-monitored patients with traumatic brain injury. Eligible study designs included randomized controlled trials, prospective observational cohort studies, retrospective observational cohort studies, registry-based cohort studies, and case-control studies. Study quality was assessed using the Strengthening the Reporting of Observational Studies in Epidemiology checklist, Risk of Bias 2.0 for randomized trials, and the Oxford Centre for Evidence-Based Medicine Levels of Evidence framework.

RESULTS: Thirty-one studies met inclusion criteria. The mortality meta-analysis included 121,799 patients. Pooled crude analysis showed lower mortality in ICP-monitored patients than in non-monitored patients (OR 0.79, 95% CI 0.68-0.91, $p = 0.002$), with high heteroge […]

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Prehospital point-of-care lactate for identification of severe injury in trauma team-activated patients: a prospective multicenter cohort study.

Faul P, Hagebusch P, Bertram E, Naujoks F, Störmann P, Schweigkofler U, Koch D

BACKGROUND: Accurate prehospital trauma triage remains challenging despite structured activation criteria, particularly in patients without overt physiologic instability. Prehospital point-of-care lactate has been proposed as a physiologic adjunct to support early identification of severely injured patients. This prospective multicenter cohort study evaluated the association between prehospital lactate and injury severity in trauma team-activated patients within a structured guideline-based trauma system.

METHODS: This prospective observational multicenter cohort study included adult trauma patients transported by emergency medical services to three Level I trauma centers and managed in the trauma room following trauma team activation (TTA). Prehospital lactate was measured by capillary earlobe sampling before or during transport and was not used for clinical decision-making. The primary outcome was severe injury, defined as Injury Severity Score (ISS) ≥16. Secondary outcomes were Abbreviated Injury Scale (AIS) ≥4 in at least one body region and intensive care unit (ICU) admission > 24 hours. Diagnostic performance was assessed using receiver operating characteristic (ROC) analysis and predefined lactate thresholds of ≥2 mmol/L and ≥4 mmol/L.

RESULTS: A total of 108 patients were included, of whom 26 (24.1%) had ISS ≥ 16. Prehospital lactate levels were higher in severely inj […]

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Surgical timing and mortality in acute traumatic subdural hematoma: is the four-hour threshold confounded by injury severity?

Rakip U, Y■ld■zhan S, Canbek ■, Boyac■ MG, Korkmaz S, Öztürk KH, Aslan MZ, Aslan A

PURPOSE: Early surgical evacuation within four hours has long been advocated for acute traumatic subdural hematoma (ASDH), yet the supporting evidence is largely observational and unadjusted for injury severity. We examined whether surgical timing is independently associated with in-hospital mortality after accounting for baseline severity.

METHODS: We retrospectively analyzed 91 consecutive patients who underwent surgical evacuation for ASDH between January 2014 and December 2024. The primary outcome was in-hospital mortality; the secondary outcome was functional status at discharge measured by the Glasgow Outcome Scale (GOS). Surgical timing (≤ 4 vs. >4 h) was assessed through unadjusted comparison, stratification by Glasgow Coma Scale (GCS) score, multivariable Firth penalized logistic regression with 1000-iteration bootstrap internal validation, and sensitivity analyses including alternative timing thresholds (1-12 h), restricted cubic spline modeling, and timing-by-severity interaction testing.

RESULTS: In-hospital mortality was 33.0% (30/91; 95% Wilson CI 24.2-43.1%). Crude analysis showed higher mortality with early surgery (50.0% vs. 15.6%, p = 0.001), but these patients had lower admission GCS and higher rates of CT-defined traumatic axonal injury (TAI) and cerebral contusion. After stratification, no statistically detectable mortality difference was observed within […]

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Association between time to transfusion and outcomes in patients with traumatic haemorrhagic shock: an analysis of a nationwide trauma registry in Japan.

Shibahashi K, Inoue K, Kato T, Sugiyama K

PURPOSE: To investigate the association of in-hospital time to transfusion on outcomes of patients with traumatic haemorrhagic shock.

METHODS: Using data from the Japan Trauma Data Bank (January 2019-December 2023), we identified adult patients (aged ≥ 18 years) with haemorrhagic shock following trauma (systolic blood pressure < 90 mmHg; shock index ≥ 1.0 upon hospital arrival) and received ≥ 10 units red blood cells within 24 h of injury. We excluded patients transferred from or to other hospitals after emergency treatment, with burn injuries or cardiac arrest, without transfusion time records, and with first transfusion > 60 min after hospital arrival. Mixed-effect multivariable logistic regression estimated the association between time to transfusion for in-hospital mortality, adjusting for patient characteristics, comorbidities, and injury severity. Heterogeneity in the association between transfusion timing and mortality was assessed through nine subgroup analyses.

RESULTS: Overall, 698 patients (median age 59 years, 66.2% male) from 132 hospitals were analysed, of whom 265 (38.0%) died during hospitalisation. Mean time to transfusion varied from 4 to 58 min across hospitals. A positive association was observed between time to transfusion and in-hospital mortality. Each 1-min delay in transfusion initiation was associated with higher in-hospital mortality (adjusted odds […]

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Neutrophil phenotyping reflects injury severity when base excess and lactate remain normal.

van Eerten FJC, Duebel LV, de Fraiture EJ, Brands SR, van Wessem KJP, Vrisekoop N, Venhuizen JH, de Groot MCH et al.

INTRODUCTION: Accurate assessment of injury severity is essential for optimal trauma care, as it dictates timing and intensity of interventional strategy. Conventional markers, such as base excess and lactate levels are usually borderline deranged in patients with midrange injury severity. This study explored whether determining circulating neutrophil phenotypes in these patients (ISS 9-35) might aid in the assessment of injury severity.

METHODS: A descriptive single-center cohort study of trauma patients ≥ 16 years between 2013 and 2023 was conducted. Data was collected from the Dutch National Trauma Registry and the Utrecht Patient-Oriented Database (< 10% missing values). Base excess, lactate levels and neutrophil phenotypes were routinely measured in emergency department blood samples. The dynamics of these parameters was analyzed in the context of increasing Injury Severity Scores (ISS).

RESULTS: 807 patients were included, of whom 591 were classified as midrange severely injured. In these patients, banded neutrophils showed a greater fold increase between ISS 9 and 35 compared with base excess and lactate (6.7, 2.7, and 1.5, respectively). Furthermore, the number of banded neutrophils exceeded the upper interquartile range of controls at lower trauma severities compared to base excess or lactate (ISS 17 vs. 30 and 39, respectively).

DISCUSSION: Although base excess and […]

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Limb occlusion pressure-based individualized tourniquet pressure management in patients undergoing internal fixation for extremity fractures: a prospective comparative feasibility study.

Yang J, Hu P, Wang L

OBJECTIVE: This study aimed to evaluate the feasibility of limb occlusion pressure (LOP)-based individualized tourniquet pressure management and to quantify the reduction in applied tourniquet pressure compared with a traditional fixed-pressure protocol in patients undergoing internal fixation for extremity fractures.

METHODS: A prospective comparative study was conducted at Civil Aviation Shanghai Hospital from January to September 2025. Patients undergoing internal fixation for extremity fractures were enrolled and allocated to either the LOP-based group (n = 171) or the traditional fixed-pressure group (n = 235) according to the attending surgical team and operative scheduling, without randomization. The LOP-based group received individualized tourniquet pressure based on preoperative LOP measurement plus a safety margin, while the control group received standardized pressures (40-50 kPa; 300-375 mmHg). The primary outcome was applied tourniquet inflation pressure. Secondary outcomes included demographic characteristics, extremity location, operative duration, and subgroup analyses stratified by extremity location, age, and sex; multivariable linear regression was performed to adjust for potential confounders.

RESULTS: The LOP-based group demonstrated significantly lower mean tourniquet pressure compared with the traditional group (32.82 +/- 3.63 kPa [246 +/- 27 mmHg] vers […]

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Impairment in activities of daily living and textbook outcome after major emergency abdominal surgery.

Mima K, Nitta H, Shikiji Y, Nakashima R, Uemura S, Morinaga T, Itoyama R, Umezaki N et al.

PURPOSE: Textbook Outcome (TO) has been a useful tool to assess benchmark patient-centred surgical outcomes in elective abdominal surgery. This study aimed to evaluate the association of preoperative activities of daily living (ADL) status with TOs following major emergency abdominal surgery.

METHODS: This retrospective study included 301 consecutive patients who underwent major emergency abdominal surgery for high-risk acute abdominal conditions, including perforated peptic ulcer, intestinal ischemia, bowel obstruction, bowel perforation, and severe intraabdominal bleeding or infection, at Kumamoto Regional Medical Center between April 2019 and March 2025. Preoperative ADL was evaluated using the Barthel Index (range, 0 to 100), with higher scores indicating greater independence. Barthel Index ≤ 85 was defined as ADL impairment. According to a recent international consensus, TO was defined as discharge to "home" without major postoperative complications (Clavien-Dindo grade ≥ III). Multivariable logistic regression analysis was conducted to assess the association of preoperative ADL with TO achievement, adjusting for clinical features and surgery-related factors.

RESULTS: Among the 301 patients, 90 (30%) had preoperative ADL impairment (Barthel Index ≤ 85) and 211 (70%) achieved TO. Multivariable analysis revealed that preoperative ADL impairment was independently associated […]

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Early calcium replacement after minimal blood transfusion is associated with adverse outcomes.

Hewgley WP, Griffin RL, Baird E, Hashmi ZG, Baker SJ, Jansen JO, Kerby J, Holcomb JB et al.

PURPOSE: Calcium disturbances are common in patients with traumatic injuries, exacerbated by shock and blood product resuscitation. Recently, empiric calcium supplementation has been recommended with the first unit of blood transfusion, despite a lack of data supporting this practice. We hypothesize that there are risks associated with calcium administration in patients receiving small volume transfusion.

METHODS: This is a retrospective review of patients admitted to our Level 1 trauma center from 2018 to 2024 who received 1-3 units of red blood cells (RBCs). Patients were stratified by calcium levels and whether they received early calcium supplementation. A logistic regression was used to estimate odds ratios for the association between calcium levels, calcium supplementation, and outcomes. The primary outcome was in-hospital mortality. Secondary outcomes included acute kidney injury, myocardial infarction, acute respiratory distress syndrome, and venous thromboembolism. The model was adjusted for age, sex, ISS and shock index.

RESULTS: A total of 2,680 patients were included. When stratified by calcium levels and supplementation status, patients were similar in terms of age, sex, mechanism and injury severity score, but shock index was higher among patients receiving calcium supplementation. After adjustment, hypocalcemic patients who received calcium supplementation had […]

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Peripheral perfusion index as a predictor of severe shock in trauma patients: a prospective observational study.

Sundar H, Purushothaman V, Mathuram SR, Chandy GM, Sharma SL, Nayak S, James JD

BACKGROUND: Early recognition of haemorrhagic shock is critical in trauma care, particularly in low- and middle-income countries where delayed referral is common. Conventional vital signs may remain normal despite ongoing hypoperfusion. The Perfusion Index (PI), derived from pulse oximetry, is a non-invasive and continuously available marker of peripheral perfusion. This study evaluated the ability of PI to identify severe haemorrhagic shock in trauma patients and compared its performance with the Shock Index (SI) and Modified Shock Index (mSI).

METHODS: This prospective, single-centre study was conducted at a Level 1 trauma centre between January and May 2024. Adult Priority 1 trauma patients were enrolled and classified as having no/mild shock (ATLS grades 1-2) or severe shock (grades 3-4). PI was recorded during initial assessment. SI, mSI, lactate, injury severity, transfusion requirement, ICU admission, and 24-hour mortality were analysed. Receiver operating characteristic (ROC) analysis and multivariable logistic regression were performed.

RESULTS: A total of 120 patients were included (mean age 40.4 ± 17.2 years; 90% male), predominantly with blunt trauma (85% road traffic injuries). Severe shock was present in 35 patients (29.2%). Median PI was significantly lower in severe shock compared with no/mild shock (0.54 [IQR 0.28-1.31] vs. 1.33 [0.71-2.16], p = 0.008). Mean […]

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A prospective study on abdominal trauma: patterns of injury, clinical presentation, organ involvement, associated injuries, and outcomes at a tertiary hospital in Mogadishu, Somalia.

Mohamud AA, Hassan AI, Ali AK, Jubur AA, Ahmed MR, Mohamed SS, Mohamed MM, Bashir AM

BACKGROUND: Trauma is a major global public health problem and one of the leading causes of death and disability worldwide, particularly among individuals younger than 44 years of age. Approximately 6 million people die each year as a result of traumatic injuries, with nearly 90% of these deaths occurring in low- and middle-income countries. Previous epidemiological studies have reported postoperative complication rates ranging from 18% to 42% and mortality rates between 8.5% and 13%.

OBJECTIVE: To evaluate the patterns of injury, clinical presentation, organ involvement, associated injuries, management strategies, and outcomes among adult patients with abdominal trauma in Mogadishu, Somalia.

METHODS: A prospective observational study was conducted from August 2023 to July 2024 at Mogadishu Somali-Türkiye Recep Tayyip Erdo■an Training and Research Hospital. A total of 146 adult patients with abdominal trauma were enrolled. Data were collected using a pretested structured questionnaire and analyzed using Stata version 16. Univariable and multivariable logistic regression analyses were performed to identify factors independently associated with in-hospital mortality.

RESULTS: The study population consisted predominantly of young adult males, with a mean age of 30.18 ± 9.2 years, and males accounted for 97.3% of all patients. Penetrating trauma was the predominant mechanism of […]

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Comparison of fluoroscopic versus electromagnetic distal locking in humeral shaft osteosynthesis using intramedullary nail: a retrospective study.

Madeja R, Szeliga J, Voves J, Pometlova J, Konde A, Janosek J, Dousa P

PURPOSE: Intramedullary nailing is a standard method for osteosynthesis of diaphyseal humeral fractures. Distal locking, however, typically requires fluoroscopic guidance, which increases radiation exposure. This study aimed to compare standard fluoroscopy-guided distal locking with the Ezy-Aim system (Austofix, Australia) based on electromagnetic pulse detection from the perspectives of surgical times, radiation exposure, and complications.

METHODS: A retrospective study included 58 patients treated for diaphyseal humeral fractures (AO/OTA 12A, 12B) between 2019 and 2023. Group A (n = 28) received osteosynthesis using the Ezy-Aim system; Group B (n = 30) underwent fluoroscopy-guided distal locking. Data on operative time, fluoroscopy time, radiation dose, and complications were analysed.

RESULTS: In all, distal locking was not successful at the first attempt in 14.3% (4/28) of screws placed with Ezy-Aim and 20.0% (6/30) when fluoroscopy was used. Time and dose analyses were performed on the remaining 48 patients. The total surgical time was significantly longer with the Ezy-Aim system (median 58 min) compared to fluoroscopy (median 52 min, p = 0.034), primarily due to the time required for the device assembly. However, distal screw placement was faster with Ezy-Aim (median 12 min vs. 14 min, p = 0.019). Radiation exposure was significantly lower in the Ezy-Aim group (median […]

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Patterns of patient worry following major emergency abdominal surgery: a 180-day follow-up study.

Parner MK, Jensen LR, Gamst-Jensen H, Burcharth J, Kokotovic D

PURPOSE: Major emergency abdominal surgery is associated with high postoperative morbidity and mortality. Beyond physical complications, patients experience psychological distress. This study aimed to describe the distribution of patient-reported worry across independent cross-sectional samples and its association with Days Alive and out of Hospital (DAOH) over 180 days postoperatively.

METHODS: Prospective cohort at Copenhagen University Hospital Herlev (March 30, 2023-March 31, 2024). Degree of Worry (DOW) was surveyed at discharge and postoperative day (POD) 30, 90, and 180, categorized as physical, mental, and physical-generic domains of worry. DAOH was computed cumulatively and incrementally for 0-30, 30-90, and 90-180 days. Associations between high DOW (hDOW) and subsequent DAOH were tested with Mann-Whitney U.

RESULTS: hDOW was reported by 36.1% at discharge, 22.4% at POD 30, 45.0% at POD 90, and 32.0% at POD 180. Across 180 days, the most frequent worries were relapse (21.3%), disease (14.1%), functional level (12.0%) and daily living (9.1%). The proportion of physical worries was significantly higher at POD 30 than at discharge and significantly lower at POD 180; mental worries varied between time points without statistically significant differences. hDOW at POD 30 was associated with fewer cumulative DAOH by POD 90 (p = 0.008).

CONCLUSION: Patient-reported worry f […]

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Outcomes and complications of metallic vs. bioabsorbable screw fixation for displaced medial epicondylar fractures in pediatric and adolescent patients.

Jhala T, Scherer S, Esser M, Lieber J, Fuchs J, Dietzel M

PURPOSE: The use of bioabsorbable screws in orthopedics and trauma surgery is increasing. The major advantage of not having to remove the implant is of particular interest in pediatric surgery. Due to drawbacks, most notably implant fracture, bioabsorbable screw osteosynthesis has not yet surpassed the gold standard of metallic screw osteosynthesis. This single-centre study compares metallic screw and bioabsorbable screw osteosynthesis of medial humeral epicondyle fractures in children and adolescents over a two-year period.

MATERIALS AND METHODS: Retrospective analysis of surgically treated (screw fixation with either bioabsorbable magnesium-based or metallic screw) patients with a medial humeral epicondyle fracture between 2020 and 2022 at the author's institution.

RESULTS: Seventeen patients with medial epicondyle fractures underwent surgical treatment using either bioabsorbable screws (n = 8) or metallic screws (n = 9). Baseline characteristics, surgical approach, and postoperative care were similar between groups. Two complications (Clavien-Dindo I) occurred in the bioabsorbable group due to implant failure (screw breakage), though without clinical sequelae. These implant failures led to a significantly increased number of postoperative radiographs in the bioabsorbable group (median 4.8 vs. 2.5; p = 0.014). Functional outcomes (LOM, MEPI) and follow-up durations were com […]

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Characteristics and temporal trends of mobility scooter accidents in the Netherlands: a multicenter cohort study.

Stensen Z, Peters A, Keereweer D, Krul I, van Osch F, de Groot B, Barten D

PURPOSE: Mobility scooters are increasingly used by older individuals. The aim of this study was to assess the characteristics and temporal trends of mobility scooter accidents (MSAs) in the Netherlands over a 10-year period using a representative dataset derived from fourteen Emergency Departments (EDs).

METHODS: In this retrospective multicenter cohort study, we analyzed MSA-related ED visits between January 2014 and December 2023, using the Dutch Injury Surveillance System (DISS). Information about age, sex, hospital admission, type and location of injury, mechanism of injury, and operational errors was analyzed. Injury severity was classified using the maximum abbreviated injury score (MAIS). Trend analyses were only performed for ED visits of patients with MAIS2+ injuries.

RESULTS: In total, 2,702 MSA-related ED visits were identified. Forty-nine percent of patients was male and the median age was 75 years (IQR 64.0-83.0). The most common mechanism of injury was a mobility scooter rollover (34.2%). From 2014 to 2023, the number of MSA-related ED visits of patients with MAIS2+ injuries increased by 38.0%. In the 70-79 age group, this rise was 91%. The total number of traumatic brain injuries (TBI) rose steeply bij 103.0%, which includes a modest (+34.0%) increase in moderate-severe TBI. The hospital admission rate was 30.1%, with a median length of stay of 7 days (IQR 3.0 […]

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"Is the effect of intraoperative 3D imaging beneficial in plate fixation of distal radius fractures: a matched-pair analysis".

Zumsteg JS, Pieringer A, Hinkelmann MA, Isenmann K, Pape HC, Allemann F, Barthel C

BACKGROUND: Distal radius fractures (DRFs) are among the most common adult fractures and are frequently treated with volar plate fixation. While intraoperative 3D imaging has been proposed to improve detection of malreduction and implant malposition, its impact on clinical and radiological outcomes remains unclear. This study aimed to evaluate whether intraoperative 3D imaging provides advantages over conventional 2D fluoroscopy. Outcome criteria were accuracy of fracture reduction, plate positioning relative to the watershed line, complication rates, operative time, and functional outcomes following volar plate fixation of distal radius fractures.

METHODS: This retrospective single-centre study included patients aged ≥ 16 years who underwent volar plate fixation for distal radius fractures between 2016 and 2024. Patients were assigned to a 3D imaging group or a conventional 2D fluoroscopy group. To ensure comparability, patients were matched 1:1 using exact AO/OTA subgroup matching and propensity score matching for age and sex. Outcomes included operative duration, radiographic reduction quality, plate positioning, complications, and wrist range of motion. Matched-pair comparisons were performed using appropriate paired statistical tests.

RESULTS: Of 1,259 patients screened, 418 patients (209 matched pairs) were included after exact AO/OTA subgroup matching and propensity sc […]

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High-velocity gunshot chest trauma: tailoring patient care through insights from a mass casualty experience.

Azulay AA, Stein M, Tabo LY, Lebenthal A, Ruderman L, Refaely Y

OBJECTIVES: This study describes the management strategies and clinical outcomes of a cohort of patients sustaining high-velocity military rifle gunshot wounds to the thorax during a mass casualty incident (MCI). The analysis focuses on damage control surgical techniques, the incidence and timeline of venous thromboembolism (VTE), regional pain management, and early psychiatric protocols.

METHODS: A descriptive retrospective analysis was conducted on patients presenting with a single penetrating high-velocity ballistic wound to the thorax on October 7, 2023. Inclusion criteria were limited to patients with a projectile trajectory traversing the lung parenchyma, managed by the cardiothoracic surgery department. Injuries to immediately adjacent structures (subclavian vessels, brachial plexus, clavicle, and scapula) were recorded as associated injuries. Patients presenting with multi-system trauma involving entirely separate, distant anatomical regions were excluded.

RESULTS: The cohort comprised 11 male patients (age range: 19-49 years; 9 soldiers, 2 civilians) injured by 7.62 mm assault rifle rounds. Six patients (54.5%) required emergency thoracotomy, sternotomy, or VATS within the first 24 h due to ongoing hemorrhage; primary hemorrhage control consisted of deep parenchymal lung suturing (n = 5) and temporary chest packing (n = 1). Four patients (36.4%) underwent delayed, el […]

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Elevated risk of wound and systemic complications in MRSA-colonized patients undergoing lower extremity orthopedic trauma surgery.

Areti AS, Hafeez H, Sangineni PV, Argao L, Deveza L, Dawson J

PURPOSE: Preoperative colonization with methicillin-resistant Staphylococcus aureus (MRSA) is a known risk factor for adverse outcomes in arthroplasty but remains under-investigated in orthopedic trauma. This study evaluates the impact of preoperative MRSA colonization on postoperative complication rates in patients undergoing lower extremity orthopedic trauma surgery.

METHODS: We conducted a retrospective cohort study using the TriNetX Research Network, identifying adult patients who underwent lower extremity orthopedic trauma procedures over a 20-year period. Patients with a documented MRSA diagnosis within three months before surgery were matched 1:1 to MRSA-negative controls based on demographics and comorbidities. Postoperative outcomes were assessed within three months of surgery, including surgical site infections (SSIs), wound dehiscence, organ dysfunction, thromboembolic events, and healthcare utilization. Odds ratios (ORs) and p-values were calculated, with statistical significance set at $p < 0.05$.

RESULTS: A total of 1,868 MRSA+ and 629,342 control patients were identified. After matching, MRSA+ patients remained at significantly increased risk for wound dehiscence (OR = 1.80, $p = 0.005$), pneumonia (OR = 2.33, $p < 0.001$), sepsis (OR = 3.71, $p < 0.001$), acute renal failure (OR = 1.45, $p < 0.001$), blood transfusion (OR = 1.74, $p < 0.001$), deep vein thrombosis (OR = 2 […]

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A global contrast shortage: a descriptive study exploring IV contrast utilization and outcomes in trauma patients at a level I trauma center.

Zakaria H, Johansen C, Waldrop C, Wan B, Graham C, Fetzer D, Minei J, Bruns B et al.

PURPOSE: Computed Tomography (CT) with intravenous (IV) contrast is critical to the evaluation of the trauma patient. A pandemic-related lockdown occurred in Spring 2022 in the city where iohexol iodinated contrast media) is manufactured, resulting in a global shortage. We explored the potential impact of this shortage on contrast utilization and patient outcomes.

METHODS: A retrospective study was performed of all trauma patients who underwent CT between January and December 2022 at a level I trauma center. Pre-contrast conservation (PRE) period was defined as January to April 2022, and post-conservation (POST) as May to December 2022. Demographics, utilization rates (bolus/patient) and outcomes were evaluated. Chi-square tests were performed for categorical variables and Mann-Whitney U tests for continuous variables. Statistical significance was set at $p < 0.05$.

RESULTS: 1,097 patients were included; 509 in the pre-shortage period, 588 post-shortage. 857 CTs with contrast (1.68 contrast bolus/patient) were performed pre-conservation and 979 post-conservation (1.66 contrast bolus/patient) ($p = 0.04$). Maximum Abbreviated Injury Scale (AIS) head, head/neck, and extremity were higher post-shortage ($p < 0.05$); however, mortality was higher in the pre-conservation group ($p < 0.0012$). There were 9 cases of Acute Kidney Injury (AKI, 8 PRE, 1 POST), 22 delayed diagnoses (14 PRE, 8 P […]

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Trauma stabilization points in armed conflict: operational evolution and strategic reflections from mosul to Gaza.

Marta C, Luca P, Kai-Hsun H, Areej AF, Athanasios G, Wessam EG, Jonas Z, Johan VS et al.

BACKGROUND: Forward trauma care remains a critical gap in emergency responses to armed conflict, particularly where conventional medical evacuation is disrupted. Trauma Stabilization Points (TSPs) were introduced in previous conflicts to bring life-saving care closer to the point of injury. To improve standards and adaptability, the World Health Organization convened a Technical Working Group to develop operational guidance for TSPs, drawing on diverse field experiences, including the ongoing Gaza deployments.

CASE PRESENTATION: Between February and July 2024, three TSPs were established in Gaza, managing over 4,000 consultations. Initially focused on trauma stabilization and referral, these sites quickly adapted to minor injuries and non-traumatic conditions, reflecting population needs and access challenges. Referral rates varied across sites due to hospital proximity, ambulance availability, and shifting frontlines. Security threats limited forward deployment and safe patient access, requiring high mobility and rapid relocation. Experience from Gaza highlighted key operational principles: locating TSPs near the point of injury; integrating within a functioning trauma referral pathway supported by evacuation capacity and hospital readiness; maintaining clear clinical functions and staffing standards; and using standardized documentation for quality assurance and continuity. […]

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Acute-on-chronic versus isolated acute subdural hematoma in complicated mild traumatic brain injury: association with radiological mass effect markers in a multicenter cohort.

Crombé A, Benhamed A, Seux M, Frassin L, Matichard R, L'Huillier R, Millon D, Gorincour G

PURPOSE: The clinical impact of acute-on-chronic subdural hematoma (acSDH) in mild traumatic brain injury (mTBI) remains incompletely characterized. We assessed whether acSDH, compared with isolated acute SDH (aSDH), is associated with worse neurological presentation and increased radiological markers of mass effect.

METHODS: We conducted a retrospective multicenter cohort study using the IMADIS teleradiology head trauma workflow (103 emergency departments in France, January 2020 to December 2022). Adult patients with complicated mTBI (i.e., GCS 13-15 plus intracranial hemorrhage or skull fracture) and acute SDH on non-contrast head CT were eligible. Exposure was acSDH versus aSDH based on structured radiology reports. Outcomes were (1) GCS < 15 at presentation, (2) SDH maximal thickness ≥ 7 mm, (3) radiological brain herniation, and (4) intermediate or high QueBIC risk category. Univariable and multivariable logistic regression models were used to estimate associations.

RESULTS: Among 935 included patients, 107 (11.4%) had acSDH. Patients with acSDH were older (≥ 75 years: 57.0% vs. 39.7%, $p < .0001$) and more frequently had dementia (8.4% vs. 3.7%, $p = .0465$). Compared with aSDH, acSDH was associated with GCS < 15 (37.4% vs. 21.7%, $P = .0010$) and confusion (41.0% vs. 25.8%, $p = .0016$). On CT, acSDH was associated with SDH thickness ≥ 7 mm (54.2% vs. 20.4%, OR 4.50, 95%CI 2.9 […]

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Persistent neuropathic pain symptoms after FPV drone-related thoracic blast injury without rib fractures: a pilot observational study from the Ukraine War.

Battle C, Baker E, Giamello JD, Melchio R, Dmytriiev D

PURPOSE: The aim of this exploratory pilot study was to describe First Person View drone (FPV) related, blast-mechanism thoracic trauma with no rib fractures in Ukrainian military patients sustained during the Russia-Ukraine Conflict, and report the prevalence of persistent neuropathic pain symptoms at three months among patients admitted to a rehabilitation and pain medicine service.

METHODS: A retrospective, single-centre observational, exploratory study design was used. Military patients (≥ 16 years old) with blast-mechanism thoracic injuries with no rib fractures, admitted to a hospital's rehabilitation and pain medicine service in Ukraine for ≥ 24 h were included. The PainDETECT survey was completed at three months post-injury. Participant characteristics were summarised using descriptive statistics. Exploratory univariate and multivariate analyses were performed to explore possible associations between variables and the development of persistent neuropathic pain symptoms.

RESULTS: 217 patients were included in the final analysis. Median age was 39 years (IQR: 32-47.5). 131 (60.4%) patients were admitted to critical care with 91 (41.9%) patients requiring mechanical ventilation. The median hospital length of stay was 22 days (IQR: 16-25) and critical care length of stay was 4 days (IQR: 2-5). Persistent neuropathic pain symptoms were reported in 173 out of 217 patients ([…]

PubMed · DOI

The association between pancreatic enzyme serum concentrations and blunt pancreatic trauma: a systematic review.

Sykes B, Gracias D, Bendinelli C, Balogh ZJ

PURPOSE: Blunt pancreatic trauma is uncommon. When suspected, serum amylase and lipase are measured, but their diagnostic value remains uncertain. This systematic review evaluated the diagnostic performance of serum amylase and serum lipase in detecting blunt pancreatic trauma in adults.

METHODS: MEDLINE, Embase, Scopus and Cochrane Central were searched for studies evaluating serum amylase and/or lipase in adults with blunt pancreatic trauma. Observational studies reporting diagnostic accuracy or clinical outcomes were included. Two reviewers independently screened studies and assessed risk of bias using the Newcastle-Ottawa Scale.

RESULTS: Four observational studies met inclusion criteria. A total of 4937 patients with blunt abdominal trauma, including 150 with pancreatic injury were investigated. Two studies evaluated serum lipase alone and two compared amylase and lipase. Pancreatic injuries were graded according to the American or Japanese Association for the Surgery of Trauma. Early serum amylase measurement (< 90 min post-injury) demonstrated low sensitivity. In contrast, testing performed more than 3 h after injury has shown improved diagnostic accuracy with one study suggesting combined serum amylase and lipase testing can achieve a sensitivity of 85% and specificity of 100%.

CONCLUSION: Early serum pancreatic enzymes measurement does not appear to aid initial manag […]

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Extension of the fluidics NTF-Biobank of polytraumatized patients with extracellular vesicles: improved vesicle preservation through local processing.

Weber B, Leppik L, Han J, Becker N, Wang T, Groven RVM, Balmayor ER, Zhao Q et al.

PURPOSE: Extracellular vesicles (EVs) show diagnostic relevance in trauma, prompting the development of the NTF■EV■Biobank as an extension of the established Network Trauma Research (NTF) plasma/serum biobank. After optimizing the EV isolation and characterization standard operating procedure (SOP) at a single center, this multicenter study evaluates its transferability and examines how sample transport affects EV quality before large■scale implementation in a nationwide biobank setting.

METHODS: Plasma and serum from polytrauma patients and healthy controls (n=10) were processed under the standardized NTF■Biobank protocol and distributed to three centers. EVs were isolated using the unified NTF■EV■Biobank workflow, exchanged between centers, and analyzed for protein markers, particle size and concentration (NTA), total protein content, and morphology (TEM) to compare EV quality across sites.

RESULTS: Implementation of the EV biobank protocol across participating centers, as well as inter-center transport on dry ice, was successfully performed without complications. All samples arrived at the analyzing centers in a frozen state. EVs exhibited typical vesicular morphology as confirmed by transmission electron microscopy and remained positive for CD9 and CD81, indicating that transportation did not affect morphology or surface marker expression. However, transportation did redu […]

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Whole blood first resuscitation and association of blood product utilization based on mechanism of injury.

Hadley J, Conner J, Thomas C, Llorente K, Gilani A, White B, Miller Iii P, Nunn A et al.

INTRODUCTION: Studies evaluating the effect of whole blood (WB) vs. component-based resuscitation accounting for mechanism of injury are lacking. We sought to evaluate the effects of a WB first resuscitation strategy regarding mechanism of injury.

METHODS: This is a retrospective cohort study of trauma patients who received any blood product during resuscitation from January 2016 to November 2021. The primary outcomes were total blood volume utilized within 24 h and 30-day mortality. Multivariable gamma regression with a log link was used for blood volume and logistic regression for mortality. A likelihood ratio test evaluated whether injury mechanism modified the treatment effect.

RESULTS: Between January 2016 and November 2021, 1,013 injured patients received blood products (785 WB-first). We did not detect statistically significant effect modification by injury mechanism for either outcome (interaction $p = 0.44$ for blood volume, $p = 0.25$ for mortality). In multivariable gamma regression, WB-first resuscitation was associated with a ratio of mean total blood volume of 0.71 (95% CI 0.55 to 0.90, $p = 0.005$) relative to components, representing approximately 29% lower utilization. WB-first was not associated with 30-day mortality (OR 0.90, 95% CI 0.48 to 1.71, $p = 0.73$).

CONCLUSIONS: A WB-first resuscitation strategy is associated with lower blood volume utilized within the f […]

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Reported complications after spinopelvic fixation for U-type sacral fractures in older adults: a systematic review with narrative synthesis.

Giordano V, Pinheiro HRS, Pires RE, Varga P, Freitas A, Giannoudis PV

BACKGROUND: Spinopelvic fixation (SPF), alone or combined with iliosacral or transiliac-transsacral fixation, is often used for highly unstable posterior pelvic ring injuries and some fragility fracture patterns in older adults. However, the evidence specifically describing complications after SPF in older patients with U-type sacral fractures is limited.

METHODS: This systematic review was conducted according to the Cochrane Handbook and PRISMA recommendations. PubMed, Embase, and grey-literature sources were searched from January 1, 2000 to July 31, 2025. Eligible studies included adults aged 60 years or older with U-type sacral fractures or closely related spinopelvic dissociation/FFP IVb patterns treated with SPF, and reporting at least one complication. Because of the small number of eligible studies, probable cohort overlap, heterogeneity in fracture definitions, treatment indications, fixation constructs, and outcome reporting, findings were synthesized narratively rather than pooled quantitatively.

RESULTS: Three studies met the eligibility criteria, comprising 157 patients overall; however, only a subset underwent SPF, and two studies were produced by the same group with probable cohort overlap. The available cohorts mixed bilateral sacral fragility fractures, FFP IVb patterns, and comparative SPF-versus-trans-sacral constructs. Reported adverse events included wound […]

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Influence of allograft augmentation on surgical outcomes and complications in proximal humerus fractures: a matched-pair analysis.

Barthel C, Rudolph F, Kalbas Y, Russek R, Pape HC, Allemann F

BACKGROUND: Proximal humerus fractures predominantly affect elderly patients and pose a growing clinical burden. This study evaluates the impact of allograft augmentation on operative time, fracture reduction quality, and complication rates compared with standard fixation across different fracture morphologies.

METHODS: This retrospective matched-pair study included patients surgically treated for proximal humerus fractures with locking plate fixation at a single tertiary trauma center between 01.01.2014 and 31.12.2024. Patients treated with allograft augmentation (Group A) were matched 1:1 to patients without allograft (Group NA) using exact matching for Mayo FJD fracture morphology and sex, and propensity score matching for age. Radiological outcomes were assessed using pre- and postoperative neck-shaft angles, with normalization defined as correction into the physiological range (120°-150°). Complications and operative parameters were recorded. Paired statistical analyses were applied.

RESULTS: A total of 1,012 patients were treated during the study period; after selection and matching, 102 matched patient pairs (204 patients) were analyzed. Operative time was longer with allograft augmentation (141.3 ± 40.3 vs. 126.9 ± 50.0 min; p = 0.08), reaching statistical significance only in varus posteromedial fractures (144.4 ± 37.8 vs. 122.2 ± 39.6 min; p = 0.007). Although overa […]

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Impact of a comprehensive trauma pathway on workflow and clinical processes in a level III trauma center.

Neitzert L, Druetto C, Forno D, Molino P, Vitale D, Grio M

PURPOSE: Organized trauma systems improve coordination of care and may enhance the efficiency of trauma management, potentially improving outcomes and reducing hospital length of stay and costs. The aim of the study was to evaluate key process-of-care indicators and clinical outcomes before and after the introduction of a comprehensive trauma pathway.

METHODS: We conducted a single-center retrospective-prospective cohort study at Rivoli Hospital, Italy. Trauma patients admitted to the Emergency Department (ED) between 2024 and 2025 who met the criteria for trauma team activation were prospectively included. Outcomes were compared with a retrospective cohort from the period prior to the implementation of the dedicated trauma pathway (2023-2024).

RESULTS: A total of 465 patients were included. Implementation of a comprehensive trauma pathway was associated with significant improvements in several time-dependent quality indicators, including ED boarding time (888 vs. 1021 min; p = 0.02), time to imaging (48 vs. 78 min; p < 0.01), time to surgery (183 vs. 237 min; p = 0.03), time to ICU admission (226 vs. 343 min; p = 0.02), and time to interhospital transfer (238 vs. 305 min; p = 0.04). We observed a trend toward a reduction in early mortality (2% vs. 3.6%; p = 0.41) and 30-day mortality (3.7% vs. 5.9%; p = 0.62).

CONCLUSIONS: The adoption of a structured multidisciplinary trau […]

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The DRoPH score: development of a tool for predicting reduction difficulty in proximal humerus fracture osteosynthesis.

Barthel C, Rudolph F, Kraus M, Pape HC, Allemann F

BACKGROUND: Proximal humerus fractures (PHFs) are among the most common and challenging fractures due to complex anatomy and variable fracture patterns. While anatomical reduction is critical for successful outcomes, no tool currently predicts the intraoperative difficulty of achieving it. Despite advances in surgical techniques, complications persist. Existing classification systems, such as Codman's, Neer's, AO/OTA, and Mayo-FJD, focus on fracture morphology and displacement but do not address intraoperative complexity. This study aims to develop a new score for the early prediction of intraoperative reduction difficulty in proximal humerus fracture surgery, the DRoPH score (Difficulty of Reduction of Proximal Humerus Fractures).

METHODS: An online survey was conducted among experienced upper extremity surgeons in Switzerland to identify factors influencing the difficulty of reducing and retaining proximal humerus fractures. Inclusion criteria for surgeon grading comprised experience with more than 100 surgically treated proximal humerus fractures. Eligible surgeons rated the relevance of factors influencing reduction difficulty using a five-point Likert scale. A rank-based weighting system derived from expert ratings was applied, assigning two points to the three highest-ranked factors. The resulting DRoPH-Score ranged from 0 to 10, with higher values indicating greater sur […]

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Interface stability of intramedullary nail locking configurations under combined axial and torsional loading.

Marianne H, Martin W, Claus G, Anja B, Peter A

The biomechanical performance of intramedullary nails in metaphyseal regions strongly depends on distal locking configuration. This study aimed to evaluate screw-bone interface stability under combined axial and torsional cyclic loading for different distal locking configurations of intramedullary nails, focusing on the effects of screw number, spatial distribution, interscrew distance, and multiplanar versus coplanar fixation using a standardized synthetic bone model. Six distal locking configurations (n = 5 per group) were tested under stepwise cyclic axial compression, superimposed with a constant torsional moment (± 2 Nm, 2 Hz). Primary endpoints were cycles to failure, cycles to 1 mm axial displacement, and cycles to 0.5° rotation. All constructs failed by progressive screw cut-through within the synthetic bone. The three-screw multiplanar configuration exhibited the highest endurance (≈ 36,000 cycles to failure) and lowest micromotion (≈ 34,000 cycles to 1 mm displacement), significantly outperforming the two-screw configurations overall (p = 0.002). Increasing interscrew distance from 10 mm to 34 mm improved migration resistance by approximately 25%, whereas the short 10 mm spacing led to premature failure. Blocked and free-hand configurations showed comparable behavior to the standard setup, indicating no added benefit from toggle elimination alone. Locking configuratio […]

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Three-dimensional finite element analysis for internal fixation of mandibular condylar fractures: a scoping review.

Kakei Y, Sader R, Ebener P, Tanaka A, Akashi M, Kuroda R, Thoenissen P

PURPOSE: Three-dimensional finite element analysis (3D-FEA) has emerged as an essential computational tool for evaluating the biomechanical behaviors of internal fixation systems for mandibular condylar fractures. This scoping review maps and synthesizes the existing literature on 3D-FEA studies investigating the internal fixation of mandibular condylar fractures and explores, in a descriptive manner, differences in methodological focus between European and non-European research centers.

METHODS: PubMed and Cochrane Library databases were systematically searched following PRISMA-ScR guidelines. Original English-language research articles using 3D-FEA to evaluate internal fixation of mandibular condylar fractures were included. Data were extracted using a standardized charting form.

RESULTS: Thirty-six studies met the inclusion criteria. European studies (n = 12) predominantly focused on development of novel plate designs and mechanical testing protocols, whereas Asian and other nationality studies (n = 24) emphasized material comparisons between titanium and bioabsorbable systems, patient-specific modeling, and loading condition analysis. Consensus findings include: (1) double-plate fixation provides superior stability; (2) among 3D plate designs, trapezoid plates demonstrate the best performance for subcondylar and condylar base fractures, whereas alpha and lambda plates may […]

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Initial surgical management of injuries to the upper extremities for patients with multiple and/or severe injuries - a systematic review and clinical practice guideline update.

Berk T, Könsgen N, Lechler P, Imach S, Schär M, Bieler D, Horst K, Hildebrand F et al.

PURPOSE: Our objective was to update the evidence-based and consensus-based recommendations for the initial surgical management of upper extremity injuries in patients with suspected multiple and/or severe injuries based on current evidence. This guideline topic is part of the 2025 update of the German Guideline on the Treatment of Patients with Multiple and/or Severe Injuries.

METHODS: MEDLINE and Embase were systematically searched to September 2024. Further literature reports were obtained from clinical experts. Randomised controlled trials (RCTs) or observational studies reporting risk-adjusted outcomes were included if they compared early versus delayed surgical treatment for fractures, vascular injuries, or nerve injuries affecting the upper extremities in patients with multiple and/or severe injuries. Studies comparing limb salvage versus amputation or comparing amputation criteria for the upper extremities were also included. We considered patient-relevant outcomes such as mortality and limb salvage, as well as the sensitivity and specificity of scores for predicting upper extremity amputation. Risk of bias was assessed at the outcome level using ROBINS-I for observational studies and AMSTAR-2 for systematic reviews. We used available meta-analyses if possible; alternatively, we synthesised the evidence narratively. We used GRADE to rate the certainty of evidence. Expe […]

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Transosseous sutures versus vertical cerclage wire for distal pole patellar fractures: a retrospective comparative cohort study.

Laso JI, Muñoz JT, Bianchi S, Martínez D, Rojas T, Franulic N, Olivieri R

PURPOSE: To compare transosseous sutures (TOS) and vertical wire cerclage (VCW) for treating distal pole patellar fractures, evaluating clinical outcomes, patient-reported outcome measures (PROMs), radiographic outcomes, and complications.

METHODS: We conducted a retrospective comparative cohort study of patients surgically treated for patellar distal pole fractures (AO/OTA 34A1b) with either TOS or VCW between 2015 and 2023. Patient characteristics, surgical technique, postoperative data, patellar height, and complications were recorded. Functional outcomes were evaluated at a minimum of 12 months postoperatively using PROMs (Kujala, Lysholm, and Bostman scores). The primary outcome of this study was functional recovery, assessed via the Bostman score at 12 months.

RESULTS: A total of 46 patellar distal pole fractures were included (30 TOS and 16 VCW). Baseline characteristics showed no statistically significant differences between groups. No significant differences were found regarding clinical or functional outcomes. Regarding radiographic outcomes, patellar height remained stable; however, both groups showed significant patellar length variation, with elongation being significantly more pronounced in the TOS group (mean 3.77 mm) compared to the VCW group (mean 0.44 mm; p = 0.001). While overall complication rates showed no statistically significant difference, all five ca […]

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Re■fracture risk following implant removal in consolidated hip fractures: a multicenter retrospective cohort study of 575 patients.

Mohammed J, Mili-Schmidt V, Wadsten M, Juto H, Wolf O, Crnalic S, Sköldenberg O, Mukka S et al.

PURPOSE: The risk of re-fracture after implant removal in healed hip fractures remains a clinical concern. This study aimed to determine the incidence of re-fracture following implant removal after osteosynthesis of a hip fracture.

METHODS: We conducted a retrospective multicenter cohort study including patients aged ≥ 50 years who underwent implant removal between 2003 and 2023 after radiologically confirmed consolidation of a hip fracture. Patients were identified using procedural codes. Baseline variables included age, sex, ASA classification, fracture type (femoral neck or trochanteric), and implant type (pins/screws, sliding hip device [SHD], or short/long cephalomedullary nail [CMN]). Patients were followed from implant removal until re-fracture, conversion to arthroplasty, death, or end of follow-up, with a minimum follow-up of 1 year. Cox proportional hazards regression was used to assess associations between implant type and re-fracture risk, adjusting for age and sex. Because the proportional hazards assumption was violated, a time-stratified Cox regression and restricted mean survival time analyses were applied.

RESULTS: A total of 575 patients (median age 73 years, IQR 65-81) were included, with a median follow-up of 53 months (IQR 18-100). Lateral hip pain was the most common indication for implant removal (72.5%). The overall re-fracture incidence was 10.4%. Ris […]

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Physical functionality two years after major trauma: predictors from a single center cohort study.

Fetz K, Koetsenruijter J, Rodrigues Recchia D, Rutetzki J, Lefering R, Kaske S

PURPOSE: Traumatic injuries are a major global public health concern, often leading to impaired physical function and disability. Long-term physical function impairments can greatly affect a patient's health related quality of life, influencing their emotional state and social interactions. This study examines patient-reported physical functionality before and after major trauma, identifying key predictors of poor functional recovery.

METHODS: This retrospective single-center cohort study investigates physical functionality 23 months after a traumatic injury using standardized questionnaires combined with clinical data. Inclusion criteria were Injury Severity Score (ISS) ≥ 9, requiring ICU treatment, and age ≥ 18 resulting in an eligible sample of 515 patients. Health outcomes were assessed using the Trauma Outcome Profile (TOP), the Abbreviated Injury Scale (AIS), and the ISS.

RESULTS: Physical functionality significantly decreased from 94.8 (SD 11.5) to 72.9 (SD 26.7) 23 months post-trauma with almost half of participants still scoring below the critical cutoff of 80 points. The strongest predictor for functional physical impairment was pre-trauma physical functionality (OR 8.21, $p < 0.001$). Other determinants included age (age 50-59 versus 18-29 OR 2.75, $p = 0.001$), as well as an impacted lower extremity including pelvic (OR 1.60, $p = 0.018$).

CONCLUSION: The most importan […]

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Association between fresh frozen plasma transfusion and mortality stratified by Glasgow Coma Scale scores in isolated traumatic brain injury: a nationwide cohort study in Japan.

Nagamura T, Aoki M, Yamada K, Terayama T, Seno S, Tomura S, Kiyozumi T

PURPOSE: Fresh frozen plasma (FFP) is widely used to manage trauma-induced coagulopathy. However, its effectiveness in isolated traumatic brain injury (TBI) management remains controversial, particularly when stratified by severity based on the Glasgow Coma Scale (GCS) scores. This study aimed to examine the association between FFP administration within 24 h and mortality in patients with isolated TBI, overall and stratified by GCS on arrival.

METHODS: We conducted a multicenter retrospective cohort study using the Japan Trauma Data Bank (2019-2023). Adults with isolated TBI (intracranial AIS ≥ 3 and extracranial AIS < 3) were included. Patients were classified into FFP and non-FFP groups based on transfusion within 24 h of admission. Subgroup analyses were performed by TBI severity (severe: GCS ≤ 8; mild-to-moderate: GCS > 8), craniotomy status, and FFP dose. The primary outcome was 28-day mortality. Propensity score matching was applied to adjust for confounding.

RESULTS: Among 12,480 patients (median age 72 years [IQR: 55, 82]; 66.6% male), 513 received FFP. After matching, FFP was not significantly associated with 28-day mortality (OR 0.80; 95% CI 0.59-1.08). In subgroup analyses, FFP was associated with higher mortality in mild-to-moderate TBI (OR 2.38; 95% CI 1.07-5.31) and lower mortality in severe TBI (OR 0.61; 95% CI 0.42-0.88). No significant differences were observ […]

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Percutaneous fixation of posterior pelvic ring lesions in the elderly: current evidence and recent advances.

Liodakis E, Schreiber S, Giannoudis VP, Fritz T, Giannoudis PV

Transsacral percutaneous fixation is increasingly seen as a safe, effective minimally invasive treatment for geriatric posterior ring fractures The use of Navigation appears a safe and efficient method for percutaneous screw placement Multiple new implants have been described for management of these injuries. Level I evidence is still lacking for treatment options in the management of posterior pelvic ring lesions with percutaneous fixation

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Infrared thermography and dual-layer spectral CT for the detection of femoral vascular compromise: an experimental large animal study.

Hübner CT, Ricklin J, Müller AL, Klingebiel FK, Stoeck CT, Weisskopf M, Carneiro Gomes A, Cinelli P et al.

BACKGROUND: Rapid identification of vascular compromise remains a clinical challenge in the acute trauma setting. Conventional diagnostic tools such as CT angiography (CTA) and Doppler-ultrasound might provide decent anatomical and functional assessment. However, they are limited by ionizing radiation, contrast exposure or time requirements. Although not validated, infrared thermography (IRT) might offer a non-invasive alternative capable of detecting surface temperature asymmetries that may reflect underlying perfusion deficits. This study aimed to evaluate the feasibility of thermographic imaging in detecting vascular lesions.

METHODS: As part of a porcine polytrauma model (n = 28), vascular lesion of the right femoral artery was defined as a ≥ 50% diameter reduction on CTA. Temperature differences between both extremities were measured using infrared thermography. SDCT based iodine concentrations distal to the stenosis were quantified to assess perfusion changes. Correlation and regression analyses were performed between stenosis degree, temperature difference, and tissue iodine uptake.

RESULTS: Six animals fulfilled the criteria for relevant vascular compromise. This resulted in a mean vessel diameter reduction behind the lesion of 68% ±18%. The vascular lesion group demonstrated a mean inter-limb temperature difference of 6.93 °C ± 1.78 °C compared to 0.71 °C ± 1.27 °C i […]

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Fixation strategies for periprosthetic femur fractures around knee replacements: a comprehensive single-center analysis of 102 consecutive cases.

Nomdedéu J, González-Morgado D, Minguell-Monyart J, Joshi-Jubert N, Teixidor-Serra J, Tomàs-Hernández J, Andrés-Peiró JV

PURPOSE: To evaluate 1 year mortality, complication rates, and functional outcomes after internal fixation of femoral periprosthetic knee fractures following total knee arthroplasty, and to identify predictors of adverse outcomes and increased healthcare resource utilization.

METHODS: This retrospective cohort study included 102 consecutive patients with femoral periprosthetic knee fractures classified as UCPF VB3 and VC3 treated with internal fixation at a single tertiary center between 2010 and 2023. Interprosthetic fractures and combined fixation techniques were excluded. Primary outcomes were 1 year mortality and postoperative complications. Secondary outcomes included length of stay, discharge destination, and ambulatory status at follow up.

RESULTS: Patients were markedly frail, with a mean age of 82.4 ± 10.8 years and mean Charlson Comorbidity Index of 5.6. One year mortality was 16.7% and was associated with older age, higher ASA class, higher Charlson index, impaired pre fracture ambulation, and fracture related infection. Overall complication rate was 29.4%, including 15.7% infections and 13.7% mechanical failures. Open reduction was more frequent in plate fixation and resulted in longer operative times, without increasing complication rates. Median hospital stay was 18 days, 66.3% required discharge to nursing facilities, and only 7.8% regained independent ambulati […]

PubMed · DOI

Epidemiology and management of pediatric femoral fractures at a level I trauma center: a 10-year retrospective monocentric study.

Dietz SO, Traub F, Schmidt E, Schippers P, Wegner E, Schierjott L, Gercek E, Nowak TE

PURPOSE: Pediatric femoral fractures are relatively uncommon but represent severe injuries frequently requiring inpatient trauma care. Epidemiological characteristics, fracture patterns, and treatment strategies vary between institutions and regions. This study aimed to analyze the epidemiology, fracture distribution, treatment modalities, complications, and fracture-predisposing comorbidities of pediatric femoral fractures treated at a German level I pediatric trauma center over a 10-year period.

METHODS: All patients aged 0-14 years treated for femoral fractures between January 2012 and December 2021 were retrospectively reviewed. Fractures were categorized using the AO Pediatric Comprehensive Classification of Long-Bone Fractures (PCCF). Demographic variables, fracture characteristics, management approaches, time to consolidation, complications, and relevant comorbidities predisposing to fracture were analysed descriptively in accordance with STROBE recommendations.

RESULTS: A total of 183 patients met the inclusion criteria (67.2% male; mean age 5.2 years). Children aged 2-5 years constituted the largest group (42.1%). Femoral shaft fractures were the most frequent femoral fracture type (70%). Surgical management, most frequently elastic stable intramedullary nailing (ESIN), was increasingly utilized with advancing age. The overall complication rate was 14.2%, with the ma […]

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Blood-based biomarkers GFAP/UCH-L1 for the diagnosis of mild traumatic brain injury (mTBI): a single-center implementation experience.

Maegele M, Breidenbach L, Kaufmann J, Keles Y, Uhl E, Schroeer P

INTRODUCTION: Mild traumatic brain injuries (mTBI) affect millions of people worldwide every year as one of the most common clinical presentations in the emergency department. Diagnosis is mainly based on clinical criteria and computed tomography scans. The use of computed tomography causes high costs, long waiting times in daily clinical practice and radiation exposure. GFAP (glial fibrillary acidic protein) and UCH-L1 (ubiquitin carboxyl-terminal hydrolase-L1) turned out to be potential biomarkers for the diagnosis of mTBI. This study retrospectively evaluates the possible use of these biomarkers combined as negative predictors for excluding brain injuries in patients with suspected mTBI in the emergency department.

METHODS: Adult patients (n = 320) registered in the emergency department at a level 1 trauma emergency center in Germany (Cologne Merheim Medical Center/CMMC) between 11/2023 and 04/2024, with suspected mTBI, Glasgow Coma Scale (GCS) score 13-15 and within 12 h after trauma were considered. All evaluable patients underwent cranial CT (cCT) scans and blood tests for GFAP and UCH-L1 serum concentrations.

RESULTS: Biomarkers GFAP and UCH-L1 were tested positive in 261 patients (82%) while CT detected intracranial injuries in only 29 patients (9%). Biomarkers combined had a sensitivity of 97% and a negative predictive value (NPV) of 98% in mTBI diagnosis with a nega […]

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Resuscitative endovascular occlusion of the aorta restores cerebral metabolic markers of ischaemia induced by haemorrhagic shock.

Bader SE, Magnuson A, Brorsson C, Wallin G, Löfgren N, Löfgren F, Blind PJ, Öman M et al.

BACKGROUND: Resuscitative endovascular balloon occlusion of the aorta (REBOA) is used as an adjunct in haemorrhagic shock (HS) to restore proximal perfusion. Its cerebral metabolic effects during haemorrhagic shock, particularly in the presence of elevated intracranial pressure (ICP), remain incompletely characterised.

METHODS: In a controlled porcine model, HS was induced by controlled bleeding to a mean arterial pressure of approximately 40 mmHg. Animals were allocated to a normal ICP group or an experimentally elevated ICP group. Total REBOA (zone 1) was applied for 90 min. Cerebral metabolism was assessed using intracerebral microdialysis with measurements of lactate, pyruvate, and lactate-pyruvate ratio (LPR). Cerebral haemodynamics and ICP were continuously monitored.

RESULTS: HS was associated with a statistically significant increase in LPR in both groups, indicating cerebral metabolic disturbance. Following aortic occlusion, LPR gradually decreased toward baseline in both groups. Animals with elevated ICP demonstrated a transient delay in metabolic normalisation during the early post-occlusion phase. Statistically significant differences between groups were limited to the first 10 min following occlusion. Overall metabolic trajectories were similar thereafter.

CONCLUSIONS: Total REBOA restored cerebral metabolic markers of ischaemia during haemorrhagic shock, as ref […]

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Delayed single-stage surgical management using a peroneus brevis muscle flap for fracture-related infection after lateral malleolus fixation: a retrospective case series.

Dyreborg K, Jørgensen AI, Peterlin AAN, Gottlieb H

BACKGROUND: Fracture-related infection (FRI) following lateral malleolus fracture fixation complicated by soft tissue breakdown represents a challenging clinical problem with no established standard of care. This study evaluates a treatment strategy consisting of prolonged antibiotic suppression until fracture union, followed by delayed single-stage surgical management.

METHODS: This retrospective consecutive case series included 11 patients treated between 2020 and 2023 at a multidisciplinary orthopaedic infection unit. All patients were managed according to a standardized protocol involving oral antibiotic suppression until radiologically confirmed fracture consolidation. This was followed by a single-stage surgical procedure including implant removal, radical debridement, deep tissue sampling, local antibiotic therapy, and soft tissue reconstruction using a distally based peroneus brevis muscle flap with split-thickness skin grafting.

RESULTS: Eleven consecutive patients (median age 58 years) were treated according to the protocol. Six patients underwent antibiotic suppression for a median duration of 147 days prior to surgery. Among patients available for outcome assessment, infection control was achieved in 10 of 10 surviving patients. One patient died of unrelated causes within 14 days postoperatively and was not classified as treatment failure. Soft tissue complication […]

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Mixed reality improves agreement on surgical approach selection and patient positioning in tibial plateau fracture planning compared to CT, 3DCT and 3D printing.

Dust T, Henneberg JE, Hartel M, Reiter A, Kylies J, Korthaus A, Streckenbach A, Keller J et al.

PURPOSE: Tibial plateau fractures require precise preoperative planning, particularly regarding treatment concept, patient positioning, and surgical approach. Mixed reality (MR) may enhance three-dimensional understanding of fracture morphology beyond conventional imaging and physical 3D models. This study investigated whether MR improves agreement on key preoperative planning decisions compared to computed tomography (CT), 3D computed tomography (3DCT), and 3D-printed fracture models.

METHODS: Twelve orthopaedic trauma surgeons (6 junior, 6 senior) evaluated 22 surgically treated tibial plateau fractures (AO/OTA type B or C). Each case was assessed in four steps using (1) CT, (2) 3DCT reconstructions, (3) physical 3D-printed models, and (4) MR visualization using Microsoft Hololens 2. Raters selected (i) treatment concept, (ii) patient positioning, and (iii) surgical approach. Interobserver agreement was analysed using Fleiss' kappa (κ) and percentage match (PM).

RESULTS: Mixed Reality (MR) yielded the highest interobserver agreement for both surgical approach (PM 32%, κ = 0.30) and patient positioning (PM 57%, κ = 0.35), particularly among junior surgeons (approach PM 39%, κ = 0.28; positioning PM 57%, κ = 0.39). Compared to CT, 3DCT, and 3D printing, MR demonstrated consistent improvements, while agreement on treatment concepts remained high across all modalities (PM > 82% […]

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Timing of fracture fixation for femur and pelvis fractures in patients with severe traumatic brain injury - an analysis of the TraumaRegister DGU®.

Bayer TP, Teuben MPJ, Halvachizadeh S, Shehu A, Lefering R, Pfeifer R, Pape HC, Jensen KO

PURPOSE: Fracture fixation timing and strategy in polytrauma patients with traumatic brain injury (TBI) remain controversial. This study investigates treatment patterns and outcomes for femoral and/or pelvic fractures stratified by TBI severity.

METHODS: Patients in the TraumaRegister DGU® (2016-2022) with pelvic and/or femoral fractures (AIS ≥3) and TBI (head AIS ≥3) were included. Strategies were non-operative management (NOM), early total care (ETC), and damage-control orthopedics (DCO). Outcomes included treatment allocation, fixation timing, and in-hospital mortality.

RESULTS: 985 patients were included (mean age 52.5, SD 26.3 years; ISS 27.8, SD 8.1). Allocation was NOM in 320 (32.5%), ETC in 336 (34.1%), and DCO in 329 (33.4%) patients. Head AIS was 3 in 48.5%, 4 in 31.1%, and 5 in 20.3%. NOM patients were older, had the highest ISS and estimated mortality, and showed the largest proportion of critical TBI (AIS 5: NOM 30.9%, ETC 14.3%, DCO 16.1%). Femoral ETC was mainly performed within the first day (median 0, IQR 0-1 days), whereas pelvic ETC was delayed with increasing TBI severity (median 3, IQR 0-5 days for head AIS 3; 5, IQR 0-7 days for AIS 4). Observed mortality was 37.2% after NOM, 9.2% after ETC, and 10.3% after DCO.

CONCLUSION: ETC in patients with moderate TBI (AIS 3) was associated with reduced observed mortality relative to NOM and matching DCO. Increasi […]

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From single to dual platelet inhibition FIBTEM assay: fibrinogen replacement thresholds for critical bleeding after trauma.

Latona A, Winearls J, Ho A, Mitra B

PURPOSE: The FIBTEM assay in Rotational Thromboelastometry (ROTEM®) measures fibrin-based clot formation. The reagent transitioned from single to dual-platelet inhibition with addition of tirofiban to cytochalasin D, offering more complete platelet suppression. The aim of this study was to assess the impact of this change on transfusion thresholds in trauma.

METHODS: In this retrospective cohort study, we included adult patients with major trauma across Queensland who presented between 2022 and 2025 and had results available for paired ROTEM and conventional coagulation tests on arrival. Patients were sub-grouped into single and dual-platelet inhibition assay groups. Correlations between FIBTEM-A5 and Clauss fibrinogen were analysed, comparing correlation strength and diagnostic accuracy for fibrinogen levels of < 2.0 g/L.

RESULTS: The relationship between FIBTEM-A5 and Clauss fibrinogen was similar using single platelet inhibition assays (R^2 0.61;95%CI: 0.51-0.70) and dual-platelet inhibition assays (R^2 0.70;95%CI: 0.62-0.77; p = 0.94). Predicted FIBTEM-A5 amplitudes at 2 g/L fibrinogen were 7.83 mm (95%CI: 7.42-8.23) and 7.54 mm (95%CI: 7.22-7.87). Diagnostic performance for detecting fibrinogen < 2.0 g/L was comparable between assays, for both FIBTEM-A5 ≤ 10 mm and FIBTEM-A5 ≤ 8 mm (all p > 0.05). For the dual-platelet inhibition assay, sensitivity and specificity were 97. […]

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Pattern of blast injuries. A systematic review: Part 2 - Landmines, unexploded ordnance and terrorism.

Neubert A, Gaeth C, Seidelmann M, Bieler D

PURPOSE: This systematic review analyzes the patterns and distribution of blast injuries from landmine explosions, UXOs and terror attacks including suicide bombing, in civilians and military personnel.

METHODS: A search was conducted in two databases on 26. March 2025 for observational studies reporting blast injuries in civilian and military populations of all ages. Studies were clustered by explosion typ. A narrative data synthesis was performed. The JBI tool for case series was used.

PROSPERO: CRD42023391736.

RESULTS: 126 studies were included in the overall systematic review. 45 studies on terror & suicide bombings and 12 studies on landmines & UXOs are analyzed in the present paper. Studies on landmines from 1980 to 2006 included 13,270 individuals (age: 24.0 years (Range 8.2 years - 32.2 years); mainly male (88.9%)) in post-conflict low- and middle-income countries. Lower extremity injuries including amputations were most frequent, followed by eye injuries. Studies on terrorist attacks with a total of 12,518 mainly civilian victims (average age: 33,1 years; 61.6% males) reported on incidents in post-conflict regions and high-income cities mainly from the early 2000s. The injury pattern was multidimensional, with extremity injuries most frequent (28%-30%), including fractures, lacerations, and traumatic amputations (18%-20%).

CONCLUSION: This Systematic Review Part 2 […]

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Is nail plate fixation the way forward for distal femur fractures with long segmental metaphyseal comminution? - A study of 56 patients.

Mahesan H, Sundaram VP, Devendra A, Ramesh P, Dheenadhayalan J, Rajasekaran S

BACKGROUND: Mechanical instability due to long metaphyseal comminution and medial bone gap after lateral locking plate results in varus collapse and implant failure. We believe that fixation in the form of Nail and plate combination provides better stability and prevents complications.

MATERIALS: We performed a retrospective analysis of 56 distal femur fractures with long metaphyseal comminution treated in our institution with nail and plate construct from 2017 to 2023. The mean age was 43.1 years and the mean follow up was 24 Months (4-36 months). Of these 56 patients, 16 were closed injuries underwent primary fixation, and 40 were open injuries requiring staged reconstruction. At the final follow up, patients were assessed for shortening, knee range of movements, Sander et al. functional score, radiological union and loss in alignment.

RESULTS: The mean length of metaphyseal comminution was 10.15 cm(5-18 cm). Primary Iliac crest bone grafting was done in 25 patients. Of the 56 patients, 51 achieved primary union and the mean time to union was 6 months. Of the remaining, 3 patients united after secondary bone grafting and 2 patients had nonunion. The mean loss in aLDFA at the final follow up was 1.75° in 7 patients. Shortening of the limb was noted in 16 patients, and 4 patients had significant shortening of 2 cm. Three patients developed deep SSI, of which one patient requi […]

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Impact of modified ERAS protocol implementation on length of stay and complications in elderly hip fracture patients: a before-and-after study.

Liu Y, Ou G, Xie M, Wu F, Wang X, Yang B

AIM: Hip fractures in elderly patients pose significant healthcare challenges with high complication rates and prolonged hospital stays. Enhanced Recovery After Surgery (ERAS) protocols have shown promise in various surgical specialties, but their application in elderly hip fracture patients, particularly in Traditional Chinese Medicine (TCM) hospital settings, remains limited.

METHODS: This retrospective before-and-after study was conducted at a TCM hospital from July 2022 to July 2025. A total of 300 elderly hip fracture patients (≥ 65 years) were included: 146 in the control group (July 2022 - October 2023) receiving conventional care and 154 in the ERAS group (February 2024 - July 2025) receiving modified ERAS protocol incorporating TCM elements. Primary outcomes were length of hospital stay and perioperative complications within 30 days. Secondary outcomes included functional recovery [Harris Hip Score (HHS), Modified Barthel Index (MBI)], time to first mobilization, pain scores, and healthcare costs.

RESULTS: The ERAS group demonstrated significantly shorter length of stay compared with controls (12.38 ± 3.75 vs. 16.24 ± 5.10 days; mean difference 3.86 days, 95% CI: 2.84-4.88; P < 0.001). Total perioperative complications were reduced (7.79% vs. 15.07%; risk difference: -7.28%, 95% CI: -14.5% to - 0.1%; P = 0.047). The ERAS group showed superior functional outcomes with […]

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Prehospital endotracheal intubation and 30-day survival in severe trauma: a matched cohort study from Navarra, Spain.

Zulet Murillo D, Ferraz Torres M, Fortún Moral M, Belzunegui Otano T, Tamayo Rodríguez I

PURPOSE: Prehospital endotracheal intubation (ETI) is frequently used in major trauma, but its association with 30-day survival remains debated. We evaluated the association between prehospital ETI and 30-day survival and assessed whether sex modifies the age-related increase in mortality risk.

METHODS: We conducted a retrospective cohort study using the Navarra Major Trauma Registry (2010-2019). Patients aged ≥15 years with New Injury Severity Score (NISS) ≥15 were included; patients who died before hospital admission or had incomplete records were excluded. To mitigate confounding by indication, intubated and non-intubated patients were matched 1:1 using prehospital Glasgow Coma Scale (GCS), respiratory distress, and prehospital cardiac arrest as matching variables. Because the proportional hazards assumption for Cox regression was not met, parametric Weibull survival models were fitted, including a sex-by-age interaction term.

RESULTS: Overall, 1,909 patients were included; 212 (11.1%) underwent prehospital ETI. Matching yielded 190 pairs (n=380). In the matched cohort, prehospital ETI was not independently associated with 30-day survival (adjusted HR 0.91, 95% CI 0.65-1.27). Mortality risk was independently associated with lower GCS (per 1-point increase: HR 0.86, 95% CI 0.82-0.91) and respiratory distress (HR 7.63, 95% CI 4.34-13.4). A significant sex-by-age interaction […]

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Mechanical and manual cardiopulmonary resuscitation in traumatic cardiac arrest: a retrospective observational study.

El-Menyar A, Ahmed K, Howland IR, Mollazehi M, Mekkodathil A, Rizoli S, Cameron P, de Oliveira Manoel AL et al.

BACKGROUND: Prehospital Traumatic cardiac arrest has a high fatality rate despite advances in trauma systems and resuscitation strategies. Mechanical chest compression devices have been increasingly adopted to optimize cardiopulmonary resuscitation (CPR) during Emergency Medical Services (EMS) transportation. We aimed to evaluate the impact of CPR (mechanical during transportation versus manual CPR only at the scene) on survival among trauma patients.

METHODS: A retrospective analysis was conducted for patients who received CPR at the scene and during transportation to the hospital between 2016 and 2024.

RESULTS: A total of 610 patients sustaining blunt traumatic cardiac arrest were included. The mean age was 34 ± 12.6 years; 94.6% were male, and 5.4% were female patients. Two hundred twenty (36.1%) received mechanical CPR during transportation to the hospital, and 390 (63.9%) received manual CPR only at the scene. Compared with the manual CPR group, the mechanical CPR group had higher rates of primary chest injury (30.5% vs. 20.5%; p = 0.006), bystander CPR (15.9% vs. 6.7%; p = 0.001), adrenaline administrations (95% vs. 26.4%; p = 0.001), and initial non-shockable rhythm (80% vs. 27.2%; p = 0.001). The two groups were comparable in median Injury Severity Score (ISS), head Abbreviated Injury Score (AIS), and total transport time. Mechanical CPR was associated with a markedly […]

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Cervical spine clearance in pediatric trauma patients - Consensus algorithm of the Pediatric Spinal Trauma Group of the Spine Section of the German Society for Orthopedic and Trauma Surgery (DGOU).

Bolte J, Rüther H, Brecht P, Wiersbicki DW, Youssef Y, Jarvers JS, Disch AC

PURPOSE: Although cervical spine injuries in pediatric patients are rare, clinical and radiological examinations are clinical routine. Currently, no standardized decision algorithm does exist for detection of these injuries in the European region. The proposed algorithm is aiming to safely identify significant injuries through consistent diagnostics while minimizing immobilization and radiation exposure at the same time.

METHODS: The Pediatric Spinal Trauma Group of the Spine Section of the German Society for Orthopedic and Trauma Surgery (DGOU) developed an algorithm based on own scientific investigations and considering primary and secondary literature. The algorithm was developed through a formal consensus process and thereby aligns with international recommendations.

RESULTS: The algorithm uses the initial pediatric Glasgow Coma Scale (pGCS) to group patients to three treatment pathways. Resulting recommendations for imaging, clinical re-assessment, and further management include the child's age, trauma mechanism, and clinical evaluations.

CONCLUSION: Despite their rarity, potential cervical spine injury in children requires accurate diagnosis. Clinical examinations can be challenging, especially in children with compromised consciousness. Increased radiation exposure from X-rays and computer tomography (CT) scans must be considered, particularly as magnetic resonance im […]

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Development of a low-cost external fixation system: an off-label solution for resource-limited settings.

Seiser M, Deininger C, Fierlbeck J, Hollensteiner M, Coles P, Reuter C, Tempfer H, Traweger A et al.

PURPOSE: External fixators are essential for the surgical management of long-bone fractures, particularly in resource-limited settings where access to commercial systems is often restricted by cost and logistics. This study aimed to develop a low-cost external fixation concept using widely available construction materials and evaluate its biomechanical performance under axial loading.

METHODS: An external fixator composed of M10 threaded rods, washers, and nuts was developed. Five configurations were tested: SP (Steinmann pins), KT (threaded K-wires), KS (smooth K-wires), and BKS (bilaterally applied, smooth K-wires). A commercial external fixator served as control (CG). Juvenile bovine ulnae with a standardized fracture gap were subjected to cyclic axial loading. Construct stiffness, elastic deformation, and plastic deformation were assessed.

RESULTS: CG showed the highest stiffness (107.80 ± 14.12 N/mm), followed by SP (87.73 ± 17.92 N/mm), BKS (83.81 ± 10.01 N/mm), KS (49.84 ± 3.80 N/mm), and KT (46.66 ± 2.39 N/mm). Post-hoc analyses showed no significant differences between CG vs. SP and CG vs. BKS, whereas KT and KS were significantly less stiff. Elastic deformation followed a similar pattern. Plastic deformation was greatest in CG, while SP and BKS showed significantly lower values.

CONCLUSIONS: The threaded-rod fixator provides relevant mechanical stability at extreme […]

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Supplementary fixation in multifragmentary midshaft clavicle fractures: do interfragmentary screws or suture cerclage improve outcomes compared with plate-only fixation?

Barthel C, Hinkelmann MA, Schibler D, Frei F, Pape HC, Allemann F

BACKGROUND: Clavicle fractures are common injuries. As displaced or multifragmentary midshaft patterns carry a substantial risk of malunion and nonunion with nonoperative treatment, they have contributed to the increasing use of surgical fixation. Modern operative techniques include plate fixation, intramedullary devices, and hybrid strategies to address comminution and improve fragment stability. Supplementary interfragmentary screws and cerclage constructs have been proposed to enhance fragment control and optimize load transfer in multifragmentary fracture patterns. This study aimed to compare three fixation strategies for operatively treated midshaft clavicle fractures managed with plate osteosynthesis: Group IS (interfragmentary screw), Group CA (cerclage augmentation), and Group POF (plate-only fixation), with respect to operative time, complication rates, and potential biological and mechanical differences.

METHODS: This retrospective study included patients aged ≥ 16 years who underwent plate fixation for multifragmentary clavicle fractures between 2016 and 2024. Patients were categorized into three groups: Group IS (supplementary interfragmentary screw), Group CA (suture cerclage augmentation), and Group POF (plate-only fixation). Demographic, surgical, and radiographic data were collected. Callus formation was assessed by three independent observers. Statistical anal […]

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Cerebral glycerol during haemorrhagic shock in normal and raised intracranial pressure resuscitated with total REBOA: an experimental porcine study.

Bader S, Magnuson A, Brorsson C, Wallin G, Löfgren N, Löfgren F, Blind PJ, Öman M et al.

BACKGROUND: Haemorrhagic shock (HS) is frequently associated with secondary cerebral injury due to compromised cerebral perfusion. Resuscitative endovascular balloon occlusion of the aorta (REBOA) is an effective haemorrhage control strategy; however, concerns persist regarding its cerebral effects, particularly in the presence of elevated intracranial pressure (ICP). Cerebral microdialysis (CMD) derived glycerol (CGly) is a recognised marker of cellular membrane stress and injury.

OBJECTIVE: To investigate CGly dynamics during prolonged total REBOA (tREBOA) in HS and to compare cerebral cellular responses between animals with normal and elevated ICP.

METHODS: In this experimental porcine study, eighteen pigs were subjected to controlled HS and resuscitated with tREBOA for 90 min. Animals were allocated to either a normal ICP group (NICPG, n = 9) or an elevated ICP group (EICPG, n = 9). Continuous monitoring of proximal arterial pressure, ICP, and cerebral perfusion pressure was performed. CGly concentrations were measured using CMD throughout the experimental protocol.

RESULTS: tREBOA effectively restored proximal arterial pressure and cerebral perfusion pressure in both groups. CGly concentrations remained relatively stable during the haemorrhage phase and increased progressively during prolonged aortic occlusion. Importantly, CGly responses did not differ significantly be […]

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Transport and destination hospital for patients with suspected multiple and/or severe injuries - a systematic review and clinical practice guideline update.

Pavlu F, Könsgen N, Bieler D, Goossen K, Gliwitzky B, Häske D, Faul P, Wölfl C et al.

PURPOSE: Our aim was to update the evidence-based and consensus-based recommendations on criteria for transport and destination hospital for patients with suspected multiple and/or severe injuries based on available evidence. This guideline topic is part of the 2025 update of the German Guideline on the Treatment of Patients with Multiple and/or Severe Injuries.

METHODS: MEDLINE and Embase were systematically searched to January/February 2024. Further literature reports were obtained from clinical experts. Randomised controlled trials (RCTs) or observational studies reporting risk-adjusted outcomes were included if they compared different transport strategies or destination hospitals for patients with (suspected) multiple and/or severe injuries. We considered patient-relevant outcomes such as mortality and neurological outcomes. Risk of bias was assessed at outcome level using ROBINS-I for observational studies and RoB 2 for RCTs. We conducted meta-analyses if possible; alternatively, we synthesised the evidence narratively. We used GRADE to rate the certainty of evidence. Expert consensus was used to develop recommendations and determine their strength.

RESULTS: A total of 127 studies were identified. They addressed helicopter vs. ground emergency medical services (EMS) transport, ground EMS with vs. without a physician, on-scene time, and handover on arrival at the hospital […]

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Does the addition of peripheral nerve blocks improve analgesia during distal radius fracture reduction? A prospective, randomized, multicenter clinical trial.

Guerrero-Serrano JA, Santamaría-López A, García-Lamas L, Jiménez-Díaz V, González-Fernández Á, Cecilia D

PURPOSE: This study aimed to evaluate whether adding a median nerve and a superficial branch of the radial nerve block to hematoma block improves analgesia during distal radius fracture reduction and to identify factors associated with increased pain perception.

METHODS: A prospective, randomized, controlled study was conducted in two tertiary hospitals, including 180 adult patients with displaced distal radius fractures requiring closed reduction. Patients were allocated either to an isolated hematoma block group or to a combined block group receiving hematoma block with median nerve and superficial branch of the radial nerve blocks. Within the combined group, half of the median nerve blocks were performed under ultrasound guidance and half using anatomical landmarks. The primary endpoint was digital pain during traction and fracture reduction, assessed using a visual analog scale (VAS, 0-10). Secondary outcomes included wrist pain at different procedural stages and the influence of demographic and clinical variables including age, sex, laterality, participating hospital, body mass index (BMI), fracture type (AO/OTA), and baseline psychiatric or chronic analgesic medication use.

RESULTS: Digital pain was significantly lower in combined group compared with isolated hematoma block during traction (4.54 ± 2.53 vs. 7.39 ± 1.76; $p < 0.001$) and reduction (5.36 ± 2.56 vs. 8.33 ± 1. […]

PubMed · DOI

Under pressure - a sensor-based analysis of simulated prehospital pressure dressing application for hemorrhage management.

Brauckmann V, Menzel J, Welke B, Decker S, Macke C

BACKGROUND: The concept of "x-ABCDE", with "X" representing exsanguinating hemorrhage, highlights the priority of immediate bleeding control in trauma care. Despite being a fundamental intervention, pressure dressing techniques lack standardization. Studies show high variability in achieved pressures, often below therapeutic targets or exceeding harmful levels, with no correlation between subjective estimates and objective measurements.

OBJECTIVES: To objectively assess current practices of pressure bandage application and identify influencing factors including material type, technique, and provider experience.

METHODS: This prospective, single-center, simulation-based observational study was conducted from August 2024 to January 2025 at a Level I Trauma Center. Emergency medical providers applied pressure dressings according to personal preference using available materials. Pressure distribution was measured via two calibrated capacitive pressure sensors (61 measurement fields). Analysis included correlation tests, group comparisons, and multiple regression.

RESULTS: A total of 124 emergency medical providers (75% male; 36% paramedics, 44% emergency medical technicians, 13% trainees, 7% emergency physicians) completed pressure dressing applications, with 116 datasets included in final analysis. Mean maximum pressure was 169.35 ± 84.33 mmHg (range 50-412 mmHg) with applicati […]

PubMed · DOI

Clinical impact of early postoperative routine radiology in isolated pelvic injuries: a retrospective study.

Bolierakis E, Ioannou P, Schubert SL, Groven RVM, Bläsius FM, Bode M, Alabdulrahman H, Hildebrand F et al.

PURPOSE: The clinical value of routine immediate postoperative conventional radiography following surgical fixation of pelvic fractures remains debated. This study assessed the impact of early postoperative X-rays on management decisions and revision surgery in patients with isolated pelvic ring or acetabular fractures.

METHODS: We conducted a retrospective cohort study at a Level 1 trauma center, including adults undergoing surgical treatment for isolated pelvic ring or acetabular fractures between 2020 and 2023. All patients received high-quality intraoperative fluoroscopy and postoperative conventional radiographs. The primary outcome was any change in standard postoperative care based on radiographic findings. The secondary outcome was revision surgery prompted by conventional postoperative imaging.

RESULTS: A total of 216 patients were included (median age 63 years), with 156 pelvic ring injuries and 60 acetabular fractures. Postoperative radiographs were obtained at a median of 3 days post-surgery. Changes in postoperative care based on radiographs occurred in 2 patients (3.3%) within the acetabular fracture group. Revision surgery triggered by conventional imaging was required in 1 patient (0.6%) among pelvic ring injuries. In all cases, additional CT imaging preceded definitive management changes.

CONCLUSION: Routine immediate postoperative conventional radiography a […]

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Older age and post-traumatic organ failure a TraumaRegister DGU® analysis of 34,469 elderly major trauma patients.

Bewersdorf TN, Lefering R, Stein S, Streblow J, Findeisen S, Schmidmaier G, Grossner T

PURPOSE: As the number of elderly major trauma patients (EMTP) increases continuously, this study sought to determine the incidence of organ failure (OF) and multiple organ failure (MOF) and their impact on mortality and recovery status in this cohort.

METHODS: Based on the German TraumaRegister DGU® database this study included EMTP aged between 65 and 99 years with an Injury Severity Score (ISS) of ≥ 16 between 2014 and 2023. OF was defined as Sequential Organ Failure Assessment (SOFA) score of ≥ 3 and MOF as failure of ≥ 2 organs.

RESULTS: The database revealed 34,469 EMTP, here 37% suffered from OF (16.0%) or MOF (21.0%). Mortality was significantly higher in OF and MOF patients compared to non-OF patients (non-OF: 7.4%; OF: 45.0%; MOF: 59.0%; $p < 0.001$). 80.9% of all deaths were associated with failure of at least one organ and 51.1% of non-survivors suffered from MOF. Highest odds ratios (OR) for mortality were found for liver (OR: 5.13, $p < 0.001$), central nervous system (OR: 3.63, $p < 0.001$) and renal failure (OR: 3.25, $p < 0.001$), followed by cardiovascular (OR: 1.95, $p < 0.001$) and lung failure (OR: 1.16, $p = 0.010$). Haemostatic failure had no impact on mortality.

CONCLUSION: OR and mortality differ significantly between distinct OFs due to varying degrees of success in treatment options and systemic impact of these OFs. Clinicians should be aware that OF and MOF o […]

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Denis pattern-dependent biomechanical behavior of posterior trans-iliac plate fixation compared with bilateral triangular osteosynthesis in unilateral sacral fractures: a finite element study.

Yang J, Böker KO, Li C, Schilling AF, Ritter Z, Acharya M, Lehmann W

OBJECTIVE: Posterior trans-iliac plate osteosynthesis (TPO) is commonly used for the stabilization of unilateral vertically unstable sacral fractures. However, its biomechanical performance across different Denis fracture patterns remains unclear. This study aimed to investigate the fracture pattern-dependent biomechanical behavior of posterior trans-iliac plate osteosynthesis (TPO) in unilateral vertically unstable sacral fractures, using bilateral triangular osteosynthesis (BTO) as a high-stability reference construct.

METHODS: A validated three-dimensional finite element model of the lumbopelvic complex was developed to simulate unilateral vertically unstable sacral fractures corresponding to Denis type I, II, and III patterns. Two posterior fixation strategies, TPO and BTO, were constructed for each fracture type. Seven physiological loading conditions were applied, including standing, flexion, extension, axial rotation, and lateral bending. Von Mises stress distribution and fracture displacement were evaluated. Fracture micromotion was quantified by calculating relative displacement changes between paired points along the fracture gap.

RESULTS: When interpreted relative to the high-stability BTO reference construct, TPO showed fracture pattern-dependent variation in fracture displacement. The difference was most pronounced in Denis type I fractures and progressively decr […]

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Factors associated with in-hospital mortality in necrotising soft tissue infections. a multicentre retrospective cohort study.

Podda M, Ceresoli M, Virdis F, Rosa F, Pilia T, Murzi V, Dessì A, Tedesco S et al.

PURPOSE: Necrotising soft tissue infections (NSTIs) are rare but life-threatening conditions associated with high mortality rates. This multicentre study aimed to identify admission variables associated with in-hospital mortality.

METHODS: This retrospective study included adult patients with surgically confirmed NSTIs treated at four high-volume academic referral centres in Italy between 2010 and 2024. Demographic, clinical, physiological, and laboratory variables available at hospital admission were analysed. Categorical variables were compared between survivors and non-survivors using the chi-square test or Fisher's exact test, while quantitative variables were compared using the Student's t test or Mann-Whitney U test. Multivariable logistic regression analyses were performed to identify independent predictors of in-hospital mortality. Results were reported as ORs with 95%CI. A prognostic nomogram was developed from the final multivariable model, including variables independently associated with mortality.

RESULTS: A total of 379 patients were included. In-hospital mortality was 16.7%. In subgroup comparisons, mortality was 17.0% in necrotising fasciitis and 22.4% in Fournier's gangrene (P = 0.275). In the overall NSTI population, age (aOR 1.061, 95%CI 1.037-1.087) and serum creatinine at admission (aOR 1.301, 95%CI 1.040-1.629) were independently associated with mortalit […]

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Influence of frailty on short-term mortality in older patients with multiple trauma in the emergency department.

Çelik ■, Toker A, Biçer S, Cebecio■lu ■K

PURPOSE: To investigate the association between frailty and 28-day mortality in older patients presenting to the emergency department with multiple trauma and to assess the discriminative performance of PRISMA-7 and the Trauma-Specific Frailty Index (TSFI) in frailty assessment in relation to commonly used trauma severity scores.

METHODS: This prospective observational study included patients aged ≥ 65 years with multiple trauma admitted to the emergency department between July 1 and August 31, 2024. Frailty was assessed using PRISMA-7 and TSFI. Patients were stratified into frail and non-frail groups and further categorized according to age groups (65-74, 75-84, and ≥ 85 years). Clinical characteristics, trauma severity scores, and outcomes, including 28-day mortality, were compared. Receiver operating characteristic (ROC) analysis was used to evaluate discriminative performance, and logistic regression analysis was performed to assess the association between scores and mortality.

RESULTS: A total of 144 patients were included. Frailty was significantly associated with increased 28-day mortality, higher ICU admission rates, and longer hospital stays (p < 0.001). Mortality increased with age, reaching 30.0% in patients aged ≥ 85 years (p = 0.006). Frailty scores (PRISMA-7 and TSFI) increased significantly with age, whereas ISS did not differ significantly across age groups. R […]

PubMed · DOI

Mortality prediction in older adults with earthquake-related trauma: a model combining MEWS and laboratory biomarkers.

Ozen A, Acehan S, Satar S, Bulut EA, Gulen M, N■g■z K, Sevd■mbas S, Sonmez GO et al.

PURPOSE: This study aimed to identify laboratory parameters and clinical scores that can predict in-hospital mortality among patients aged 65 years and older who presented to the emergency department (ED) with earthquake-related trauma following the February 6 and 20, 2023 earthquakes.

METHODS: In this retrospective observational study, 362 patients with earthquake-related trauma between February 6-21, 2023, were included. Demographic characteristics, laboratory results, and clinical scores such as the Modified Early Warning Score (MEWS), Injury Severity Score (ISS), and Shock Index (SI) were analyzed.

RESULTS: Of the included patients, 53.9% (n = 195) were female, and the median age was 73 years (IQR: 69-80). The overall mortality rate was 9.4%. Multivariable logistic regression analysis identified MEWS (OR: 3.535; 95% CI: 2.397-5.212; p < 0.001), urea (OR: 1.009; 95% CI: 1.002-1.015; p = 0.008), and hs-Troponin I (OR: 1.001; 95% CI: 1.000-1.002; p = 0.016) as independent predictors of mortality. Receiver operating characteristic (ROC) analysis demonstrated an AUC of 0.936 for the combined model including MEWS, urea, and hs-Troponin I (95% CI: 0.875-0.996; p < 0.001). Among the variables, MEWS showed the highest AUC (0.903; 95% CI: 0.829-0.976). The optimal cutoff value for MEWS was determined as 1.5, with a sensitivity of 70.6% and a specificity of 97.6%.

CONCLUSIONS: In o […]

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The economic burden in terms of cost of illness and generic health-related quality of life of posttraumatic long bone non-unions among the adult population of the Netherlands from a societal perspective.

van der Broeck LCA, Shchurov DG, Gabrio A, Lodewijks AJL, Poeze M, Blokhuis TJ, Evers S

PURPOSE: This study assessed the societal economic burden in terms of cost of illness and health-related quality of life (HRQoL) of posttraumatic long bone non-unions in the Netherlands.

METHODS: An incidence-based bottom-up approach was used, focusing on adult patients with posttraumatic long bone non-unions. The analysis included healthcare costs, patient and family costs, and productivity losses, measured using the iMCQ and iPCQ questionnaires. Cost evaluations followed Dutch costing guidelines, and productivity losses were calculated using the friction cost method. HRQoL was assessed with the EQ-5D-5L. A deterministic one-way sensitivity analysis varied baseline characteristics by ± 10%. Scenario analyses were conducted from healthcare and patient perspectives, as well as for patient subgroups.

RESULTS: Average costs per patient during the three months before the initial visit at a non-union clinic were €8,928 (healthcare, N = 78), €1,360 (patient and family, N = 58), and €4,313 (productivity losses, N = 59), the average of the observed data calculates to 10,831 (N = 44). The mean EQ-5D-5L utility score was 0.390 (± 0.29 SD). Subgroup analysis showed no significant cost increase for patients with infections or open fractures. However, a higher Non-Union Severity Score and a lower HRQoL were significantly associated with higher total costs.

CONCLUSION: Posttraumatic long […]

PubMed · DOI

Nonoperative treatment of Maisonneuve fractures.

Frühauf J, Bartoní■ek J, Fojtík P, Hor■áková A, Tu■ek M

PURPOSE: The aim of this study is to describe the basic pathoanatomical characteristics of a stable Maisonneuve fracture and mid-term results of its nonoperative treatment.

METHODS: The study included 17 prospectively collected patients with a mean age of 59 years. The postinjury ankle CT had to meet the following criteria: nondisplaced or minimally displaced (up to 1 mm) fracture of medial malleolus, medial clear space less than 3 mm, nondisplaced or minimally displaced (up to 2 mm) fracture of posterior malleolus, anatomical position or minimal malposition of the distal fibula in the fibular notch (widening of the tibiofibular space up to 2 mm or external rotation of the distal fibula up to 10°). The average follow-up period was 34 months, the final follow-up included CT examination and functional evaluation based on AOFAS and FAOS scores.

RESULTS: A medial malleolus fracture was recorded in 12% cases, a posterior malleolus fracture in 29% patients and a Tillaux-Chaput tubercle fracture in 18% cases. All fractures of the proximal fibula, medial and posterior malleolus healed within 3 months. The position of the distal fibula in the fibular notch did not change in 11 cases compared to the post-injury CT scan, improved slightly in 5 cases, and worsened slightly in 1 case. The average final AOFAS hindfoot score was 96.9 points and the average final FAOS score was 98.7%.

CONCL […]

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Age-related differences in prehospital triage and critical interventions: a multicenter TraumaRegister DGU® analysis.

Koch DA, Hagebusch P, Hackenberg L, Pavlu F, Jakobi T, Münzberg M, Lefering R, Schweigkofler U et al.

PURPOSE: Older trauma patients are known to be undertriaged in the prehospital setting. However, it remains unclear whether these disparities translate into differences in the delivery of critical prehospital interventions. This study aimed to evaluate age-related differences in prehospital triage, allocation, and treatment within a physician-staffed emergency medical service system.

METHODS: A retrospective multicenter analysis of the TraumaRegister DGU® was conducted, including trauma patients aged ≥ 20 years between 2020 and 2023 with trauma team activation and subsequent intensive care unit admission. Patients were stratified into two age groups, 20-59 vs. ≥60 years, with additional subgroups up to 90-99 years. Prehospital interventions, transport modality, and trauma center allocation were analyzed. Furthermore, predefined high-risk scenarios (severe motor vehicle trauma, traumatic brain injury with GCS ≤ 8, and pneumothorax) were evaluated.

RESULTS: A total of 67,069 patients were included, with comparable injury severity across age groups (mean ISS 20.2). Older patients were less frequently transported by air and less often allocated to supra-regional trauma centers. While overall intervention rates were similar, differences emerged in high-risk scenarios. Older patients received endotracheal intubation and tranexamic acid less frequently despite comparable injury seve […]

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Prediction of risk of death in severely injured patients: the revised injury severity classification score, version 3 (RISC III).

Lefering R, Imach S, Bieler D

PURPOSE: Trauma Registries with a focus on severely injured patients use survival as their primary outcome. In order to compare hospitals, interventions, and changes over time, a precise risk of death prediction is mandatory. The German TraumaRegister DGU® uses the RISC II model (Revised Injury Severity Classification, version II) since 2013. The aging trauma population and an improved handling of missing data required the present revision.

METHODS: A total of 53,738 seriously injured trauma patients documented in 2022-2023 served as basis for development and validation (3:1 ratio). Missing values should now be considered to be within normal physiological range except in patients with specific findings indicative for an altered physiology. These findings were suggested by clinical experts and validated by registry data. The increasing age of trauma patients was addressed by additional categories and higher point weights. A logistic regression analysis provided new point weights for all predictors. Precision (observed versus predicted mortality) and discrimination (area under the receiver operating characteristic curve, AUROC) were calculated in the development and validation dataset.

RESULTS: Patients in both datasets were well comparable, with a mean age of 55 years, 69% males, and an average Injury Severity Score (ISS) of 18 points. The rate of missing values ranged from 0% […]

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Dislodgement forces and cost effectiveness of prehospital securement of peripheral intravenous catheters on moist skin: a randomized controlled clinical trial in healthy volunteers.

Vogeler J, Schumann S, Heinrich S, Hell J, Schmutz A

OBJECTIVES: Peripheral intravenous catheters (PIVC) are widely used clinical devices for administering fluids or essential drugs to patients. The failure rate of PIVC remains high and Emergency Medicine has been shown to be a risk factor for dislodgement. When the skin is moist from sweat or fluids, standard intra-hospital dressings and securements fail. In emergency situations, a failed catheter can then critically delay intravenous therapies. The most effective dressing to prevent accidental removal of a prehospital PIVC remains unclear. It was the aim of this study to compare the force required to dislodge a PIVC with four different methods of securing PIVCs used in emergency medicine. In addition, the costs were calculated.

METHODS: Artificial sweat was applied to the skin of 180 volunteers. PIVCs were attached onto the forearm using four different securements (elastic gauze, cohesive gauze, clingfilm and a velcro securing device). Continuously increasing traction force was applied until dislodgement of the respective securement. For statistical tests, either Friedman's test or repeated measures ANOVA was used.

RESULTS: Clingfilm showed the greatest resistance to increasing pulling force with cohesive bandages as close second. The velcro securing device was strongest at resisting low level of forces but fell off sharply at higher force. Elastic bandages were the weakest i […]

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Injury to the thorax is a dominant contributor to post-traumatic neutrophil mediated inflammatory response.

van Eerten FJC, Duebel LV, de Fraiture EJ, Hanlo MF, Breuking EA, Brands SR, van den Bosch TD, van Wessem KJP et al.

PURPOSE: Traumatic injury induces a neutrophil associated inflammatory response mediated by the release of damage-associated molecular patterns (DAMPs). This response can be maladaptive, increasing risk of infection. Since blunt thoracic injuries are associated with high infection rates, we hypothesized that these injuries elicit a stronger neutrophil associated inflammatory response than injuries to other body regions.

METHODS: A cohort study was conducted in adult patients presenting at the trauma bay of a level 1 trauma center between March 2020 and November 2022. These patients underwent routine analysis of their neutrophil associated inflammatory response by phenotyping blood neutrophils using automated flow cytometry. Injury per body region was counted when the Abbreviated Injury Severity (AIS) scores were ≥3 to focus on substantial injuries. The primary outcome was an extensive post-traumatic inflammatory neutrophil response.

RESULTS: 861 patients were included. Blunt internal thoracic injuries showed the strongest association with severe neutrophil associated inflammation (OR 5.7), followed by abdominal (OR 2.6), pelvis (OR 2.5) and lower extremities (OR 2.3), even when corrected for ISS, age and hemodynamic instability. In multivariable analysis, only thoracic internal injuries remained a statistically significant contributor to excessive neutrophil associated inflam […]

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Major improvement using peroperative 3D compared to 2D imaging for SI screw fixation in traumatic pelvic fractures.

van Eerten FJC, Benders KEM, van Baal MCPM, Hietbrink F

INTRODUCTION: Pelvic fractures, though accounting for only 3% of skeletal injuries, carry a high mortality rate, primarily due to the associated risk of hemodynamic instability. Sacroiliac (SI) screw fixation is a common surgical intervention, which can be technically challenging due to the complex pelvic anatomy. Imaging techniques have evolved from 2D to 3D imaging for its improved accuracy in screw placement. The impact of this transition for SI screw placement was evaluated.

METHOD: A single-center observational retrospective study was performed between 2013 and 2023. Patients ≥ 18 years who underwent SI screw placement following pelvic injuries were analyzed. 2D imaging was used until 2021, after which 3D imaging was implemented. Outcomes included the need for revision surgery during initial admission and < 18 months after trauma.

RESULTS: 2D imaging was used in 42 patients and 3D imaging in 44 patients. Baseline characteristics and injury severity of both groups were comparable. In the 2D group, 18 patients (42.9%) had suboptimal placement of the SI screw. Of these, 3 (7.1%) required revision surgery due to malpositioned screws. In contrast, no patients (0.0%) in the 3D group had suboptimally positioned screws or underwent revision surgery, ($p < 0.001$, $p = 0.22$).

DISCUSSION: Intraoperative 3D imaging enhanced the accuracy of SI screw placement, leading to an encouragin […]

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Minimally invasive versus open fixation for flail chest in elderly patients: a prospective cohort study on perioperative safety and long-term outcomes.

Tian Y, Zheng Z, Ji Z, Lu W, Song N, Wang J

OBJECTIVE: To systematically evaluate the perioperative safety and short- and long-term efficacy of minimally Invasive fixation using a reversed shape-memory alloy plate versus conventional open fixation for the treatment of flail chest in elderly patients.

METHODS: This prospective cohort study enrolled 130 elderly patients (≥ 60 years) with flail chest between January 2022 and January 2026. After 1:1 propensity score matching, 100 patients were included (minimally invasive group, n = 50; open group, n = 50). Perioperative parameters (operative time, blood loss, C-reactive protein(CRP) at 24 h/48 h/72 h, complications), short-term outcomes (resting/cough-evoked Numerical Rating Scale(NRS) pain scores, non-steroidal analgesic drug duration, rescue analgesic frequency, Patient-Controlled Analgesia Pump(PCA) consumption, time to ambulation, hospital stay), and long-term outcomes (implant dislodgement, 6-month pulmonary function, Douleur Neuropathique 4 Questions Naire(DN4), Medical Outcomes Study 36- Item Short Form Health Survey(SF-36)) were compared.

RESULTS: The minimally invasive group had significantly fewer incision-related complications (P = 0.046), lower resting NRS scores on postoperative days 1 and 3 (P < 0.01; P = 0.041), lower cough-evoked NRS score on day 1 (P = 0.011), shorter non-steroidal analgesic duration (P < 0.01), and lower PCA consumption (P = 0.027). At 6 […]

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Epidemiology and outcomes of traumatic sternal fractures and associated blunt cardiac injury: a nationwide cohort study in the Netherlands.

Klei DS, Benders KEM, Leenen LPH, van Wessem KJP

PURPOSE: Comprehensive data on epidemiology, trauma mechanisms, associated injuries, and outcomes of traumatic sternal fractures are scarce. This study analysed nationwide data to improve diagnosis and management within the Dutch healthcare system.

METHODS: This nationwide retrospective cohort study using the Dutch National Trauma Registry included adult patients admitted with traumatic sternal fractures between 2015 and 2023. Patients with prehospital cardiopulmonary resuscitation or penetrating trauma were excluded. Incidence, patient characteristics, trauma mechanisms, associated injuries, and in-hospital outcomes were analysed. Subgroup analyses evaluated patients with concomitant blunt cardiac injury (BCI).

RESULTS: Of 568,399 adult trauma admissions, 4,765 patients (0.84%) sustained traumatic sternal fractures. Median age was 62 years; 60% were male. Motor vehicle accidents (48%) and falls (28%) were the leading mechanisms. 35% were severely injured (ISS ≥ 16). Associated injuries included rib fractures (51%), spinal fractures (36%), and lung contusions (18%). Critical care unit admission was 40%, with median mechanical ventilation duration of 4 days; median hospital stay was 5 days. In-hospital mortality was 5.7%, and 30-day mortality 6.0%. BCI occurred in 9.5% of patients and was associated with a higher number of injuries and increased injury severity, emergency inte […]

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Trauma patients ≥ 65 years old constitute 40% of severely injured adult trauma patients in Sweden - a 10-year analysis of geriatric trauma prevalence.

Bredberg S, Modalen E, Jöneby C, Nilsson D, Lapidus O

BACKGROUND: The elderly population in Sweden is steadily increasing. Age is an independent predictor of mortality in trauma patients. There is growing evidence that geriatric patients constitute an increasing proportion of the Swedish trauma population. No studies have analyzed temporal trends in the proportion of geriatric trauma patients in the context of the national trauma population in Sweden.

AIM: To determine the proportion of geriatric patients in the Swedish trauma population and analyze temporal trends in geriatric trauma prevalence.

METHODS: The study cohort was 11,969 severely injured trauma patients (NISS > 15) treated in Sweden during 2013-2022, obtained from the Swedish Trauma Registry. The proportion of geriatric patients was compiled annually. Temporal trends were analyzed using weighted linear regression models. Two subgroups were defined based on injury severity (NISS ≥ 25 and NISS ≥ 40).

RESULTS: The proportion of patients ≥ 65 years in the adult trauma population increased from 30% to 40% during 2013-2022 ($p = 0.003$), while the proportion of patient ≥ 80 years also increased, from 8.6% to 17% ($p = 0.009$). Similar increases were also seen in the NISS ≥ 25 and NISS ≥ 40 subgroups.

CONCLUSION: There is likely an increase in geriatric trauma prevalence in Sweden, but the full extent of this temporal trend remains uncertain. Trauma patients ≥ 65 years old co […]

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A totally laparoscopic strategy combining microwave ablation and autologous blood transfusion for patients with initial WSES grade IV traumatic splenic rupture in selected transient responders: a proof-of-concept study.

Feng Z, Xie D, Hong W, Zheng C

PURPOSE: Splenic artery embolization (SAE) is the standard intervention for transient responders with high-grade splenic trauma. When SAE is unavailable, contraindicated, or declined, alternative minimally invasive approaches may be warranted. This study evaluates a totally laparoscopic strategy combining microwave ablation (MWA) for hemostasis with intraoperative autologous blood transfusion (ABT) in selected transient responders with initial WSES Grade IV splenic rupture.

METHODS: This retrospective study included 10 consecutive transient responders (October 2021-June 2024) with initial WSES Grade IV splenic rupture who underwent totally laparoscopic MWA combined with intraoperative ABT. All patients exhibited transient hemodynamic response to limited resuscitation (≤ 1500 mL fluids) and preoperative shock index > 0.9. Intraoperative injury was graded by the AAST system. Outcomes included technical success, transfusion requirements, complications, and follow-up.

RESULTS: All 10 patients underwent successful spleen-preserving laparoscopy without conversion. AAST grades: II ($n = 2$), III ($n = 4$), IV ($n = 4$). Mean operative time was 191.3 ± 101.7 min. Mean blood salvage was 1587.5 ± 663.9 mL, yielding 490.8 ± 80.2 mL autologous red cells. No patient required postoperative allogeneic transfusion. Hemoglobin, platelets, and lactate improved significantly by postoperative day 3 (P […]

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Pediatric pancreatic trauma: a 10-year retrospective analysis from two high-complexity centers in Medellín, Colombia.

Barrientos ML, Calderon SP, Zapata CAL, Lopez CAD

PURPOSE: Pediatric pancreatic trauma is uncommon, but it is associated with significant morbidity and an ongoing debate regarding its management. Hemodynamic stability and pancreatic duct involvement are key factors in treatment decisions.

METHODS: A retrospective study analyzed 25 pediatric patients with pancreatic trauma treated between 2010 and 2020 at two high-complexity hospitals in Medellin. Clinical variables, trauma mechanism, imaging findings, management, and complications were analyzed.

RESULTS: Blunt trauma was the predominant mechanism (64%). Preoperative computed tomography was performed in 12 patients (48%); radiological AAST grading was not retrievable in 13 (52%). Immediate surgery was performed in 13 patients (52%), predominantly driven by associated intra-abdominal injuries rather than the pancreatic injury itself. Non-operative management was initially attempted in 9 patients (36%), with a 56% success rate; failures (44%) were related to clinical deterioration and progression of peripancreatic collections. Clinically relevant postoperative pancreatic fistula (ISGPS Grade B) occurred in 3 of 13 operated patients (23%), with a mean closure time of 55.6 days (range 18-110); no Grade A or Grade C fistulas were recorded. Other complications included post-traumatic pancreatitis (24%), pancreatic pseudocyst (8%), and organ/space surgical site infection (8%); most […]

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Influence of Antithrombotics on Severely Injured Patients Over 50 With Blunt Abdominal Trauma.

Eibinger N, Limberger B, Hörlesberger N, Puchwein P, Lefering R, Bayer TP, Zumsteg JS, Pape HC et al.

INTRODUCTION: Severe abdominal trauma represents a critical subset of injuries, particularly in the aging European population, where the prevalence of preexisting anticoagulant or antiplatelet therapy is increasing. While the impact of such medications on outcomes in traumatic brain injury is well studied, limited data exist regarding their influence in patients with abdominal trauma.

MATERIALS AND METHODS: This retrospective cohort study used data from the TraumaRegister DGU® between January 2015 and December 2023. Inclusion criteria were patients aged ≥ 50 years with severe blunt abdominal trauma (AIS Abdomen ≥ 3) without relevant head injury (AIS Head ≤ 3) from Austria, Germany, or Switzerland. Patients were grouped according to pre-injury antithrombotic medication (no medication [NM] vs. antithrombotic medication [AM], with subgroups). Statistical analysis included descriptive comparisons, standardized mortality ratios (SMR) using RISC III, and multivariate logistic regression to identify independent predictors of hospital mortality.

RESULTS: Of 328,281 patients, 4,069 met inclusion criteria (2,831 NM; 1,238 ATM). Patients in the AM group were older (74.1 vs. 62.3 years) and had higher mortality (25.0% vs. 10.4%), particularly in the DOAC subgroup (30.3%). SMR analysis demonstrated increased mortality in the AM group compared to expected rates. In multivariate logistic re […]

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Predicting limb amputation in trauma-related necrotizing fasciitis of the extremities: insights from a 10-year cohort.

Yang S, Ren Z, Li Y, Zhao Z, Sun C, Liu L, Jin L, Hou Z

BACKGROUND: Necrotizing fasciitis (NF) is a rapidly progressive, life-threatening soft tissue infection that in most cases requires aggressive surgical management. Major limb amputation is a devastating limb-related outcome. We retrospectively analyzed NF cases to quantify the rate of amputation and identify associated risk factors.

METHODS: We retrospectively reviewed the medical records of all patients with trauma-related extremity NF treated at Hebei Medical University Third Hospital from Jan 2014 to Dec 2024. Demographics, comorbidities, admission laboratories and key time intervals were compared between patients who did and did not undergo major limb amputation. Multivariable logistic regression identified independent predictors, and receiver operating characteristic (ROC) analysis assessed the discriminative power of continuous variables.

RESULTS: Of 134 patients, 22 (16.4%) required amputation. Compared with the non-amputation group, amputees had a markedly longer pre-hospital delay (median 242 h vs. 65 h), more multiple injuries (63.6% vs. 13.4%) and a higher prevalence of diabetes (59.1% vs. 21.4%). Admission lactate dehydrogenase (LDH) was substantially higher (median 589 vs. 182 U/L). In multivariable analysis, elevated LDH, prolonged time from traumatic injury to hospital admission, multiple injuries and diabetes were independently associated with amputation. LDH […]

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Is 18 F-FDG PET-CT-guided geographic debridement superior to conventional debridement in appendicular chronic osteomyelitis? a retrospective comparative cohort study.

Elsheikh A, Elseidy H, Saad H

BACKGROUND: Determining the surgical extent of chronic osteomyelitis and fracture-related infection (FRI) remains challenging. Adequate resection margins prevent recurrence; however, no standardised preoperative tool objectively defines resection borders.

OBJECTIVE: To compare PET■CT-guided surgical planning versus the local standard■of■care imaging pathway (predominantly plain radiographs, with selective CT or MRI in a minority of patients) for chronic limb osteomyelitis and FRI outcomes.

METHODS: Retrospective comparative cohort study of 126 patients treated 2019-2024. Group 1: 84 patients underwent 18 F-FDG PET-CT imaging and map-based geographic debridement. Group 2: 42 patients received surgery after local standard■of■care imaging pathway (predominantly plain radiographs, with selective CT or MRI).

PRIMARY OUTCOME: infection recurrence at 12 months. Multivariable logistic regression controlled for confounders.

RESULTS: PET-CT group achieved 3.6% recurrence versus 16.7% in the conventional group ($p = 0.011$), representing 78.5% relative risk reduction and a number-needed-to-treat of 8. Critically, the PET-CT cohort had significantly worse baseline complexity: 3.3-fold longer infection duration (43.1 vs. 13.1 months, $p < 0.001$) and 3.6-fold more prior surgeries (2.48 vs. 0.69, $p < 0.001$). Multivariable analysis confirmed PET-CT-guided surgery as an independent protective […]

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Digital coordination in mass casualty incidents: a retrospective analysis of prehospital distribution times before and after IVENA-MANV implementation.

Brauckmann V, Ringe B, Caviglia M

BACKGROUND: Digital coordination tools have been proposed to improve management during mass casualty incidents (MCIs). In 2021, the IVENA-MANV system was introduced in the Hannover region as an extension of the established IVENA eHealth platform to enable digital patient tracking and hospital capacity management. However, no real-world evaluations have yet quantified their operational impact.

OBJECTIVES: This study aimed to provide the first real-world evaluation of the IVENA-MANV digital coordination system and to determine how its implementation affected prehospital distribution processes and time intervals during mass casualty incidents.

METHODS: A retrospective observational study to assess the impact of the IVENA-MANV digital coordination platform on prehospital times during MCIs in Germany was conducted using dispatch records and IVENA-MANV logs from all MCIs between January 2018 and July 2025 in the Hanover-Region in Germany. Primary outcomes were the triage-to-evacuation (TtE) and total prehospital time (TPHT) per patient. Mann-Whitney U tests compared pre- and post-implementation groups, and multiple linear regression models examined associations between IVENA-MANV use, MCI-level, and time intervals.

RESULTS: Seventy-three MCIs met inclusion criteria, including 188 documented individual casualty characteristics. 41.1% of incidents used IVENA-MANV. Most MCIs were tra […]

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The real-world management of acute mesenteric ischemia in Spain: results from a multicenter national survey.

González-Castillo AM, Calsina Juscafresa L, Gelabert A, Velescu A, Marrano E, Vikatmaa P, Tolonen M

BACKGROUND: Acute mesenteric ischemia (AMI) remains one of the most lethal vascular emergencies, with in-hospital mortality rates frequently exceeding 50%. Although early diagnosis and timely revascularization are critical, clinical practice remains highly variable. This study aimed to evaluate the real-world management of AMI in Spain, identifying gaps in resources, protocols, and interhospital coordination.

METHODS: A national cross-sectional survey was conducted between March and August 2024 using the Survio® platform. The questionnaire was distributed to general surgeons through national surgical societies and included 27 items covering hospital infrastructure, clinical protocols, diagnostic and therapeutic availability, personal experience, and perceived system-level barriers.

RESULTS: A total of 291 surgeons responded. The median age was 40 years (IQR: 17). Most were consultants (76.3%) working in tertiary (42.6%), secondary (33%), or community hospitals (24.4%). While 97.6% reported access to 24/7 multiphasic CT and 90.4% to round-the-clock radiology, only 51.9% had 24/7 interventional radiology and 60.1% vascular surgery. Just 26.8% had institutional AMI protocols. The median distance to a referral center was 25 km (range: 2-250 km), and 68.4% reported difficulty in patient transfers. While 96.9% felt competent managing AMI, only 36.4% were familiar with the term "int […]

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Changes in operative and in-hospital outcome of proximal femoral fractures before and after the implementation of work hour restrictions in Switzerland.

Tewes A, Canal C, Schöb O, Neuhaus V

PURPOSE: Surgeon fatigue is a recognized risk factor for patient safety; however, reduced working hours may limit surgical training exposure. In 2005, Switzerland introduced a maximum 50-hour workweek for resident surgeons. In view of current discussions on further reductions (42 + 4 h), concerns persist regarding training quality and workforce capacity. This study evaluates the impact of the 2005 Swiss labor law on patient outcomes, operative characteristics, and intraoperative teaching utilizing a standardized trauma procedure.

METHODS: A retrospective analysis of anonymized data from a national database was performed. All patients undergoing operative fixation of trochanteric femur fractures (ICD-10 S72.1) were included. Two four-year periods were compared: pre-regulation (2001-2004) and post-regulation (2016-2019). Primary endpoints were in-hospital complications, mortality, and length of stay. Secondary endpoints included patient characteristics, surgeon seniority, teaching status, operative duration, and time to surgery.

RESULTS: Post-regulation patients were older and exhibited higher American Society of Anesthesiologists classifications despite fewer comorbidities. Complications increased from 9.1% to 17.7%, and mortality from 1.6% to 3.5%. Length of stay decreased from 14 to 9 days. Operative duration decreased by 10 min across all surgeon levels. Resident surgeons i […]

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Early discharges in severely injured patients by ISS score: exploring injury patterns and coding practices.

Parouei F, Krüsemann EJZ, Poeze M, de Jongh MAC, Hietbrink F, Landelijke Beraadsgroep Traumachirurgen (LBTC)

PURPOSE: This study examines the prevalence of early discharge (ED) among patients classified as severely injured in the Dutch National Trauma Registry (DNTR) and evaluates whether identifiable patient-, injury-, and system-related factors are associated with the occurrence of ED in this population.

METHODS: This retrospective cohort study analyzed DNTR data from 2015 to 2022, focusing on severely injured patients (Injury Severity Score [ISS] ≥ 16). Patients were grouped as early discharge (ED; discharged within 48 h without in-hospital mortality) or non-ED (NED). Patients' characteristics were compared using descriptive statistics, and a mixed-effect model identified factors associated with ED.

RESULTS: From 2015 to 2022, a total of 37,626 severely injured patients were registered including 1,454 (3.9%) ED patients. This proportion increased from 3.2% in 2015 to 4.8% in 2022. ED was more common in younger patients, males, general practitioner (GP) or self-referred cases, those with severe head injury (Abbreviated Injury Scale [AIS] ≥ 3) and fewer registered injury codes. In the mixed-effects model, younger age (OR up to 2.0), increasing head injury severity (OR: 1.16 per AIS point), less than 3 registered AIS codes (OR: 0.60), referral by a general practitioner (OR 1.68) or self-referral (OR 1.92), and admission year (OR: 1.08 per year) independently predicted early discharg […]

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Trauma team activation for pediatric patients in Denmark: a multicenter study of criteria, organization, and injury severity.

Nordestgaard CH, Gude MF, Leth SV, Raaber N

BACKGROUND: Pediatric trauma team activation (TTA) is intended to ensure timely management of severely injured children, yet both prehospital visitation practices and in-hospital TTA criteria vary across trauma systems. The aim of this study was to describe and compare pediatric TTA criteria, organizational models, and injury severity across all Danish level I trauma centers.

METHODS: We conducted a retrospective multicenter cohort study including all pediatric trauma patients (< 18 years) admitted via TTA at the four Danish level I trauma centers between 2014 and 2024. Center-specific prehospital and in-hospital TTA protocols were obtained through structured inquiries. Injury severity was assessed using Injury Severity Score (ISS) and Abbreviated Injury Scale (AIS). Overtriage was defined as TTA in patients with ISS < 15.

RESULTS: A total of 3,452 pediatric trauma patients were included. Two distinct TTA models (single-criterion and point-based) and four organizational structures were identified. Overall, 14% of patients had ISS ≥ 15, corresponding to high overtriage across all centers. Overtriage rates ranged from 81% to 95% and were highest at centers using point-based TTA models. Mortality was low and did not differ significantly between centers. Among the most severely injured patients (ISS ≥ 25), head and thoracic injuries predominated.

CONCLUSION: Pediatric trauma tri […]

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Does the operative technique (MIPO vs. ORIF) influence surgical outcomes and complications in proximal humerus fractures? A matched-pair analysis.

Barthel C, Rudolph F, Kraus M, Kalbas Y, Russek R, Pape HC, Allemann F

INTRODUCTION: Proximal humerus fractures account for approximately 4-6% of all fractures and are among the most common fractures in the elderly. Surgical treatment is increasingly applied for displaced and unstable fracture patterns. The two most commonly used techniques are open reduction and internal fixation (ORIF) via a deltopectoral approach and minimally invasive plate osteosynthesis (MIPO) using a deltoid-split or anterolateral approach. ORIF allows direct fracture visualization and anatomical reduction but is associated with more extensive surgical exposure. MIPO aims to preserve soft tissues and vascular supply through indirect reduction techniques. This study compares ORIF and MIPO in proximal humerus fractures with regard to surgical parameters, radiographic outcomes, and complications.

METHODS: A retrospective single-center cohort of 392 proximal humerus fractures was analyzed. Seventy-nine patients treated with MIPO were matched to 79 patients treated with ORIF using exact matching for fracture type (Mayo FJD classification) and propensity score matching for age and sex. Primary outcomes included operative time, reduction quality, and complication rates.

RESULTS: The MIPO group showed significantly shorter operative times (97.3 ± 30.8 min vs. 112.2 ± 34.0 min, $p = 0.004$). No significant differences were observed in reduction quality or overall complication rates […]

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